

Algebra

Important Formulas :-

- $(a + b)^2 = a^2 + b^2 + 2ab$
- $(a - b)^2 = a^2 + b^2 - 2ab$
- $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
- $(a + b)^2 - (a - b)^2 = 4ab$
- $(a + b)^2 - 4ab = (a - b)^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $a^4 - b^4 = (a^2 + b^2)(a + b)(a - b)$
- $a^2 + \frac{1}{a^2} = (a + \frac{1}{a})^2 - 2$
 $= (a - \frac{1}{a})^2 + 2$
- $a^2 + b^2 + c^2 - ab - bc - ca =$
 $\frac{1}{2}\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$
 If $a = b = c = k$ then,
 $a^2 + b^2 + c^2 = ab + bc + ca,$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$
- $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$
- $a^3 + \frac{1}{a^3} = (a + \frac{1}{a})^3 - 3(a + \frac{1}{a})$
- $a^3 - \frac{1}{a^3} = (a - \frac{1}{a})^3 + 3(a - \frac{1}{a})$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)$
 $(a^2 + b^2 + c^2 - ab - bc - ca)$
 Or, $\frac{1}{2}(a + b + c)\{(a - b)^2 + (b - c)^2 + (c - a)^2\}$ Or,
 $(a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$
 Or, $\frac{1}{2}(a + b + c)$
 $[3(a^2 + b^2 + c^2) - (a + b + c)^2]$
- If $(a + b + c) = 0$, then
 $a^3 + b^3 + c^3 = 3abc$
- If $(a - b - c) = 0$, then
 $a^3 - b^3 - c^3 = 3abc$
- If $(a + b - c) = 0$, then
 $a^3 + b^3 - c^3 = -3abc$
- If $a^3 + b^3 + c^3 = 3abc$,
 then either $a = b = c$ or $a + b + c = 0$
- $(a + b + c)^2 =$
 $a^2 + b^2 + c^2 + 2(ab + bc + ac)$
- $(a + b + c)^3 = a^3 + b^3 + c^3 +$
 $3(a + b)(b + c)(a + c)$

- $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)$
 $(a^2 - ab + b^2)$
- If $(x^2 + y^2 + z^2) = 0$,
 then $x = 0, y = 0, z = 0$
- If $x + \frac{1}{x} = a$, then $x^3 + \frac{1}{x^3} = a^3 - 3a$
- If $x - \frac{1}{x} = a$, then $x^3 - \frac{1}{x^3} = a^3 + 3a$
- If $x + \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 - 2$
- If $x - \frac{1}{x} = a$, then $x^2 + \frac{1}{x^2} = a^2 + 2$
- If $x + \frac{1}{x} = a$, then $x - \frac{1}{x} = \sqrt{a^2 - 4}$
- If $x - \frac{1}{x} = a$, then $x + \frac{1}{x} = \sqrt{a^2 + 4}$
- If $x + \frac{1}{x} = a$, then
 $x^2 - \frac{1}{x^2} = a\sqrt{a^2 - 4}$
- If $x - \frac{1}{x} = a$, then
 $x^2 - \frac{1}{x^2} = a\sqrt{a^2 + 4}$
- If $x + \frac{1}{x} = a$, then
 $x^4 + \frac{1}{x^4} = (a^2 - 2)^2 - 2$
- If $x - \frac{1}{x} = a$, then
 $x^4 + \frac{1}{x^4} = (a^2 + 2)^2 - 2$
- If $x + \frac{1}{x} = a$, then
 $x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$
- If $x - \frac{1}{x} = a$, then
 $x^5 - \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$
- If $x + \frac{1}{x} = \pm\sqrt{3}$, then $x^6 = -1$
- If $x + \frac{1}{x} = 1$, then $x^3 = -1$
- If $x + \frac{1}{x} = -1$, then $x^3 = 1$
- If $x + \frac{1}{x} = 2$, then $x = 1$, similarly if
 $x + \frac{1}{x} = -2$, then $x = -1$

Some Special Case :-

- If $a^2 - ab + b^2 = 0$ then $a^3 + b^3 = 0$
- If $\frac{1}{a} - \frac{1}{b} = \frac{1}{a - b}$ then $a^3 + b^3 = 0$
- If $\frac{1}{b} + \frac{1}{a} = \frac{1}{a + b}$ then $a^3 - b^3 = 0$
- If $\frac{a}{b} + \frac{b}{a} = -1$ then $a^3 - b^3 = 0$
- If $\frac{a}{b} + \frac{b}{a} = 1$ then $a^3 + b^3 = 0$

- If $a^2 + a + 1 = 0$ then $a^3 - 1 = 0$ or $a^3 = 1$
- If $ab(a + b) = 1$ then $\frac{1}{a^3 b^3} - a^3 - b^3 = 3$

Dividendo - Componendo rule:

$$\text{If } \frac{a}{b} = \frac{c}{d} \text{ then, } \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$$\text{If } \frac{a+b}{a-b} = \frac{a}{b} \text{ then } \frac{c}{d} = \frac{c+d}{c-d}$$

• If $(x - a)^2 + (y - b)^2 + (z - c)^2 = 0$
 then by putting $x = a, y = b, z = c$ in
 an expression we can find the value of
 the expression.

• If $x + y = a$, then xy will be
 maximum when $x = y$.

• If $xy = a$, then $(x + y)$ will be
 minimum when $x = y = \sqrt{a}$

Law of Indices :

$$(1) a^m \times a^n = a^{m+n} \quad (2) \frac{a^m}{a^n} = a^{m-n}$$

$$(3) (a^m)^n = a^{mn} \quad (4) \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(5) a^0 = 1 \quad (6) a^{\frac{1}{n}} = \sqrt[n]{a}$$

If a is negative number and n is an odd
 number then $(-a)^n = -a^n$

Laws of surds :

$$(1). \sqrt[n]{a} = a^{\frac{1}{n}}$$

$$(2). \sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$$

$$(3). \frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$$

$$(4). \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

$$(5). (\sqrt[n]{a})^m = \sqrt[n]{a^m}$$

Ex.1. If $3\sqrt{5}$ is given, find out the
 rationalizing factor.

Sol : $3\sqrt{5} \times \sqrt{5} = 3 \times 5 = 15$, so the
 rationalizing factor of $3\sqrt{5}$ is $\sqrt{5}$.

Rules for making a perfect square :

1. Middle term =
 $\pm \sqrt{\text{First term} \times \text{Third term}}$
2. First term = $\frac{(\text{Middle term})^2}{4 \times \text{Third term}}$
3. Third term = $\frac{(\text{Middle term})^2}{4 \times \text{First term}}$

Quadratic and cubic equation :

- For quadratic equation :
 $ax^2 + bx + c = 0$, then
 Sum of its zeros $(\alpha + \beta) = -\frac{b}{a}$
 Product of its zeros $(\alpha\beta) = \frac{c}{a}$
 where, α and β are roots of a



given quadratic equation.

• **For cubic equation :**

$$ax^3 + bx^2 + cx + d = 0, \text{ then}$$

$$\text{Sum of its zeros } (\alpha + \beta + \gamma) = -\frac{b}{a}$$

$$\text{Product of its zeros } (\alpha\beta\gamma) = -\frac{d}{a}$$

Sum of its zeros taken two at a

$$\text{time } (\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{c}{a}$$

where α , β and γ are roots of a given cubic equation.

Types of solution and their conditions :-

Types of solutions	Conditions
Unique solutions	$\frac{a1}{a2} \neq \frac{b1}{b2}$
No solutions	$\frac{a1}{a2} = \frac{b1}{b2} \neq \frac{c1}{c2}$
Infinite number of solutions	$\frac{a1}{a2} = \frac{b1}{b2} = \frac{c1}{c2}$

Variety Questions

Q.1. If the cubic equation $a^3 - 8a^2 + 10a - 4 = 0$ has the root x, y, z then find the value of $x + y + z$.

SSC CGL 18/09/2025 (Shift 1)

- (a) 5 (b) 6 (c) 7 (d) 8

Q.2. If $(4x + \frac{3}{x} - 4) = 15$, then what is the value of $(16x^2 + \frac{9}{x^2} + 20)$?

Graduate Level 29/08/2025 (Shift - 1)

- (a) 337 (b) 357 (c) 346 (d) 318

Q.3. If $X^Y = 144$, what will $(X - 1)^{(Y-2)} = ?$

Graduate Level 25/07/2025 (Shift - 3)

- (a) 0 (b) 5 (c) 3 (d) 1

Q.4. Which of the following is true for all positive integers x and y , where $x \neq y$?

Higher Secondary 30/07/2025 (Shift - 2)

- (a) $\sqrt{x} + \sqrt{y} = \sqrt{(x + y)}$
 (b) $\sqrt{x} - \sqrt{y} = \sqrt{(x - y)}$
 (c) $(\sqrt{x} + \sqrt{y})^2 = x + y + 2\sqrt{xy}$
 (d) $(x + \sqrt{y})^2 = x + y - 2\sqrt{xy}$

Q.5. If $x = \sqrt{6} + 2$ and $y = \sqrt{6} - 2$, then what is the value of $(\frac{x}{y} + \frac{y}{x})^2 - 3$?

- (a) 42 (b) 97 (c) 35 (d) 22

Q.6. 'x' varies directly with the cube of 'y' and inversely with the square of 'z'. If $x = \frac{1}{36}$ when $y = 2$ and $z = 3$, then what is the value of $800x$ when $y = 3$ and $z = 5$?

- (a) $\frac{9}{800}$ (b) 27 (c) $\frac{800}{9}$ (d) 9

Q.7. If $0.4x + 0.16y = 1.7$ and $0.3x + 0.12y$

$= 3.4$, then which of the following is correct?

- (a) The system has finitely many solutions but not unique.
 (b) The system has infinitely many solutions.
 (c) The system has no solution.
 (d) The system has unique solution.

Q.8. If the equations $4x + (k - 2)y + 3 = 0$ and $(k - 2)x + 9y - 5 = 0$ ($k > 0$) are parallel, then find the value of $k^2 + 6$.

- (a) 68 (b) 70 (c) 72 (d) 64

Q.9. If the polynomial $(2x^3 + ax^2 + 3x - 5)$ and $(x^3 + x^2 - 2x + a)$ leave the same remainder when divided by $(x - 2)$, find the remainder.

- (a) 4 (b) 3 (c) 2 (d) 5

Q.10. Simplify the following expression: $(x + y)(x - y)(x^2 + y^2)(x^4 + y^4)(x^8 + y^8)(x^{16} + y^{16})$

- (a) $x^{32} - y^{32}$ (b) $x^{128} - y^{128}$
 (c) $x^{64} - y^{64}$ (d) $x^{256} - y^{256}$

Q.11. If $x^4 + y^4 = x^2y^2$, then the value of $x^6 + y^6$ is:

- (a) 1 (b) 3 (c) 2 (d) 0

Q.12. If $ax + by = 1$ and $bx + ay = \frac{2ab}{a^2 + b^2}$, then the value of x (in terms of a and b) is:

- (a) $\frac{2b}{a^2 + b^2}$ (b) $\frac{a}{a^2 + b^2}$
 (c) $\frac{b}{a^2 + b^2}$ (d) $\frac{2a}{a^2 + b^2}$

Q.13. If $x = (7 + 3\sqrt{5})$ then find the value of $x^2 + \frac{1}{x^2}$

- (a) $\frac{580 + 315\sqrt{5}}{8}$ (b) $\frac{799 + 328\sqrt{5}}{8}$
 (c) $\frac{799 + 315\sqrt{5}}{12}$ (d) $\frac{799 + 315\sqrt{5}}{8}$

Q.14. If $x^2 - 11x + 1 = 0$ what is the value of $x^8 - 14159x^4 + 11$?

- (a) 9 (b) 10 (c) 12 (d) 11

Q.15. Simplify:

$$\frac{(1 - m)^3 + (m - n)^3 + (n - 1)^3}{12(1 - m)(m - n)(n - 1)} = ?$$

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $\frac{1}{4}$

Q.16. For what positive value of k , the equation $3x^2 + 2kx + 3 = 0$ has real and equal roots?

- (a) 12 (b) 1 (c) 3 (d) 9

Q.17. Simplify the following expression.

$$\frac{(a^2 + b^2 - c^2)^2 - (a^2 - b^2 + c^2)^2}{b^2 - c^2}$$

- (a) $3a^2$ (b) $4a^2$ (c) $5a^2$ (d) $2a^2$

Q.18. Find the value of $(a^3 + b^3 + c^3 - 3abc)$, where $a = 335$, $b = 215$ and $c = 180$.

- (a) 14472250 (b) 15421320
 (c) 14502230 (d) 15452630

Q.19. Simplify the expression :

$$\frac{(u - v)^3 + (v - w)^3 + (w - u)^3}{(u^2 - v^2)^3 + (v^2 - w^2)^3 + (w^2 - u^2)^3}$$

(a) $\frac{1}{(u + v)(v + w)(w + u)}$ (b) 1
 (c) $\frac{3}{(u + v)(v + w)(w + u)}$ (d) 0

Q.20. If $x^3 = 270 + y^3$ and $x = (6 + y)$, then What is the value of $(x + y)$?

(given that $x > 0$ and $y > 0$)

- (a) $2\sqrt{3}$ (b) $\sqrt{3}$ (c) $4\sqrt{3}$ (d) $4\sqrt{2}$

Q.21. If $p + q + r = 0$, then what is the simplified value of the expression

$$\left(\frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - pr} + \frac{r^2}{r^2 - pq} \right)$$

(a) 0 (b) 2 (c) 1 (d) -1

Q.22. If $(x + \frac{1}{x}) = 5\sqrt{2}$, and $x > 1$, what is

the value of $(x^6 - \frac{1}{x^6})$?

- (a) $22970\sqrt{23}$ (b) $23030\sqrt{23}$
 (c) $23060\sqrt{23}$ (d) $22960\sqrt{23}$

Q.23. The difference between the cubes of two given natural numbers is 6272, while the positive difference between the two given numbers is 8. What is the sum of the cubes of the two given numbers?

- (a) 9728 (b) 9684 (c) 8000 (d) 9600

Q.24. If $a = 101$, $b = 102$ and $c = 103$, then $a^2 + b^2 + c^2 - ab - bc - ca =$ _____.

- (a) 2 (b) 4 (c) 3 (d) 6

Q.25. If $(x + \frac{1}{x}) = 5$, and $x > 1$, what is

the value of $(x^8 - \frac{1}{x^8})$?

- (a) $60605\sqrt{21}$ (b) $60615\sqrt{21}$
 (c) $60705\sqrt{21}$ (d) $60725\sqrt{21}$

Q.26. If $X - \frac{1}{X} = -6$, What will be the value of $X^5 - \frac{1}{X^5}$?

- (a) -8898 (b) -8896
 (c) -8886 (d) -8892

Q.27. If $(x^2 + \frac{1}{x^2}) = 7$, and $0 < x < 1$, find

the value of $x^2 - \frac{1}{x^2}$.

- (a) $3\sqrt{5}$ (b) $4\sqrt{3}$ (c) $-4\sqrt{3}$ (d) $-3\sqrt{5}$



Q.28. If $x > 0$, and $x^4 + \frac{1}{x^4} = 254$, what is

the value of $x^5 + \frac{1}{x^5}$?

- (a) $717\sqrt{2}$ (b) $723\sqrt{2}$ (c) $720\sqrt{2}$ (d) $726\sqrt{2}$

Q.29. If $\frac{2x}{x+2} + \frac{1}{2x+4} - \frac{12-x}{2x^2+4x} = a + \frac{b}{x}$, where a and b are integers, then the

value of $ab =$ _____.

- (a) 3 (b) -6 (c) 6 (d) -3

Q.30. If $6^x = 3^y = 2^z$, then what is the value of $\frac{1}{y} + \frac{1}{z} - \frac{1}{x}$?

- (a) 1 (b) 0 (c) 3 (d) 6

Q.31. For what value of m will the system of equations $17x + my + 102 = 0$ and $23x + 299y + 138 = 0$ have an infinite number of solutions?

- (a) 221 (b) 223 (c) 220 (d) 219

Q.32. If $a + b + c = 5$, $a^3 + b^3 + c^3 = 85$ and $abc = 25$, then find the value of $a^2 + b^2 + c^2 - ab - bc - ca$.

- (a) 2 (b) 4 (c) 6 (d) 8

Q.33. If α, β are the roots of $6x^2 + 13x + 7 = 0$, then the equation whose roots are α^2, β^2 is:

- (a) $36x^2 - 87x + 49 = 0$
 (b) $36x^2 - 85x + 49 = 0$
 (c) $36x^2 - 85x - 49 = 0$
 (d) $36x^2 + 87x - 49 = 0$

Q.34. If $x = 222$, $y = 223$ and $z = 224$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

- (a) 2007 (b) 2004 (c) 2006 (d) 2005

Q.35. What is the value of

$$(K - \frac{1}{K})(K^2 + \frac{1}{K^2})(K^4 + \frac{1}{K^4})$$

$$(K^8 + \frac{1}{K^8})(K^{16} + \frac{1}{K^{16}})(K^{32} + \frac{1}{K^{32}})?$$

$$(a) \frac{K^{64} - \frac{1}{K^{64}}}{K + \frac{1}{K}} \quad (b) \frac{K^{32} - \frac{1}{K^{32}}}{K - \frac{1}{K}}$$

$$(c) \frac{K^{32} - \frac{1}{K^{32}}}{K + \frac{1}{K}} \quad (d) \frac{K^{32} + \frac{1}{K^{32}}}{K + \frac{1}{K}}$$

Q.36. Which of the following statements is correct?

I. The value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 2^2 - 1^2$ is 5050.

II. If $8x + \frac{8}{x} = -16$ and $x < 0$, then the value of $x^{197} + x^{-197}$ is 2.

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

Q.37. If $mx^m - nx^n = 0$ then what is the value of $\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n}$ in terms of x^n ?

$$(a) \frac{2mn}{x^n(n^2 + m^2)} \quad (b) \frac{2m}{x^n(m^2 - n^2)}$$

$$(c) \frac{2mn}{x^n(n^2 - m^2)} \quad (d) \frac{2mn}{x^n(m^2 - n^2)}$$

Q.38. $a^2 + b^2 = 25$, $x^2 + y^2 = 17$ and $ax + by = 8$ then find the value of $(ay - bx)$

- (a) 33 (b) 25 (c) 21 (d) 19

Q.39. If $2 = x + \frac{1}{1 + \frac{1}{5 + \frac{1}{2}}}$, then the value of x is equal to:

- (a) $\frac{14}{13}$ (b) 1 (c) $\frac{15}{13}$ (d) $\frac{13}{15}$

Q.40. If $x^3 + 2x^2 - ax - b$ is exactly divisible by $(x^2 - 1)$, then the values of a and b are:

- (a) $a = -1$, and $b = 2$ (b) $a = 1$ and $b = -2$
 (c) $a = 1$ and $b = 2$ (d) $a = 2$ and $b = 2$

Q.41. If $x = \sqrt[3]{5} + 2$, then the value of $x^3 - 6x^2 + 12x - 12$ is equal to:

- (a) 0 (b) 2 (c) 1 (d) -1

Q.42. $(ax + by)$ is a factor of:

- (a) $a^2x^2 + 2ab - b^2y^2$
 (b) $a^2x^2 + 2abxy + b^2y^2$
 (c) $a^2x^3 + 2abx + b^2y^2x$
 (d) $a^2x^2 + 2ab + b^2y^2$

Q.43. If 'a' is a natural number, then $(7a^2 + 7a)$ is always divisible by:

- (a) 7 and 14 both (b) 7 only
 (c) 14 only (d) 21 only

Q.44. The value of $(a^{\frac{2}{3}} + 2a^{\frac{1}{2}} + 3a^{\frac{1}{3}} + 2a^{\frac{1}{6}} + 1)(a^{\frac{1}{3}} - 2a^{\frac{1}{6}} + 1) - a^{\frac{1}{2}}(a^{\frac{1}{2}} - 2)$, when $a = 7$, is:

- (a) 7 (b) 0 (c) 1 (d) $\sqrt{7}$

Q.45. If $a + b + c + d = 2$, then the maximum value of $-(1+a)(1+b)(1+c)(1+d)$

- (a) $\frac{91}{9}$ (b) $\frac{81}{16}$ (c) $\frac{63}{22}$ (d) $\frac{54}{13}$

Q.46. The coefficient of x^2 in $(2x + y)^3$ is:

- (a) $12y^2$ (b) 12 (c) $12y$ (d) 8

Q.47. If $x^{2a} = y^{2b} = z^{2c} \neq 0$ and $x^2 = yz$, then the value of $\frac{ab+bc+ca}{bc}$ is:

- (a) 3 (b) $3ac$ (c) $3bc$ (d) $3ab$

Q.48. If $x = 5.51$, $y = 5.52$ and $z = 5.57$, then what is the value of $x^3 + y^3 + z^3 - 3xyz$?

- (a) 5.146 (b) 51.46
 (c) 0.05146 (d) 0.5146

Q.49.

The value of $\frac{1}{\sqrt{7} - 4\sqrt{3}}$ is closest to:

- (a) 4.1 (b) 4.2 (c) 1.2 (d) 3.7

Q.50. If $(2x + 3y + 4)(2x + 3y - 5)$ is equivalent to $(ax^2 + by^2 + 2hxy + 2gx + 2fy + c)$, then what is the value of $\{(g + f - c)/ab\}$

- (a) $\frac{35}{432}$ (b) $\frac{19}{108}$ (c) $\frac{19}{316}$ (d) $\frac{37}{216}$

Q.51. $(4x^3y - 6x^2y^2 + 4xy^3 - y^4)$ can be expressed as:

- (a) $(x - y)^4 - x^4$ (b) $(x + y)^4 - y^4$
 (c) $(x + y)^4 - x^4$ (d) $x^4 - (x - y)^4$

Q.52. Let $x = \sqrt[6]{27} - \sqrt{6\frac{3}{4}}$ and $y =$

$$\frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}, \text{ then the value of } x^2 + y^2 \text{ is:}$$

- (a) $\frac{223}{36}$ (b) $\frac{221}{36}$ (c) $\frac{221}{9}$ (d) $\frac{227}{9}$

Q.53. If $\sqrt{86} - 60\sqrt{2} = a - b\sqrt{2}$, then what will be the value of $\sqrt{a^2 + b^2}$, correct to one decimal place?

- (a) 8.4 (b) 8.2 (c) 7.8 (d) 7.2

Q.54. If $a = \sqrt{8} - \sqrt{7}$ and $a = \frac{1}{b}$, then

$$\frac{a^2 + b^2 - 3ab}{a^2 + ab + b^2} \text{ is equal to:}$$

- (a) $\frac{27}{31}$ (b) $\frac{27}{32}$ (c) $\frac{29}{33}$ (d) $\frac{29}{31}$

Q.55. If $(a + b + 4)\{ab + 4(a + b)\} - 4ab = 0$, $a \neq -4$, $b \neq -4$, then,

$$\left\{ \frac{1}{(a + b + 4)^{117}} - 2^{-234} \right\} \text{ is equal to:}$$

- (a) $\frac{1}{4^{117}}$ (b) $\frac{1}{2^{117}}$ (c) $-\frac{1}{2^{234}}$ (d) 0

Q.56. $(x + y)^{\frac{1}{3}} + (z + y)^{\frac{1}{3}} = -(x + z)^{\frac{1}{3}}$ then $(x^3 + y^3 + z^3)$ can be expressed as:

- (a) $\frac{1}{8}xyz$
 (b) $(x + y)(y + z)(z + x)$
 (c) $\frac{3}{8}(x + y)(y + z)(z + x)$
 (d) $3xyz$

Q.57. The value of the expression

$$\frac{1}{4}\{(a + \frac{1}{a})^2 - (a - \frac{1}{a})^2\} \text{ is:}$$

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) 1 (d) 4

Q.58. If $3\sqrt[4]{x} + 4\sqrt[4]{x} = 5\sqrt[4]{x}$, then the value



of x is :

- (a) 4 (b) 2 (c) 8 (d) 16

Q.59. If $x = 2 - \sqrt{3}$ then the value of $x^3 - x^{-3}$ is :

- (a)
- $-30\sqrt{3}$
- (b)
- $30\sqrt{3}$
- (c)
- $-30\sqrt{2}$
- (d)
- $30\sqrt{2}$

Q.60. If $x \neq -1, 2$ and 5 , then the simplified value of $\left\{\frac{2(x^3-8)}{x^2-x-2}\right\} \times$ $\frac{x^2+2x+1}{x^2-4x-5} \div \frac{x^2+2x+4}{3x-15}$ is equal to :

- (a)
- $\frac{1}{6}$
- (b) 6 (c)
- $\frac{3}{2}$
- (d)
- $\frac{2}{3}$

Q.61. If x is real, and $x^4 - 5x^2 - 1 = 0$, then the value of $(x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6} + 1)$ is :

- (a) 126 (b) 110 (c) 116 (d) 96

Q.62. If $x + y = 7$ and $xy = 10$, then the value of $\frac{1}{x^3} + \frac{1}{y^3}$ is :

- (a) 0.543 (b) 0.131 (c) 0.133 (d) 0.453

Q.63. If $(3x - 1)^3 + (4x - 3)^3 + (2x + 1)^3 = 3(3x - 1)(4x - 3)(2x + 1)$ and $x \neq \frac{1}{3}$ then $x = ?$

- (a)
- $\frac{1}{2}$
- (b) 2 (c)
- $\frac{1}{4}$
- (d) 1

Q.64. If $x + y + z = 0$, then the value of $(x^2 + y^2 + z^2) \div (z^2 - xy)$ is :

- (a) 1 (b) 2 (c) -2 (d) -1

Q.65. If $x^2 - 3x - 1 = 0$, then the value of $(x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$ is :

- (a)
- $\frac{3}{8}$
- (b) 8 (c) 1 (d) 3

Q.66. If $x = 2 - p$, then $x^3 + 6xp + p^3$, is equal to :

- (a) 12 (b) 6 (c) 8 (d) 4

Q.67. If $2x^2 + y^2 + 6x - 2xy + 9 = 0$, then the value of $4x^3 - y^3 + x^2y^2$ is :

- (a) 0 (b) 9 (c) -3 (d) -9

Q.68. If $ab + bc + ca = 8$ and $a^2 + b^2 + c^2 = 20$, then a possible value of $\frac{1}{2}(a + b +$ $c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$ is :

- (a) 72 (b) 56 (c) 84 (d) 80

Q.69. If $x + y = 1$ and $xy(xy - 2) = 12$, then the value of $x^4 + y^4$ is :

- (a) 19 (b) 25 (c) 20 (d) 23

Q.70. If $16x^2 + 9y^2 + 4z^2 = 24(x - y + z) - 61$, then the value of $(xy + 2z)$ is :

- (a) 1 (b) 2 (c) 3 (d) 5

Q.71. If $(5\sqrt{5}x^3 - 81\sqrt{3}y^3) \div \sqrt{5}x - 3\sqrt{3}y = Ax^2 + By^2 + Cxy$, then the value of $(6A + B - \sqrt{15}C)$ is :

- (a) 10 (b) 9 (c) 15 (d) 12

Q.72. If $x + y + z = 19, x^2 + y^2 + z^2 = 133$ and $xz = y^2$, then the difference between z and x is :

- (a) 5 (b) 3 (c) 6 (d) 4

Q.73. If $x^4 + x^{-4} = 194, x > 0$ then the value of $(x - 2)^2$ is :

- (a) 1 (b) 6 (c) 2 (d) 3

Q.74. If roots of $x^2 - 4x + a = 0$ are equal, then $a = ?$

- (a) -4 (b) 4 (c) 8 (d) -8

Examination wise Questions

SSC CGL 2025 Tier - 1

Q.75. If $x = \sqrt{3}$, determine the value of $x + \frac{1}{x}$.

SSC CGL 12/09/2025 (Shift 2)

- (a)
- $\frac{8\sqrt{3}}{3}$
- (b)
- $\frac{4\sqrt{3}}{7}$
- (c)
- $\frac{4\sqrt{3}}{3}$
- (d)
- $\frac{5\sqrt{3}}{3}$

Q.76. If $x = \sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}}}$, then determine the value of $x^2 - x - 7$?

SSC CGL 12/09/2025 (Shift 2)

- (a)
- $-2 + 3\sqrt{3}$
- (b)
- $2 + 3\sqrt{3}$
-
- (c)
- $-3 + 3\sqrt{3}$
- (d)
- $-5 + 3\sqrt{3}$

Q.77. If $x = \sqrt{7} - 3$, find $x^2 + 6\sqrt{7}$.

SSC CGL 13/09/2025 (Shift 1)

- (a) 16 (b) 17 (c) 19 (d) 13

Q.78. Given, $x + \frac{1}{x} = 6$, then determine the value of $x^2 + \frac{1}{x^2}$.

SSC CGL 13/09/2025 (Shift 2)

- (a) 35 (b) 34 (c) 36 (d) 33

Q.79. What is the value of $\frac{(0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04)}{(0.4 \times 0.4 \times 0.4 + 0.08 \times 0.08 \times 0.08)}$?

SSC CGL 14/09/2025 (Shift 1)

- (a) 0.037 (b) 0.05 (c) 0.125 (d) 0.025

Q.80. Given, $x + \frac{1}{x} = 5$ find the value of $x^3 + \frac{1}{x^3}$

SSC CGL 14/09/2025 (Shift 1)

- (a) 110 (b) 120 (c) 100 (d) 130

Q.81. Given, $b = \sqrt{7}$, what is $(b + 1)^2 + (b -$ $1)^2$?

SSC CGL 14/09/2025 (Shift 3)

- (a) 13 (b) 12 (c) 14 (d) 16

Q.82. Given, $x - \frac{1}{x} = 4$, find the value of $x^4 + \frac{1}{x^4}$.

SSC CGL 15/09/2025 (Shift 3)

- (a) 322 (b) 318 (c) 320 (d) 325

Q.83. What is the value of: $(\sqrt{9} + \sqrt{4})^2 - (\sqrt{8} + \sqrt{2})^2$

SSC CGL 16/09/2025 (Shift 2)

- (a) 4 (b) 5 (c) 6 (d) 7

Q.84. If $a + b = 7$ and $ab = 12$,Find: $(a^3 + b^3)^2 - 8a^2b^2(a + b)^2$.

SSC CGL 17/09/2025 (Shift 1)

- (a) -48167 (b) 68752
-
- (c) -38761 (d) 38767

Q.85. If $a = 0.04, b = 0.06, c = -0.10$, and $a + b + c = 0$, find $(a^3 + b^3 + c^3) \div (3abc)$

SSC CGL 17/09/2025 (Shift 1)

- (a) -1 (b) 0 (c) 1 (d) -2

Q.86. What is the value of $(0.8 \times 0.8 \times 0.8 + 0.04 \times 0.04 \times 0.04) \div (1.6 \times 1.6 \times 1.6 + 0.08 \times 0.08 \times 0.08)$?

SSC CGL 17/09/2025 (Shift 2)

- (a)
- $\frac{1}{9}$
- (b)
- $\frac{1}{7}$
- (c)
- $\frac{1}{8}$
- (d)
- $\frac{1}{6}$

Q.87. Given $a + b + c = 0$, and $a^3 + b^3 + c^3 = 3abc$, evaluate $(a - b)^3 + (b - c)^3 + (c - a)^3$.

SSC CGL 17/09/2025 (Shift 3)

- (a)
- $27abc$
- (b)
- $9(a - b)(b - c)(c - a)$
-
- (c) 0 (d)
- $3(a - b)(b - c)(c - a)$

Q.88. Given, $\sqrt{x + 5} + \sqrt{x - 4} = 9$, then find the value of x .

SSC CGL 18/09/2025 (Shift 1)

- (a) 18 (b) 20 (c) 16 (d) 12

Q.89. Given, $y + \frac{1}{y} = 3$, Find the value of: $y^5 + \frac{1}{y^5} - 3(y^3 + \frac{1}{y^3}) + 4(y + \frac{1}{y})$

SSC CGL 18/09/2025 (Shift 2)

- (a) 81 (b) 91 (c) 88 (d) 96

Q.90. Given, $\sqrt{m} + \frac{1}{\sqrt{m}} = 7$, then what is $m + \frac{1}{m}$?

SSC CGL 18/09/2025 (Shift 2)

- (a) 55 (b) 29 (c) 47 (d) 36

Q.91. If $(x + \frac{2}{x} = 5)$, find the value of $x^3 + \frac{8}{x^3}$.

SSC CGL 18/09/2025 (Shift 3)

- (a) 95 (b) 105 (c) 115 (d) 100



Q.92. Given, $\frac{\sqrt{5+x} + \sqrt{5-x}}{\sqrt{5+x} - \sqrt{5-x}} = 2$,

then what is the value of x ?

SSC CGL 18/09/2025 (Shift 3)

- (a) 7 (b) 4 (c) 6 (d) 2

Q.93. If $x + \frac{1}{x} = -2$, then compute:

$$x^4 + \frac{1}{x^4} + 3x^2 + \frac{3}{x^2}.$$

SSC CGL 19/09/2025 (Shift 1)

- (a) -3 (b) 3 (c) 8 (d) -4

Q.94. If $\sqrt{a} + \sqrt{b} = 7$ and $\sqrt{a} - \sqrt{b} = 1$, what is the value of a and b ?

SSC CGL 19/09/2025 (Shift 2)

- (a) a = 4, b = 3 (b) a = 9, b = 16
(c) a = 16, b = 9 (d) a = 3, b = 4

Q.95. If $a^3 + b^3 = 152$ and $ab = 15$, find the value of a + b.

SSC CGL 19/09/2025 (Shift 2)

- (a) 9 (b) 6 (c) 10 (d) 8

Q.96. If $x + y + z = 0$ and $x = 0.5, y = 0.4, z = -0.9$, then what is $(x^3 + y^3 + z^3) \div (3xyz)$?

SSC CGL 19/09/2025 (Shift 2)

- (a) -3 (b) 1 (c) 0 (d) -2

Q.97. Given: $m + \frac{1}{m} = -3$ then determine

$$\text{the value of } m^5 + \frac{1}{m^5} - 2(m^3 + \frac{1}{m^3}) +$$

$$4(m + \frac{1}{m}).$$

SSC CGL 19/09/2025 (Shift 3)

- (a) -99 (b) -98 (c) -95 (d) 99

Q.98. If $x = 4 + \sqrt{15}$, find $x - \frac{1}{x}$.

SSC CGL 20/09/2025 (Shift 1)

- (a) $\sqrt{15}$ (b) $2\sqrt{15}$ (c) $\frac{3}{\sqrt{15}}$ (d) $\frac{5}{\sqrt{15}}$

Q.99. If $x = \sqrt{7} + 3$, find the value of $x^2 - 4\sqrt{7}$

SSC CGL 20/09/2025 (Shift 3)

- (a) $16 + 2\sqrt{7}$ (b) $9 + 2\sqrt{7}$
(c) $8 + 2\sqrt{7}$ (d) $10 + 2\sqrt{7}$

Q.100. If $x = 3 + \sqrt{2}$, find the value of $x^2 - 6x + 7$.

SSC CGL 21/09/2025 (Shift 1)

- (a) $3 + 2\sqrt{2}$ (b) $4 + \sqrt{2}$ (c) 0 (d) 2

Q.101. Given, $x + \frac{1}{x} = 9$, then determine

$$\text{the value of } \frac{2x}{x^2 - 1}.$$

SSC CGL 22/09/2025 (Shift 1)

- (a) $\frac{2\sqrt{77}}{77}$ (b) $\frac{2\sqrt{71}}{77}$ (c) $\frac{7\sqrt{21}}{71}$ (d) $\frac{9\sqrt{66}}{77}$

Q.102. Given, $y = \sqrt{5}$, what is $(y+1)^2 + (y-1)^2$?

SSC CGL 26/09/2025 (Shift 3)

- (a) 12 (b) 17 (c) 9 (d) 16

Q.103. If $a^3 + b^3 = 407$ and $a + b = 11$, find ab.

SSC CGL 14/10/2025 (Shift 2)

- (a) 24 (b) 26 (c) 28 (d) 22

Q.104. Given $x + y + z = 0$, and $x^3 + y^3 + z^3 = 3xyz$, evaluate $(x-y)^3 + (y-z)^3 + (z-x)^3$.

SSC CGL 14/10/2025 (Shift 3)

- (a) 0 (b) $9(x-y)(y-z)(z-x)$
(c) $27xyz$ (d) $3(x-y)(y-z)(z-x)$

Q.105. If $a^3 + b^3 = 189$ and $ab = 20$, find the value of a + b.

SSC CGL 14/10/2025 (Shift 3)

- (a) 9 (b) 6 (c) 7 (d) 8

SSC Selection Post (XIII)

Q.106. If $\sqrt{y} = 13$, what is the value of $y + \sqrt{y}$?

Higher Secondary 24/07/2025 (Shift - 2)

- (a) 181 (b) 182 (c) 172 (d) 183

Q.107. If, $x + y = 9$ and $xy = 20$ then the value of $x^2 + y^2$ is:

Higher Secondary 24/07/2025 (Shift - 2)

- (a) 41 (b) 39 (c) 49 (d) 11

Q.108. The difference between two numbers is 12 & their product is 45. What is the square of their sum?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) 324 (b) 376 (c) 555 (d) 225

Q.109. If $\sqrt{(x^2 + 12x + 36)} + \sqrt{(x^2 - 8x + 16)} = 10$, find the value of x.

Higher Secondary 26/07/2025 (Shift - 2)

- (a) 1 (b) 3 (c) 4 (d) 5

Q.110. If $\frac{(2-\sqrt{3})}{(2+\sqrt{3})} = p + q\sqrt{3}$, then the values of p and q are respectively:

Higher Secondary 26/07/2025 (Shift - 2)

- (a) p = 7, q = 4 (b) p = 3, q = 8
(c) p = 7, q = -4 (d) p = 8, q = -3

Q.111. If $a - b = 6$ and $a^2 + b^2 = 70$, then the value of ab is:

Higher Secondary 26/07/2025 (Shift - 2)

- (a) 17 (b) 24 (c) 37 (d) 28

Q.112. If $a^2 + b^2 + c^2 = 30$ and $ab + bc + ca = 20$, then the value of $(a + b + c)^2$ is:

Higher Secondary 26/07/2025 (Shift - 2)

- (a) 74 (b) 50 (c) 70 (d) 60

Q.113. If $x - \frac{1}{x} = 5$, then the value of $x^2 + \frac{1}{x^2}$ is:

$$\frac{1}{x^2} \text{ is:}$$

Higher Secondary 28/07/2025 (Shift - 2)

- (a) 27 (b) 20 (c) 10 (d) 5

Q.114. The expression $(a + b + c)^2 - (a^2 + b^2 + c^2)$ is equal to:

Higher Secondary 28/07/2025 (Shift - 2)

- (a) $2(ab + bc + ca)$ (b) 0
(c) 1 (d) $(a + b + c)$

Q.115. If $x + y + z = 0$, then $x^3 + y^3 + z^3$ is equal to:

Higher Secondary 29/07/2025 (Shift - 2)

- (a) $x^2 + y^2 + z^2$ (b) $3xyz$ (c) 0 (d) 1

Q.116. If $y = \frac{(\sqrt{7} + \sqrt{5})}{(\sqrt{7} - \sqrt{5})} + \frac{(\sqrt{7} - \sqrt{5})}{(\sqrt{7} + \sqrt{5})}$,

then the value of y is:

Higher Secondary 30/07/2025 (Shift - 2)

- (a) 5 (b) 6 (c) 12 (d) 10

Q.117. If the sum of a number and its reciprocal is 18. What is the sum of the square of the number and the square of its reciprocal?

Higher Secondary 31/07/2025 (Shift - 2)

- (a) $161 \times 2 + 0$ (b) $161 \times 2 + 1$
(c) $161 \times 2 + 2$ (d) $161 \times 2 + 3$

Q.118. Given that $(\sqrt{a} + \frac{1}{\sqrt{a}})^2 = 15$, find

$$\text{the value of } a^3 + \frac{1}{a^3}.$$

Higher Secondary 02/08/2025 (Shift - 2)

- (a) 2197 (b) 2358 (c) 2000 (d) 2158

Q.119. Given: $\frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 4$ and $\frac{p}{a} + \frac{q}{b} + \frac{r}{c} = 0$, where a, b, c and p, q, r are

$$\text{non-zero, then the value of } \frac{a^2}{p^2} + \frac{b^2}{q^2} + \frac{c^2}{r^2} ?$$

Graduate Level 24/07/2025 (Shift - 3)

- (a) 8 (b) 12 (c) 16 (d) 64

Q.120. If $x + \frac{1}{x+3} = 3$, determine the

value of $(x+3)^3 + \frac{1}{(x+3)^3}$

Graduate Level 25/07/2025 (Shift - 3)

- (a) 198 (b) 190 (c) 234 (d) 300

Q.121. If $x + \frac{1}{x+2} = 4$, determine the

value of $(x+2)^3 + \frac{1}{(x+2)^3}$

Graduate Level 26/07/2025 (Shift - 3)

- (a) 198 (b) 186 (c) 189 (d) 155

Q.122. If $A^b = 144$, what will $(A-1)^{(b-2)}$ = ?

Graduate Level 26/07/2025 (Shift - 3)

- (a) 2 (b) 11 (c) 0 (d) 1

Q.123. If $a^2 + b^2 + c^2 + 48 = 8(a + b + c)$, then the value of $\sqrt[3]{a^3 - b^3 + c^3}$?



Graduate Level 28/07/2025 (Shift - 3)

- (a) 6 (b) 4 (c) 3 (d) 2

Q.124. If $4^n = 1024$, what will be the value of $(2)^{n-2}$?

Graduate Level 28/07/2025 (Shift - 3)

- (a) 0 (b) 8 (c) 6 (d) 5

Q.125. If $k^2 + \frac{1}{k^2} = 12$, find the value of

$$k^4 + \frac{1}{k^4}$$

Graduate Level 29/07/2025 (Shift - 3)

- (a) 146 (b) 142 (c) 144 (d) 156

Q.126. If $\sqrt{(1-x^2)(1-y^2)} = \frac{4}{5}$ and $y = \frac{3}{5}$, then evaluate:

$$[(5x^2 + 5y^2 + 10xy) + (5x^2 + 5y^2 - 10xy)]$$

Graduate Level 31/07/2025 (Shift - 3)

- (a)
- $\frac{9}{5}$
- (b)
- $\frac{14}{5}$
- (c)
- $\frac{18}{5}$
- (d)
- $\frac{12}{5}$

Q.127. Determine the value of an expression $(\frac{3}{a} + 3a)(\frac{9}{a^2} - 9 + 9a^2)$ if $a = 2$.

Graduate Level 02/08/2025 (Shift - 3)

- (a)
- $\frac{1755}{8}$
- (b)
- $\frac{82650}{27}$
- (c)
- $\frac{9800}{27}$
- (d)
- $\frac{17550}{8}$

Q.128. If $3x + 4y = 70$ and $x, y \in \mathbb{N}$, then what is the value of x for which y is minimum?

Graduate Level 29/08/2025 (Shift - 1)

- (a) 18 (b) 22 (c) 20 (d) 24

Q.129. If $y + \frac{1}{y} = 2$, determine the value of $y^6 + \frac{1}{y^6}$.

Graduate Level 29/08/2025 (Shift - 1)

- (a) 1 (b) 2 (c) 3 (d) 0

eduquity-based pattern (ebp) Questions

Q.130. Simplify $\frac{x^3 + 15x^2 + 75x + 125}{x^2 - 25}(x - 5)$

- (a)
- $x^2 + 25$
- (b)
- $x^2 + 5x + 25$
-
- (c)
- $x^2 + 10x + 25$
- (d)
- $x^2 - 25$

Q.131. Which of the following can be the value of k , if $\frac{(88 \div 8 \times k - 3 \times 3)}{(6^2 - 7 \times 5 + k^2)} = 1$?

- (a) 1, 10 (b) 4, 7 (c) 3, 10 (d) 2, 7

Q.132. Let $a + b = 2c$, then which of the following expressions is true?

- (a)
- $a^2 + 2ac = b^2 + 2bc$
-
- (b)
- $a^2 - 2bc = b^2 - 2ac$
-
- (c)
- $a^2 + 2bc = b^2 + 2ac$
-
- (d)
- $a^2 - ac = b^2 - bc$

Q.133. If x, y and z are positive numbers and $x + y + z = 1$, then the least value of

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$
 is:

- (a) 7 (b) 5 (c) 9 (d) 11

Q.134. If $a + \frac{1}{a} = 12$, then find the value

$$\text{of } a^2 + \frac{1}{a^2}.$$

- (a) 144 (b) 146 (c) 142 (d) 140

Q.135. If the sum of three numbers is 18 and the sum of their squares is 36, then find the difference between the sum of their cubes and three times of their product.

- (a) 1449 (b) -1944 (c) -1494 (d) 4149

Q.136. Find the values of x, y and z , so as to satisfy the equations given below:

$$x + y + z = 12; x + y - z = 6; x - y + z = 4$$

- (a)
- $x = 5, y = 4, z = -3$
-
- (b)
- $x = 5, y = -4, z = 3$
-
- (c)
- $x = 5, y = 4, z = 3$
-
- (d)
- $x = 5, y = -4, z = -3$

Q.137. If $\sqrt{x} + \frac{1}{\sqrt{x}} = 7$, then the value of

$$x + \frac{1}{x}$$
 is equal to:

- (a) 47 (b) 49 (c) 51 (d) 45

Q.138. If $x^2 + y^2 = 280$ and $xy = 120$, then find the value of $\frac{x-y}{x+y}$.

- (a)
- $\frac{1}{\sqrt{13}}$
- (b)
- $\sqrt{15}$
- (c)
- $\sqrt{13}$
- (d)
- $\frac{1}{\sqrt{15}}$

Q.139. For what value of k will the lines $2x + 7ky - 8 = 0$ and $x + y - 9 = 0$ have no solution?

- (a)
- $\frac{2}{7}$
- (b)
- $\frac{9}{7}$
- (c)
- $\frac{8}{9}$
- (d)
- $\frac{1}{2}$

Q.140. If $x + \frac{1}{x} = 2\sqrt{10}$, where $x > 1$,then the value of $(x^3 - \frac{1}{x^3})$ is:

- (a) 221 (b) 234 (c) 198 (d) 216

Q.141. Solve the given set of three equations to get the value of the unknowns:

$$x + 2z = 3$$

$$x + 2y + 3z = 5$$

$$3x - 5z = -13$$

- (a)
- $x = 0, y = -1, z = 2$
- (b)
- $x = -1, y = 0, z = 2$
-
- (c)
- $x = -1, y = 2, z = 0$
- (d)
- $x = 2, y = 0, z = -1$

Q.142. If $x + \frac{1}{9x} = 3$, then the value of

$$9x^2 + \frac{1}{9x^2}$$
 is:

- (a) 87 (b) 81 (c) 77 (d) 79

Q.143. Let $t = \frac{2}{5}$, then the value of theexpression $t^3 + (\frac{3}{5})^3 + \frac{9}{5}t$ is:

- (a)
- $\frac{1}{5}$
- (b) 1 (c) 2 (d)
- $\frac{1}{2}$

Q.144. If $x + \frac{1}{x} = 15$, then the value of

$$\frac{7x^2 - 9x + 7}{x^2 - x + 1}$$
 is:

- (a)
- $-\frac{48}{7}$
- (b)
- $-\frac{22}{7}$
- (c)
- $\frac{48}{7}$
- (d)
- $\frac{22}{7}$

Q.145. For the following equations, what are the values of a and b to have infinitely many solutions?

$$ax + by = 2$$

$$3x - (5 - 2a)y = 6$$

- (a)
- $a = -1, b = -1$
- (b)
- $a = -1, b = 1$
-
- (c)
- $a = 1, b = -1$
- (d)
- $a = 1, b = 1$

Q.146. If $m + n = 24$, then $(m - 16)^3 + (n - 8)^3$ is _____.

- (a) 0 (b) 40 (c) 576 (d) 320

Q.147. Which of the following is the sum of the values of a and b if the equations $2x + y = a$, $8x + by = 12$ have infinite solutions?

- (a) 16 (b) 9 (c) 7 (d) 18

Q.148. If $a^2 + b^2 + c^2 = 2(a + c - 1)$, then the value of $a^3 + b^3 + c^3$ is?

- (a) 0 (b) 2 (c) 4 (d) 1

Q.149. Find the values of x, y and z , so as to satisfy the equations given below:

$$5x - 3y + 7z = 22; 3x - 5y - 2z = -46; 2x - 2y + 5z = 24$$

- (a)
- $x = -5, y = 3, z = -8$
-
- (b)
- $x = 5, y = -8, z = 3$
-
- (c)
- $x = -5, y = 3, z = 8$
-
- (d)
- $x = -5, y = -3, z = 8$

Q.150. The solution of the following equations is:

$$5x + 4y - 8z = 1$$

$$7x - 9y + z = -1$$

$$2x + 3y - 4z = 1$$

- (a)
- $x = -1, y = 1, z = 1$
- (b)
- $x = 1, y = 1, z = 1$
-
- (c)
- $x = 1, y = -1, z = 1$
- (d)
- $x = 1, y = 1, z = -1$

Q.151. Let $a + b = 1$, then the value of $(\frac{1}{a^2} + \frac{1}{b^2} + \frac{2}{ab})$ is:

- (a)
- $\frac{1}{a^2 b^2}$
- (b)
- $\frac{1}{ab}$
- (c)
- $\frac{1}{a^2 b}$
- (d)
- $\frac{1}{ab^2}$

Q.152. For which of the following values of a and b do the given equations, have no solution?

$$x - ay = 2 - a$$

$$(1 - a)x + 6y = a + b$$

- (a)
- $a = -3, b \neq 1$
- (b)
- $a = 3, b \neq -1$
-
- (c)
- $a = -3, b \neq -1$
- (d)
- $a = 3, b \neq 1$

Q.153. If $m = a \sec A$ and $y = b \tan A$ then

find the value of $b^2m^2 - a^2y^2 + \frac{a^2y^2}{b^2m^2} + \cos^2 A$.

- (a) a^2b^2 (b) $1 - a^2b^2$
(c) $a^2b^2 + 2$ (d) $a^2b^2 + 1$

Q.154. For what value of k , the system of equations $kx - 15y + 7 = 0$ and $7x - 21y - 19 = 0$ has NO solution?

- (a) 5 (b) 12 (c) 25 (d) 6

Q.155. If $X > 1$ and $x^2 + \frac{1}{x^2} = 83$, then $x^3 - \frac{1}{x^3}$ is:

- (a) 884 (b) 876 (c) 754 (d) 756

Q.156. If x is an integer such that $x + \frac{1}{x} = \frac{65}{8}$, then find the value of $x - \frac{1}{x}$.

- (a) $\frac{63}{6}$ (b) $\frac{63}{8}$ (c) $\frac{36}{8}$ (d) $\frac{32}{6}$

Q.157. The sum of five consecutive numbers is 80. Find the largest number.

- (a) 18 (b) 15 (c) 19 (d) 14

Q.158. If $x = 7 - 4\sqrt{3}$, then find the value of $(x^2 + \frac{1}{x^2})$.

- (a) 188 (b) 194 (c) 186 (d) 196

Q.159. If the system of the following equations has the value of the variables as three consecutive integers, then the value of a is ____.

$$x - y + z = 2a \quad x + 4y - 2z = 3(4 - a)$$

$$2x - 3y + 4z = 6 - 2a$$

- (a) 2 (b) 4 (c) 1 (d) 3

Q.160. If $x + \frac{1}{x} = 2$, then find the value of $(x^{2723} + \frac{1}{x^{3356}})$.

- (a) 1 (b) 2 (c) 0 (d) -1

Q.161. If $3a + 2b = 27$ and $27a^3 + 8b^3 = 1458$, then find $2ab$.

- (a) 72 (b) 70 (c) 77 (d) 75

Q.162. If $(a + \frac{1}{a}) = 7\sqrt{3}$ then what is the value of $(a^6 + a^6)$?

- (a) 3048190 (b) 3048542
(c) 3048132 (d) 3048625

Q.163. If $(x + \frac{1}{x}) = 7$, $x > 0$. The positive value of $(x - \frac{1}{x})$ is:

- (a) $5\sqrt{4}$ (b) $5\sqrt{5}$ (c) $3\sqrt{5}$ (d) $4\sqrt{5}$

Q.164. For the equations $ax + (a^2 + 1)y = 4$ and $4x + ay = a^2$, which of the following statements is TRUE?

(a) If $a = 6$, then $x = \frac{325}{28}$, $y = -\frac{25}{28}$ are the solutions.

(b) If $a = 6$, then $x = 48$, $y = 4$ are the solutions.

(c) If $a = -12$, then $x = \frac{325}{28}$, $y = -\frac{25}{28}$ are the solutions.

(d) If $a = -12$, then $x = 48$, $y = 4$ are the solutions

Q.165. If $a + b + c = 0$, then the value of $(a^2 + b^2 + 2ab)$ is equal to:

- (a) c^2 (b) $-c^2$ (c) c (d) $-c$

Q.166. Let $3t - \frac{1}{3t} = 3$, then which of the following expressions has the value equal to 12?

(a) $9t^2 + \frac{1}{9t^2} + 2$ (b) $9t^2 + \frac{1}{9t^2} - 2$

(c) $9t^2 + \frac{1}{9t^2} + 1$ (d) $9t^2 + \frac{1}{9t^2}$

Q.167. If $3x + 2y = 10$ and $2xy = 7$, then find the value of $3x - 2y$ (if $3x - 2y > 0$).

- (a) 4 (b) 10 (c) 8 (d) 6

Q.168. What conclusion can be drawn about the solution of the following system of linear equations in two variables:

$$3x + 2y = 7, \quad 2x + 3y = 7$$

- (a) No solution
(b) Unique solution
(c) Infinite solutions
(d) More than two solution

Q.169. Simplify the following expression.

$$(3x + 17 - 2y)(3x - 17 - 2y)$$

$$(a) 9x^2 - 12xy + 4y^2 + 289$$

$$(b) 9x^2 + 12xy + 4y^2 - 289$$

$$(c) 9x^2 - 12xy + 4y^2 - 289$$

$$(d) 9x^2 - 6xy + 4y^2 - 289$$

Q.170. If $a + b = c$, then the expression $a^3 + b^3 - c^3 + 3abc$ will be equal to:

- (a) 0 (b) 3 (c) 2 (d) 1

Q.171. If $a^3 + \frac{1}{a^3} = 2$ ($a > 0$), then the value of $a + \frac{1}{a}$ is:

- (a) 1 (b) 3 (c) 2 (d) 4

Q.172. If $x^2 + 7x + 8 = 0$, then find the value of $\frac{4x}{x^2 - 5x + 8}$

- (a) $\frac{1}{2}$ (b) $-\frac{1}{4}$ (c) $-\frac{1}{3}$ (d) $\frac{1}{6}$

Q.173. Find the values of 'a' and 'b' for which the system of equations $3x + y = 3$ and $(a - b)x + (a + b)y = 3a + b - 3$ has infinite solutions.

- (a) $a = 3, b = -\frac{2}{3}$ (b) $a = -\frac{3}{2}, b = 2$

(c) $a = 3, b = -\frac{3}{2}$ (d) $a = 2, b = -\frac{3}{2}$

Q.174. The value of $(x + y)^2 + (x - y)^2$ is equal to:

$$(a) 2(x + y)^2 \quad (b) 2(x^2 + y^2)$$

$$(c) (2x + 2y)^2 \quad (d) 4xy$$

Q.175. Find the value of 'k' for which the system of equations $4x + 6y = 7$ and $6x + (k + 4)y - 21 = 0$ has a unique solution.

- (a) $k = 5$ (b) $k \neq 5$ (c) $k = 7$ (d) $k \neq 7$

Q.176. If $a^2 + b^2 = 148$ and $ab = 54$, then find the value of $\frac{a+b}{a-b}$.

- (a) $\frac{8}{\sqrt{10}}$ (b) $\frac{2}{\sqrt{10}}$ (c) $8\sqrt{7}$ (d) $5\sqrt{3}$

Q.177. Given a linear equation in two variables: $5x + 7y - 8 = 0$, which of the following linear equations, along with the given equation, forms a system of linear equations having no solution?

- (a) $5x + 7y - 16 = 0$ (b) $7x + 5y - 8 = 0$
(c) $5x - 7y - 8 = 0$ (d) $10x + 14y - 16 = 0$

Q.178. If $2a - b = 3$ and $8a^3 - b^3 = 999$, then find the value of $4a^2 - b^2$?

- (a) 63 (b) 65 (c) 61 (d) 67

Q.179. There are 160 multiple choice questions in a test. 4 marks are allotted for a correct answer and 1 mark is deducted for a wrong answer or un-attempted question from the total score of correct answers. If a candidate scored 400 marks in the test, how many questions did he answer correctly?

- (a) 110 (b) 112 (c) 115 (d) 120

Q.180. Find the remainder, when we divide $2x^3 - 3x^2 + 6x - 4$ by $2x - 3$.

- (a) 6 (b) -6 (c) 5 (d) -5

Q.181. If $A = \frac{9}{16} \div \frac{25}{36}$, $B = 9 \div [(16 \div 25) \div 36]$, $C = [9 \div (16 \div 25)] \div 36$, $D = (9 \div 16) \div (25 \div 36)$ then which of the following is true?

- (a) B and D are equal.
(b) A and B are equal.
(c) C and D are equal.
(d) A and D are equal.

Q.182. If $A = 60 \left[\frac{6}{25} \div \left(\frac{16}{15} \text{ of } \frac{3}{8} \right) + 5 - (1 - 2\frac{1}{12}) \right]$ then the sum of the digits of A is:

- (a) 3 (b) 7 (c) 5 (d) 6

Q.183. If $640 \div 8 + 930 \div 15 - k + 32 \times 5 = 1104 \div 16 \times 148 \div 37$, then the value of k is:

- (a) 276 (b) 26 (c) 35 (d) 302



Q.184. For what value of k ($k > 0$) will the lines $5x + ky - 2 = 0$ and $(k + 1)x + 4y + 7 = 0$ be parallel?

- (a) 2 (b) 5 (c) 3 (d) 4

Q.185. If $\frac{15}{8} + \left\{ \frac{4}{5} \text{ of } \frac{5}{12} \div \left(1\frac{1}{15} \right) \right\} \times 1\frac{7}{12}$ is expressed in the form of $\frac{a}{b}$

Where a and b are integers and relatively prime to each other, the value $4b - a$ is :

- (a) 164 (b) 313 (c) 189 (d) 225

Q.186. If the sum of two numbers x and y is 12 and their product is 27, then find the sum of their cubes.

- (a) 657 (b) 576 (c) 765 (d) 756

Q.187. The sum of two numbers x and y (where x is greater than y) is equal to four times of their difference. Find the value of $\frac{5xy}{3(x^2 - y^2)}$

- (a) $\frac{25}{16}$ (b) 16 (c) $2\frac{3}{16}$ (d) 25

Q.188. If $x + y + z = 1$, $xy + yz + zx = -1$ and $xyz = -1$, then $(x^3 + y^3 + z^3)$ is equal to _____.

- (a) 0 (b) 1 (c) 2 (d) 3

Q.189. If $\alpha + \beta + \gamma = 0$, then $\frac{3\beta^2 + \alpha^2 + \gamma^2}{2\beta^2 - \alpha\gamma}$ = ?

- (a) $\frac{1}{2}$ (b) 1 (c) 2 (d) $\frac{1}{4}$

Q.190. If $x = 3y + 4$, then what is the value of $(x^3 - 27y^3 - 36xy)$?

- (a) 27 (b) 64 (c) 1 (d) 8

Q.191. If $x + \frac{1}{x} = 2$ then find the value of $(x^4 + \frac{4}{(x^4)})(x^3 - \frac{4}{(x^3)})$.

- (a) -15 (b) 15 (c) -10 (d) 10

Q.192. If $16x^4 + \frac{1}{16x^4} = 14159$, then find the value of $2x + \frac{1}{2x}$

- (a) 9 (b) 11 (c) 12 (d) 15

Q.193. If $a + 2b + 3c = 0$, then what is the value of $(a^3 + 8b^3 + 27c^3)$?

- (a) 18 abc (b) 12 abc (c) 0 (d) 6 abc

Q.194. If $x = 110$, $y = 111$, $z = 112$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.

- (a) 999 (b) 991 (c) 997 (d) 995

Q.195. If $(a - 18)^2 + (b - 12)^2 + (c - 6)^2 = 0$, then find the value of $(a + b + c)^{\frac{1}{2}}$.

- (a) ± 3 (b) ± 6 (c) ± 4 (d) ± 2

Q.196. The sum of two numbers is 28

and the sum of their squares is 528. Find the square root of the product of both the numbers.

- (a) $8\sqrt{2}$ (b) $2\sqrt{2}$ (c) $4\sqrt{2}$ (d) $6\sqrt{2}$

Q.197. If $\frac{a}{b} + \frac{b}{a} = 1$, $a \neq 0$, $b \neq 0$ then

find the value of $\frac{a^3 + b^3}{3ab}$

- (a) 3 (b) 0 (c) 2 (d) -3

Q.198. If $x = 11$, then the value of $(x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1)$

- (a) 10 (b) 8 (c) 9 (d) 14

Q.199. If $a + b + c = 15$ and $ab + bc + ca = 35$ then find the value of $a^3 + b^3 + c^3 - 3abc$.

- (a) 2100 (b) 1500 (c) 1200 (d) 1800

Q.200. What must be added to $(x^3 - 3x^2 + 4x - 75)$ to obtain a polynomial that is exactly divisible by $(x - 5)$?

- (a) 5 (b) 6 (c) 4 (d) 2

Q.201. If $a + b + c = 13$ and $ab + bc + ca = 22$, then the value of $a^3 + b^3 + c^3 - 3abc$ is _____.

- (a) 1911 (b) 1225 (c) 1625 (d) 1339

Q.202. If $x^2 + 4y^2 + 2x + 1 = 0$, then find the value of $x^{39} + y^{36}$

- (a) -1 (b) -2 (c) 2 (d) 1

Q.203. If $x^2 - 6x - 1 = 0$, then find the value of $x^2 + \frac{1}{x^2} - 8$.

- (a) 30 (b) 32 (c) 28 (d) 34

Q.204. If $2a + 2b + c = 0$, then find the value of $\frac{4a^2 + 4b^2 + 4c^2}{5c^2 - 8ab}$

- (a) -1 (b) $-2\sqrt{3}$ (c) 1 (d) $2\sqrt{3}$

Q.205. If $a = (\sqrt{2} - 1)^{\frac{1}{3}}$, then the value of $(a - \frac{1}{a})^3 + 3(a - \frac{1}{a})$ is:

- (a) 2 (b) -2 (c) $\sqrt{2}$ (d) $-\sqrt{2}$

Q.206. If $x^3 = 184 + y^3$ and $x = 4 + y$, then the value of $(x + y)$ is (given that $x > 0$ and $y > 0$):

- (a) $-2\sqrt{11}$ (b) $2\sqrt{14}$ (c) $-2\sqrt{14}$ (d) $2\sqrt{11}$

Q.207. Simplify the following.

$$(x - 3y)^3 + 27(y^3 - xy^2)$$

$$(a) x^3 - 3x^2y$$

$$(b) x^3 - 27y^3 - 9x^2y$$

$$(c) x^3 - 27y^3 - 9x^2y + 27xy^2 + 27y^2 - 27y^2$$

$$(d) x^3 - 9x^2y$$

Q.208. If $x^2 + y^2 + z^2 = xy + yz + zx$, then

the value of $(\frac{17x^4 + 9y^4 + 16z^4}{8x^2y^2 + 6y^2z^2 + 10z^2x^2})$ is:

- (a) 2.00 (b) 3.25 (c) 2.50 (d) 1.75

Q.209. If for a non-zero x , $7x^2 + 5x + 7 = 0$, then the $x^3 + \frac{1}{x^3}$ is _____.

- (a) $\frac{343}{610}$ (b) $\frac{420}{343}$ (c) $\frac{610}{420}$ (d) $\frac{610}{343}$

Q.210. Simplify the following expression. $49a^2 + 70ab + 25b^2$

$$(a) (5a + 7b)^2 \quad (b) (5a - 7b)^2$$

$$(c) (7a - 5b)^2 \quad (d) (7a + 5b)^2$$

Q.211. If $x + y = 10$ and $xy = 12$, find the value of $x^2 - xy + y^2$

- (a) 74 (b) 82 (c) 64 (d) 78

Q.212. If $(x - \frac{1}{x-1}) = 3$, then the value of $(x-1)^2 + \frac{1}{(x-1)^2}$ is:

- (a) 3 (b) 5 (c) 6 (d) 4

Q.213. If $p = 106$, $q = 105$ and $r = 104$, then value of $p^2 + q^2 + r^2 - pq - qr - rp$ is:

- (a) 0 (b) 1 (c) 2 (d) 3

Q.214. If $\sqrt{1 + \frac{x^3}{9}} = 5$, then the value of x is equal to:

- (a) 9 (b) 8 (c) 6 (d) 4

Q.215. If $x + \frac{1}{x} = 15$, then the value of $(x^2 + \frac{1}{x^2})$ is:

- (a) 225 (b) 223 (c) 17 (d) 13

Q.216. If $(a - 2b) = 8$ and $2ab = 9$, then find the value of $(a + 2b)$.

- (a) 12 (b) 6 (c) 10 (d) 8

Q.217. A and B each have a certain number of oranges. A says to B, "If you give me 10 of your oranges, I will have twice the number of oranges left with you." B replies, "If you give me 10 of your oranges, I will have the same number of oranges left with you." Find the number of oranges that A has.

- (a) 65 (b) 50 (c) 55 (d) 70

Q.218. If $\frac{x}{y} + \frac{y}{x} = 1$ and $x + y = 2$, then the value of $x^3 + y^3$ is:

- (a) 0 (b) 1 (c) 3 (d) 2

Q.219. If $\frac{A}{L} + \frac{M}{B} = 1$ and $\frac{B}{M} + \frac{N}{C} = 1$,

then the value of $\frac{L}{A} + \frac{C}{N}$ is :

- (a) 1 (b) $\frac{B}{M}$ (c) $\frac{M}{B}$ (d) 0



Q.220. If $a \propto b$, $b \propto \frac{1}{c}$ and $c \propto d$, then the relation between a and b is :

- (a) $a \propto \sqrt{d}$ (b) $a \propto d$ (c) $a \propto \frac{1}{d}$ (d) $a \propto d^2$

Q.221. If $A + \frac{1}{1 + \frac{1}{2 + \frac{1}{3}}} = \frac{9}{10}$, then the value of A is:

- (a) $\frac{3}{10}$ (b) $\frac{2}{5}$ (c) $\frac{1}{10}$ (d) $\frac{1}{5}$

Q.222. If $x(5 - \frac{2}{x}) = \frac{5}{x}$, then the value of $x^2 + \frac{1}{x^2}$ is:

- (a) $\frac{54}{23}$ (b) $\frac{53}{28}$ (c) $\frac{53}{27}$ (d) $\frac{54}{25}$

Q.223. The value of

$$\frac{(157 \times 157) + (157 \times 133) + (133 \times 133)}{(157 \times 157 \times 157) - (133 \times 133 \times 133)}$$

is :

- (a) 24 (b) $\frac{1}{290}$ (c) 290 (d) $\frac{1}{24}$

Q.224. Simplify the given expression.

$$\frac{(3.254 \times 3.254 \times 3.254) - (0.746 \times 0.746 \times 0.746)}{(3.254 \times 3.254) + (0.746 \times 0.746) + (3.254 \times 0.746)}$$

- (a) 2.508 (b) 4.508 (c) 2.146 (d) 3.156

Q.225. The value of $\frac{a^2 - (b - c)^2}{(a + c)^2 - b^2} +$

$$\frac{b^2 - (a - c)^2}{(a + b)^2 - c^2} + \frac{c^2 - (a - b)^2}{(b + c)^2 - a^2}$$
 is :

- (a) 3 (b) -1 (c) 1 (d) 2

Q.226. Simplify the given expression.

$$(1 + x)^3 + (1 - x)^3 + (-2)^3$$

- (a) $3(1 - x^2)$ (b) $6(1 - x^2)$
(c) $-6(1 - x^2)$ (d) $-3(1 - x^2)$

Q.227. Simplify the given expression.

$$(1 - 2x)^2 - (1 + 2x)^2$$

- (a) $-8x$ (b) $8x$
(c) $2 + 8x^2$ (d) $-(2 + 8x^2)$

Q.228. If $a + b + c = 5$ and $ab + bc + ca = 7$, then the value of $a^3 + b^3 + c^3 - 3abc$ is:

- (a) 20 (b) 25 (c) 15 (d) 30

Q.229. Simplify the given expression. $(x - 2y)(y - 3x) + (x + y)(x - y) + (x - 3y)(2x + y)$

- (a) $2x(x - 3y)$ (b) $2y(x + 3y)$
(c) $2x(x + 3y)$ (d) $2y(x - 3y)$

Q.230. If $(x - 2)$ is a factor of $(2x^2 + 12kx - 25k)$, then what is the value of k ?

- (a) 2 (b) 8 (c) 4 (d) 6

Q.231. If $2x + \frac{2}{x} = 5$, then the value of $(x^3 + \frac{1}{x^3} + 2)$ will be :

- (a) $\frac{81}{11}$ (b) $\frac{81}{7}$ (c) $\frac{71}{8}$ (d) $\frac{81}{8}$

Q.232. If $(a - b) = 9$ and $(a^3 - b^3) = 4401$, find the value of ab .

- (a) 190 (b) 112 (c) 162 (d) 136

Q.233. If $(a + \frac{1}{a}) = 6$, then what is the value of $\frac{3}{4}(a^2 + \frac{1}{a^2})$?

- (a) 22.5 (b) 34 (c) 25.5 (d) 36

Q.234. If $a = \frac{1}{a - \sqrt{6}}$ ($a > 0$), then the value of $(a + \frac{1}{a})$ is :

- (a) $\sqrt{6}$ (b) $\sqrt{10}$ (c) $\sqrt{15}$ (d) $\sqrt{7}$

Q.235. If $\sqrt{\frac{a}{b}} = \frac{8}{3} + \sqrt{\frac{b}{a}}$ and $(a + b) = 30$, then what is the value of ab ?

- (a) 64 (b) 28 (c) 81 (d) 36

Q.236. If $2x + 3y = 9$, and $xy = 3$, what is $8x^3 + 27y^3$?

- (a) 235 (b) 244 (c) 243 (d) 234

Q.237. If $a^2 + \frac{1}{a^2} = \frac{7}{3}$, then what is the value of $(a^3 - \frac{1}{a^3})$?

- (a) $\frac{5}{3\sqrt{3}}$ (b) $\frac{10}{3\sqrt{3}}$ (c) $\frac{7}{3\sqrt{3}}$ (d) $\frac{8}{3\sqrt{3}}$

Q.238. If $a + b = 10$ and $a^2 + b^2 = 58$, find the value of ab .

- (a) 24 (b) 27 (c) 30 (d) 21

Q.239. If $(a + b + c) \neq 0$, then $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ is equal to:

- (a) $a^3 + b^3 + c^3 - 3abc$ (b) $a^3 - b^3 + c^3 - 3abc$
(c) $a^3 + b^3 + c^3 - 3abc$ (d) $a^3 + b^3 + c^3 + 3abc$

Q.240. If $2a + 3b = 14$ and $2a - 3b = 10$, then find the value of 'ab'.

- (a) 5 (b) 3 (c) 4 (d) 2

Q.241. If $\sqrt{\frac{a}{b}} = \frac{8}{3} - \sqrt{\frac{b}{a}}$ and $a - b = 10$ then the value of ab is :

- (a) $32\frac{1}{7}$ (b) $32\frac{3}{7}$ (c) $32\frac{4}{7}$ (d) $32\frac{2}{7}$

Q.242. Which of the following is an identity ?

- (a) $(a - b)^2 = a^2 - b^2$
(b) $(a + b)^2 = a^2 + b^2 + 2ab$
(c) $(a + b)^2 = a^2 + b^2$
(d) $(a - b)^2 = a^2 - b^2 - 2ab$

Q.243. If $x + \frac{2}{x} = 1$ then the value of $\frac{x^2 + 7x + 2}{x^2 + 13x + 2}$ is:

- (a) $\frac{5}{7}$ (b) $\frac{3}{7}$ (c) $\frac{4}{7}$ (d) $\frac{2}{7}$

Q.244. If $(x + \frac{1}{x}) = 7$ and $x > 1$, what is the value of $(x^3 - \frac{1}{x^3})$?

- (a) $174\sqrt{5}$ (b) $164\sqrt{5}$ (c) $154\sqrt{5}$ (d) $144\sqrt{5}$

Q.245. If $(y - \frac{1}{y}) = 9$, find the value of $(y^3 - \frac{1}{y^3})$

- (a) 766 (b) 756 (c) 702 (d) 729

Q.246. If $2x + 3y = 4$ and $4x^2 + 9y^2 = 64$, then what is the value of $8x^3 + 27y^3$.

- (a) 253 (b) 235 (c) 352 (d) 325

Q.247. If $a + b + c = 15$ and $ab + bc + ca = 22$, then find the value of $a^2 + b^2 + c^2$.

- (a) 131 (b) 141 (c) 161 (d) 181

Q.248. Simplify the following expression.

$$\{[(x - 5)(x - 1)] - [(9x - 5)(9x - 1)]\} \div 16x$$

- (a) $2x(5x - 3)$ (b) $-(5x - 3)$
(c) $x(5x - 3)$ (d) $-6x(5x - 3)$

Q.249. If $x + \frac{1}{x} = 2$, then find the value of $x^{1823} + \frac{1}{x^{1929}}$

- (a) 2 (b) 1 (c) 0 (d) -1

Q.250. The algebraic expression $2x^2 - 32$ is equal to_____.

- (a) $2(x - 4)^2$ (b) $(x + 4)(x - 4)$
(c) $2(x + 4)^2$ (d) $2(x + 4)(x - 4)$

Q.251. If $(x - \frac{1}{x}) = 0.4$, and $x > 0$, what is the value of $(x^2 + \frac{1}{x^2})$?

- (a) 4.16 (b) 3.32 (c) 1.84 (d) 2.16

Q.252. If $(x + \frac{1}{x}) = 5$, then the value of $(x^5 + \frac{1}{x^5})$ is:

- (a) 2535 (b) 2540 (c) 2515 (d) 2525

Q.253. If $x^2 + y^2 + z^2 = 628$ and $xy + yz + zx = 24$, then find the value of $(x + y + z)$.

- (a) 22 (b) 28 (c) 26 (d) 24

Q.254. if $x + \frac{1}{x} = -1$, then the value of $x^{15} + \frac{1}{x^{15}}$ will be:

- (a) 3 (b) 2 (c) -1 (d) 0

Q.255. Simplify the following expression.

$$\frac{x^3 + y^3 + z^3 - 3xyz}{(x - y)^2 + (y - z)^2 + (z - x)^2}$$

- (a) $\frac{1}{3}(x + y + z)$ (b) $(x + y + z)$



(c) $\frac{1}{4}(x + y + z)$ (d) $\frac{1}{2}(x + y + z)$

Q.256. If $(x - \frac{1}{x-2}) = 7$, then the value of

$(x - 2)^3 - \frac{1}{(x-2)^3}$ is:

- (a) 130 (b) 115 (c) 140 (d) 125

Q.257. If $a + b + c = 16$, $ab + bc + ac = 81$, what is the value of $a^2 + b^2 + c^2$

- (a) 94 (b) 98 (c) 89 (d) 87

Q.258. Simplify the following expression
 $(a + b + c)^2 - (a - b + c)^2 + 4ac$

- (a) $4(bc + ac)$ (b) $2(ab + bc + ac)$
 (c) $4(ab + bc + ac)$ (d) $4(bc + ab)$

Q.259. If $x^2 - 5\sqrt{5}x + 1 = 0$, and $x > 0$, what is the value of $(x^3 - \frac{1}{x^3})$?

- (a) 1331 (b) 1364 (c) 1296 (d) 1244

Q.260. If $x^4 + \frac{1}{x^4} = 194$, $x > 0$, then find

the value of $x^3 + \frac{1}{x^3} + x + \frac{1}{x}$.

- (a) 76 (b) 66 (c) 56 (d) 46

Q.261. If $a + b + c = 5$ and $a^2 + b^2 + c^2 = 15$, then find the value of $a^3 + b^3 + c^3 - 3abc - 27$.

- (a) 23 (b) 27 (c) 25 (d) 21

Q.262. If $x^2 + y^2 = 427$ and $xy = 202$, then find the value of $\frac{x+y}{x-y}$.

(a) $\sqrt{\frac{835}{23}}$ (b) $\sqrt{\frac{830}{29}}$

(c) $\sqrt{\frac{831}{23}}$ (d) $\sqrt{\frac{830}{23}}$

Q.263. The algebraic expression $4x^2 - y^2 + 6y - 9$ is equal to _____.

- (a) $(2x + y - 3)(2x - y + 3)$
 (b) $(2x - y - 3)(2x - y + 3)$
 (c) $(2x + y - 3)^2$
 (d) $(2x + y + 3)^2$

Q.264. If $x = (\sqrt{6} - 1)^{\frac{1}{3}}$, then the value of

$(x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$ is:

(a) $\frac{2\sqrt{6}-6}{5}$ (b) $\frac{4\sqrt{6}-6}{5}$

(c) $\frac{4\sqrt{6}-6}{3}$ (d) $\frac{4\sqrt{3}-6}{5}$

Q.265. If $(a + b + c) = 17$, and $(a^2 + b^2 + c^2) = 101$, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.

- (a) 12 (b) 14 (c) 10 (d) 16

Q.266. If $\frac{x^2+1}{x} = 5$, then find the value

of $x^4 + \frac{1}{x^4} - 36$.

- (a) 491 (b) 149 (c) 419 (d) 194

Q.267. If $a^2 + b^2 + c^2 = ab + bc + ca$, then

the value of $\frac{11a^4 + 13b^4 + 15c^4}{16a^2b^2 + 19b^2c^2 + 17c^2a^2}$ is:

- (a) $1\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $1\frac{3}{4}$

Q.268. Simplify the expression: $(c + d)^2 - (c - d)^2$

- (a) $4cd$ (b) $(c^2 + d^2)$ (c) $2(c^2 + d^2)$ (d) $2cd$

Q.269. Which of the following expressions is equal to the expression

$\frac{x^2 - 3x + 2}{x^2 - 4}$?

(a) $\frac{x+1}{x-2}$ (b) $\frac{x-1}{x+2}$

(c) $\frac{x+1}{x+2}$ (d) $\frac{x-1}{x-2}$

Q.270. If $(l + m + n) = 0$, then what will

be the value of $(\frac{l^2}{mn} + \frac{m^2}{nl} + \frac{n^2}{lm})^2$?

- (a) 9 (b) 16 (c) 4 (d) 1

Q.271. If $2x + 5y = 15$ and $xy = 6$, then the value of $4x^2 + 25y^2$ is _____.

- (a) 105 (b) 95 (c) 100 (d) 90

Q.272. If $x^2 - \sqrt{9.76}x + 1 = 0$ and $x > 1$, the value of $(x^3 - \frac{1}{x^3})$ is:

- (a) 21.042 (b) 24.024
 (c) 21.024 (d) 24.042

Q.273. If $(5a + \frac{4}{a} - 2) = 13$ and $a > 0$,

what is the value of $(25a^2 + \frac{16}{a^2})$?

- (a) 158 (b) 157 (c) 185 (d) 175

Q.274. Simplify the given expression.

$\frac{(x^3 - y^3)(x + y)}{x^2 + xy + y^2}$

- (a) $x - y$ (b) $x^2 - y^2$ (c) $x + y$ (d) $x^2 + y^2$

Q.275. If $(a + b + c) = 17$, and $(a^2 + b^2 + c^2) = 115$, find the value of

$(a + b)^2 + (b + c)^2 + (c + a)^2$.

- (a) 402 (b) 408 (c) 394 (d) 404

Q.276. Simplify:

$(2a - 3b - c)^2 - (a + 2b + c)^2$

(a) $3a^2 + 5b^2 - 16ab + 2bc - 8ac$

(b) $3a^2 + 5b^2 - 8ab + 2bc - 6ac$

(c) $3a^2 + 5b^2 - 16ab - 6ac$

(d) $3a^2 + 5b^2 - 16ab + 2bc - 6ac$

Q.277. If $x^2 - 3.2x + 1 = 0$ and $x > 1$,

the value of $(x^2 - \frac{1}{x^2})$ is:

(a) $16.8\sqrt{0.39}$ (b) $12.8\sqrt{0.39}$

(c) $16.8\sqrt{0.32}$ (d) $12.8\sqrt{0.32}$

Q.278. If $x^2 + y^2 + z^2 = 2(x + z - 1)$, then the value of $x^3 + y^3 + z^3$ is equal to:

- (a) 6 (b) 1 (c) 2 (d) 8

Q.279. If $x + 3y = 6$, what is the value of $x^3 + 27y^3 + 54xy$?

- (a) 264 (b) 258 (c) 216 (d) 220

Q.280. If $p + q = 12$ and $pq = 14$, then find the value of $p^2 - pq + q^2$.

- (a) 192 (b) 181 (c) 102 (d) 144

Q.281. Simplify the given expression,

$\frac{(x+3)^3 + (x-3)^3}{x^2 + 27}$

- (a) $3x$ (b) x (c) $4x$ (d) $2x$

Q.282. Simplify the expression:

$\frac{1}{8} [\frac{1}{b-1} - \frac{1}{b+1} - \frac{2}{b^2+1} - \frac{4}{b^4+1}]$

(a) $\frac{1}{b^8-1}$ (b) $\frac{8}{b^8+1}$

(c) $\frac{8}{b^8-1}$ (d) $\frac{1}{b^8+1}$

Q.283. If $(d + e + f) = 14$, $(d^2 + e^2 + f^2) = 96$, then find the value of $(de + ef + fd)$.

- (a) 75 (b) 25 (c) 50 (d) 100

Q.284. If $g = (2 - \sqrt{3})$, what will be the value of $(g^3 - \frac{1}{g^3})$?

- (a) $-30\sqrt{3}$ (b) 52 (c) $30\sqrt{3}$ (d) -52

Q.285. Simplify the expression:

$(2x + 13 - y)(2x - 13 - y)$

(a) $4x^2 - y^2 - 4xy - 169$

(b) $4x^2 + y^2 + 4xy - 169$

(c) $4x^2 + y^2 - 4xy - 169$

(d) $4x^2 + y^2 - 4xy + 169$

Q.286. Simplify:

$(3x - 2y)(3x + 2y)(9x^2 + 4y^2)$

(a) $81x^4 + 16y^4$ (b) $81y^4 - 16x^4$

(c) $81y^4 + 16x^4$ (d) $81x^4 - 16y^4$

Q.287. Simply the given expression:

$(5p + 3q)(5p - 3q)$.

(a) $25p^2 - 9q^2 + 30pq$ (b) $25p^2 - 9q^2$

(c) $25p^2 + 9q^2$ (d) $25p^2 + 9q^2 - 30pq$

Q.288. If $p^2 - 4p - 1 = 0$, then the value of

$p^2 + 3p + \frac{1}{p^2} - \frac{3}{p}$ is:

- (a) 40 (b) 50 (c) 65 (d) 30

Q.289. If $3x - 2y = 3$ and $xy = 9$, then find the value of $3x + 2y$.

- (a) 16 (b) 13 (c) 14 (d) 15

Q.290. If $px + qy = 1$ and $qx + py =$



$\frac{2pq}{p^2 + q^2}$, then $(x^2 + y^2)(p^2 + q^2)$ is equal to:

- (a) 2 (b) 3 (c) 0 (d) 1

Q.291. If $7b - \frac{1}{4b} = 7$, then what is the value of $16b^2 + \frac{1}{49b^2}$?

- (a) $\frac{80}{49}$ (b) $\frac{104}{7}$ (c) $\frac{120}{7}$ (d) $\frac{7}{2}$

Q.292. What is the value of $(3x^3 + 5x^2y + 12xy^2 + 7y^3)$, when $x = -4$ and $y = -1$?

- (a) -329 (b) -359 (c) -361 (d) -327

Q.293. If $(a + b + c) = 16$, and $(a^2 + b^2 + c^2) = 90$, find the value of $(ab + bc + ca)$.

- (a) 84 (b) 83 (c) 82 (d) 81

Q.294. What is the value of $a^3 + b^3 + c^3$, if $(a + b + c) = 0$?

- (a) $a^2 + b^2 + c^2 - 3abc$
(b) 0
(c) $3abc$
(d) $a^2 + b^2 + c^2 - ab - bc - ca$

Q.295. If $x + \frac{1}{x} = 7$, then the value of

$x^6 + \frac{1}{x^6}$ is:

- (a) 113682 (b) 103682
(c) 103882 (d) 103862

Q.296. If $a = 17$, $b = 13$, then find the value of the expression $(a^3 - b^3 - 3a^2b + 3ab^2)$.

- (a) -64 (b) -27000 (c) 27000 (d) 64

Q.297. Simplify, $\frac{x^4 - 2x^2 + 1}{x^2 - 2x + 1}$.

- (a) $x^2 - 2x + 1$ (b) $x^2 + 2x + 2$
(c) $x^2 + 2x + 1$ (d) $x^2 + x + 1$

Q.298. If $(x - \frac{1}{x}) = \sqrt{6}$, and $x > 1$, what is the value of $(x^8 - \frac{1}{x^8})$?

- (a) $1024\sqrt{15}$ (b) $992\sqrt{15}$
(c) $998\sqrt{15}$ (d) $1012\sqrt{15}$

Q.299. What is the value of the given expression?

- $(a + b + c)^2 - a^2 - b^2 - c^2$
(a) $2abc$ (b) $2(ab + bc + ca)$
(c) $2(a + b + c)$ (d) $2ab + bc - 2ca$

Q.300. What is the value of $\frac{X}{Y}$ if

$$\frac{X - 5Y}{X + 5Y} = \frac{7}{13}$$

- (a) $\frac{23}{7}$ (b) $\frac{24}{9}$ (c) $\frac{50}{3}$ (d) $\frac{100}{7}$

Q.301. The cube of the difference between two given natural numbers is

1728, while the product of these two given numbers is 108. Find the positive difference between the cubes of these two given numbers.

- (a) 4104 (b) 5616 (c) 2160 (d) 5626

Q.302. If $(a + b + c) = 19$, and $(a^2 + b^2 + c^2) = 155$, find the value of $(a - b)^2 + (b - c)^2 + (c - a)^2$.

- (a) 104 (b) 108 (c) 100 (d) 98

Q.303. If $a^2 + b^2 + c^2 = ab + bc + ac$, then the value of $\frac{11a^4 + 13b^4 + 17c^4}{17a^2b^2 + 9b^2c^2 + 15c^2a^2}$ is:

- (a) 1 (b) 2 (c) 11 (d) 4

Q.304. The square of the difference between two given natural numbers is 324, while the product of these two given numbers is 144. Find the positive difference between the squares of these two given numbers.

- (a) 630 (b) 540 (c) 450 (d) 360

Q.305. If $p = \frac{\sqrt{2} + 1}{\sqrt{2} - 1}$ and $q = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$

then find the value of $\frac{p^2}{q} + \frac{q^2}{p}$

- (a) 200 (b) 196 (c) 198 (d) 188

Q.306. If $(x + \frac{1}{x}) = 2\sqrt{2}$, and $x > 1$, what

is the value of $(x^6 - \frac{1}{x^6})$?

- (a) $140\sqrt{2}$ (b) $116\sqrt{2}$ (c) $144\sqrt{2}$ (d) $128\sqrt{2}$

Q.307. If $(a + b + c) = 12$, and $(a^2 + b^2 + c^2) = 50$, find the value of $(a^3 + b^3 + c^3 - 3abc)$.

- (a) 36 (b) 24 (c) 42 (d) 48

Q.308. If $(3y - \frac{3}{y}) = 5$, find the value of

$(y^2 + \frac{1}{y^2})$.

- (a) $\frac{47}{9}$ (b) $\frac{49}{9}$ (c) $\frac{41}{9}$ (d) $\frac{43}{9}$

Q.309. If $x^2 - 7x + 1 = 0$, and $0 < x < 1$, what is the value of $x^2 - \frac{1}{x^2}$?

- (a) $21\sqrt{5}$ (b) $-21\sqrt{5}$ (c) $28\sqrt{5}$ (d) $-28\sqrt{5}$

Q.310. If $(x + \frac{1}{x}) = 2$, then $x^7 + \frac{1}{x^{117}} =$

_____.

- (a) 1 (b) 2 (c) 4 (d) 3

Q.311. If $(a + b + c) = 20$, and $a^2 + b^2 + c^2 = 152$, find the value of $a^3 + b^3 + c^3 - 3abc$.

- (a) 560 (b) 640 (c) 480 (d) 720

Q.312. Simplify the expression $\frac{l^2 - m^2}{(l + m)^2}$,

provided $(l + m) \neq 0$.

- (a) $\frac{(l + m)}{(l - m)}$ (b) $\frac{(l - m)}{(l + m)}$ (c) 0 (d) 1

Q.313. If $x > 0$ and $x^4 + \frac{1}{x^4} = 142$, what is the value of $x^7 + \frac{1}{x^7}$?

- (a) $1561\sqrt{14}$ (b) $1563\sqrt{14}$
(c) $1560\sqrt{14}$ (d) $1562\sqrt{14}$

Q.314. If $a + b + c = 7$, $ab + bc + ca = 11$ and $abc = -1$, then $a^3 + b^3 + c^3$ is equal to:

- (a) 101 (b) 107 (c) 109 (d) 111

Q.315. If $(x - \frac{1}{x}) = 6$, and $x > 0$, find the value of $(x^2 - \frac{1}{x^2})$.

- (a) $12\sqrt{10}$ (b) $18\sqrt{10}$ (c) $24\sqrt{2}$ (d) $24\sqrt{10}$

Q.316. If $(x + \frac{1}{x}) = 3\sqrt{2}$, and $x > 1$, what is the value of $(x^8 - \frac{1}{x^8})$?

- (a) $24364\sqrt{7}$ (b) $24384\sqrt{11}$
(c) $24384\sqrt{7}$ (d) $12192\sqrt{7}$

Q.317. What is the value of $(\frac{1}{a} - \frac{1}{b} - \frac{1}{c})$

if, $\frac{2a-5}{a} - \frac{4b-5}{b} + \frac{6c+5}{c} = 0$?

- (a) $\frac{4}{5}$ (b) $-\frac{8}{5}$ (c) $\frac{2}{5}$ (d) $-\frac{12}{5}$

Q.318. Simplify the following expression. $[(1 + P)(1 + P^2)(1 + P^4)(1 + P^8)(1 + P^{16})(1 - P) - 1]$

- (a) $-P^{32}$ (b) P^{32} (c) $(1 + P^{32})$ (d) $(1 - P^{32})$

Q.319. If x and y are positive numbers such that $x - y = 5$ and $xy = 150$, then the value of $(x + y)$ is:

- (a) 45 (b) 25 (c) 35 (d) 15

Q.320. If $(a - b) = 1$ then what is the value of $(a^3 - b^3)$?

- (a) $a^2 + ab + b^2$ (b) $(a + b)^2 + 3ab$
(c) $a^2 - ab + b^2$ (d) $a^2 + 2ab + b^2$

Q.321. If $(x^2 + \frac{1}{x^2}) = 18$, and $x > 0$, what

is the value of $(x^3 + \frac{1}{x^3})$?

- (a) $36\sqrt{5}$ (b) $40\sqrt{5}$ (c) $46\sqrt{5}$ (d) $34\sqrt{5}$

Q.322. The value of $\frac{p^2 - (q - r)^2}{(p + r)^2 - q^2} +$

$\frac{q^2 - (p - r)^2}{(p + q)^2 - r^2} + \frac{r^2 - (p - q)^2}{(q + r)^2 - p^2}$

- (a) 1 (b) 2 (c) 0 (d) 3

Q.323. If $\frac{a}{b} = \frac{7}{6}$, then find the value of the expression $\frac{6a + 13b}{6a - 13b}$.



(a) $-\frac{3}{10}$ (b) $\frac{10}{3}$ (c) $-\frac{10}{3}$ (d) $\frac{3}{10}$

Q.324. Simplify $\frac{1}{2+2p} + \frac{1}{2+2q} + \frac{1}{2+2r}$, where $p = \frac{x}{y+z}$, $q = \frac{y}{z+x}$ and $r = \frac{z}{x+y}$.

(a) 1 (b) $x+y+z$ (c) 2 (d) $\frac{1}{2}$

Q.325. If $(y - \frac{1}{y}) = 4$, find the value of $(y^6 + \frac{1}{y^6})$.

(a) 5774 (b) 4096 (c) 5776 (d) 5778

Q.326. If $x^2 - 15x + 1 = 0$, what is the value of $x^4 - 223x^2 + 6$?

(a) 9 (b) 5 (c) 6 (d) 0

Q.327. If $p + q + r = pqr = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$, then find $p^3 + q^3 + r^3$.

(a) 1 (b) -1 (c) 5 (d) -5

Q.328. Simplify $\frac{256x^4 - 16y^4}{(80x^2 - 20y^2)(16x^2 + 4y^2)}$

(a) 5 (b) $\frac{1}{20}$ (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

Q.329. If $7a - \frac{7}{a} + 4 = 0$, then find

$a^3 - \frac{1}{a^3} - 1$.

(a) $-\frac{995}{343}$ (b) $-\frac{875}{248}$ (c) $-\frac{694}{315}$ (d) $-\frac{765}{262}$

Q.330. Simplify the expression $\frac{s^2 + t^2 + 2st - u^2}{s^2 - t^2 - 2tu - u^2}$, provided $(s + t + u) \neq 0$.

(a) $\frac{s+t-u}{s-t-u}$ (b) $\frac{s+t+u}{s-t+u}$

(c) $\frac{s-t-u}{s+t-u}$ (d) $\frac{s-t+u}{s+t+u}$

Q.331. If $x^2 - 8x - 1 = 0$, what is the value of $x^2 + \frac{1}{x^2}$?

(a) 68 (b) 62 (c) 64 (d) 66

Q.332. If $(x^2 + \frac{1}{x^2}) = 6$, and $0 < x < 1$,

what is the value of $x^4 - \frac{1}{x^4}$?

(a) $24\sqrt{2}$ (b) $-24\sqrt{2}$
(c) $-12\sqrt{10}$ (d) $12\sqrt{10}$

Q.333. If $x = \frac{1}{x-3}$, ($x > 0$), then the value of $x + \frac{1}{x}$ is:

(a) $\sqrt{11}$ (b) $\sqrt{17}$ (c) $\sqrt{15}$ (d) $\sqrt{13}$

Q.334. If $x + \frac{1}{x} = 1$, then the value of

$\frac{x^2 + 7x + 1}{x^2 + 11x + 1} = ?$

(a) $\frac{3}{4}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Q.335. If $(x + \frac{1}{x}) = \sqrt{6}$, and $x > 1$, what is the value of $(x^8 - \frac{1}{x^8})$?

(a) $120\sqrt{3}$ (b) $128\sqrt{3}$ (c) $112\sqrt{3}$ (d) $108\sqrt{3}$

Q.336. If $b = 5$, determine the value of an expression $(\frac{5}{b} + 5b)(\frac{25}{b^2} - 25 + 25b^2)$ using an identity.

(a) 25726 (b) 15626 (c) 15438 (d) 25636

Q.337. If $(x - \frac{1}{x}) = 10$, what is the value of $(x^4 + \frac{1}{x^4})$?

(a) 10404 (b) 10402 (c) 10406 (d) 10400

Q.338. Expand and simplify the algebraic expression: $(x-5)^2 + (x+3)^2 + 4x$

(a) $2(x^2 - 17)$ (b) $2(x^2 + 17)$
(c) $(x^2 + 17)$ (d) $2(x^2 - 5x + 17)$

Q.339. If $2x + 3y = 16$ and $xy = 9$, then find $8x^3 + 27y^3$.

(a) 1980 (b) 2980 (c) 1504 (d) 2189

Q.340. If $x + y = 25$ and $xy = 20$, then find the value of $x^3 + y^3$.

(a) 13152 (b) 13125 (c) 14125 (d) 14152

Q.341. If $x^2 - \frac{1}{x^2} = 4\sqrt{2}$ what is the value of $x^4 - \frac{1}{x^4}$?

(a) $16\sqrt{2}$ (b) $8\sqrt{2}$ (c) $24\sqrt{2}$ (d) $32\sqrt{2}$

Q.342. If $(x^2 + \frac{1}{x^2}) = 18$, and $x > 1$, what is the value of $(x^3 - \frac{1}{x^3})$?

(a) 52 (b) 76 (c) 80 (d) 64

Q.343. What is the solution to the following system of linear equations?
 $3x - 9y + 4z = 5$
 $2x + 7y + z = 12$
 $3x - z = 0$

(a) $x = \frac{143}{150}$, $y = \frac{31}{37}$, $z = \frac{143}{50}$

(b) $x = \frac{143}{150}$, $y = \frac{31}{30}$, $z = \frac{143}{50}$

(c) $x = \frac{137}{150}$, $y = \frac{31}{30}$, $z = \frac{143}{50}$

(d) $x = \frac{143}{150}$, $y = \frac{31}{30}$, $z = \frac{143}{51}$

Q.344. If $2a + b = 10$ and $2ab = 9$, then one of the values of $2a - b$ is:

(a) 8 (b) 4 (c) 6 (d) 10

Q.345. If $x > 1$, and $x^2 + \frac{1}{x^2} = 2\sqrt{5}$, what

is the value of $x^4 - \frac{1}{x^4}$?

(a) $4\sqrt{5}$ (b) $4\sqrt{30}$ (c) $8\sqrt{5}$ (d) $8\sqrt{6}$

Q.346. If $x > 1$, and $x^4 + \frac{1}{x^4} = 79$, what is the value of $x - \frac{1}{x}$?

(a) $2\sqrt{2}$ (b) $\sqrt{7}$ (c) $\sqrt{11}$ (d) $\sqrt{10}$

Q.347. If $x + y = \sqrt{13}$ and $x - y = \sqrt{11}$, then the value of $xy(x^2 + y^2)$ is:

(a) 10 (b) 6 (c) $\sqrt{6}$ (d) $\sqrt{10}$

Q.348. If $ab + bc + ca = 0$, then find the value of $\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$.

(a) 0 (b) $\frac{1}{abc}$ (c) 1 (d) ab

Q.349. A varies directly with B. When $A = 52$, $B = 91$. Find the value of A when $B = 126$.

(a) 76 (b) 84 (c) 72 (d) 70

Q.350. For what value of q does the system of equations $38x + qy + 171 = 0$ and $46x + 414y + 207 = 0$ have infinite number of solutions?

(a) 380 (b) 345 (c) 342 (d) 350

Q.351. If $x = 14$, $y = 15$ and $z = 17$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is:

(a) 333 (b) 322 (c) 312 (d) 222

Q.352. For what value of p does the system of equations $18x + 36y + 45 = 0$ and $px - 54y + 67 = 0$ have no solution?

(a) 54 (b) -27 (c) -36 (d) 27

Q.353. Solve $(\frac{9}{8} + \frac{8}{9})^2 - (\frac{9}{8} - \frac{8}{9})^2$.

(a) 4 (b) 1 (c) 8 (d) 2

Q.354. What is the value of $\frac{1+x^2}{1-x} \div \frac{1-x^4}{x-1} \times \frac{x(1-x)}{(1+x)}$?

(a) $\frac{-x}{1+2x^2}$ (b) $\frac{-x}{1+x}$

(c) $\frac{-x}{(1+x)^2}$ (d) $\frac{-x}{1+x^2}$

Q.355. If $K + \frac{1}{K} - 2 = 0$ and $K > 0$, then what is the value of $K^{17} + \frac{1}{K^{12}}$?

(a) 2 (b) 1 (c) 0 (d) 12

Q.356. If $a^4 + \frac{1}{a^4} = 194$, then what is the value of $\frac{a^6 + 1}{a^3}$?

(a) 46 (b) 50 (c) 52 (d) 48

Q.357. What is the value of $(a-b)(b+a)(a^2+b^2)(2b^4+2a^4)$?



- (a) $a^4 - b^4$ (b) $a^8 - b^8$
 (c) $2(a^8 - b^8)$ (d) $\frac{(a^8 - b^8)}{2}$

Q.358. If $\frac{(15)^3 + (4)^3}{(15^2 + 4^2 - k)} = 19$, then what is the value of K?
 (a) - 60 (b) 120 (c) 60 (d) - 120

Q.359. If $x^2 + \frac{1}{x^2} = 38$, then what is the value of $x - \frac{1}{x}$?
 (a) 6 (b) 7 (c) 8 (d) 10

Q.360. Find the value of $p^3 - r^3$, if $p + r = 21$, and $pr = 38$.
 (a) 6538 (b) 6113 (c) 6102 (d) 6851

Q.361. What is the value of $(a + b + c) \{(a - b)^2 + (b - c)^2 + (c - a)^2\}$?
 (a) $2a^3 + 2b^3 + 2c^3$
 (b) $2a^3 + 2b^3 + 2c^3 - 6abc$
 (c) $3abc$
 (d) $6abc$

Q.362. If $x + \frac{1}{x} = -2$, then what is the value of $x^7 + x^{-7} + x^2 + x^{-2}$ ($x < 0$)
 (a) 4 (b) 2 (c) 1 (d) 0

Q.363. What is the value of $\frac{4x^2 + 9y^2 + 12xy}{144}$?
 (a) $(\frac{x}{6} + \frac{y}{4})^2$ (b) $(\frac{x}{3} + y)^2$
 (c) $(\frac{x}{3} + \frac{y}{4})^2$ (d) $(\frac{x}{4} + \frac{y}{6})^2$

Q.364. What is the expansion of $(a - 4)^3$?
 (a) $a^3 - 12a^2 + 48a + 64$
 (b) $a^3 + 12a^2 + 48a + 64$
 (c) $a^3 - 12a^2 + 48a - 64$
 (d) $a^3 + 12a^2 + 48a - 64$

Q.365. If $x - y = 8$ and $8xy = 8$, then what is the value of $2x^3 - 2y^3$?
 (a) 536 (b) 1092 (c) 1072 (d) 1608

Q.366. If $a^3 = b^3 + 988$ and $a - b = 4$, then what is the value of $2a^2 + 2ab + 2b^2$?
 (a) 516 (b) 494 (c) 500 (d) 247

Q.367. If $a^3 + 3a^2 + 3a = 63$, then the value of $a^2 + 2a$ is:
 (a) 22 (b) 19 (c) 15 (d) 8

Q.368. If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 40$, then what is the value of $a^3 + b^3 + c^3 - 3abc$?
 (a) 212 (b) 252 (c) 232 (d) 206

Q.369. If $9a + 9b + 9c = 81$ and $4ab + 4bc + 4ca = 160$, then what is the value of $6a^2 + 6b^2 + 6c^2$?

- (a) 1 (b) 3 (c) 4 (d) 6

Q.370. If $N + \frac{1}{N} = \sqrt{3}$, then find the value of $N^6 + \frac{1}{N^6} + 11$.
 (a) - 11 (b) 9 (c) - 2 (d) 11

Q.371. What is the value of $\frac{(a+b)^2 - (a-b)^2}{ab}$?
 (a) 1 (b) 4 (c) 8 (d) 2

Q.372. What is the value of $(a - b)^3 - a^3 + b^3$?
 (a) $3ab(a - b)$ (b) $-3ab(a + b)$
 (c) $3ab(a + b)$ (d) $-3ab(a - b)$

Q.373. Compute the expression $(299)^2$ using algebraic identities.
 (a) 89301 (b) 89402 (c) 88401 (d) 89401

Q.374. What is the value of $(k - \frac{1}{k})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})$
 (a) $k^{64} - \frac{1}{k^{64}}$ (b) $\frac{k^{32} - \frac{1}{k^{32}}}{k - \frac{1}{k}}$
 (c) $(k^{32} - \frac{1}{k^{32}})$ (d) $\frac{k^{32} - \frac{1}{k^{32}}}{k + \frac{1}{k}}$

Q.375. If $x + \frac{1}{x} = \frac{K}{2}$, then what is the value of $\frac{x^8 + 1}{x^4}$?
 (a) $\frac{K^4 - 16K^2 + 32}{16}$ (b) $\frac{K^4 - 8K^2 + 36}{32}$
 (c) $\frac{K^4 - 8K^2 - 32}{16}$ (d) $\frac{K^4 - 8K^2 + 32}{16}$

Q.376. If $(p - q) = 6$, $(r - q) = 5$ and $(r - p) = 3$, then find the value of $(p^3 + q^3 + r^3 - 3pqr)/(p + q + r)$.
 (a) 35 (b) 45 (c) 30 (d) 40

Q.377. If $A + \frac{1}{A} = -1$, then find the value of $\frac{A^6 + A^3 - 1}{A^9 + A^3 - 1}$?
 (a) 0 (b) - 1 (c) 2 (d) 1

Q.378. If $(x + 1)^2 + (x + 2)^2 = 16$, then what is the value of $4x^2 + 12x + 40$?
 (a) 52 (b) 62 (c) 56 (d) 74

Q.379. What is the value of $a^2 + 4b^2 + 9c^2 - 4ab + 12bc - 6ca$?
 (a) $(a + 2b - 3c)^2$ (b) $(2a - b - 3c)^2$
 (c) $(2b - a - 3c)^2$ (d) $(a - 2b - 3c)^2$

Q.380. If $a + b + c = 1$, $ab + bc + ca = -1$ and $abc = -1$, then what is the value

- $a^3 + b^3 + c^3$?
 (a) 3 (b) 5 (c) 2 (d) 1

Q.381. What is the value of $(a + b)^3 - a^3 - b^3$?
 (a) $3ab(a - b)$ (b) $-3ab(a - b)$
 (c) $3ab(a + b)$ (d) $-3ab(a + b)$

Q.382. If $\frac{1}{(x - 2)} + x = 8$, then what is the value of $\frac{1}{(x - 2)^2} + (x - 2)^2$?
 (a) 38 (b) 36 (c) 40 (d) 34

Q.383. What is the value of $\frac{1 + X}{1 - X^2} \div \frac{1 + X}{1 - X^4} \cdot \frac{1 - X^4}{1 - X} \times \frac{1 + X}{1 - X^2}$?
 (a) $\frac{2X(1 + X^2)}{(1 - X)}$ (b) $\frac{2X(1 + X^2)}{(X - 1)}$
 (c) $(1 - X^2)$ (d) $(1 + X^2)$

Q.384. If $x + y + z = 0$, then what is the value of $\frac{x}{(yz)^2} + \frac{y}{(xz)^2} + \frac{z}{(xy)^2}$?
 (a) $\frac{xyz}{3}$ (b) $\frac{3}{xyz}$ (c) xyz (d) $\frac{1}{xyz}$

Q.385. If $\frac{K}{K^2 - 2K + 1} = \frac{1}{6}$, then what is the value of $\frac{1}{K^3} + K^3$?
 (a) 472 (b) 492 (c) 512 (d) 488

Q.386. If $(x - \frac{1}{x} = 4)$, then what is the value of $(x^6 + \frac{1}{x^6})$?
 (a) 4689 (b) 4786 (c) 5832 (d) 5778

Q.387. For what value of m will the system of equations $18x - 72y + 13 = 0$ and $7x - my - 17 = 0$ have no solution?
 (a) 28 (b) 24 (c) 9 (d) 12

Q.388. If $2^x + 3^y = 17$ and $2^{x+2} - 3^{y+1} = 5$, then:
 (a) $x = 1, y = 3$ (b) $x = 3, y = 3$
 (c) $x = 3, y = 2$ (d) $x = 1, y = 2$

Q.389. If $x^2 - 5x + 1 = 0$, then the value of $\frac{x^6 + x^4 + x^2 + 1}{5x^3}$.
 (a) 30 (b) 25 (c) 23 (d) 28

Q.390. x is a negative number such that $k + k^{-1} = -2$, then what is the value of $\frac{k^2 + 4k - 2}{k^2 + k - 5}$?
 (a) 7 (b) 1 (c) - 7 (d) - 1

Q.391. If $(x + \frac{1}{x}) = 5\sqrt{2}$, then what is the value of $(x^4 + x^{-4})$?



(a) 2542 (b) 2650 (c) 2452 (d) 2302

Q.392. If $a + b = 11$ and $ab = 35$, then what is the value of $(a^4 + b^4)$?

(a) 151 (b) 261 (c) 124 (d) 102

Q.393. If $x + y = 36$, then find

$$(x - 27)^3 + (y - 9)^3.$$

(a) 1 (b) 81 (c) 2y (d) 0

Q.394. If $\frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$ and

$a + b = 5c$, where $c > 0$, then the value of c is _____.

(a) 8 (b) 4 (c) 12 (d) 5

Q.395. If $x = \frac{\sqrt{5} - 2}{\sqrt{5} + 2}$, then the value of

$$x^2 + x^{-2} \text{ is:}$$

(a) 350 (b) 345 (c) 284 (d) 322

Q.396. Simplify $\frac{x^2 + 2x + y^2}{x^3 - 5x^2}$ if $x + \frac{y^2}{x} = 5$

(a) $\frac{5}{y^2}$ (b) $\frac{7}{y^2}$ (c) $-\frac{5}{y^2}$ (d) $-\frac{7}{y^2}$

Q.397. If $x + \frac{1}{x} = 8$, then find the value of

$$\frac{5}{x^2 - 8x + 2}$$

(a) 3 (b) 4 (c) 0 (d) 5

Q.398. If $(4a + \frac{5}{a} + 5) = 14$, what is the

value of $(16a^2 + \frac{25}{a^2})$?

(a) 25 (b) 36 (c) 41 (d) 40

Q.399. If $x - \frac{1}{x} = 13$ then, what will be

the value of $x^4 + \frac{1}{x^4}$?

(a) 28561 (b) 29243 (c) 27887 (d) 29239

Q.400. The factors of $x^4 + x^2 + 25$ are:

(a) $(x^2 + 3x - 5)(x^2 - 3x + 5)$
 (b) $(x^2 + 3x + 5)(x^2 - 3x + 5)$
 (c) $(x^2 - 3x + 5)(x^2 - 3x + 5)$
 (d) $(x^2 + 3x + 5)(x^2 + 3x + 5)$

Q.401. If $x = 3 + 2\sqrt{2}$, $x > 0$, then the value

of $\sqrt{x} - \frac{1}{\sqrt{x}}$ is:

(a) 1 (b) $\sqrt{2}$ (c) 2 (d) $2\sqrt{2}$

Q.402. If $x + \frac{1}{x} = -2$, then what is the

value of $x^{17} + x^{-17} + x^{12} + x^{-12}$? ($x < 0$)

(a) -2 (b) -1 (c) 1 (d) 0

Q.403. $(mx + n)$ is a factor of:

(a) $m^2 x^2 + 2nx + n^2$ (b) $m^2 x^2 + 2mnx + n^2$
 (c) $m^2 x^2 + 2mx + n^2$ (d) $m^2 x^2 + 2mn + n^2$

Q.404. If $x + y + z = 0$, then what is the

value of $\frac{x^2}{(yz)} + \frac{y^2}{(xz)} + \frac{z^2}{(xy)}$?

(a) 1 (b) 0 (c) 2 (d) 3

Q.405. What is the possible value of $(a + b + c) - 3$, if $a^2 + b^2 + c^2 = 9$ and $ab + bc + ca = 8$?

(a) 5 (b) 3 (c) 9 (d) 2

Q.406. If $a - b = 2$ and $a^3 - b^3 = 80$, then what will be the value of ab ?

(a) 12 (b) -24 (c) 24 (d) -12

Q.407. If $\frac{1}{x^2 + a^2} = x^2 - a^2$, then the value of x is:

(a) $(1 - a^4)^{\frac{1}{4}}$ (b) a (c) $(a^4 - 1)^{\frac{1}{4}}$ (d) $(a^4 + 1)^{\frac{1}{4}}$

Q.408. Which of the following statement is correct?

I. The value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots + 22^2 - 21^2$ is 4840.

II. The value of $(k^2 + \frac{1}{k^2})(k - \frac{1}{k})(k^4 + \frac{1}{k^4})(k + \frac{1}{k})(k^4 - \frac{1}{k^4})$ is $k^{16} - \frac{1}{k^{16}}$.

(a) Neither I nor II (b) Only II
 (c) Only I (d) Both I and II

Q.409. Select the correct algebraic expression.

(a) $ab - a - b + 1 = (a - 1)(b - 1)$ (b) $ab + a - b + 1 = (1 - a)(1 - b)(1 - a)$ (c) $ab - a - b + 1 = (1 - a)(b - 1)$ (d) $ab - a - b + 1 = (a - 1)(1 - b)$ (a) $ab - a - b + 1 = (a - 1)(b - 1)$ (b) $ab + a - b + 1 = (1 - a)(1 - b)(1 - a)$ (c) $ab - a - b + 1 = (1 - a)(b - 1)$ (d) $ab - a - b + 1 = (a - 1)(1 - b)$

Q.410. If $x^2 + y^2 + z^2 = xy + yz + zx$ and $x = 1$, then find the value of

$$\frac{10x^4 + 5y^4 + 7z^4}{13x^2 y^2 + 6y^2 z^2 + 3z^2 x^2}.$$

(a) 2 (b) 0 (c) -1 (d) 1

Q.411. If $x^2 - 2xy = 84$ and $x - y = -10$, then the value of y is:

(a) 2 (b) 1 (c) 4 (d) 3

Q.412. If $x + \frac{1}{x} = 1$, then the value of

$$x^{12} + x^9 + x^6 + x^3 + 1 \text{ is:}$$

(a) 1 (b) -1 (c) 0 (d) -2

Q.413. If $\frac{a}{b} + \frac{b}{a} = 1$ and $a + b = 2$, then

the value of $a^3 + b^3$ is:

(a) 0 (b) 3 (c) 1 (d) 2

Q.414. If $4x^2 + y^2 = 40$ and $xy = 6$, find the positive value of $2x + y$.

(a) 8 (b) 6 (c) 5 (d) 4

Q.415. If $2a + 3b = 10$ and $ab = 3$, then find the value of $4a^2 + 9b^2$.

(a) 60 (b) 62 (c) 64 (d) 66

Q.416. If $(4x - 7y) = 11$ and $xy = 8$, what is the value of $16x^2 + 49y^2$, given that x and y are positive numbers?

(a) 596 (b) 484 (c) 569 (d) 448

Q.417. If $mx^m - nx^n = 0$, then what is the value of $\frac{1}{x^m + x^n} + \frac{1}{x^m - x^n}$ terms of x^n is:

Where $x, m, n > 0$ (a) $\frac{2mn}{\{(n^2 + m^2)\}}$ (b) $\frac{2mn}{\{x^n(m^2 - n^2)\}}$ (c) $\frac{-2mn}{\{x^n(m^2 + n^2)\}}$ (d) $\frac{2mn}{\{x^n(n^2 - m^2)\}}$

Q.418. If, for a non-zero x , $5x^2 + 7x + 5 = 0$ then the value of $x^3 + \frac{1}{x^3}$ is:

(a) $\frac{496}{125}$ (b) $\frac{532}{343}$ (c) $\frac{125}{532}$ (d) $\frac{182}{125}$

Q.419. If $x^2 + y^2 + 2y + 4x + 5 = 0$, then

$$\frac{(x + y)}{(x - y)} = \text{_____}.$$

(a) -3 (b) $\frac{1}{3}$ (c) -1 (d) 3

Q.420. The value of

$$\frac{(x - y)^3 + (y - z)^3 + (z - x)^3}{6(x - y)(y - z)(z - x)} \text{ where } x \neq$$

$y \neq z$ is equal to:

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{9}$

Q.421. If $k + \frac{1}{k} + 2 = 0$ and $K < 0$, then

what is the value of $k^{10} + \frac{1}{k^{11}}$?

(a) 1 (b) 0 (c) -1 (d) 2

Q.422. $3[a - \frac{1}{a}] + [a - \frac{1}{a}]^3 = ?$

(a) $a^2 - \frac{1}{a^3}$ (b) $a^3 - \frac{1}{a^3}$ (c) $a^3 + \frac{1}{a^3}$ (d) $a^2 - \frac{1}{a^2}$

Q.423. If $x^4 + \frac{1}{x^4} = 194$ and $x^3 + \frac{1}{x^3} = ?$

(a) 52 (b) 58 (c) 76 (d) 67

Q.424. $(a + 1)^2 + (a + 2)^2 = 16$ then what is the value of $40 + 12a + 4a^2$?

(a) 56 (b) 62 (c) 74 (d) 52

Q.425. Which of the following statements is correct?

I. If $K + \frac{1}{K} = 12$, then $K^2 + \frac{1}{K^2} = 142$

II. The value of $(K^2 + \frac{1}{K^2})(K - \frac{1}{K})$

$$(K^4 + \frac{1}{K^4})(K + \frac{1}{K}) \text{ is } K^{16} - \frac{1}{K^{16}}$$



- (a) Only I (b) Neither I nor II
(c) Both I and II (d) Only II

Q.426. Find the remainder when we divide $3x^4 - 2x^2 + 4x - 1$ by $2x - 1$.

- (a) 2 (b) 3 (c) $\frac{11}{16}$ (d) $\frac{15}{16}$

Q.427. What is the value of $(a + b)^2 - (a - b)^2$?

- (a) $2(a^2 + b^2)$ (b) $4(a^2 + b^2)$ (c) $8ab$ (d) $4ab$

Q.428. If m and n are two positive real numbers such that $9m^2 + n^2 = 40$ and $mn = 4$, then the value of $3m + n$ is:

- (a) 160 (b) 64 (c) 10 (d) 8

Q.429. The factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$ are:

- (a) $(x - 2y - 4)(x - 2y + 2)$
(b) $(x^2 - 2y - 4)(x^2 - 2y + 2)$
(c) $(x + 2y - 4)(x + 2y + 2)$
(d) $(x^2 - 2y - 4)(x^2 + 2y + 2)$

Q.430. If $y = 1 + \sqrt{3} + \sqrt{4}$, then the value of $2y^4 - 8y^3 - 6y^2 + 28y - 84$ is:

- (a) $40\sqrt{3}$ (b) $80\sqrt{3}$ (c) $20\sqrt{3}$ (d) $60\sqrt{3}$

Q.431. If $x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}$ and $y = \frac{\sqrt{5} + \sqrt{4}}{\sqrt{5} - \sqrt{4}}$

then the value of $\frac{x^2 - xy + y^2}{x^2 + xy + y^2}$?

- (a) $\frac{361}{363}$ (b) $\frac{341}{343}$ (c) $\frac{384}{387}$ (d) $\frac{321}{323}$

Q.432. If $\frac{2p}{p^2 - 5p + 1} = \frac{1}{10}$, $p \neq 0$, then

the value of $(p + \frac{1}{p})$ is:

- (a) 10 (b) 25 (c) 1 (d) 15

Q.433. Which of the following will satisfy $a^2 = b^2 + (ab)^2$ for the values a and b ?

- (a) $a = \sin x$, $b = \cot x$
(b) $a = \cos x$, $b = \tan x$
(c) $a = \cot x$, $b = \cos x$
(d) $a = \sin x$, $b = \tan x$

Q.434. If $abc = 5$, what is the value of

$$\left(\frac{1}{1+a+b^{-1}} + \frac{1}{1+b+5c^{-1}} + \frac{1}{1+\frac{c}{5}+a^{-1}} \right)?$$

- (a) 5 (b) 1 (c) $\frac{1}{5}$ (d) $(a + b + c)$

Q.435. If $4(z + 7)(2z - 1) = Az^2 + Bz + C$, then the value of $A + B + C$ is:

- (a) 28 (b) 32 (c) 74 (d) 81

Q.436. Factorize the following expression: $p^3 + 27$

- (a) $(p - 3)(p^2 - 3p + 9)$
(b) $(p + 3)(p^2 + 3p + 9)$
(c) $(p + 3)(p^2 - 3p + 9)$

(d) $(p + 3)(p^2 - 3p - 9)$

Q.437. What is the value of $a^2 + b^2 + c^2$, if $a + b + c = 9$ and $ab + bc + ca = 23$?

- (a) 22 (b) 32 (c) 49 (d) 35

Q.438. If $x + \frac{1}{x} = 2\sqrt{5}$ where $x > 1$, then

the value of $x^3 - \frac{1}{x^3}$ is:

- (a) 82 (b) 76 (c) 86 (d) 78

Q.439. If $x + y + z = 0$, then the value of

$$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$$
 is:

- (a) 3 (b) 1 (c) 0 (d) 2

Q.440. What is the value of $(a + b)^3 - a^3 + b^3$?

- (a) $-3ab(a - b) + 2b^3$ (b) $3ab(a + b) + 2b^3$
(c) $3ab(a - b)b^3$ (d) $-3ab(a + b) + b^3$

Q.441. If $x^2 + \frac{1}{x^2} = 66$, the value of $x - \frac{1}{x}$ is.

- (a) 10 (b) 8 (c) 9 (d) 6

Q.442. If $P - \frac{1}{P} = 6$, then what is the

value of $P^4 + \frac{1}{P^4}$?

- (a) 1562 (b) 1432 (c) 1442 (d) 1444

Q.443. Simplify $x^4 - 15x^3 + 15x^2 - 15x + 40$; given $x = 14$.

- (a) 0 (b) 40 (c) 14 (d) 26

Q.444. If $y + \frac{1}{y} = 11$, then the value of $y^3 - \frac{1}{y^3}$ is:

- (a) $345\sqrt{13}$ (b) $360\sqrt{13}$
(c) $352\sqrt{13}$ (d) $368\sqrt{13}$

Q.445. If $(x - \frac{1}{x})^2 = 12$, what is the value

of $(x^2 - \frac{1}{x^2})$, given that $x > 0$?

- (a) $6\sqrt{2}$ (b) $8\sqrt{3}$ (c) $6\sqrt{3}$ (d) $8\sqrt{2}$

Q.446. $a^3 + b^3 + c^3 = 3abc$ if _____.

- (a) $a + b + c = -3$ (b) $a + b + c = 0$
(c) $a + b + c = 1$ (d) $a + b + c = -1$

Q.447. If $m + \frac{1}{m-2} = 4$, then find the

value of $(m - 2)^2 + \frac{1}{(m - 2)^2}$.

- (a) -2 (b) 4 (c) 0 (d) 2

Q.448. If $5x - \frac{5}{x} + 6 = 0$, then $x^2 + \frac{1}{x^2}$ is

- (a) $\frac{86}{25}$ (b) $\frac{43}{12}$ (c) $\frac{81}{10}$ (d) $\frac{86}{11}$

Q.449. If $a = 26$ and $b = 22$, then the

value of $\frac{a^3 - b^3}{a^2 - b^2} - \frac{3ab}{a + b}$ is _____.

- (a) $\frac{5}{3}$ (b) $\frac{13}{11}$ (c) $\frac{1}{3}$ (d) $\frac{11}{13}$

Q.450. What is the value of $a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$?

- (a) $(2a + b + c)^2$ (b) $(a - b + c)^2$
(c) $(a - b - 2c)^2$ (d) $(a + 2b - c)^2$

Q.451. If $\frac{x}{8} + \frac{8}{x} = 1$, then the value of x^3 is:

- (a) -256 (b) -512 (c) 256 (d) 512

Q.452. If $p = 8.15$, $q = 9.06$ and $r = -17.21$, then the value of $p^3 + q^3 + r^3 - 3pqr$ is:

- (a) -3.81 (b) -5.62 (c) 4.75 (d) 0

Q.453. If $x^4 + \frac{16}{x^4} = 15617$, $x > 0$, then

find the value of $x + \frac{2}{x}$.

- (a) $\sqrt{121}$ (b) $\sqrt{129}$ (c) $\sqrt{123}$ (d) $\sqrt{127}$

Q.454. If $x + y + z = 0$ and $x^2 + y^2 + z^2 = 40$, then what is the value of $xy + yz + zx$?

- (a) -20 (b) 5 (c) -5 (d) -10

Q.455. Which of the following statement is correct?

I. If $x = 12$, $y = -2$ and $z = -10$, then $x^3 + y^3 + z^3 = 360$.

II. If $x + y = 48$ and $4xy = 128$, then $4x^2 + 4y^2 = 4480$.

- (a) Neither I nor II (b) Only I
(c) Both I and II (d) Only II

Q.456. if $x + \frac{1}{x} = 6$, then find the value of

$$\frac{3x}{2x^2 - 5x + 2}.$$

- (a) 1 (b) $\frac{3}{7}$ (c) $\frac{2}{3}$ (d) 0

Q.457. If $2p + q = 19$ and $8p^3 + q^3 = 361$, then find the value of pq .

- (a) 56 (b) 59 (c) 58 (d) 57

Q.458. If $\frac{a}{b} + \frac{b}{a} = -1$ and $a - b = 2$, then the value of $a^3 - b^3$ is:

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) -1

Q.459. If $x = 2 - 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$ then find the value of $x^3 - 6x^2 + 18x$.

- (a) 40 (b) 33 (c) 45 (d) 22

Q.460. If $a = -12$, $b = -6$ and $c = 18$, then what is the value of $\frac{2abc}{9}$.

- (a) 554 (b) 288 (c) -288 (d) 144



Q.461. If $x^4 + x^{-4} = 47$, $x > 0$, then what is the value of $x + \frac{1}{x} - 2$?

- (a) 1 (b) 0 (c) 5 (d) 3

Q.462. The value of

$\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x^2-y^2)^3 + (y^2-z^2)^3 + (z^2-x^2)^3}$ where $x \neq y \neq z$, is:

- (a) 0 (b) $\frac{1}{(x+y+z)}$
(c) $\frac{1}{(x+y)(y+z)(z+x)}$ (d) 1

Q.463. If $a + b + c = 10$; $a^2 + b^2 + c^2 = 38$, what is the value of $(a-b)^2 + (b-c)^2 + (c-a)^2$?

- (a) 15 (b) 12 (c) 14 (d) 13

Q.464. If $16y^2 - k = (4y + \frac{3}{2})(4y - \frac{3}{2})$, then the value of k is:

- (a) $\frac{9}{4}$ (b) $\frac{11}{4}$ (c) $\frac{6}{4}$ (d) $\frac{7}{4}$

Q.465. The value of

$\frac{\{(m^2 + n^2)(m - n) - (m - n)^3\}}{(m^2n - mn^2)}$ is:

- (a) $m + n$ (b) $m - n$ (c) 2 (d) $\frac{m}{n}$

Q.466. If $x + y + z = 8$, and $x^2 + y^2 + z^2 = 20$ then the value of $x^3 + y^3 + z^3 - 3xyz$ is

- (a) 16 (b) 10 (c) 15 (d) -16

Q.467. If $\{(1 \div x) \times (x^3 - 2x^2) + (4^3 \div 2^5) \times x\} \div 25 = 1$, then the possible values of x is/are:

- (a) -5 (b) 5 (c) 0 (d) Both 5 and -5

Q.468. The simplified form of $(7x + 4y)^2 + (7x - 4y)^2$ is:

- (a) $98x^2 + 32y^2$ (b) $98x^2 + 32y^2 - 5xy$
(c) $32x^2 + 98y^2$ (d) $98x^2 - 32y^2$

Q.469. What are the values of x and y , respectively, from the following equations?

$$6x + 7y = 5xy$$

$$10y - 4x = 4xy$$

- (a) 3 and 4 (b) 4 and 5
(c) 2 and 4 (d) 2 and 5

Q.470. If $x + y = 10$, $2xy = 48$ and $x > y$, then find $2x - y$.

- (a) 6 (b) 8 (c) 4 (d) 3

Q.471. If $a^2 + b^2 + 1 = 2a$, then the value of $a^4 + b^7$ is:

- (a) 1 (b) 0 (c) 2 (d) 4

Q.472. If a, b, c are all non-zero and $a + b + c = 0$, find the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?

- (a) 3 (b) 4 (c) 1 (d) $\frac{1}{2}$

Q.473. Simplify: $(x + y)^3 - (x - y)^3 - 6y(x^2 - y^2)$

- (a) $8y^3$ (b) x^3 (c) $8x^3$ (d) y^3

Q.474. If $x^2 + 6x + 1 = 0$ then the value of

$$(x + 6)^3 + \frac{1}{(x + 6)^3} = ?$$

- (a) 245 (b) 216 (c) 186 (d) 198

Q.475. If a, b and c are positive integers such that $a^2 + b^2 = 82$ and $b^2 + c^2 = 65$, then the value of $2a + 7b - 3c$ is:

- (a) 2 (b) 5 (c) 49 (d) 1

Q.476. If $x(x - 5) = -1$, then the value of $x^3(x^3 - 110) = ?$

- (a) 0 (b) -1 (c) 1 (d) 2

Q.477. If $a + \frac{1}{a} = p^2$, then find $a^2 + \frac{1}{a^2}$.

- (a) $p^2 - 2$ (b) $p^2 + 2$ (c) $p^4 - 2$ (d) $p^4 + 2$

Q.478. If $a = -9.6$, $b = 4.44$ and $c = 5.16$, then the value of $a^3 + b^3 + c^3 - 3abc$ is:

- (a) 0 (b) -1 (c) 2 (d) 1

Q.479. If $\sqrt{1 + \frac{x}{529}} = \frac{24}{23}$, then the value of x is:

- (a) 15 (b) 27 (c) 47 (d) 30

Q.480. If the equation $k(21x^2 + 24) + rx + (14x^2 - 9) = 0$ and $k(7x^2 + 8) + px + (2x^2 - 3) = 0$ have both roots common, then

the value of $\frac{p}{r}$ is:

- (a) $\frac{1}{3}$ (b) $\frac{2}{5}$ (c) $\frac{4}{3}$ (d) $\frac{7}{5}$

Q.481. What is the value of $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \div$

$$\frac{\sqrt{14} + \sqrt{10}}{\sqrt{14} - \sqrt{10}} + \frac{\sqrt{10}}{\sqrt{5}} ?$$

- (a) $\sqrt{2} + 2$ (b) $2\sqrt{2} + 2$
(c) $\sqrt{2} + 1$ (d) $2\sqrt{2} + 1$

Q.482. If $(x + \frac{1}{x})^2 = 3$, then what is the value of $x^6 + x^{-6}$?

- (a) 6 (b) 2 (c) -6 (d) -2

Q.483. What is the solution of the following equations?

$$2X + 3Y = 12 \text{ and } 3X - 2Y = 5$$

- (a) $X = 3, Y = 2$ (b) $X = 2, Y = 3$
(c) $X = -2, Y = 3$ (d) $X = 3, Y = -2$

Q.484. If $xy = -6$ and $x^3 + y^3 = 19$ (x and y are integers), then what is the value of

$$\frac{1}{x^{-1}} + \frac{1}{y^{-1}} ?$$

- (a) -1 (b) -2 (c) 1 (d) 2

Q.485. if $2^x = 3^y = 6^z$, then $\frac{1}{x} + \frac{1}{y} - \frac{1}{z}$

is equal to:

- (a) 1 (b) $-\frac{1}{2}$ (c) 0 (d) $\frac{3}{2}$

Q.486. If $(a + b) = \sqrt{11}$ and $(a - b) = \sqrt{7}$ then find the value of $a^2b^2(a^2 + b^2)$.

- (a) 11 (b) 7 (c) 5 (d) 9

Q.487. If arithmetic and geometric mean of x and y is 8 and $3\sqrt{7}$, respectively, then the value of $x^3 + y^3$ is:

- (a) 1072 (b) 945 (c) 559 (d) 855

Q.488. The algebraic expression $25x^2 + 16y^2 + 9z^2 - 40xy + 24yz - 30zx$ is equal to:

- (a) $(4y + 3z - 5x)(4x + 5y - 3z)$
(b) $(4y - 3z - 5x)(4x + 5y + 3z)$
(c) $(4y + 3z - 5x)^2$
(d) $(4y - 3z - 5x)^2$

Q.489. If $xy = 6$ and $x^2y + xy^2 + x + y = 63$, then the value of $x^2 + y^2$ is:

- (a) 69 (b) 59 (c) 70 (d) 79

Q.490. Factorize the given algebraic expression. $x^3 + 27y^3 + 64z^3 - 36xyz$

- (a) $(x - 3y - 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$
(b) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 3xy - 12yz - 4xz)$
(c) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 - 12xy - 3yz - 4xz)$
(d) $(x + 3y + 4z)(x^2 + 9y^2 + 16z^2 + 3xy + 12yz + 4xz)$

Q.491. Simplify $(5x + 4y)^2 + (5x - 4y)^2$

- (a) $32x^2 + 50y^2$ (b) $50x^2 + 32y^2$
(c) $-80xy$ (d) $80xy$

Q.492. What must be subtracted from $x^3 - 3x^2y + 3xy^2 - y^3$ to get $x^3 - y^3$?

- (a) $-3x^2y + 3xy^2$ (b) $3x^2y - 3xy^2$
(c) $x^3 - y^3$ (d) $-x^3 + y^3$

Q.493. If $a - b = 10$ and $ab = 4$, then the value of $a^3 - b^3 + 4(a + b)^2$ is:

- (a) 1500 (b) 1584 (c) 1280 (d) 1623

Q.494. Find the value of $a^2 + b^2 + c^2 - 2ab + 2ac - 2bc$, if $a = x + y$, $b = x - y$ and $c = 2x - 1$.

- (a) $(x - y - 1)^2$ (b) 0
(c) $(2x + 2y - 1)^2$ (d) $(2x - 2y - 1)^2$

Q.495. If $y + \frac{1}{y} = \sqrt{5}$, then the value of $y^6 + \frac{1}{y^6}$ is:

- (a) 18 (b) 15 (c) 10 (d) 12

Q.496. If $P = \frac{x^2 - 100}{x^2 - 16}$ and $Q = \frac{x + 10}{x - 4}$,

then the value of $\frac{P}{Q}$ is:



(a) $\frac{x+10}{x+4}$ (b) $\frac{x-10}{x+4}$
 (c) $\frac{x+10}{x-4}$ (d) $\frac{x-10}{x-4}$

Q.497. If $\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{2a-3b+5c}{k}$, then find the value of K.
 (a) 15 (b) 18 (c) 10 (d) 12

Q.498. if $\frac{x-1}{x} = 5$, then the value of $1 + \frac{1}{x^2}$ is:
 (a) 16 (b) 18 (c) 17 (d) 27

Q.499. If $2x + \frac{1}{3x} = 12$, find $9x^2 + \frac{1}{4x^2}$.
 (a) 142 (b) 324 (c) 321 (d) 327

Q.500. What is the value of x .
 if $\frac{4}{x} - \frac{3}{2x} = 20$?
 (a) $\frac{1}{4}$ (b) 8 (c) $\frac{1}{8}$ (d) 4

Q.501. If $x - y = 5$ and $x^3 - y^3 = 520$, then the value of xy is :
 (a) $\frac{79}{3}$ (b) $\frac{36}{5}$ (c) $\frac{34}{3}$ (d) $\frac{76}{5}$

Q.502. If $x + y + z = 0$, then what will be the value of $(\frac{x^2}{yz}) + (\frac{y^2}{zx}) + (\frac{z^2}{xy})$?
 (a) $\frac{3(x^2 + y^2 + z^2)}{xyz}$ (b) $x^2 + y^2 + z^2$
 (c) $\frac{x^2 y^2 z^2}{x}$ (d) 3

Q.503. If $2a + b = 10$, $2ab = 9$, and $2a > b$, then find the value of $2a - b$.
 (a) 12 (b) 6 (c) 8 (d) 10

Q.504. If $a + b = 12$ and $ab = 35$, find the value of $a^4 + b^4 - 10^3$?
 (a) 2226 (b) 2026 (c) 2126 (d) 1226

Q.505. If $a + 2b = 27$ and $a^3 + 8b^3 = 5427$, then find the value of $2ab$
 (a) 176 (b) 156 (c) 172 (d) 149

Q.506. If $x^2 = b - ax$ and $x^3(x^3 + c) = d$, then which of the following can be the values of c and d ?
 (a) $c = a^3 - 3ab$ and $d = b^3$
 (b) $c = -a^3 - 3ab$ and $d = b^3$
 (c) $c = a^3 + 3ab$ and $d = b^3$
 (d) $c = a^3 + 3ab$ and $d = -b^3$

Q.507. If $x + y + z = 0$, then find the value of $x^3 + y^3 + z^3 - 3xyz$.
 (a) 1 (b) $x^3 + y^3 + z^3$ (c) 0 (d) $3xyz$

Q.508. If $r + \frac{64}{r} = 16$, then the value of

$r^4 + \frac{1}{r^3}$ is _____.

(a) $4096\frac{1}{512}$ (b) 512
 (c) $512\frac{1}{4096}$ (d) 4096

Q.509. If $p = 11$, then the value of $p(p^2 + 3p + 3)$ is:
 (a) 1629 (b) 1225 (c) 1727 (d) 1111

Q.510. If $\frac{X^2}{y^2} + \frac{y^2}{X^2} = 7$, A possible of $\frac{X^3}{y^3} + \frac{y^3}{X^3}$ is:
 (a) 18 (b) 15 (c) 16 (d) 17

Q.511. If $a^2 - 4a + 1 = 0$, then the value of $a^2 + a + \frac{1}{a} + \frac{1}{a^2}$ is :
 (a) 10 (b) 1 (c) 18 (d) 16

Q.512. If $(x - \frac{1}{x}) = \frac{7}{3}$, what is the value of $(x^3 - \frac{1}{x^3})$?
 (a) $19\frac{20}{27}$ (b) $19\frac{19}{27}$ (c) $19\frac{7}{9}$ (d) $19\frac{2}{3}$

Q.513. If $(x + \frac{1}{x}) = \frac{11}{5}$, what is the value of $(x^3 + \frac{1}{x^3})$?
 (a) $4\frac{6}{125}$ (b) $5\frac{101}{125}$
 (c) $10\frac{81}{125}$ (d) $17\frac{31}{125}$

Q.514. If $1 + x^2 + x^4 = 585$ and $1 - x + x^2 = 65$, then find the value of $1 + x + x^2$.
 (a) 8 (b) 7 (c) 6 (d) 9

Q.515. If $a = 2022$, $b = 2021$ and $c = 2020$, then value of $a^2 + b^2 + c^2 - ab - bc - ca$ is :
 (a) 2 (b) 4 (c) 3 (d) 1

Q.516. If $x^2 + 4y^2 = 40$, $xy = 6$, and $x > 2y$, then the value of $x - 2y$ is:
 (a) 1 (b) 3 (c) 2 (d) 4

Q.517. If $x^2 - 8x + 1 = 0$, what is the value of $x^8 - 3842x^4 + 1$?
 (a) -1 (b) 0 (c) 2 (d) 1

Q.518. $144x^2 - 36x + \frac{9}{4}$ can be expressed as the square of _____.
 (a) $14x - \frac{3}{2}$ (b) $12x - \frac{9}{4}$
 (c) $12x - \frac{3}{2}$ (d) $12x - 9$

Q.519. The sum of two numbers is 17 and their product is 34. Find the sum of their squares.

(a) 224 (b) 221 (c) 225 (d) 222

Q.520. If $8x^3 + 27y^3 + 64z^3 = 72xyz$, then the relation between x, y and z can be :
 (a) $2x + 3y = -4z$ (b) $2x + y + z = 0$
 (c) $2x - 3y + 4z = 0$ (d) $2x + 3y = 4z$

Q.521. The value of $\frac{[(5.4)^3 - 0.064]}{[(5.4)^2 + 2.16 + 0.16]}$ is _____.
 (a) 5 (b) 7 (c) 3 (d) 1

Q.522. If $x = 9$, then the value of $x^5 - 10x^4 + 10x^3 - 10x^2 + 10x - 1$ is :
 (a) 10 (b) 8 (c) 9 (d) 1

Q.523. If $p^3 + 3p^2 + 3p = 26$, then the value of $p^2 + 2p$ is ?
 (a) 8 (b) 12 (c) 10 (d) 15

Q.524. If $\frac{a}{b} + \frac{b}{a} = -1$, $a \neq 0$, $b \neq 0$, then the value of $6(a^3 - b^3)$ is:
 (a) 6 (b) 3 (c) 0 (d) 1

Q.525. If $p^4 = 4354 - \frac{1}{p^4}$, then the value of $p^3 - \frac{1}{p^3}$ can be :
 (a) 536 (b) 436 (c) 416 (d) 516

Q.526. If $x + y = 14$, then the value of $x^3 + y^3 + 42xy$ is:
 (a) 2744 (b) 2644 (c) 2742 (d) 2714

Q.527. If $x = 5t$ and $y = \frac{1}{3}(t + 1)$, then the value of t for which $x = 3y$ is:
 (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$

Q.528. If $x > 0$ and $x^4 + \frac{1}{x^4} = 2207$, what is the value of $(x^5 + \frac{1}{x^5})$?
 (a) 15127 (b) 15134 (c) 15141 (d) 15130

Q.529. $(2x + 3y + 4z)(4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz) = ?$
 (a) $8x^3 + 27y^3 + 64z^3 + 72xyz$
 (b) $8x^3 + 27y^3 + 64z^3 - 72xyz$
 (c) $8x^3 + 27y^3 - 64z^3 + 72xyz$
 (d) $8x^3 - 27y^3 - 64z^3 - 72xyz$

Q.530. Simplify : $3ab \times (a + b)^{-1} \times (a^{-1} + b^{-1})$
 (a) $\frac{1}{(a + b)}$ (b) 1 (c) $(a + b)$ (d) 3

Q.531. If $x^4 + \frac{1}{x^4} = 14159$, then a possible value of $x + \frac{1}{x}$ is:
 (a) 69 (b) 121 (c) 81 (d) 11



Q.532. If $8x^2 + 9x + 8 = 0$, then the value

of $x^3 + \frac{1}{x^3}$ is :

- (a) $\frac{999}{212}$ (b) $\frac{199}{212}$ (c) $\frac{999}{512}$ (d) $\frac{199}{512}$

Q.533. Expand $x^2 + 2x + 3$ about $x = -2$

- (a) $(x - 2)^2 - 2(x + 2) + 3$
 (b) $(x + 2)^2 - 2(x + 2) + 3$
 (c) $(x + 2)^2 + 2(x + 2) + 3$
 (d) $(x - 2)^2 - 2(x - 2) - 3$

Q.534. If $a + b = 2c$ then the value of

$\frac{a}{a-c} + \frac{c}{b-c}$ is.

- (a) $\frac{1}{2}$ (b) 1 (c) 0 (d) -1

Q.535. If $x + y + z = 25$, $x^3 + y^3 + z^3 = 85$ and $xyz = 20$, then find the value of $x^2 + y^2 + z^2 - xy - yz - zx$.

- (a) 2 (b) 1 (c) 4 (d) 3

Q.536. Which among the below options can be a square root of?

$(x^2 - 14x + 49)(x^2 + 6x + 9)$

- (a) $(x - 4)(x + 9)$ (b) $(x - 1)(x + 17)$
 (c) $(x - 7)(x + 3)$ (d) $(x - 3)(x + 8)$

Q.537. If $(x^3 + y^3)^{\frac{2}{3}} = (x^2 + y^2)$, then

the value of $\frac{x^2 + y^2}{xy} x \neq 0, y \neq 0$, is:

- (a) $\frac{2}{3}$ (b) $x^2 + y$ (c) 1 (d) $x + y^2$

Q.538. The factors of $a^2 - 1 - 2x - x^2$ are

- (a) $(a - x - 1)(a - x - 1)$
 (b) $(a + 1 + x)(a - 1 - x)$
 (c) $(a - x + 1)(a - x + 1)$
 (d) $(a - x + 1)(a - x - 1)$

Q.539. If $x - 2y = 3$ and $2xy = 5$, then find

the value of $(x + 2y)^2$

- (a) 29 (b) 27 (c) 30 (d) 28

Q.540. If $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$, $a \neq 0$,

$b \neq 0$, then the value of $a^4 + b^4$ is:

- (a) 32 (b) 256 (c) 64 (d) 2

Q.541. If $a + b + c = 0$, then find the value

of $\frac{a^2 + b^2 + c^2}{a^2 - bc}$

- (a) -1 (b) 1 (c) 2 (d) -2

Q.542. If $a - b = 3$ and $a^3 - b^3 = 999$, then

find the value of $a^2 - b^2$

- (a) 60 (b) 62 (c) 64 (d) 63

Q.543. If $(a - 1)^2 + (b + 2)^2 + (c - 1)^2$

$= 0$, then the value of $a^2 + b^2 + c^2$

- (a) 6 (b) 0 (c) 2 (d) 1

Q.544. $16a^2 + 4b^2 + 9c^2 - 16ab + 12bc - 24ca = ?$

- (a) $(4a - 2b + 3c)^2$ (b) $(4a - 2b - 3c)^2$
 (c) $(4a + 2b - 3c)^2$ (d) $(4a + 2b + 3c)^2$

Q.545. If $p = 38$, then the value of $p(p^2 + 3p + 3)$ is _____.

- (a) 39313 (b) 59319 (c) 39318 (d) 59318

Q.546. If $a + b = \sqrt{7}$ and $a - b = \sqrt{5}$, then

the value of $8ab(a^2 + b^2) - (a - b)^2$

- (a) 19 (b) 23 (c) 27 (d) 21

Q.547. If $P(x) = (x^3 - 8)(x + 1)$ and $Q(x) = (x^3 + 1)(x - 2)$ the LCM of $P(x)$ and $Q(x)$ is:

- (a) $(x^2 + 2x + 4)(x^2 + 4x + 1)$
 (b) $(x + 1)(x - 2)(x^2 + 2x + 4)(x^2 - x + 1)$
 (c) $(x + 1)^2(x - 2)^2(x^2 + 2x + 4)(x^2 + 4x + 1)$
 (d) $(x - 2)(x + 1)$

Q.548. If $x + y + z = 0$, then find the value

of $\frac{x^2}{x^2 - yz} + \frac{y^2}{y^2 - xz} + \frac{z^2}{z^2 - xy}$

- (a) 2 (b) 1 (c) 3 (d) 0

Q.549. If $x = \sqrt{10} + \sqrt{11}$, $y = \sqrt{10} - \sqrt{11}$

, then value of $7x^2 - 50xy + 7y^2$ _____.

- (a) 344 (b) 704 (c) 1360 (d) 386

Q.550. What is the value of the following expression?

$2^2 \left(\frac{x^a}{x^b}\right)^{(a+b)} \times 3^2 \left(\frac{x^b}{x^c}\right)^{(b+c)} \times 6^{-2} \left(\frac{x^c}{x^a}\right)^{(a+c)}$

- (a) 1 (b) 0 (c) 4 (d) 9

Q.551. If $x - \frac{1}{x} = 11$ and $x > 0$, what is

the value of $(x^2 - \frac{1}{x^2})$?

- (a) $-11\sqrt{123}$ (b) $55\sqrt{5}$
 (c) $11\sqrt{123}$ (d) $-55\sqrt{5}$

Q.552. If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the values of p and q are:

- (a) $p = -\frac{3}{4}$ and $q = 4$
 (b) $p = \frac{3}{4}$ and $q = 4$
 (c) $p = \frac{3}{4}$ and $q = -4$
 (d) $p = -\frac{3}{4}$ and $q = -4$

Q.553. If $x^3 + y^3 = 416$ and $x + y = 8$, then find $x^4 + y^4$

- (a) 3002 (b) 3204 (c) 3004 (d) 3104

Q.554. If $2z = x + y$, then the value of

$\frac{x}{x-z} + \frac{y}{y-z}$

- (a) 0 (b) 1 (c) 2 (d) 5

Q.555. If $(ab + bc + ca) = 0$, then what is the value of

$(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ac} + \frac{1}{c^2 - ab})$?

- (a) 2 (b) 0 (c) 1 (d) $a + b + c$

Q.556. If $(x + 2)$ and $(x - 3)$ are the factors of $x^2 + k_1x + k_2$, then :

- (a) $k_1 = 1$ and $k_2 = -6$ (b) $k_1 = -1$ and $k_2 = -6$
 (c) $k_1 = -1$ and $k_2 = 6$ (d) $k_1 = 1$ and $k_2 = 6$

Q.557. If $k + \frac{1}{k} = 2$, find the value of

$8k \times k \times k$

- (a) 8 (b) 1 (c) 4 (d) 2

Q.558. For what value(s) of k will the expression $p + \frac{1}{9}\sqrt{p} + k^2$ be a perfect square?

- (a) $k = \pm \frac{1}{8}$ (b) $k = \pm \frac{1}{9}$
 (c) $k = \pm \frac{1}{21}$ (d) $k = \pm \frac{1}{18}$

Q.559. If $x + y + 3 = 0$, then find the value of $x^3 + y^3 - 9xy + 9$.

- (a) -18 (b) -36 (c) 18 (d) 36

Q.560. If $(4x + 2y)^3 + (4x - 2y)^3 = 16$

$(Ax^3 + Bxy^2)$, then what is the value of

$\frac{1}{2}(\sqrt{A^2 + B^2})$?

- (a) 8 (b) 3 (c) 5 (d) 7

Q.561. If $x = 4 + \sqrt{15}$, what is the value of

$(x^2 + \frac{1}{x^2})$?

- (a) 48 (b) 54 (c) 72 (d) 62

Q.562. If $x + y + z = 11$, $xy + yz + zx = -6$,

and $x^3 + y^3 + z^3 = 1604$, then the value of xyz is:

- (a) 25 (b) 4 (c) 1 (d) 5

Q.563. If $a^2 + b^2 + 49c^2 + 18 = 2(b - 28c - a)$ then the value of $(a + b - 7c)$ is

- (a) 4 (b) 3 (c) 2 (d) 1

Q.564. If $8k^6 + 15k^3 - 2 = 0$, then the

positive value of $(k + \frac{1}{k})$ is:

- (a) $2\frac{1}{2}$ (b) $2\frac{1}{8}$ (c) $8\frac{1}{2}$ (d) $8\frac{1}{8}$

Q.565. If $x - y + z = 0$ then find the value

of $\frac{y^2}{2xz} - \frac{x^2}{2yz} - \frac{z^2}{2xy}$

- (a) $\frac{3}{2}$ (b) $\frac{1}{2}$ (c) -6 (d) $-\frac{3}{2}$



Q.566. If $2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)(Ax^2 - Bxy + Cy^2)$, then the value of $(A^2 + B^2 + C^2)$ is:

- (a) 16 (b) 11 (c) 19 (d) 18

Q.567. If $x + y + z = 18$, $xyz = 81$ and $xy + yz + zx = 90$, then find the value

of $\sqrt[4]{x^3 + y^3 + z^3 + xyz}$

- (a) 6 (b) 12 (c) 9 (d) 10

Q.568. If $a + b - c = 5$ and $ab - bc - ac = 10$, then find the value of $a^2 + b^2 + c^2$

- (a) 40 (b) 5 (c) 45 (d) 15

Q.569. If $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{3}$, then what is the

value of $x^4 + \frac{1}{x^4}$?

- (a) 531 (b) 721 (c) 623 (d) 527

Q.570. If $a^3 + b^3 = 218$ and $a + b = 2$, then the value of $\sqrt{1 - ab}$ is:

- (a) 5 (b) 3 (c) 4 (d) 6

Q.571. If $x^2 - 5x - 1 = 0$, what is the value of $\frac{x^6 - x^4 + x^2 - 1}{x^3}$?

- (a) 145 (b) 140 (c) 130 (d) 135

Q.572. If $(x + y)^3 - (x - y)^3 - 3y(2x^2 - 3y^2) = ky^3$ then find the value of k

- (a) 10.5 (b) 8 (c) 11 (d) 10

Q.573. If $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{5}$, $x \neq 0$ then what

is the value of $\frac{(x^4 + \frac{1}{x^2})}{(x^2 + 1)}$.

- (a) 42 (b) 44 (c) 48 (d) 46

Q.574. If $a + b + c = 6$, $a^2 + b^2 + c^2 = 32$, and $a^3 + b^3 + c^3 = 189$, then the value of $4abc$ is:

- (a) 12 (b) 16 (c) 9 (d) 8

Q.575. If $x^4 - 79x^2 + 1 = 0$, then a value of $x + x^{-1}$ can be:

- (a) 9 (b) 5 (c) 7 (d) 8

Q.576. If $4x - 3y = 12$ and $xy = 5$, the find

the value of $\frac{16x^2 + 9y^2}{8}$

- (a) 33 (b) 18 (c) 3 (d) 44

Q.577. If $(a + \frac{1}{a} + 3)^2 = 16$, where a is a non-zero real number, then find the value of $a^2 + \frac{1}{a^2}$.

- (a) 3 (b) 47 (c) 49 (d) 7

Q.578. If $a + b + c = 1$, $ab + bc + ca = -22$

and $abc = -40$, then what is the value of $a^3 + b^3 + c^3$?

- (a) -51 (b) 27 (c) 67 (d) -53

Q.579. If $2x^2 + 5x + 1 = 0$, then one of the values of $x - \frac{1}{2x}$ is:

- (a) $\frac{\sqrt{13}}{2}$ (b) $\frac{\sqrt{17}}{2}$ (c) $\frac{5}{2}$ (d) $\frac{13}{2}$

Q.580. $\frac{22\sqrt{2}}{4\sqrt{2} - \sqrt{3} + \sqrt{5}} = a + \sqrt{5}b$, with a ,

$b > 0$, then what is the value of $(ab) : (a + b)$?

- (a) 7 : 8 (b) 4 : 7 (c) 7 : 4 (d) 8 : 7

Q.581. If $2x - y = 2$ and $xy = \frac{3}{2}$, then what

is the value of $x^3 - \frac{y^3}{8}$?

- (a) $\frac{13}{4}$ (b) $-\frac{5}{4}$ (c) $\frac{9}{2}$ (d) $\frac{5}{2}$

Q.582. If $x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$

then the value of $\frac{\sqrt{3} - x}{\sqrt{3} + x}$ is:

(corrected to two decimal places)

- (a) 0.25 (b) 0.19 (c) 0.27 (d) 0.17

Q.583. If $x^2 - \sqrt{7}x + 1 = 0$, then what is the value of $x^5 + \frac{1}{x^5}$?

- (a) $25\sqrt{7}$ (b) $21\sqrt{7}$ (c) $27\sqrt{7}$ (d) $19\sqrt{7}$

Q.584. If $x = 32.5$, $y = 34.6$ and $z = 30.9$, then the value of $x^3 + y^3 + z^3 - 3xyz$ is $0.98k$, where k is equal to:

- (a) 1,033 (b) 933 (c) 1,026 (d) 921

Q.585. If $x + \frac{1}{x} = 3$, $x \neq 0$, then the

value of $x^7 + \frac{1}{x^7}$ is:

- (a) 749 (b) 843 (c) 746 (d) 849

Q.586. If $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ and

$x^2 - xy + y^2 = 5\frac{1}{4}$, then one of the values of $(x - y)$ is:

- (a) $\frac{5}{2}$ (b) $\frac{3}{4}$ (c) $\frac{5}{4}$ (d) $\frac{3}{2}$

Q.587. If x and y are two numbers having opposite signs, and $x^2 : y^2 = 16 : 81$, then

the value of $\frac{3x + 4y}{2x - 3y}$?

- (a) $\frac{12}{25}$ (b) $-\frac{27}{40}$ (c) $-\frac{24}{35}$ (d) $\frac{48}{35}$

Q.588. If $(11a^3 - 15b^3) : (10a^3 + 4b^3) = 5 : 7$, then $(9a + 5b) : (5a - 3b)$ is equal to:

- (a) 10 : 7 (b) 13 : 5 (c) 15 : 4 (d) 14 : 9

Q.589. If $x + y = 4$ and $\frac{1}{x} + \frac{1}{y} = \frac{16}{15}$,

then what is the value of $(x^3 + y^3)$?

- (a) 18 (b) 19 (c) 21 (d) 16

Q.590. If $x - y = 11$ and $\frac{1}{x} - \frac{1}{y} = \frac{11}{24}$,

then what is the value of $x^3 - y^3 + x^2y^2$?

- (a) 1331 (b) 1115 (c) 1105 (d) 1307

Q.491. If $16x^2 + y^2 = 48$ and $xy = 2$, $x, y > 0$, then the value of $(64x^3 + y^3)$ is:

- (a) 320 (b) 340 (c) 300 (d) 240

Q.592. If $x - \frac{1}{x} = 5$, $x \neq 0$, then what is

the value of $\frac{x^6 - 5x^3 - 1}{x^6 + 7x^3 - 1}$?

- (a) $\frac{41}{45}$ (b) $\frac{45}{41}$ (c) $\frac{45}{49}$ (d) $\frac{49}{45}$

Q.593. If $x^4 + \frac{1}{x^4} = 727$, $x > 1$, then

what is the value of $(x - \frac{1}{x})$?

- (a) 6 (b) -6 (c) -5 (d) 5

Q.594. If $x - \frac{1}{x} = 1$, then what is the

value of $x^8 + \frac{1}{x^8}$?

- (a) 3 (b) 119 (c) 47 (d) -1

Q.595. If $a^3 + b^3 = 405$ and $a + b = 9$, then the value of ab is:

- (a) 15 (b) 10 (c) 12 (d) 8

Q.596. If $(2x - \frac{3}{x}) = 2$, then what is the

value of $(16x^4 + \frac{81}{x^4})$?

- (a) 184 (b) 328 (c) 180 (d) 220

Q.597. If $a - \frac{12}{a} = 1$, where $a > 0$, then

the value of $a^2 + \frac{16}{a^2}$ is:

- (a) 15 (b) 17 (c) 11 (d) 19

Q.598. If $4x^4 - 37x^2 + 9 = 0$, $x > \sqrt{\frac{3}{2}}$,

then what is the value of $8x^3 - \frac{27}{x^3}$?

- (a) 215 (b) -215 (c) 35 (d) -35

Q.599. If $x + y = 2$ and $\frac{1}{x} + \frac{1}{y} = \frac{18}{5}$,

then the value of $(x^3 + y^3)$

- (a) $4\frac{2}{3}$ (b) $4\frac{3}{5}$ (c) $3\frac{1}{3}$ (d) $3\frac{1}{5}$

Q.600. If $x^2 + 4y^2 = 53$ and $x - 2y = 5$, then what is the value of $x^3 - 8y^3$?

- (a) -85 (b) 335 (c) 155 (d) 85

Q.601. If $a^2 + c^2 + 17 = 2(a - 8b - 2b^2)$, then what is the value of $(a^3 + b^3 + c^3)$?

- (a) 9 (b) -7 (c) 10 (d) -4



Answer Key :-

1.(d)	2.(b)	3.(d)	4.(c)
5.(b)	6.(b)	7.(c)	8.(b)
9.(d)	10.(a)	11.(d)	12.(b)
13.(d)	14.(b)	15.(d)	16.(c)
17.(b)	18.(a)	19.(a)	20.(c)
21.(b)	22.(b)	23.(a)	24.(c)
25.(a)	26.(c)	27.(d)	28.(a)
29.(b)	30.(b)	31.(a)	32.(a)
33.(b)	34.(a)	35.(a)	36.(a)
37.(c)	38.(d)	39.(c)	40.(c)
41.(c)	42.(b)	43.(a)	44.(c)
45.(b)	46.(c)	47.(a)	48.(c)
49.(d)	50.(a)	51.(d)	52.(a)
53.(c)	54.(a)	55.(d)	56.(c)
57.(c)	58.(d)	59.(a)	60.(b)
61.(a)	62.(c)	63.(b)	64.(b)
65.(c)	66.(c)	67.(a)	68.(a)
69.(b)	70.(d)	71.(d)	72.(a)
73.(d)	74.(b)	75.(c)	76.(a)
77.(a)	78.(b)	79.(c)	80.(a)
81.(d)	82.(a)	83.(d)	84.(a)
85.(c)	86.(c)	87.(d)	88.(b)
89.(a)	90.(c)	91.(a)	92.(d)
93.(c)	94.(c)	95.(d)	96.(b)
97.(a)	98.(b)	99.(a)	100.(c)
101.(a)	102.(a)	103.(c)	104.(d)
105.(a)	106.(b)	107.(a)	108.(a)
109.(c)	110.(c)	111.(a)	112.(c)
113.(a)	114.(a)	115.(b)	116.(c)
117.(a)	118.(d)	119.(c)	120.(a)
121.(a)	122.(d)	123.(b)	124.(b)
125.(b)	126.(c)	127.(a)	128.(b)
129.(b)	130.(c)	131.(a)	132.(c)
133.(c)	134.(c)	135.(b)	136.(c)
137.(a)	138.(a)	139.(a)	140.(b)
141.(b)	142.(d)	143.(b)	144.(c)
145.(c)	146.(a)	147.(c)	148.(b)
149.(c)	150.(b)	151.(a)	152.(b)
153.(d)	154.(a)	155.(d)	156.(b)
157.(a)	158.(b)	159.(c)	160.(b)
161.(d)	162.(a)	163.(c)	164.(d)
165.(a)	166.(c)	167.(a)	168.(b)
169.(c)	170.(a)	171.(c)	172.(c)
173.(c)	174.(b)	175.(b)	176.(a)
177.(a)	178.(a)	179.(b)	180.(c)

181.(d)	182.(c)	183.(b)	184.(d)
185.(b)	186.(d)	187.(a)	188.(b)
189.(c)	190.(b)	191.(a)	192.(b)
193.(a)	194.(a)	195.(b)	196.(a)
197.(b)	198.(a)	199.(d)	200.(a)
201.(d)	202.(a)	203.(a)	204.(c)
205.(b)	206.(b)	207.(d)	208.(d)
209.(d)	210.(d)	211.(c)	212.(c)
213.(d)	214.(c)	215.(b)	216.(c)
217.(d)	218.(a)	219.(a)	220.(c)
221.(d)	222.(d)	223.(d)	224.(a)
225.(c)	226.(c)	227.(a)	228.(a)
229.(d)	230.(b)	231.(d)	232.(d)
233.(c)	234.(b)	235.(c)	236.(c)
237.(b)	238.(d)	239.(c)	240.(c)
241.(a)	242.(b)	243.(c)	244.(d)
245.(b)	246.(c)	247.(d)	248.(b)
249.(a)	250.(d)	251.(d)	252.(d)
253.(c)	254.(b)	255.(d)	256.(c)
257.(a)	258.(c)	259.(b)	260.(c)
261.(a)	262.(c)	263.(a)	264.(b)
265.(b)	266.(a)	267.(c)	268.(a)
269.(b)	270.(a)	271.(a)	272.(c)
273.(c)	274.(b)	275.(d)	276.(d)
277.(b)	278.(c)	279.(c)	280.(c)
281.(d)	282.(a)	283.(c)	284.(a)
285.(c)	286.(d)	287.(b)	288.(d)
289.(d)	290.(d)	291.(c)	292.(d)
293.(b)	294.(c)	295.(b)	296.(d)
297.(c)	298.(b)	299.(b)	300.(c)
301.(b)	302.(a)	303.(a)	304.(b)
305.(c)	306.(a)	307.(a)	308.(d)
309.(b)	310.(b)	311.(a)	312.(b)
313.(a)	314.(c)	315.(a)	316.(c)
317.(a)	318.(a)	319.(b)	320.(a)
321.(d)	322.(a)	323.(c)	324.(a)
325.(d)	326.(b)	327.(a)	328.(c)
329.(a)	330.(a)	331.(d)	332.(b)
333.(d)	334.(b)	335.(c)	336.(b)
337.(b)	338.(b)	339.(c)	340.(c)
341.(c)	342.(b)	343.(b)	344.(a)
345.(c)	346.(b)	347.(b)	348.(a)
349.(c)	350.(c)	351.(b)	352.(b)
353.(a)	354.(c)	355.(a)	356.(c)
357.(c)	358.(c)	359.(a)	360.(d)
361.(b)	362.(d)	363.(a)	364.(c)

365.(c)	366.(b)	367.(c)	368.(b)
369.(d)	370.(b)	371.(b)	372.(d)
373.(d)	374.(d)	375.(a)	376.(a)
377.(d)	378.(b)	379.(d)	380.(d)
381.(c)	382.(d)	383.(b)	384.(b)
385.(d)	386.(d)	387.(a)	388.(c)
389.(c)	390.(b)	391.(d)	392.(a)
393.(d)	394.(a)	395.(d)	396.(d)
397.(d)	398.(c)	399.(d)	400.(b)
401.(c)	402.(d)	403.(b)	404.(d)
405.(d)	406.(a)	407.(d)	408.(c)
409.(a)	410.(d)	411.(c)	412.(a)
413.(a)	414.(a)	415.(c)	416.(c)
417.(d)	418.(d)	419.(d)	420.(b)
421.(b)	422.(b)	423.(a)	424.(b)
425.(a)	426.(c)	427.(d)	428.(d)
429.(a)	430.(a)	431.(d)	432.(b)
433.(c)	434.(b)	435.(b)	436.(c)
437.(d)	438.(b)	439.(a)	440.(b)
441.(b)	442.(c)	443.(d)	444.(b)
445.(b)	446.(b)	447.(d)	448.(a)
449.(c)	450.(b)	451.(b)	452.(d)
453.(b)	454.(a)	455.(a)	456.(b)
457.(d)	458.(a)	459.(d)	460.(b)
461.(a)	462.(c)	463.(c)	464.(a)
465.(c)	466.(d)	467.(d)	468.(a)
469.(c)	470.(b)	471.(a)	472.(a)
473.(a)	474.(d)	475.(d)	476.(b)
477.(c)	478.(a)	479.(c)	480.(a)
481.(c)	482.(d)	483.(a)	484.(c)
485.(c)	486.(d)	487.(a)	488.(c)
489.(a)	490.(b)	491.(b)	492.(a)
493.(b)	494.(c)	495.(a)	496.(b)
497.(a)	498.(c)	499.(c)	500.(c)
501.(a)	502.(d)	503.(c)	504.(b)
505.(a)	506.(c)	507.(c)	508.(a)
509.(c)	510.(a)	511.(c)	512.(b)
513.(a)	514.(d)	515.(c)	516.(d)
517.(b)	518.(c)	519.(b)	520.(a)
521.(a)	522.(b)	523.(a)	524.(c)
525.(a)	526.(a)	527.(c)	528.(a)
529.(b)	530.(d)	531.(d)	532.(c)
533.(b)	534.(b)	535.(b)	536.(c)
537.(a)	538.(b)	539.(a)	540.(d)
541.(c)	542.(d)	543.(a)	544.(b)
545.(d)	546.(a)	547.(b)	548.(a)



549.(a)	550.(a)	551.(b)	552.(d)
553.(d)	554.(c)	555.(b)	556.(b)
557.(a)	558.(d)	559.(a)	560.(c)
561.(d)	562.(a)	563.(a)	564.(a)
565.(a)	566.(c)	567.(a)	568.(b)
569.(d)	570.(d)	571.(d)	572.(c)
573.(d)	574.(a)	575.(a)	576.(a)
577.(b)	578.(d)	579.(b)	580.(a)
581.(a)	582.(c)	583.(d)	584.(a)
585.(b)	586.(a)	587.(c)	588.(c)
589.(b)	590.(b)	591.(a)	592.(c)
593.(d)	594.(c)	595.(c)	596.(a)
597.(b)	598.(a)	599.(a)	600.(b)
601.(b)			

Solutions :-**Sol.1.(d)** Sum of the roots

$$= - \frac{\text{coefficient of } a^2}{\text{coefficient of } a^3} = - \frac{(-8)}{1} = 8$$

Sol.2.(b) Given, $(4x + \frac{3}{x} - 4) = 15$

$$4x + \frac{3}{x} = 15 + 4 = 19$$

Square on both side

$$\Rightarrow (4x + \frac{3}{x})^2 = (19)^2$$

$$\Rightarrow 16x^2 + \frac{9}{x^2} + 2 \times 4x \times \frac{3}{x} = 361$$

$$\Rightarrow 16x^2 + \frac{9}{x^2} = 361 - 24 = 337$$

$$\text{Now, } 16x^2 + \frac{9}{x^2} + 20 = 337 + 20 = 357$$

Sol.3.(d) $X^Y = 144$

$$\Rightarrow X^Y = 12^2$$

On comparing, we get

$$\Rightarrow Y = 2 \text{ and } X = 12$$

Now,

$$\Rightarrow (X - 1)^{(Y-2)} = (12 - 1)^{(2-2)} = 1$$

Sol.4.(c) On checking all options one by one, only option (c) satisfies,

$$\Rightarrow (\sqrt{x} + \sqrt{y})^2 = x + y + 2\sqrt{xy}$$

$$\text{RHS} = (\sqrt{x} + \sqrt{y})^2 = (\sqrt{x})^2 + (\sqrt{y})^2 + 2 \times \sqrt{x} \times \sqrt{y} = x + y + 2\sqrt{xy} = \text{LHS}$$

Sol.5.(b) Given: $x = \sqrt{6} + 2$ and $y = \sqrt{6} - 2$

$$= \left(\frac{x}{y} + \frac{y}{x}\right)^2 - 3 = \left(\frac{\sqrt{6}+2}{\sqrt{6}-2} + \frac{\sqrt{6}-2}{\sqrt{6}+2}\right)^2 - 3$$

$$= \left[\frac{(\sqrt{6}+2)^2 + (\sqrt{6}-2)^2}{(\sqrt{6})^2 - (2)^2}\right]^2 - 3$$

$$= \left[\frac{6+4+4\sqrt{6}+6+4-4\sqrt{6}}{6-4}\right]^2 - 3$$

$$= \left[\frac{20}{2}\right]^2 - 3 = 100 - 3 = 97$$

Sol.6.(b) $x \propto \frac{y^3}{z^2}$

$$\frac{1}{36} = k \frac{2^3}{3^2} \Rightarrow k = \frac{9}{36 \times 8} = \frac{1}{32}$$

Now,

$$800x \text{ when } y = 3 \text{ and } z = 5$$

$$x \propto \frac{y^3}{z^2} \Rightarrow x = k \times \frac{3^3}{5^2} = \frac{1}{32} \times \frac{27}{25}$$

$$800x = 800 \times \frac{1}{32} \times \frac{27}{25} = 27$$

Sol.7.(c) $0.4x + 0.16y - 1.7 = 0$

$$0.3x + 0.12y - 3.4 = 0$$

Comparing these two equations with

$$a_1x + b_1y + C_1 = 0 \text{ \& } a_2x + b_2y + C_2 = 0.$$

We have ;

$$\frac{a_1}{a_2} = \frac{0.4}{0.3} = \frac{4}{3}, \frac{b_1}{b_2} = \frac{0.16}{0.12} = \frac{4}{3},$$

$$\frac{c_1}{c_2} = \frac{-1.7}{-3.4} = \frac{1}{2} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{1}{2}$$

Clearly, we can see that the equations has no solution.

Sol.8.(b)

$$\text{For the two parallel lines:- } \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

According to the question,

$$a_1 = 4, a_2 = (k-2), b_1 = (k-2), b_2 = 9$$

$$\Rightarrow \frac{4}{(k-2)} = \frac{(k-2)}{9} \Rightarrow 36 = (k-2)^2$$

$$\Rightarrow 6 = k - 2 \quad (\text{as } k > 0)$$

$$\Rightarrow k = 8$$

$$\text{Now, } k^2 + 6 = 64 + 6 = 70$$

Sol.9.(d) $p(x) = (2x^3 + ax^2 + 3x - 5)$

$$\text{and } g(x) = (x^3 + x^2 - 2x + a)$$

$$\text{Put } x - 2 = 0 \Rightarrow x = 2$$

On putting value of x in $p(x)$ and $g(x)$

$$p(2) = 16 + 4a + 6 - 5 = 17 + 4a \quad \dots (i)$$

$$\text{and } g(2) = 8 + 4 - 4 + a = 8 + a \quad \dots (ii)$$

 $p(x)$ and $g(x)$ leaves the same remainder.

Hence, from (i) and (ii)

$$17 + 4a = 8 + a \Rightarrow 3a = -9 \Rightarrow a = -3$$

$$\text{Now, Remainder} = 8 + a = 5$$

Sol.10.(a) Formula used :

$$a^2 - b^2 = (a-b)(a+b)$$

$$(x+y)(x-y)(x^2+y^2)(x^4+y^4)(x^8+y^8)(x^{16}+y^{16})$$

$$(x^2-y^2)(x^2+y^2)(x^4+y^4)(x^8+y^8)(x^{16}+y^{16})$$

$$(x^4-y^4)(x^4+y^4)(x^8+y^8)(x^{16}+y^{16})$$

$$(x^8-y^8)(x^8+y^8)(x^{16}+y^{16})$$

$$(x^{16}-y^{16})(x^{16}+y^{16}) = x^{32} - y^{32}$$

Sol.11.(d) Given,

$$x^4 + y^4 = x^2 y^2 \Rightarrow x^4 + y^4 - x^2 y^2 = 0$$

$$x^6 + y^6 = (x^2)^3 + (y^2)^3$$

$$= (x^2 + y^2)(x^4 + y^4 - x^2 y^2)$$

$$x^6 + y^6 = (x^2 + y^2) \times 0 = 0$$

Sol.12.(b) $ax + by = 1 \dots (1)$

$$\text{And, } bx + ay = \frac{2ab}{a^2 + b^2} \dots (2)$$

Put $b = 0$ in equation (1) we get;

$$ax + 0 \times y = 1 \Rightarrow a = x = 1$$

Now, put $b = 0$ in equation (2) we get;

$$0 + ay = \frac{2a \times 0}{a^2 + 0^2} \Rightarrow ay = 0 \Rightarrow y = 0$$

So, put ($a = 1$ and $b = 0$) in the optionsValue of $x = 1$ obtained from option b **Sol.13.(d)** $x = (7 + 3\sqrt{5}) \Rightarrow x^2 = 94 + 42\sqrt{5}$

$$\frac{1}{x} = \frac{1}{(7 + 3\sqrt{5})} \times \frac{(7 - 3\sqrt{5})}{(7 - 3\sqrt{5})}$$

$$= \frac{(7 - 3\sqrt{5})}{4} \Rightarrow \frac{1}{x^2} = \frac{94 - 42\sqrt{5}}{16}$$

$$\text{Now, } x^2 + \frac{1}{x^2}$$

$$= (94 + 42\sqrt{5}) + \frac{94 - 42\sqrt{5}}{16}$$

$$= (94 + 42\sqrt{5}) + \frac{47 - 21\sqrt{5}}{8}$$

$$= \frac{94 \times 8 + 42\sqrt{5} \times 8 + 47 - 21\sqrt{5}}{8}$$

$$= \frac{799 + 315\sqrt{5}}{8}$$

Sol.14.(b) $x + \frac{1}{x} = 11$

$$x^2 + \frac{1}{x^2} = 11^2 - 2 = 119$$

$$x^4 + \frac{1}{x^4} = 119^2 - 2 = 14159$$

$$\text{Now, } x^8 - 14159x^4 + 11$$

$$x^8 - (x^4 + \frac{1}{x^4})x^4 + 11$$

$$x^8 - x^8 - \frac{x^4}{x^4} + 11 = 10$$

Sol.15.(d) If $(a + b + c) = 0$, Then $a^3 +$

$$b^3 + c^3 = 3abc$$

$$\frac{(1-m)^3 + (m-n)^3 + (n-1)^3}{12(1-m)(m-n)(n-1)}$$

$$= \frac{3(1-m)(m-n)(n-1)}{12(1-m)(m-n)(n-1)} = \frac{1}{4}$$

Sol.16.(c) $3x^2 + 2kx + 3 = 0$

$$D = b^2 - 4ac$$

For real roots, D must be equal to zero.

$$0 = (2k)^2 - 4 \times 3 \times 3$$

$$k^2 = \frac{36}{4} = 9 \Rightarrow k = 3$$

Sol.17.(b) $\frac{(a^2 + b^2 - c^2)^2 - (a^2 - b^2 + c^2)^2}{b^2 - c^2} =$

$$\frac{(a^2 + b^2 - c^2 + a^2 - b^2 + c^2) \times (a^2 + b^2 - c^2 - a^2 + b^2 - c^2)}{b^2 - c^2}$$

$$= \frac{2a^2 \times 2(b^2 - c^2)}{b^2 - c^2} = 4a^2$$



Sol.18.(a) $a = 335$, $b = 215$ and $c = 180$

$$a^3 + b^3 + c^3 - 3abc = \left(\frac{a+b+c}{2}\right)$$

$$\{(a-b)^2 + (b-c)^2 + (c-a)^2\} \\ = \left(\frac{335+215+180}{2}\right)$$

$$\{(120)^2 + (35)^2 + (155)^2\}$$

Using digital sum :-

$$= \frac{10}{2} \{(3)^2 + (8)^2 + (11)^2\}$$

$$= 5\{9 + 64 + 121\}$$

$$= 5\{9 + 1 + 4\} = 5\{5\} = 25 = 7$$

digital sum

Only option (a) has 7 as a digital sum.

Sol.19.(a) If $a + b + c = 0$, then $a^3 + b^3$

$$+ c^3 = 3abc$$

$$\frac{(u-v)^3 + (v-w)^3 + (w-u)^3}{(u^2-v^2)^3 + (v^2-w^2)^3 + (w^2-u^2)^3}$$

$$= \frac{3(u-v)(v-w)(w-u)}{3(u^2-v^2)(v^2-w^2)(w^2-u^2)}$$

$$= \frac{1}{(u+v)(v+w)(w+u)}$$

Sol.20.(c) $x^3 - y^3 = 270$, and $x - y = 6$

$$(x^3 - y^3) = (x - y)\{(x - y)^2 + 3xy\}$$

$$270 = 6(36 + 3xy) \Rightarrow xy = 3$$

$$\text{Now, } (x + y) = \sqrt{(6)^2 + 4xy}$$

$$= \sqrt{36 + 4 \times 3} = 4\sqrt{3}$$

Sol.21.(b) $p + q + r = 0$ (given)

Let, $p = 0$ then $q = -r$

$$\text{Now, } \left(\frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - pr} + \frac{r^2}{r^2 - pq}\right)$$

$$\left(0 + \frac{q^2}{q^2 - 0} + \frac{r^2}{r^2 - 0}\right) = 2$$

$$\text{Sol.22.(b)} \quad x + \frac{1}{x} = 5\sqrt{2}$$

$$x^3 + \frac{1}{x^3} = (5\sqrt{2})^3 - 3(5\sqrt{2})$$

$$= 250\sqrt{2} - 15\sqrt{2} = 235\sqrt{2}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \sqrt{\left(x^3 + \frac{1}{x^3}\right)^2 - 4}$$

$$= \sqrt{(235\sqrt{2})^2 - 4}$$

$$= \sqrt{110450 - 4} = \sqrt{110446} = 49\sqrt{46}$$

$$\text{Now, } x^6 - \frac{1}{x^6} = \left(x^3 + \frac{1}{x^3}\right)\left(x^3 - \frac{1}{x^3}\right)$$

$$= 235\sqrt{2} \times 49\sqrt{46}$$

$$= 235 \times 49 \times 2 \times \sqrt{23} = 23030\sqrt{23}$$

Sol.23.(a) $a^3 - b^3 = 6272$ and $a - b = 8$

$$a^3 - b^3 = (a - b)[(a - b)^2 + 3ab]$$

$$6272 = 8[8^2 + 3ab]$$

$$784 = 64 + 3ab$$

$$720 = 3ab \Rightarrow ab = 240$$

$$(a + b)^2 = (a - b)^2 + 4ab$$

$$(a + b)^2 = 8^2 + 4 \times 240 = 1024$$

$$\Rightarrow a + b = 32$$

$$\text{Now, } a^3 + b^3 = (a + b)[(a + b)^2 - 3ab]$$

$$a^3 + b^3 = 32[32^2 - 3 \times 240]$$

$$= 32[1024 - 720] \Rightarrow 32 \times 304 = 9728$$

Short trick :-

$$a^3 - b^3 = 6272, \text{ and } a - b = 8,$$

By hit and trial,

$a = 20$ and $b = 12$ satisfy the given equation.

Now,

$$a^3 + b^3 = 8000 + 1728 = 9728$$

Sol.24.(c) $a^2 + b^2 + c^2 - ab - bc - ca$

$$= \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2}$$

$$= \frac{(101-102)^2 + (102-103)^2 + (103-101)^2}{2}$$

$$= \frac{1+1+4}{2} = 3$$

$$\text{Sol.25.(a)} \quad \left(x + \frac{1}{x}\right) = 5$$

$$x^2 + \frac{1}{x^2} = (5)^2 - 2 = 23$$

$$\text{And, } x^4 + \frac{1}{x^4} = (23)^2 - 2 = 527 \text{ ---- (1)}$$

$$\text{Now, } x - \frac{1}{x} = \sqrt{5^2 - 4} = \sqrt{21}$$

$$x^2 - \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right) = 5\sqrt{21}$$

$$\text{And, } x^4 - \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)\left(x^2 - \frac{1}{x^2}\right)$$

$$= 23 \times 5\sqrt{21} = 115\sqrt{21} \text{ ---- (2)}$$

$$\text{Now, } x^8 - \frac{1}{x^8} = \left(x^4 + \frac{1}{x^4}\right)\left(x^4 - \frac{1}{x^4}\right)$$

$$= 527 \times 115\sqrt{21} = 60605\sqrt{21}$$

$$\text{Sol.26.(c)} \quad x - \frac{1}{x} = -6$$

$$x^2 + \frac{1}{x^2} = (-6)^2 + 2 = 38$$

$$x^3 - \frac{1}{x^3} = (-6)^3 + 3 \times (-6)$$

$$= -216 - 18 = -234$$

$$x^5 - \frac{1}{x^5} = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 - \frac{1}{x^3}\right) - \left(x - \frac{1}{x}\right)$$

$$= 38 \times (-234) - (-6) = -8892 + 6 = -8886$$

$$\text{Sol.27.(d)} \quad x^2 + \frac{1}{x^2} = 7$$

$$x + \frac{1}{x} = \sqrt{7 + 2} = 3$$

As $0 < x < 1$, so value of $\left(x - \frac{1}{x}\right)$ will always be -ve.

$$x - \frac{1}{x} = -\sqrt{9 - 4} = -\sqrt{5}$$

$$x^2 - \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)$$

$$= (3)(-\sqrt{5}) = -3\sqrt{5}$$

$$\text{Sol.28.(a)} \quad x^4 + \frac{1}{x^4} = 254$$

$$x^2 + \frac{1}{x^2} = \sqrt{254 + 2} = 16$$

$$x + \frac{1}{x} = \sqrt{16 + 2} = 3\sqrt{2} \text{ And,}$$

$$x^3 + \frac{1}{x^3} = (3\sqrt{2})^3 - 3 \times 3\sqrt{2}$$

$$= 54\sqrt{2} - 9\sqrt{2} = 45\sqrt{2}$$

Now,

$$x^5 + \frac{1}{x^5} = \left(x^3 + \frac{1}{x^3}\right)\left(x^2 + \frac{1}{x^2}\right) - \left(x + \frac{1}{x}\right)$$

$$= 45\sqrt{2} \times 16 - 3\sqrt{2} = 717\sqrt{2}$$

Sol.29.(b)

$$\frac{2x}{x+2} + \frac{1}{2x+4} - \frac{12-x}{2x^2+4x} = a + \frac{b}{x}$$

$$\Rightarrow \frac{4x^2 + x - (12 - x)}{2x^2 + 4x} = \frac{4x^2 + 2x - 12}{2x^2 + 4x}$$

$$= \frac{2x^2 + x - 6}{x^2 + 2x} = \frac{(x+2)(2x-3)}{x(x+2)} = 2 + \frac{-3}{x}$$

On comparing we get,

$$a = 2 \text{ and } b = -3$$

$$\text{So, } ab = -6$$

Sol.30.(b) $6^x = 3^y = 2^z = k$ (let)

$$\text{Then, } k^{\frac{1}{z}} = 2, k^{\frac{1}{y}} = 3, k^{\frac{1}{x}} = 6$$

$$\Rightarrow k^{\frac{1}{x}} \rightarrow 2 \times 3 \Rightarrow k^{\frac{1}{x}} = k^{\frac{1}{z}} \times k^{\frac{1}{y}}$$

$$\text{therefore, } k^{\frac{1}{x}} = k^{\frac{1}{y}} \times k^{\frac{1}{z}}$$

$$\Rightarrow \frac{1}{x} = \frac{1}{z} + \frac{1}{y} \Rightarrow \frac{1}{y} + \frac{1}{z} - \frac{1}{x} = 0$$

Sol.31.(a) $17x + my + 102 = 0$

$$23x + 299y + 138 = 0$$

For infinite number of solutions,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{17}{23} = \frac{m}{299}$$

$$m = \frac{17 \times 299}{23} = 221$$

Sol.32.(a) Formula : $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 85 - 3 \times 25 = 5(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 85 - 75 = 5(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 10 = 5(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = \frac{10}{5} = 2$$

Sol.33.(b) Since, α, β are the roots of

$$6x^2 + 13x + 7 = 0. \text{ then,}$$

$$\alpha + \beta = \frac{-13}{6} \text{ and } \alpha\beta = \frac{7}{6}$$

$$\alpha^2 + \beta^2 = \left(\frac{-13}{6}\right)^2 - 2 \times \frac{7}{6} = \frac{169}{36} - \frac{7}{3} = \frac{85}{36}$$

So, the equation whose roots are α^2, β^2 is given by :

$$x^2 - \left(\frac{85}{36}\right)x + \frac{49}{36} = 0 \Rightarrow 36x^2 - 85x + 49 = 0$$

Sol.34.(a) Formula used :

$$x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x + y + z)$$



$$\begin{aligned} & [(x-y)^2 + (y-z)^2 + (z-x)^2] \\ &= \frac{(222+223+224)}{2} [(222-223)^2 \\ &+ (223-224)^2 + (224-222)^2] \\ &= \frac{669}{2} [(-1)^2 + (-1)^2 + (2)^2] \\ &= \frac{669}{2} \times 6 = 2007 \end{aligned}$$

Sol.35.(a) Dividing and multiplying by

$(k + \frac{1}{k})$, we get

$$\begin{aligned} & \frac{(k+\frac{1}{k})(k-\frac{1}{k})(k^2+\frac{1}{k^2})(k^2-\frac{1}{k^2})(k^4+\frac{1}{k^4})(k^4-\frac{1}{k^4})(k^{16}+\frac{1}{k^{16}})(k^{32}+\frac{1}{k^{32}})}{(k+\frac{1}{k})} \\ &= \frac{(k^2-\frac{1}{k^2})(k^2+\frac{1}{k^2})(k^4+\frac{1}{k^4})(k^4-\frac{1}{k^4})(k^8+\frac{1}{k^8})(k^{16}+\frac{1}{k^{16}})(k^{32}+\frac{1}{k^{32}})}{(k+\frac{1}{k})} \\ &= \frac{(k^4-\frac{1}{k^4})(k^4+\frac{1}{k^4})(k^8+\frac{1}{k^8})(k^{16}+\frac{1}{k^{16}})(k^{32}+\frac{1}{k^{32}})}{(k+\frac{1}{k})} \\ &= \frac{(k^8-\frac{1}{k^8})(k^8+\frac{1}{k^8})(k^{16}+\frac{1}{k^{16}})(k^{32}+\frac{1}{k^{32}})}{(k+\frac{1}{k})} \\ &= \frac{(k^{16}-\frac{1}{k^{16}})(k^{16}+\frac{1}{k^{16}})(k^{32}+\frac{1}{k^{32}})}{(k+\frac{1}{k})} \\ &= \frac{(k^{32}-\frac{1}{k^{32}})(k^{32}+\frac{1}{k^{32}})}{(k+\frac{1}{k})} = \frac{(k^{64}-\frac{1}{k^{64}})}{(k+\frac{1}{k})} \end{aligned}$$

Sol.36.(a) Statement 1.

Sum of squares of n even natural no.

$$= \frac{2}{3} \times (n)(n+1)(2n+1)$$

Sum of squares of n odd natural no.

$$\begin{aligned} &= \frac{1}{3} \times n \times (2n+1) \times (2n-1) \\ &= (100^2 + 98^2 + 96^2 + \dots + 2^2) - \\ & \quad (99^2 + 97^2 + 95^2 + \dots + 1^2) \\ &= \frac{2}{3} \times (50)(50+1)(2 \times 50+1) - \frac{1}{3} \\ & \quad \times 50 \times (2 \times 50+1) \times (2 \times 50-1) \\ &= \frac{1}{3} \times 50 \times 101 \times (102-99) = 5050 \end{aligned}$$

Statement 1 is correct.

Statement 2 :-

$$8x + \frac{8}{x} = -16 \Rightarrow x + \frac{1}{x} = -2 \Rightarrow x = -1$$

$$\text{Now, } x^{197} + x^{-197} = -2$$

Statement 2 is incorrect.

Sol.37.(c) $mx^m - nx^n = 0 \Rightarrow mx^m = nx^n$

$$\Rightarrow \frac{m}{n} = \frac{x^n}{x^m}$$

$$\begin{aligned} & \frac{1}{x^m + x^n} + \frac{1}{x^m - x^n} = \frac{1}{n+m} + \frac{1}{n-m} \\ &= \frac{2n}{n^2 - m^2} = \frac{2mn}{m(n^2 - m^2)} = \frac{2mn}{x^n(n^2 - m^2)} \end{aligned}$$

Sol.38.(d) $(ax+by)^2 + (ay-bx)^2 = 8^2 + p^2$
 $\Rightarrow a^2x^2 + b^2y^2 + 2abxy + a^2y^2 + b^2x^2 - 2abxy$

$$\begin{aligned} &= 64 + p^2 \\ &\Rightarrow a^2x^2 + b^2y^2 + a^2y^2 + b^2x^2 = 64 + p^2 \\ &\Rightarrow a^2(x^2 + y^2) + b^2(x^2 + y^2) = 64 + p^2 \\ &\Rightarrow (x^2 + y^2)(a^2 + b^2) = 64 + p^2 \\ &\Rightarrow 25 \times 17 = 425 = 64 + p^2 \\ &\Rightarrow p^2 = 425 - 64 = 361 \Rightarrow p = 19 \end{aligned}$$

Short-Trick: Value putting $a = 4, b = 3, x = -1$ and $y = 4$

$$\text{Sol.39.(c)} \quad 2 = x + \frac{1}{1 + \frac{1}{5 + \frac{1}{2}}}$$

$$2 = x + \frac{1}{1 + \frac{2}{11}} \Rightarrow 2 = x + \frac{11}{13} \Rightarrow x = \frac{15}{13}$$

Sol.40.(c) Given:- $x^3 + 2x^2 - ax - b$ is divisible by $x^2 - 1$.

Therefore, $x^2 = 1$ satisfies the equation

$$\text{and } x^3 + 2x^2 - ax - b = 0 \text{ at } x^2 = 1$$

Or $x = +1$ or -1

$$\Rightarrow x^3 + 2x^2 - ax - b = 0 \quad (\because x = 1)$$

$$\Rightarrow 1 + 2 - a - b = 0 \Rightarrow 3 - a - b = 0$$

$$\Rightarrow a + b = 3 \dots (i)$$

$$\Rightarrow x^3 + 2x^2 - ax - b = 0 \quad (\because x = -1)$$

$$\Rightarrow -1 + 2 + a - b = 0 \Rightarrow 1 + a - b = 0$$

$$\Rightarrow a - b = -1 \dots (ii)$$

From eq. (i) and (ii):- $a = 1$ and $b = 2$

$$\text{Sol.41.(c)} \quad x = \sqrt[3]{5} + 2 \Rightarrow x - 2 = \sqrt[3]{5}$$

Cubing both sides, we get

$$(x-2)^3 = 5 \Rightarrow x^3 - 6x^2 + 12x - 8 = 5$$

$$\Rightarrow x^3 - 6x^2 + 12x - 12 = 1$$

Sol.42.(b) $(ax+by)$, on squaring it,

$$(ax+by)^2 = a^2x^2 + 2abxy + b^2y^2$$

So,

$(ax+by)$ is a factor of $a^2x^2 + 2abxy + b^2y^2$

Sol.43.(a) $7a^2 + 7a = 7a(a+1)$

a and $a+1$ are two consecutive number and out of two consecutive number one is odd and one is even and even is always divisible by 2

So, the complete number is divisible by 14 and 7 both.

Short-Trick:

Put the value of $a = 1$

$$7a^2 + 7a = 7(1)^2 + 7(1) = 14$$

So, number is divisible by both 7 and 14

Sol.44.(c) $(a^{\frac{2}{3}} + 2a^{\frac{1}{2}} + 3a^{\frac{1}{3}} + 2a^{\frac{1}{6}} + 1)$

$$(a^{\frac{1}{3}} - 2a^{\frac{1}{6}} + 1) - a^{\frac{1}{2}}(a^{\frac{1}{2}} - 2)$$

Let $a^{\frac{1}{6}} = p$ then,

$$= (p^4 + 2p^3 + 3p^2 + 2p + 1)(p^2 - 2p + 1) - p^3(p^3 - 2)$$

$$= p^6 + 2p^5 + 3p^4 + 2p^3 + p^2 - 2p^5 - 4p^4 - 6p^3 - 4p^2 - 2p + p^4 + 2p^3 + 3p^2 + 2p + 1 - p^6 + 2p^3 = 1$$

Short-Trick:

As the equation is independent of a , so

you can put $a = 0$ for the solution

Which gives 1 as answer

Sol.45.(b) $a + b + c + d = 2 \dots$ (given)

$$\text{Let } a = b = c = d = \frac{1}{2}$$

Therefore, $(1+a)(1+b)(1+c)(1+d)$

$$= (1 + \frac{1}{2})(1 + \frac{1}{2})(1 + \frac{1}{2})(1 + \frac{1}{2})$$

$$= (\frac{3}{2})(\frac{3}{2})(\frac{3}{2})(\frac{3}{2}) = \frac{81}{16}$$

Sol.46.(c)

$$(2x+y)^3 = 8x^3 + y^3 + 12x^2y + 6xy^2$$

Coefficient of $x^2 = 12y$

Sol.47.(a) Given, $x^{2a} = y^{2b} = z^{2c} \neq 0 \dots (1)$ and $x^2 = yz \dots (2)$

Let $x = y = z = m$... (where m is any integer value) put these value in eq (2)

$$x^2 = yz \Rightarrow m^2 = m \times m$$

.....(condition satisfied)

put these value in eq (1)

$$x^{2a} = y^{2b} = z^{2c} \Rightarrow m^{2a} = m^{2b} = m^{2c} \neq 0$$

.....(3)....(m is an integer value)

in equation (3) since base is same then power must be same

$$\Rightarrow 2a = 2b = 2c \Rightarrow a = b = c$$

$$\text{let } a = b = c = k \Rightarrow \frac{ab+bc+ca}{bc}$$

$$\Rightarrow \frac{(k \times k) + (k \times k) + (k \times k)}{k \times k} = \frac{3k^2}{k^2} = 3$$

Short trick : On putting the values as, $a = b = c = 1$ and $x = y = z = 2$, all equations are satisfied.

Sol.48.(c)

$$x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x+y+z)$$

$$[(x-y)^2 + (y-z)^2 + (z-x)^2]$$

$$= \frac{1}{2}(5.51 + 5.52 + 5.57) [(5.51 - 5.52)^2$$

$$+ (5.52 - 5.57)^2 + (5.57 - 5.51)^2]$$

$$= \frac{16.6}{2} \times [(-0.01)^2 + (-0.05)^2 + (0.06)^2]$$

$$= 8.3 \left[\frac{1+25+36}{10000} \right] = \frac{83 \times 62}{100000} = 0.05146$$

$$\text{Sol.49.(d)} \quad \frac{1}{\sqrt{7-4\sqrt{3}}} \dots (i)$$

We know that, $7 - 4\sqrt{3} = (2 - \sqrt{3})^2$

Putting the values in eq (i), We get

$$= \frac{1}{\sqrt{(2-\sqrt{3})^2}} = \frac{1}{2-\sqrt{3}}$$

On solving we get,

$$= \frac{1 \times (2 + \sqrt{3})}{(2 - \sqrt{3})(2 + \sqrt{3})} = \frac{(2 + \sqrt{3})}{4 - 3}$$

$$= 2 + \sqrt{3} = 2 + 1.73 = 3.73$$

Sol.50.(a) On solving the equation

$$(2x + 3y + 4)(2x + 3y - 5)$$

$$= 4x^2 + 6xy - 10x + 6xy + 9y^2 - 15y + 8x + 12y - 20$$

$$= 4x^2 + 9y^2 + 12xy - 2x - 3y - 20 \dots \dots \text{eq(1)}$$

On comparing equation (1) with

$$(ax^2 + by^2 + 2hxy + 2gx + 2fy + c)$$

We get,

$$a = 4, b = 9, h = 6, g = -1, f = -\frac{3}{2}, c = -20$$

Now, putting these values in equation

$$\left\{ \frac{(g+f-c)}{abh} \right\} = \left(\frac{-1 - (-\frac{3}{2}) + 20}{216} \right) = \frac{35}{432}$$

Sol.51.(d)

Let us assume the value of $x = 3$ and $y = 2$

Now, putting the value in equation

$$(4x^3y - 6x^2y^2 + 4xy^3 - y^4)$$

$$= 4 \times 27 \times 2 - 6 \times 9 \times 4 + 4 \times 3 \times 8 - 16$$

$$= 216 - 216 + 96 - 16 = 80$$

Now, check for these values in the four options.

Only option (d) is satisfied.

$$= x^4 - (x - y)^4 = 3^4 - (3 - 2)^4 = 80$$

$$\text{Sol.52.(a) Given, } x = \sqrt[6]{27} - \sqrt{6\frac{3}{4}}$$

$$\text{and } y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$$

$$\Rightarrow x^2 = (\sqrt[6]{27} - \sqrt{6\frac{3}{4}})^2$$

$$= \sqrt[3]{27} + \frac{27}{4} - 2 \times \sqrt[6]{27} \times \sqrt{6\frac{3}{4}}$$

$$= 3 + \frac{27}{4} - 9 = \frac{3}{4}$$

$$y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$$

$$y = \frac{3\sqrt{5} + 11\sqrt{5} + 7\sqrt{5}}{4\sqrt{5} + 5\sqrt{5}} = \frac{7}{3} \Rightarrow y^2 = \frac{49}{9}$$

$$x^2 + y^2 = \frac{3}{4} + \frac{49}{9} = \frac{223}{36}$$

Sol.53.(c) Given,

$$\sqrt{86} - 60\sqrt{2} = a - b\sqrt{2} \dots (1)$$

$$\text{But } \sqrt{86} - 60\sqrt{2}$$

$$\Rightarrow \sqrt{36 + 50 - 2 \times 6 \times 5\sqrt{2}}$$

$$\Rightarrow \sqrt{(6 - 5\sqrt{2})^2} \Rightarrow (6 - 5\sqrt{2})$$

Comparing this with equation (1)

$$a = 6, b = 5$$

Now,

$$\sqrt{a^2 + b^2} = \sqrt{6^2 + 5^2} = \sqrt{61} \approx 7.8$$

Sol.54.(a) Given, $a = \sqrt{8} - \sqrt{7}$ and

$$a = \frac{1}{b} \Rightarrow b = \frac{1}{\sqrt{8} - \sqrt{7}}$$

Apply rationalization

$$\Rightarrow b = \frac{1}{\sqrt{8} - \sqrt{7}} \times \frac{\sqrt{8} + \sqrt{7}}{\sqrt{8} + \sqrt{7}} = \sqrt{8} + \sqrt{7}$$

$$\Rightarrow a + b = a + \frac{1}{a} \dots (a = \frac{1}{b})$$

$$= \sqrt{8} - \sqrt{7} + \sqrt{8} + \sqrt{7} = 2\sqrt{8}$$

$$\Rightarrow a^2 + \frac{1}{a^2} = a^2 + b^2 = (2\sqrt{8})^2 - 2 = 30$$

Now,

$$ab = (\sqrt{8} - \sqrt{7})(\sqrt{8} + \sqrt{7}) = 1$$

$$\Rightarrow \frac{a^2 + b^2 - 3ab}{a^2 + ab + b^2} = \frac{30 - 3(1)}{30 + 1} = \frac{27}{31}$$

Sol.55.(d) Short-Trick: Put $a = 1, b = -1$

$$(a + b + 4)\{ab + 4(a + b)\} - 4ab = 0$$

$$\Rightarrow (1 - 1 + 4)\{1(-1) + 4(1 - 1)\} - 4 \times 1 \times -1 = 0$$

$$\Rightarrow (4)(-1) + 4 = 0 \Rightarrow -4 + 4 = 0$$

Condition satisfied.

$$\left\{ \frac{1}{(a + b + 4)^{117}} - 2^{-234} \right\}$$

$$\Rightarrow \left\{ \frac{1}{(1 - 1 + 4)^{117}} - 2^{-234} \right\}$$

$$\Rightarrow \frac{1}{(2^2)^{117}} - 2^{-234} \Rightarrow \frac{1}{(2)^{234}} - 2^{-234}$$

$$\Rightarrow 2^{-234} - 2^{-234} = 0$$

Sol.56.(c) $(x + y)^{\frac{1}{3}} + (z + y)^{\frac{1}{3}} = -(x + z)^{\frac{1}{3}}$

$$\Rightarrow (x + y)^{\frac{1}{3}} + (z + y)^{\frac{1}{3}} + (x + z)^{\frac{1}{3}} = 0$$

$$\Rightarrow [(x + y)^{\frac{1}{3}}]^3 + [(z + y)^{\frac{1}{3}}]^3 + [(x + z)^{\frac{1}{3}}]^3$$

$$= 3(x + y)^{\frac{1}{3}}(z + y)^{\frac{1}{3}}(x + z)^{\frac{1}{3}}$$

$$\Rightarrow 2(x + y + z) = 3[(x + y)(z + y)(x + z)]^{\frac{1}{3}}$$

Cubing both sides,

$$\Rightarrow 8(x + y + z)^3 = 27[(x + y)(z + y)(x + z)]$$

$$\Rightarrow x^3 + y^3 + z^3 + 3(x + y)(y + z)(z + x)$$

$$= \frac{27}{8}[(x + y)(z + y)(x + z)]$$

$$\Rightarrow x^3 + y^3 + z^3 = \left(\frac{27}{8} - 3\right)[(x + y)(z + y)(x + z)]$$

$$(x + z)]$$

$$\Rightarrow x^3 + y^3 + z^3 = \frac{3}{8}[(x + y)(z + y)(x + z)]$$

Sol.57.(c) $\frac{1}{4}\{(a + \frac{1}{a})^2 - (a - \frac{1}{a})^2\}$

$$\Rightarrow \frac{1}{4}\{(a^2 + (\frac{1}{a})^2 + 2a \times \frac{1}{a}) - \{(a^2 +$$

$$(\frac{1}{a})^2 - 2a \times \frac{1}{a})\}$$

$$\Rightarrow \frac{1}{4}(a^2 + \frac{1}{a^2} + 2 - a^2 - \frac{1}{a^2} + 2)$$

$$\Rightarrow \frac{1}{4} \times 4 = 1$$

Short-Trick:

We know that,

$$(a + b)^2 - (a - b)^2 = 4ab$$

$$= \frac{1}{4}\{(a + \frac{1}{a})^2 - (a - \frac{1}{a})^2\}$$

$$= \frac{1}{4} \times 4a \times \frac{1}{a} = 1$$

Sol.58.(d) Given, $3\sqrt[4]{x} + 4\sqrt[4]{x} = 5\sqrt[4]{x}$

We know that 3, 4 and 5 are triplet pairs

$$\Rightarrow 3^2 + 4^2 = 5^2$$

Comparing it with the question term

$$\sqrt[4]{x} = 2 \Rightarrow (\sqrt[4]{x})^4 = 2^4 \Rightarrow x = 16$$

$$\text{Sol.59.(a) Given, } x = 2 - \sqrt{3} \Rightarrow \frac{1}{x} = \frac{1}{2 - \sqrt{3}}$$

Applying rationalization

$$\Rightarrow \frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = 2 + \sqrt{3}$$

$$\Rightarrow x - \frac{1}{x} = (2 - \sqrt{3}) - (2 + \sqrt{3}) = -2\sqrt{3}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = (-2\sqrt{3})^3 + 3(-2\sqrt{3})$$

$$(\text{If } x - \frac{1}{x} = k, \text{ then } x^3 - \frac{1}{x^3} = k^3 + 3k)$$

$$= -24\sqrt{3} - 6\sqrt{3} = -30\sqrt{3}$$

Sol.60.(b) Short-Trick: Put $x = 0$

$$\left\{ \frac{2(x^3 - 8)}{x^2 - x - 2} \times \frac{x^2 + 2x + 1}{x^2 - 4x - 5} \right\} \div \frac{x^2 + 2x + 4}{3x - 15}$$

$$\Rightarrow \left\{ \frac{2(0 - 8)}{0 - 2} \times \frac{0 + 1}{0 - 5} \right\} \div \frac{0 + 4}{0 - 15}$$

$$\Rightarrow (8 \times \frac{1}{-5}) \div (\frac{4}{-15}) = 6$$

Sol.61.(a) Given,

$$x^4 - 5x^2 - 1 = 0 \Rightarrow x^2 - \frac{1}{x^2} = 5$$

$$\Rightarrow (x^6 - 3x^2 + \frac{3}{x^2} - \frac{1}{x^6} + 1)$$

$$= \{x^6 - \frac{1}{x^6} - 3(x^2 - \frac{1}{x^2}) + 1\}$$

$$\Rightarrow \{(x^2)^3 - \frac{1}{(x^2)^3} - 3(x^2 - \frac{1}{x^2}) + 1\}$$

$$\Rightarrow (x^2 - \frac{1}{x^2})^3 + 1 \Rightarrow 5^3 + 1 = 126$$

Sol.62.(c) Given, $x + y = 7$ and $xy = 10$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

$$\Rightarrow (7)^2 = x^2 + y^2 + 2(10)$$

$$\Rightarrow x^2 + y^2 = 49 - 20 = 29$$

Also,

$$x^3 + y^3 = (x + y)(x^2 + y^2 - xy)$$

Put the values of $(x + y)$,

$$(x^2 + y^2) \text{ and } (xy)$$

$$x^3 + y^3 = (7)(29 - 10) = 133$$

Now,

$$\frac{1}{x^3} + \frac{1}{y^3} \Rightarrow \frac{x^3 + y^3}{(xy)^3} = \frac{133}{10^3} = 0.133$$

Short-Trick:

Using hit and trial method :-

put $x = 5$ and $y = 2$

$$x + y = 7 \Rightarrow 5 + 2 = 7$$

$$xy = 10 \Rightarrow 5 \times 2 = 10$$

Both conditions satisfied clearly

5 and 2 are the values of x and y respectively.

$$\frac{1}{x^3} + \frac{1}{y^3} \Rightarrow \frac{1}{5^3} + \frac{1}{2^3}$$

$$\Rightarrow \frac{1}{125} + \frac{1}{8} = \frac{8 + 125}{125 \times 8} = 0.133$$

Sol.63.(b) Given,

$$(3x - 1)^3 + (4x - 3)^3 + (2x + 1)^3$$

$$= 3(3x - 1)(4x - 3)(2x + 1)$$

$$\Rightarrow (3x - 1)^3 + (4x - 3)^3 + (2x + 1)^3 - 3(3x - 1)(4x - 3)(2x + 1) = 0$$

We know that

$$a^3 + b^3 + c^3 - 3abc = 0, \text{ Only if either } a + b + c = 0 \text{ or } a = b = c$$

Case 1:

$$(3x - 1) + (4x - 3) + (2x + 1) = 0$$

$$3x + 4x + 2x - 1 - 3 + 1 = 0 \Rightarrow 9x - 3 = 0$$

$$\Rightarrow x = \frac{1}{3} \text{ (But } x \neq \frac{1}{3} \text{)}$$

Case 2 :

$$(3x - 1) = (4x - 3) = (2x + 1)$$

$$(3x - 1) = (4x - 3) \text{ or } (4x - 3)$$

$$= (2x + 1) \text{ or } (3x - 1) = (2x + 1)$$

$$4x - 3x = 3 - 1 \text{ or } 4x - 2x = 1 + 3 \text{ or } 3x - 2x = 1 + 1$$

$$\Rightarrow x = 2 \text{ or } x = \frac{4}{2} = 2 \text{ or } x = 2$$

Sol.64.(b) Put $x = 1, y = 1$ and $z = -2$

$$x + y + z = 0 \Rightarrow 1 + 1 + (-2) = 0$$

$$0 = 0 \text{(condition satisfied)}$$

$$(x^2 + y^2 + z^2) \div (z^2 - xy)$$

$$\Rightarrow [(1)^2 + (1)^2 + (-2)^2] \div [(-2)^2 - 1 \times 1]$$

$$\Rightarrow 6 \div 3 = 2$$

Sol.65.(c) Given,

$$x^2 - 3x - 1 = 0 \Rightarrow x^2 - 1 = 3x \Rightarrow x - \frac{1}{x} = 3$$

$$\text{Now, } (x^2 + 8x - 1)(x^3 + x^{-1})^{-1}$$

$$\Rightarrow \frac{(x^2 + 8x - 1)}{(x^3 + \frac{1}{x})} \Rightarrow \frac{(x^3 + 8x^2 - x)}{(x^4 + 1)}$$

$$\Rightarrow \frac{x^2(x - \frac{1}{x} + 8)}{x^2(x^2 + \frac{1}{x^2})} \Rightarrow \frac{x - \frac{1}{x} + 8}{x^2 + \frac{1}{x^2}}$$

$$\Rightarrow \frac{3 + 8}{\frac{3^2}{2} + 2} = 1 \dots (\text{If } x - \frac{1}{x} = k, \text{ then } x^2 + \frac{1}{x^2} = k^2 + 2)$$

Sol.66.(c) $x = 2 - p \Rightarrow x + p = 2$

Cubing both sides

$$(x + p)^3 = 2^3$$

$$\Rightarrow x^3 + p^3 + 3xp(x + p) = 8$$

$$\Rightarrow x^3 + p^3 + 3xp(2) = 8$$

$$\Rightarrow x^3 + 6xp + p^3 = 8$$

Sol.67.(a) $2x^2 + y^2 + 6x - 2xy + 9 = 0$

$$\Rightarrow x^2 + 6x + 9 + x^2 + y^2 - 2xy = 0$$

$$\Rightarrow (x + 3)^2 + (x - y)^2 = 0$$

Sum of the squares of two numbers is equal to zero only if the numbers themselves are zero.

$$\Rightarrow (x + 3) = 0 \text{ and } (x - y) = 0 \Rightarrow x = y = -3$$

Now,

$$4x^3 - y^3 + x^2y^2 \Rightarrow 3x^3 + x^4 \dots (x = y)$$

$$\Rightarrow x^3(3 + x) \Rightarrow (-3)^3(3 - 3) = 0$$

Sol.68.(a) Given, $ab + bc + ca = 8$ and

$$a^2 + b^2 + c^2 = 20$$

We know that,

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) = 20 + 2(8) = 36$$

$$\Rightarrow (a + b + c) = 6$$

$$\Rightarrow \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

$$= a^3 + b^3 + c^3 - 3abc = (a + b + c)\{(a + b + c)^2 - 3(ab + bc + ca)\}$$

$$\Rightarrow 6\{(6)^2 - 3 \times 8\} = 6(36 - 24) = 6 \times 12 = 72$$

Sol.69.(b) $x + y = 1$

Squaring both sides

$$x^2 + y^2 + 2xy = 1 \Rightarrow x^2 + y^2 = 1 - 2xy$$

Again squaring both sides

$$x^4 + y^4 + 2x^2y^2 = 1 + 4x^2y^2 - 4xy$$

$$x^4 + y^4 = 1 + 2x^2y^2 - 4xy$$

$$x^4 + y^4 = 1 + 2(x^2y^2 - 2xy)$$

$$\text{Put } (x^2y^2 - 2xy = 12) = 1 + 2(12) = 25$$

Sol.70.(d)

$$16x^2 + 9y^2 + 4z^2 = 24(x - y + z) - 61$$

$$\Rightarrow (16x^2 - 24x + 9) + (9y^2 + 24y + 16)$$

$$+ (4z^2 - 24z + 36) = 0$$

$$\Rightarrow (4x - 3)^2 + (3y + 4)^2 + (2z - 6)^2 = 0$$

$$\Rightarrow x = \frac{3}{4}, y = -\frac{4}{3}, z = 3$$

Therefore,

$$xy + 2z = \frac{3}{4} \times (-\frac{4}{3}) + 2 \times 3$$

$$= (-1) + 6 = 5$$

Sol.71.(d) We know that,

$$(a^3 - b^3) = (a - b)(a^2 + b^2 + ab)$$

$$\text{So, } 5\sqrt{5}x^3 - 81\sqrt{3}y^3$$

$$= (\sqrt{5}x - 3\sqrt{3}y)[(\sqrt{5}x)^2 + (3\sqrt{3}y)^2 + \sqrt{5}x \times 3\sqrt{3}y]$$

$$= (\sqrt{5}x - 3\sqrt{3}y)(5x^2 + 27y^2 + 3\sqrt{15}xy)$$

Put this value in question term,

$$\frac{5\sqrt{5}x^3 - 81\sqrt{3}y^3}{(\sqrt{5}x - 3\sqrt{3}y)} = Ax^2 + By^2 + Cxy$$

$$\Rightarrow \frac{(\sqrt{5}x - 3\sqrt{3}y)(5x^2 + 27y^2 + 3\sqrt{15}xy)}{(\sqrt{5}x - 3\sqrt{3}y)}$$

$$= Ax^2 + By^2 + Cxy$$

$$\Rightarrow (5x^2 + 27y^2 + 3\sqrt{15}xy)$$

$$= Ax^2 + By^2 + Cxy$$

Comparing the coefficients of x^2, y^2 and xy

$$A = 5, B = 27 \text{ and } C = 3\sqrt{15}$$

$$(6A + B - \sqrt{15}C) = 6(5) + 27 - (\sqrt{15} \times 3\sqrt{15})$$

$$= 12$$

Sol.72.(a) We know that

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

Put the given values

$$19^2 = 133 + 2(xy + yz + y^2)$$

$$114 = y(x + y + z) \Rightarrow y = 6$$

$$x + z = 19 - 6 = 13 \Rightarrow xz = 6^2 = 36$$

Only 9 and 4 are such pairs whose sum is 13 and multiplication is 36

Required difference = $9 - 4 = 5$

Sol.73.(d) $x^4 + \frac{1}{x^4} = 194$

$$\Rightarrow x^2 + \frac{1}{x^2} = \sqrt{194 + 2} = 14$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{14 + 2} = 4$$

$$= x^2 + 1 = 4x \Rightarrow x^2 - 4x = -1$$

$$(x - 2)^2 = 4 - 1 = 3$$

Sol.74.(b) We know the roots of an

$$\text{equation} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

here, $a = 1, b = -4$ and $c = a$

According to question

$$\frac{-(-4) + \sqrt{(-4)^2 - 4a(1)}}{2(1)} \dots (1)$$

$$= \frac{-(-4) - \sqrt{(-4)^2 - 4a(1)}}{2(1)} \dots (2)$$

Equating eq. (1) and (2), we get

$$2\sqrt{16 - 4a} = 0$$

$$16 = 4a \text{ so, } a = 4 \text{ ans.}$$

Short trick :

For real and equal roots,

$$D = b^2 - 4ac = 0$$

$$a = 1, b = -4 \text{ and } c = a$$

$$\Rightarrow (-4)^2 - 4a = 0 \Rightarrow 4a = 16 \Rightarrow a = 4$$

Sol.75.(c) Given :- $x = \sqrt{3}$

$$x + \frac{1}{x} \rightarrow \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$$

Sol.76.(a) Given :-

$$x = \sqrt{\frac{(2 + \sqrt{3})}{(2 - \sqrt{3})}} \times \frac{(2 + \sqrt{3})}{(2 + \sqrt{3})}$$

$$x = \sqrt{(2 + \sqrt{3})^2}$$

$$x = 2 + \sqrt{3}$$

$$x^2 = 4 + 3 + 4\sqrt{3} = 7 + 4\sqrt{3}$$

$$x^2 - x - 7 = 7 + 4\sqrt{3} - 2 - \sqrt{3} - 7 = -2 + 3\sqrt{3}$$

Sol.77.(a) $x^2 + 6\sqrt{7}$ (put $x = \sqrt{7} - 3$)

$$= (\sqrt{7} - 3)^2 + 6\sqrt{7}$$

$$= (7 + 9 - 2 \times 3 \times \sqrt{7}) + 6\sqrt{7} = 16$$

Sol.78.(b) Given:- $x + \frac{1}{x} = 6$

Square both sides

$$(x + \frac{1}{x})^2 = 6^2$$

$$x^2 + \frac{1}{x^2} + 2 \times x \times \frac{1}{x} = 36 \Rightarrow x^2 + \frac{1}{x^2} = 34$$

Sol.79.(c) Given ,

$$\frac{(0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04)}{(0.4 \times 0.4 \times 0.4 + 0.08 \times 0.08 \times 0.08)}$$

$$= \frac{(0.2)^3 + (0.04)^3}{(0.4)^3 + (0.08)^3} = \frac{(0.2)^3 + (0.04)^3}{2^3[(0.2)^3 + (0.04)^3]}$$

$$= \frac{1}{8} = 0.125$$



Sol.80.(a) $x^3 + \frac{1}{x^3} = 5^3 - 3$ (5)
 $= 125 - 15 = 110$

Sol.81.(d) We know that
 $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
 $= 2(\sqrt{7}^2 + 1^2) = 16$

Sol.82.(a) Given, $x - \frac{1}{x} = 4$

$$x^2 + \frac{1}{x^2} = 16 + 2 = 18$$

Therefore, $x^4 + \frac{1}{x^4} = 324 - 2 = 322$

Sol.83.(d) $(\sqrt{9} + \sqrt{4})^2 - (\sqrt{8} + \sqrt{2})^2$
 $= (3+2)^2 - (8+2+2 \times \sqrt{8} \times \sqrt{2})$
 $= 25 - 10 - 8 = 7$

Sol.84.(a) Given: $a + b = 7$ and $ab = 12$
 Taking $a = 4$ and $b = 3$, then value of a and b satisfies, Now,
 $= (a^3 + b^3)^2 - 8a^2b^2(a+b)^2$
 $= (4^3 + 3^3)^2 - 8 \times 4^2 \times 3^2 \times (4+3)^2$
 $= (91)^2 - 8 \times 16 \times 9 \times (7)^2$
 $= 8281 - 56448 = -48167$

Sol.85.(c)
 $a + b + c = 0.04 + 0.06 - 0.10 = 0$
 when, $a + b + c = 0$ then,
 $\Rightarrow (a^3 + b^3 + c^3) = (3abc)$
 So,
 $\Rightarrow \frac{a^3 + b^3 + c^3}{3abc} = \frac{3abc}{3abc} = 1$

Sol.86.(c)
 $(0.8 \times 0.8 \times 0.8 + 0.04 \times 0.04 \times 0.04) \div$
 $(1.6 \times 1.6 \times 1.6 + 0.08 \times 0.08 \times 0.08)$
 $= \frac{(0.8^3 + 0.04^3)}{(1.6^3 + 0.08^3)} = \frac{(0.8^3 + 0.04^3)}{2^3(0.8^3 + 0.04^3)} = \frac{1}{8}$

Sol.87.(d) We know that:-
 If $(a+b+c) = 0$, then $a^3 + b^3 + c^3 = 3abc$
 $\Rightarrow (a-b)^3 + (b-c)^3 + (c-a)^3$
 $= 3(a-b)(b-c)(c-a)$

Sol.88.(b) Given :- $\sqrt{x+5} + \sqrt{x-4} = 9$
 From option (b) :-
 $\Rightarrow \sqrt{20+5} + \sqrt{20-4} = 9$
 $\Rightarrow 5 + 4 = 9$
 $\Rightarrow 9 = 9$ (L.H.S = R.H.S)

Sol.89.(a) Given: $y + \frac{1}{y} = 3$

$$(y + \frac{1}{y})^2 = y^2 + \frac{1}{y^2} + 2$$

$$y^2 + \frac{1}{y^2} = 9 - 2 = 7$$

$$(y + \frac{1}{y})^3 = y^3 + \frac{1}{y^3} + 3 \times 3$$

$$y^3 + \frac{1}{y^3} = 27 - 9 = 18$$

$$(y^3 + \frac{1}{y^3})(y^2 + \frac{1}{y^2}) = y^5 + \frac{1}{y^5} + y + \frac{1}{y}$$

$$y^5 + \frac{1}{y^5} = 18 \times 7 - 3 = 123$$

$$y^5 + \frac{1}{y^5} - 3(y^3 + \frac{1}{y^3}) + 4(y + \frac{1}{y})$$

$$= 123 - 3 \times 18 + 4 \times 3 = 81$$

Sol.90.(c) $\sqrt{m} + \frac{1}{\sqrt{m}} = 7$

Squaring on both side

$$(\sqrt{m} + \frac{1}{\sqrt{m}})^2 = 7^2$$

$$m + \frac{1}{m} + 2 = 49 \Rightarrow m + \frac{1}{m} = 47$$

Sol.91.(a)

$$x^3 + \frac{8}{x^3} = (x + \frac{2}{x})^3 - 3x \times \frac{2}{x} \times (x + \frac{2}{x})$$

$$= 5^3 - 3(2)(5) = 125 - 30 = 95$$

Sol.92.(d) $\frac{\sqrt{5+x} + \sqrt{5-x}}{\sqrt{5+x} - \sqrt{5-x}} = 2$

Substituting option (b) for $x = 4$:-

$$\frac{\sqrt{5+4} + \sqrt{5-4}}{\sqrt{5+4} - \sqrt{5-4}} = \frac{\sqrt{9} + \sqrt{1}}{\sqrt{9} - \sqrt{1}}$$

$$= \frac{4}{2} = 2 = \text{RHS}$$

Sol.93.(c) Given :- $x + \frac{1}{x} = -2$

Taking $x = -1$ then satisfies the equation,

$$\Rightarrow x^4 + \frac{1}{x^4} + 3x^2 + \frac{3}{x^2} = 1 + 1 + 3 + 3 = 8$$

Sol.94.(c) By option one by one options checking option (c) satisfy the given condition

$$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$$

$$\sqrt{16} - \sqrt{9} = 4 - 3 = 1$$

Sol.95.(d) $ab = 15 \Rightarrow ab = 5 \times 3$ or 1×15
 $a^3 + b^3 \Rightarrow 5^3 + 3^3 \Rightarrow 125 + 27 = 152$ (satisfy the given condition)

$a^3 + b^3 \Rightarrow 15^3 + 1^3 \Rightarrow 3375 + 1 = 3376$ (not satisfy the given condition)
 So, the value of $(a+b) = (3+5) = 8$

Sol.96.(b) we know that, if $x + y + z = 0$
 $\Rightarrow x^3 + y^3 + z^3 = 3xyz$
 Therefore, $(x^3 + y^3 + z^3) \div (3xyz)$
 $= (3xyz) \div (3xyz) = 1$

Sol.97.(a) Given: $m + \frac{1}{m} = -3$

Here, $a = -3$

$$\Rightarrow m^2 + \frac{1}{m^2} = a^2 - 2 = 9 - 2 = 7$$

$$\Rightarrow m^3 + \frac{1}{m^3} = a^3 - 3a = -27 + 9 = -18$$

$$\Rightarrow m^5 + \frac{1}{m^5} = (m^3 + \frac{1}{m^3})(m^2 + \frac{1}{m^2}) - (m + \frac{1}{m})$$

$$= -18 \times 7 - (-3) = -123$$

So, $m^5 + \frac{1}{m^5} - 2(m^3 + \frac{1}{m^3}) + 4(m + \frac{1}{m})$
 $= -123 - 2(-18) + 4(-3) = -99$

Sol.98.(b) Given :- $x = 4 + \sqrt{15}$,
 $\frac{1}{x} = \frac{1}{4 + \sqrt{15}} \times \frac{4 - \sqrt{15}}{4 - \sqrt{15}} = \frac{4 - \sqrt{15}}{16 - 15}$
 $= 4 - \sqrt{15}$

$$x - \frac{1}{x} = 4 + \sqrt{15} - 4 + \sqrt{15} = 2\sqrt{15}$$

Sol.99.(a) Given :- $x = \sqrt{7} + 3$
 $x^2 - 4\sqrt{7} = (\sqrt{7} + 3)^2 - 4\sqrt{7}$
 $= 7 + 9 + 2 \times 3 \times \sqrt{7} - 4\sqrt{7}$
 $= 16 + 6\sqrt{7} - 4\sqrt{7} = 16 + 2\sqrt{7}$

Sol.100.(c) Given : $x = 3 + \sqrt{2}$
 $= x^2 - 6x + 7$
 $= (3 + \sqrt{2})^2 - 6(3 + \sqrt{2}) + 7$
 $= (9 + 6\sqrt{2} + 2) - (18 + 6\sqrt{2}) + 7$
 $= 11 + 6\sqrt{2} - 18 - 6\sqrt{2} + 7 = 0$

Sol.101.(a) Given , $x + \frac{1}{x} = 9$
 $x - \frac{1}{x} = \sqrt{9^2 - 4} = \sqrt{77}$

$$\frac{2x}{x^2-1} = 2 \times \frac{1}{x - \frac{1}{x}} = \frac{2 \times 1}{\sqrt{77}} = \frac{2\sqrt{77}}{77}$$

Sol.102.(a) According to the question:
 $(\sqrt{5} + 1)^2 + (\sqrt{5} - 1)^2$
 $\Rightarrow 5 + 1 + 2\sqrt{5} + 5 + 1 - 2\sqrt{5} = 12$

Sol.103.(c)
Given, $a^3 + b^3 = 407$ and $a + b = 11$
 By checking all options one by one, only option(c) satisfies
 $ab = 28 \Rightarrow 7 \times 4$
 $a + b = 11 \Rightarrow 7 + 4 = 11$ and $7^3 + 4^3$
 $= 343 + 64 = 407$

Sol.104.(d) We know that:-
 If $(x + y + z) = 0$, then
 $x^3 + y^3 + z^3 = 3xyz$
 $\Rightarrow (x-y)^3 + (y-z)^3 + (z-x)^3$
 $= 3(x-y)(y-z)(z-x)$

Sol.105.(a) $ab = 20 \Rightarrow ab = 5 \times 4$
 $a^3 + b^3 \Rightarrow 5^3 + 4^3 \Rightarrow 125 + 64 = 189$
 (satisfy the given condition)
 So, the value of $(a+b) = (5+4) = 9$

Sol.106.(b) Given: $\sqrt{y} = 13$
 So, $y = 13^2 = 169$
 Now, $y + \sqrt{y} = 169 + 13 = 182$

Sol.107.(a) According to the question:
 $(x+y)^2 = x^2 + y^2 + 2xy$
 $(9)^2 = x^2 + y^2 + 2 \times 20$
 $x^2 + y^2 = 81 - 40 = 41$

Sol.108.(a)

Formula :- $(a + b)^2 = (a - b)^2 + 4ab$

Given: $a - b = 12$ and $ab = 45$

$$(a + b)^2 = 12^2 + 4 \times 45 = 324$$

Sol.109.(c)

$$\sqrt{x^2 + 12x + 36} = \sqrt{(x + 6)^2}$$

$$\sqrt{x^2 - 8x + 16} = \sqrt{(x - 4)^2}$$

$$\Rightarrow x + 6 + x - 4 = 10 \Rightarrow x = 4$$

Sol.110.(c) $\frac{(2 - \sqrt{3})}{(2 + \sqrt{3})} = p + q\sqrt{3}$

On rationalizing

$$\frac{(2 - \sqrt{3})}{(2 + \sqrt{3})} \times \frac{(2 - \sqrt{3})}{(2 - \sqrt{3})} = p + q\sqrt{3}$$

$$\frac{4 + 3 - 2 \times 2 \times \sqrt{3}}{4 - 3} = p + q\sqrt{3}$$

$$7 - 4\sqrt{3} = p + q\sqrt{3}$$

Comparing on both sides

$$p = 7, q = -4$$

Sol.111.(a) $(a - b)^2 = a^2 + b^2 - 2ab$

$$6^2 = 70 - 2ab$$

$$ab = \frac{70 - 36}{2} = 17$$

Sol.112.(c) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
 $= 30 + 2 \times 20 = 70$

Sol.113.(a) **Given:-** $x - \frac{1}{x} = 5$

$$(x - \frac{1}{x})^2 = (5)^2$$

$$x^2 + \frac{1}{x^2} - 2 = 25$$

$$x^2 + \frac{1}{x^2} = 27$$

Sol.114.(a)

Given:- $(a + b + c)^2 - (a^2 + b^2 + c^2)$

$$= a^2 + b^2 + c^2 + 2(ab + bc + ca) - (a^2 + b^2 + c^2) = 2(ab + bc + ca)$$

Sol.115.(b) **Given,** $x + y + z = 0$

$$\text{So, } x^3 + y^3 + z^3 - 3xyz = 0$$

$$\Rightarrow x^3 + y^3 + z^3 = 3xyz$$

Sol.116.(c) $y = \frac{(\sqrt{7} + \sqrt{5})}{(\sqrt{7} - \sqrt{5})} + \frac{(\sqrt{7} - \sqrt{5})}{(\sqrt{7} + \sqrt{5})}$

$$\Rightarrow y = \frac{(\sqrt{7} + \sqrt{5})^2 + (\sqrt{7} - \sqrt{5})^2}{(\sqrt{7})^2 - (\sqrt{5})^2} = \frac{24}{2} = 12$$

Sol.117.(a) According to question,

$$x + \frac{1}{x} = 18$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 18^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 322 = 161 \times 2 + 0$$

Sol.118.(d) $(\sqrt{a} + \frac{1}{\sqrt{a}})^2 = 15$

$$a + \frac{1}{a} + 2 = 15$$

$$a + \frac{1}{a} = 13$$

On cubing both sides

$$a^3 + \frac{1}{a^3} + 3 \times 13 = 2197$$

$$a^3 + \frac{1}{a^3} = 2158$$

Sol.119.(c) Let $x = \frac{a}{p}, y = \frac{b}{q}, z = \frac{c}{r}$

According to question,

$$x + y + z = 4$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$\frac{yz + zx + xy}{xyz} = 0$$

xyz cannot be zero

$$xy + zx + xy = 0$$

Now,

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$16 = x^2 + y^2 + z^2 + 0$$

$$\therefore \frac{a^2}{p^2} + \frac{b^2}{q^2} + \frac{c^2}{r^2} = 16$$

Sol.120.(a) $x + \frac{1}{x + 3} = 3$

$$\Rightarrow (x + 3) + \frac{1}{(x + 3)} = 3 + 3 = 6$$

Now,

$$\Rightarrow (x + 3)^3 + \frac{1}{(x + 3)^3} = 6^3 - 3 \times 6 = 198$$

Sol.121.(a) $x + \frac{1}{x + 2} = 4$

Adding 2 both sides we get,

$$(x + 2) + \frac{1}{(x + 2)} = 6$$

$$(x + 2)^3 + \frac{1}{(x + 2)^3} = 216 - 3 \times 6 = 198$$

Sol.122.(d) $A^b = 144 \Rightarrow (12)^2 = 144$

$$A = 12, b = 2$$

$$(A - 1)^{(b - 2)} = (12 - 1)^{(2 - 2)} = (11)^0 = 1$$

Sol.123.(b)

$$a^2 + b^2 + c^2 + 48 = 8(a + b + c)$$

$$\Rightarrow a^2 - 8a + 16 + b^2 - 8b + 16 + c^2 - 8c + 16 = 0$$

$$\Rightarrow (a - 4)^2 + (b - 4)^2 + (c - 4)^2 = 0$$

$$a = 4, b = 4, c = 4$$

$$\sqrt[3]{a^3 - b^3 + c^3} = \sqrt[3]{4^3 - 4^3 + 4^3} = 4$$

Sol.124.(b) $4^n = 1024$

$$4^n = (4)^5$$

$$n = 5$$

$$(2)^{5 \cdot 2} = 2^3 = 8$$

Sol.125.(b) $k^2 + \frac{1}{k^2} = 12$

$$k^4 + \frac{1}{k^4} = 144 - 2 = 142$$

Sol.126.(c) **Given :-** $y = \frac{3}{5} \Rightarrow y^2 = \frac{9}{25}$

A/Q,

$$\sqrt{(1 - x^2)(1 - \frac{9}{25})} = \frac{4}{5}$$

$$\sqrt{(1 - x^2)(\frac{16}{25})} = \frac{4}{5}$$

Squaring both sides

$$(1 - x^2) \frac{16}{25} = \frac{16}{25}$$

$$1 - x^2 = 1$$

$$x^2 = 0 \Rightarrow x = 0$$

$$[(5x^2 + 5y^2 + 10xy) + (5x^2 + 5y^2 - 10xy)]$$

$$[5 \times 0 + 5 \times \frac{9}{25} + 0 + 5 \times 0 + 5 \times \frac{9}{25} - 0]$$

$$[\frac{9}{5} + \frac{9}{5}] = \frac{18}{5}$$

Sol.127.(a)

Given, $(\frac{3}{a} + 3a)(\frac{9}{a^2} - 9 + 9a^2)$

We know that $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Put $a = 2$

$$(\frac{3}{a})^3 + (3a)^3 = (\frac{3}{2})^3 + (3 \times 2)^3 = \frac{27}{8} +$$

$$216 = \frac{1755}{8}$$

Sol.128.(b) **Given** $3x + 4y = 70$

and $x, y \in \mathbb{N}$,

And we know that, $\mathbb{N} = 1, 2, 3, 4, \dots$

Let $y = 1$

$$\Rightarrow 3x + 4 = 70$$

$$\Rightarrow 3x = 66 \Rightarrow x = 22$$

Sol.129.(b) If, $y + \frac{1}{y} = 2$ then $y = 1$

$$\text{So, } y^6 + \frac{1}{y^6} = 1 + 1 = 2$$

Sol.130.(c)

$$\frac{x^3 + 15x^2 + 75x + 125}{x^2 - 25} \times (x - 5)$$

$$= \frac{x^3 + 5^3 + 3 \times x \times 5(x + 5)}{x^2 - 5^2} \times (x - 5)$$

$$= \frac{(x + 5)^3}{(x - 5)(x + 5)} \times (x - 5) = (x + 5)^2$$

$$= x^2 + 25 + 10x \text{ or } x^2 + 10x + 25$$

Sol.131.(a) $\frac{88 \div 8 \times k - 3 \times 3}{6^2 - 7 \times 5 + k^2} = 1$

$$\Rightarrow \frac{11 \times k - 9}{36 - 35 + k^2} = 1 \Rightarrow 11k - 9 = 1 + k^2$$

$$\Rightarrow k^2 - 11k + 10 = 0 \Rightarrow k^2 - 10k - k + 10 = 0$$

$$\Rightarrow k(k - 10) - 1(k - 10) = 0$$

$$\Rightarrow (k - 10)(k - 1) = 0 \Rightarrow k = 10 \text{ and } 1$$

Sol.132.(c) **Given:** $a + b = 2c$

$$a + b = c + c \Rightarrow a - c = c - b$$

On squaring both side

$$a^2 + c^2 - 2ac = c^2 + b^2 - 2bc$$

$$a^2 + 2bc = b^2 + 2ac$$



Sol.133.(c) Let: $x = y = z = \frac{1}{3}$

Then, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{\frac{1}{3}} + \frac{1}{\frac{1}{3}} + \frac{1}{\frac{1}{3}}$
 $= 3 + 3 + 3 = 9$

Sol.134.(c) $a^2 + \frac{1}{a^2} = (a + \frac{1}{a})^2 - 2$
 $= 12^2 - 2 = 142$

Sol.135.(b)

Given that, $a+b+c = 18$ & $a^2 + b^2 + c^2 = 36$

Then, $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)$

$[3(a^2 + b^2 + c^2) - (a+b+c)^2]$

$= \frac{1}{2} \times 18[3 \times 36 - 18^2]$

$= 9[108 - 324] = 9 \times (-216) = -1944$

Sol.136.(c) After checking all the options one by one, only option (c) satisfied.

$x = 5, y = 4, z = 3$

$\Rightarrow x + y + z = 12;$

LHS = $5 + 4 + 3 = 12 = \text{RHS}$

$\Rightarrow x + y - z = 6;$

LHS = $5 + 4 - 3 = 6 = \text{RHS}$

$\Rightarrow x - y + z = 4$

LHS = $5 - 4 + 3 = 4 = \text{RHS}$

Sol.137.(a) $\Rightarrow \sqrt{x} + \frac{1}{\sqrt{x}} = 7$

On squaring both side,

$\Rightarrow x + \frac{1}{x} + 2 = 49 \Rightarrow x + \frac{1}{x} = 47$

Sol.138.(a)

$x^2 + y^2 = 280$ and $xy = 120$

$x^2 + y^2 - 2xy = 280 - 2 \times 120 = 40$

.....(i)

$x^2 + y^2 + 2xy = 280 + 2 \times 120 = 520$

.....(ii)

Now, on dividing (i) by (ii):

$\frac{x^2 + y^2 - 2xy}{x^2 + y^2 + 2xy} = \frac{40}{520}$

$\Rightarrow \frac{(x-y)^2}{(x+y)^2} = \frac{1}{13} \Rightarrow \frac{(x-y)}{(x+y)} = \frac{1}{\sqrt{13}}$

Sol.139.(a) If the lines have no solution

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}, \Rightarrow \frac{2}{1} = \frac{7k}{1} \Rightarrow k = \frac{2}{7}$

Sol.140.(b) $x + \frac{1}{x} = 2\sqrt{10}$

$x - \frac{1}{x} = \sqrt{(2\sqrt{10})^2 - 4}$

$= \sqrt{40 - 4} = \sqrt{36} = 6 \dots (x > 1)$

So, $x^3 - \frac{1}{x^3} = (6)^3 + 3 \times 6 = 234$

Sol.141.(b) Given :- $x + 2z = 3$

$x + 2y + 3z = 5$

$3x - 5z = -13$

By checking all options one by one, only option (b) satisfies,

hence, $x = -1, y = 0, z = 2$

Sol.142.(d) $x + \frac{1}{9x} = 3$

Multiplying both side by 3, we have ;

$3x + \frac{1}{3x} = 9$

Then, $9x^2 + \frac{1}{9x^2} = 9^2 - 2 = 81 - 2 = 79$

Sol.143.(b) $t^3 + (\frac{3}{5})^3 + \frac{9}{5}t$

$= (\frac{2}{5})^3 + (\frac{3}{5})^3 + \frac{9}{5} \times \frac{2}{5}$

$= \frac{8}{125} + \frac{27}{125} + \frac{90}{125} = \frac{125}{125} = 1$

Sol.144.(c)

$\frac{7x^2 - 9x + 7}{x^2 - x + 1} = \frac{7x - 9 + \frac{7}{x}}{x - 1 + \frac{1}{x}}$

$= \frac{7(x + \frac{1}{x}) - 9}{(x + \frac{1}{x}) - 1} = \frac{7 \times 15 - 9}{15 - 1} = \frac{96}{14}$

$= \frac{48}{7}$

Sol.145.(c) For many solutions

$\frac{a}{3} = \frac{b}{-(5-2a)} = \frac{2}{6}$

$\Rightarrow \frac{a}{3} = \frac{2}{6} \Rightarrow a = 1$

And $\frac{b}{-(5-2a)} = \frac{2}{6} \Rightarrow 3b = -5 + 2a$

$\Rightarrow 3b = -5 + 2 \Rightarrow b = -1$

Sol.146.(a) $m + n = 24$

Let $m = 12$

$n = 12$

$\Rightarrow (m-16)^3 + (n-8)^3 = (12-16)^3 + (12-8)^3$

$= (-4)^3 + (4)^3 = 0$

Sol.147.(c) Given Equation :

$2x + y = a$ and $8x + by = 12$

For infinite solution $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \dots (i)$

Here, $a_1 = 2, a_2 = 8, b_1 = 1, b_2 = b, c_1 = -a, c_2 = -12$

Put this value in eqn (i)

$\frac{2}{8} = \frac{1}{b} = \frac{-a}{-12}$

By equating this equation ,

$a = 3, b = 4$

Now, Sum of a and b = $4 + 3 = 7$

Sol.148.(b)

Given, $a^2 + b^2 + c^2 = 2(a + c - 1)$

Here, $a^2 + b^2 + c^2 - 2a - 2c + 2 = 0$

$\Rightarrow a^2 - 2a + 1 + b^2 + c^2 - 2c + 1 = 0$

$\Rightarrow (a - 1)^2 + b^2 + (c - 1)^2 = 0$

Now, $(a - 1)^2 = 0, a = 1$

$b^2 = 0, b = 0$ and $(c - 1)^2 = 0, c = 1$

$\therefore a^3 + b^3 + c^3 = 2$

Sol.149.(c) Given: $5x - 3y + 7z = 22; 3x - 5y - 2z = -46; 2x - 2y + 5z = 24$

By checking option one by one, option (c) satisfy

$5x - 3y + 7z = 22$

Put $x = -5, y = 3, z = 8$

$-25 - 9 + 56 = 22$

Sol.150.(b) This type of question we will go to options,

After checking all options one by one only option (b) satisfies.

$x = 1, y = 1, z = 1$

Equation (i)

$5x + 4y - 8z = 1 \text{ LHS}$

$= 5x + 4y - 8z = 5 + 4 - 8 = 1 = \text{RHS}$

Equation(ii)

$7x - 9y + z = -1 \text{ LHS}$

$= 7x - 9y + z = 7 - 9 + 1 = -1 = \text{RHS}$

Equation (iii)

$2x + 3y - 4z = 1 \text{ LHS}$

$= 2x + 3y - 4z = 2 + 3 - 4 = 1 = \text{RHS}$

Sol.151.(a) Given :- $a + b = 1$

$\frac{1}{a^2} + \frac{1}{b^2} + \frac{2}{ab} = \frac{a^2 + b^2 + 2ab}{a^2 b^2} = \frac{(a+b)^2}{a^2 b^2}$
 $= \frac{1}{a^2 b^2}$

Sol.152.(b) $x - ay = 2 - a$

$x - ay - (2 - a) = 0 \dots (i)$

$(1 - a)x + 6y - (a + b) \dots (ii)$

For No solution

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$\frac{a_1}{a_2} = \frac{b_1}{b_2}$	$\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
$\frac{1}{1-a} = \frac{-a}{6}$	$\frac{-a}{6} \neq \frac{-2+a}{-a-b}$
$(1-a)(-a) = 6$	On putting value of $a = 3$
$a^2 - a - 6 = 0$	$\frac{-3}{6} \neq \frac{-2+3}{-3-b}$
$a^2 - 3a + 2a - 6 = 0$	$-3 - b \neq -2$
$a(a-3) + 2(a-3) = 0$	$-b \neq 1$
$(a-3)(a+2) = 0$	$b \neq -1$
$a = 3 \text{ or } -2$	

Sol.153.(d)

Given: $m = a \sec A$ and $y = b \tan A$

$b^2 m^2 - a^2 y^2 + \frac{a^2 y^2}{b^2 m^2} + \cos^2 A$

$= b^2 a^2 \sec^2 A - a^2 b^2 \tan^2 A + \frac{a^2 b^2 \tan^2 A}{b^2 a^2 \sec^2 A} +$

$\cos^2 A$

$= a^2 b^2 (\sec^2 A - \tan^2 A) + \frac{\tan^2 A}{\sec^2 A} + \cos^2 A$

$= a^2 b^2 (1) + \frac{\sin^2 A \times \cos^2 A}{\cos^2 A} + \cos^2 A$

$= a^2 b^2 + 1$

Sol.154.(a) Given equation, $kx - 15y + 7 = 0$ and $7x - 21y - 19 = 0$

$$\text{There is no solution} = \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Then,

$$\Rightarrow \frac{k}{7} = \frac{-15}{-21} \Rightarrow k = 5$$

$$\text{Sol.155.(d)} \quad x^2 + \frac{1}{x^2} = 83$$

Subtract both side by 2 in above equation,

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 83 - 2 \Rightarrow (x - \frac{1}{x})^2 = 81$$

$$\Rightarrow x - \frac{1}{x} = \sqrt{81} = 9$$

We know that,

$$x^3 - \frac{1}{x^3} = a^3 + 3$$

$$= 9^3 + 3 \times 9 = 729 + 27 = 756$$

$$\text{Sol.156.(b)} \quad x + \frac{1}{x} = \frac{65}{8}$$

$$\Rightarrow x - \frac{1}{x} = \sqrt{(x + \frac{1}{x})^2 - 4}$$

$$= \sqrt{(\frac{65}{8})^2 - 4} = \sqrt{\frac{4225}{64} - 4}$$

$$= \sqrt{\frac{4225 - 256}{64}} = \sqrt{\frac{3969}{64}} = \frac{63}{8}$$

Sol.157.(a) Let the five consecutive numbers are $x-2, x-1, x, x+1, x+2$

According to question

$$x-2 + x-1 + x + x+1 + x+2 = 80$$

$$\Rightarrow 5x = 80 \Rightarrow x = 16$$

$$\text{Largest number} = x+2 = 16+2 = 18$$

$$\text{Sol.158.(b) Given } x = 7 - 4\sqrt{3}$$

$$\frac{1}{x} = 7 + 4\sqrt{3} \Rightarrow x + \frac{1}{x} = 14$$

$$x^2 + \frac{1}{x^2} = 14^2 - 2 = 196 - 2 = 194$$

Sol.159.(c) Let the three consecutive numbers x, y , and z be 3, 2 and 1 respectively,

$$x - y + z = 2a \dots (i)$$

Put the value x, y and z , we get

$$\Rightarrow 3 - 2 + 1 = 2a \Rightarrow a = 1$$

$$x + 4y - 2z = 3(4 - a) \dots (ii)$$

$$\Rightarrow 3 + 8 - 2 = 3(4 - a) \Rightarrow 9 = 3(4 - a) \Rightarrow a = 1$$

$$2x - 3y + 4z = 6 - 2a \dots (iii)$$

$$\Rightarrow 6 - 6 + 4 = 6 - 2a \Rightarrow 4 = 6 - 2a \Rightarrow a = 1$$

It is clear from all above equations satisfies the value of x, y and z , so the value of $a = 1$

$$\text{Sol.160.(b) Given: } x + \frac{1}{x} = 2$$

Put the value of $x = 1$ satisfies the equation,

$$\Rightarrow x + \frac{1}{x} = 1 + \frac{1}{1} = 2$$

$$\Rightarrow \text{LHS} = \text{RHS}$$

Now,

$$(x^{2723} + \frac{1}{x^{3356}}) = (1)^{2723} + \frac{1}{(1)^{3356}}$$

$$= 1 + 1 = 2$$

$$\text{Sol.161.(d)} \quad 3a + 2b = 27$$

On cubing both side we get

$$(3a)^3 + (2b)^3 + 3 \times 3a \times 2b(3a + 2b) = 27^3$$

$$27a^3 + 8b^3 + 18ab(3a + 2b) = 27^3$$

$$1458 + 18ab \times 27 = 19683$$

$$18ab \times 27 = 19683 - 1458$$

$$2ab = \frac{18225}{9 \times 27} = 75$$

$$\text{Sol.162.(a)} \quad (a + \frac{1}{a}) = 7\sqrt{3}$$

$$a^2 + \frac{1}{a^2} = (7\sqrt{3})^2 - 2 = 49 \times 3 - 2 = 145$$

$$(a^6 + \frac{1}{a^6}) = 145^3 - 3 \times 145$$

$$= 3048625 - 435 = 3048190$$

$$\text{Sol.163.(c)} \quad (x + \frac{1}{x}) = 7$$

$$(x - \frac{1}{x}) = \sqrt{7^2 - 4} = \sqrt{49 - 4}$$

$$= \sqrt{45} = 3\sqrt{5}$$

Sol.164.(d) Given,

$$ax + (a^2 + 1)y = 4 \text{ and } 4x + ay = a^2$$

After checking all options one by one, only option (d) satisfies,

$$a = -12, \text{ then } x = 48, y = 4$$

$$\Rightarrow ax + (a^2 + 1)y = 4$$

$$\text{LHS} = ax + (a^2 + 1)y$$

$$= -12 \times 48 + [(-12)^2 + 1]4 = -576 + 145 \times 4$$

$$= -576 + 580 = 4 = \text{RHS}$$

$$\Rightarrow 4x + ay = a^2$$

$$\text{LHS} = 4x + ay$$

$$= 4 \times 48 - 12 \times 4 = 192 - 48 = 144$$

$$\text{RHS} = a^2 = (-12)^2 = 144$$

$$\text{LHS} = \text{RHS}$$

$$\text{Sol.165.(a) Given: } a + b + c = 0$$

$$a + b = -c$$

On squaring both side we get,

$$a^2 + b^2 + 2ab = (-c)^2$$

$$a^2 + b^2 + 2ab = c^2$$

$$\text{Sol.166.(c)} \quad 3t - \frac{1}{3t} = 3$$

Taking square both side,

$$\Rightarrow 9t^2 + \frac{1}{9t^2} - 2 = 9 \Rightarrow 9t^2 + \frac{1}{9t^2} = 11$$

adding 1 to both sides

$$\Rightarrow 9t^2 + \frac{1}{9t^2} + 1 = 11 + 1$$

$$\Rightarrow 9t^2 + \frac{1}{9t^2} + 1 = 12$$

Sol.167.(a)

$$\text{Given: } 3x + 2y = 10 \text{ and } 2xy = 7$$

We know that,

$$(a - b) = \sqrt{(a + b)^2 - 4 \times a \times b}$$

$$(3x - 2y) = \sqrt{(3x + 2y)^2 - 4 \times 3x \times 2y}$$

So,

$$(3x - 2y) = \sqrt{(10)^2 - 12 \times 7}$$

$$= \sqrt{100 - 84} = \sqrt{16} = 4$$

Sol.168.(b) Given,

$$3x + 2y = 7 \dots (i)$$

$$2x + 3y = 7 \dots (ii)$$

If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then these equations have

a unique Solution

$$\text{Here, } a_1 = 3, a_2 = 2, b_1 = 2, b_2 = 3$$

$$\frac{3}{2} \neq \frac{2}{3}$$

$\therefore$ it has unique solution

$$\text{Sol.169.(c)} \quad (3x + 17 - 2y)(3x - 17 - 2y)$$

After rearranging we can write it,

$$= [(3x - 2y) + 17][(3x - 2y) - 17]$$

$$= (a + b)(a - b) = a^2 - b^2$$

$$= (3x - 2y)^2 - 17^2$$

$$= 9x^2 - 12xy + 4y^2 - 289$$

Sol.170.(a) When $a + b + c = 0$ then

$$a^3 + b^3 + c^3 - 3abc = 0$$

now

$$a + b = -c \dots (\text{given})$$

$$a + b + (-c) = 0$$

$$a^3 + b^3 + (-c)^3 - 3ab(-c)$$

$$a^3 + b^3 - c^3 + 3abc = 0$$

$$\text{Sol.171.(c)} \quad a^3 + \frac{1}{a^3} = 2$$

Let $a = 1$

$$\text{L.H.S.} = a^3 + \frac{1}{a^3} = 1^3 + \frac{1}{1^3} = 2 = \text{R.H.S.}$$

Therefore,

$$a + \frac{1}{a} = 1 + \frac{1}{1} = 2$$

$$\text{Sol.172.(c)} \quad x^2 + 7x + 8 = 0$$

On dividing by x we get,

$$x + 7 + \frac{8}{x} = 0 \Rightarrow x + \frac{8}{x} = -7 \dots (i)$$

Now,

$$\frac{4x}{x^2 - 5x + 8}$$

On dividing by x we get,

$$\frac{4}{x + \frac{8}{x} - 5} \Rightarrow \frac{4}{-7 - 5} = \frac{4}{-12} = -\frac{1}{3}$$

$$\text{Sol.173.(c)} \quad a_1 = 3, b_1 = 1, c_1 = -3$$

$$a_2 = a - b, b_2 = a + b, c_1 = -3a - b + 3$$

For infinite solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Hence,

$$\frac{3}{a - b} = \frac{1}{a + b} = \frac{-3}{-3a - b + 3}$$



$$\frac{1}{a+b} = \frac{-3}{-3a-b+3}$$

$$-3a - b + 3 = -3a - 3b$$

$$2b = -3 \text{ or } b = \frac{-3}{2} \dots (i)$$

Now,

$$\frac{3}{a-b} = \frac{1}{a+b} \Rightarrow 3a + 3b = a - b$$

$$2a + 4b = 0 \Rightarrow 2a + 4 \times \frac{-3}{2} = 0$$

$$2a = 6 \Rightarrow a = 3$$

$$\text{hence, } a = 3 \text{ and } b = \frac{-3}{2}$$

Sol.174.(b) $(x+y)^2 + (x-y)^2$
 $x^2 + y^2 + 2xy + x^2 + y^2 - 2xy$
 $2(x^2 + y^2)$

Sol.175.(b)

$$a_1 = 4, b_1 = 6, c_1 = -7$$

$$a_2 = 6, b_2 = K + 4, c_1 = -21$$

For Unique solution

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \frac{4}{6} \neq \frac{6}{K+4}$$

$$K + 4 \neq 9 \Rightarrow K \neq 5$$

Sol.176.(a) $a^2 + b^2 = 148$

$$\text{Then } a + b = \sqrt{148 + 2ab}$$

$$= \sqrt{148 + 108} = \sqrt{256} = 16$$

and

$$a - b = \sqrt{148 - 2ab} = \sqrt{148 - 108}$$

$$= \sqrt{40} = 2\sqrt{10}$$

now,

$$\frac{a+b}{a-b} = \frac{16}{2\sqrt{10}} = \frac{8}{\sqrt{10}}$$

Sol.177.(a) Equation :- $5x + 7y - 8 = 0$

For no solution -

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Therefore, the correct option is (a).

Sol.178.(a) $2a - b = 3 \dots (i)$

On cubing both side we get,

$$8a^3 - b^3 - 3 \times 2ab(2a - b) = 27$$

$$8a^3 - b^3 - 6ab \times 3 = 27$$

$$999 - 18ab = 27 \Rightarrow -18ab = -972$$

$$ab = 54 \Rightarrow ab = 9 \times 6$$

$$\text{Take the value of } a = 6 \text{ and } b = 9$$

$$2a - b = 3$$

$$\text{LHS} = 2a - b = 2 \times 6 - 9 = 3 = \text{RHS}$$

$$8a^3 - b^3 = 999$$

$$\text{LHS} = 8a^3 - b^3$$

$$= 8 \times 216 - 729 = 999$$

Value of a and b satisfies the given equation,

So,

$$4a^2 - b^2 = 4 \times 36 - 81 = 63$$

Sol.179.(b) Let correct answers be x and wrong or un-attempted answer be y
 According to the question,

$$x + y = 160$$

$$y = 160 - x \dots (i)$$

$$4x - y = 400 \dots (ii)$$

Put the value of y in equation (ii)

$$4x - (160 - x) = 400 \Rightarrow 4x - 160 + x = 400$$

$$5x = 560 \Rightarrow x = 112$$

Hence, he did 112 answers correctly.

Sol.180.(c) $2x - 3 = 0 \Rightarrow x = \frac{3}{2}$

Putting the value of x in $2x^3 - 3x^2 + 6x - 4$

$$\Rightarrow 2\left(\frac{3}{2}\right)^3 - 3\left(\frac{3}{2}\right)^2 + 6 \times \frac{3}{2} - 4$$

$$\Rightarrow 2 \times \frac{27}{8} - 3 \times \frac{9}{4} + 6 \times \frac{3}{2} - 4$$

$$\Rightarrow \frac{27}{4} - \frac{27}{4} + 9 - 4 = 5$$

Sol.181.(d) $A = \frac{9}{16} \div \frac{25}{36}$

$$= \frac{9}{16} \times \frac{36}{25} = \frac{81}{100}$$

$$B = 9 \div [(16 \div 25) \div 36]$$

$$= 9 \div \left[\frac{16}{25} \times \frac{1}{36}\right] = 9 \div \left[\frac{4}{225}\right]$$

$$= 9 \times \left[\frac{225}{4}\right] = \frac{2025}{4}$$

$$C = [9 \div (16 \div 25)] \div 36$$

$$= [9 \div \frac{16}{25}] \div 36 = [9 \times \frac{25}{16}] \div 36$$

$$= \frac{25}{16} \times \frac{1}{4} = \frac{25}{64}$$

$$D = (9 \div 16) \div (25 \div 36)$$

$$= \left(\frac{9}{16}\right) \div \left(\frac{25}{36}\right) = \left(\frac{9}{16}\right) \times \left(\frac{36}{25}\right) = \frac{81}{100}$$

Clearly, value of A and D are equal.

Sol.182.(c)

$$A = 60 \left[\frac{6}{25} \div \left(\frac{16}{15} \text{ of } \frac{3}{8} \right) + 5 - \left(1 - 2\frac{1}{12} \right) \right]$$

$$= 60 \left[\frac{6}{25} \div \left(\frac{2}{5} \right) + 5 - \left(1 - \frac{25}{12} \right) \right]$$

$$= 60 \left[\frac{6}{25} \times \left(\frac{5}{2} \right) + 5 - \left(-\frac{13}{12} \right) \right]$$

$$= 60 \left[\frac{3}{5} + 5 + \frac{13}{12} \right] = 60 \left[\frac{28}{5} + \frac{13}{12} \right]$$

$$= 60 \left[\frac{28 \times 12 + 13 \times 5}{60} \right] = 60 \left[\frac{336 + 65}{60} \right]$$

$$= 60 \left[\frac{401}{60} \right] = 401$$

Sum of the digits of A = 4 + 0 + 1 = 5

Sol.183.(b) $640 \div 8 + 930 \div 15 - k + 32$

$$\times 5 = 1104 \div 16 \times 148 \div 37$$

$$\Rightarrow 80 + 62 - k + 160 = 69 \times 4$$

$$\Rightarrow 80 + 62 - k + 160 = 276$$

$$\Rightarrow 302 - k = 276 \Rightarrow -k = -26 \Rightarrow k = 26$$

Sol.184.(d) For system of linear

$$\text{equations to be parallel, } \frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\text{Here, } a_1 = 5, b_1 = k, c_1 = -2 \text{ \& } a_2 = k + 1,$$

$$b_2 = 4, c_2 = 7$$

$$\frac{5}{k+1} = \frac{k}{4} \Rightarrow k^2 + k = 20$$

$$k^2 + k - 20 = 0 \Rightarrow k^2 + 5k - 4k - 20 = 0$$

$$k(k+5) - 4(k+5) = 0$$

$$(k+5)(k-4) = 0 \Rightarrow k = 4$$

Sol.185.(b)

$$\frac{15}{8} + \left[\left\{ \frac{4}{5} \text{ of } \frac{5}{12} \div \left(1 - \frac{1}{15} \right) \right\} \times 1\frac{7}{12} \right]$$

$$\frac{15}{8} + \left[\left\{ \frac{1}{3} \div \frac{16}{15} \right\} \times \frac{19}{12} \right]$$

$$\frac{15}{8} + \left[\left\{ \frac{1}{3} \times \frac{15}{16} \right\} \times \frac{19}{12} \right]$$

$$\frac{15}{8} + \left[\frac{5}{16} \times \frac{19}{12} \right]$$

$$\frac{15}{8} + \frac{95}{192} = \frac{360 + 95}{192} = \frac{455}{192}$$

According to question,

$$\frac{a}{b} = \frac{455}{192}$$

Hence,

$$4b - a = 4 \times 192 - 455$$

$$= 768 - 455 = 313$$

Sol.186.(d) Given:- $x + y = 12$ and $xy = 27$ As we know, $x^3 + y^3 = (x + y)$

$$[(x + y)^2 - 3xy] = 12[12^2 - 3 \times 27]$$

$$= 12[144 - 81] = 12 \times 63 = 756$$

Sol.187.(a) $x + y = 4(x - y)$

$$x + y = 4x - 4y \Rightarrow 5y = 3x \Rightarrow \frac{y}{x} = \frac{3}{5}$$

Let $x = 5a$ and $y = 3a$

$$\frac{5xy}{3(x^2 - y^2)} = \frac{5 \times 5a \times 3a}{3(25a^2 - 9a^2)}$$

$$= \frac{75a^2}{3 \times 16a^2} = \frac{25}{16}$$

Sol.188.(b) Formula used :

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$[(x + y + z)^2 - 3(xy + yz + zx)]$$

$$\text{Now, } x^3 + y^3 + z^3 - 3 \times (-1) = 1$$

$$[1^2 - 3 \times (-1)]$$

$$x^3 + y^3 + z^3 + 3 = 4 \Rightarrow x^3 + y^3 + z^3 = 1$$

Sol.189.(c) Put $\alpha = 0, \beta = 1, \gamma = -1$.

we have ;

$$\frac{3\beta^2 + \alpha^2 + \gamma^2}{2\beta^2 - \alpha\gamma} = \frac{3 + 0 + 1}{2 - 0} = \frac{4}{2} = 2$$

Sol.190.(b) Put $y = 0$, then $x = 4$

$$\text{Now, } x^3 - 27y^3 - 36xy = 4^3 - 0 - 0 = 64$$

Sol.191.(a) Given: $x + \frac{1}{x} = 2$ Putting the value of $x = 1$

$$\text{LHS} = x + \frac{1}{x} = 1 + 1 = 2$$

LHS = RHS

$$\text{Now, } (x^4 + \frac{4}{x^4})(x^3 - \frac{4}{x^3})$$

$$= [(1)^4 + \frac{4}{(1)^4}][(1)^3 - \frac{4}{(1)^3}]$$

$$= 5 \times -3 = -15$$



Sol.192.(b) $16x^4 + \frac{1}{16x^4} = 14159$

$$4x^2 + \frac{1}{4x^2} = \sqrt{14159 + 2}$$

$$= \sqrt{14161} = 119 \Rightarrow 4x^2 + \frac{1}{4x^2} = 119$$

$$\text{and } 2x + \frac{1}{2x} = \sqrt{119 + 2} = \sqrt{121} = 11$$

Sol.193.(a) $a + 2b + 3c = 0$

We know that when

$$x + y + z = 0$$

$$\text{then } x^3 + y^3 + z^3 = 3xyz$$

Hence,

$$a^3 + 8b^3 + 27c^3 = 3 \times a \times 2b \times 3c = 18abc$$

Sol.194.(a) $x = 110, y = 111, z = 112$

$$x^3 + y^3 + z^3 - 3xyz$$

$$= \frac{1}{2}(x + y + z) \{(x - y)^2 + (y - z)^2 + (z - x)^2\}$$

$$= \frac{1}{2}(110 + 111 + 112)\{(110 - 111)^2 +$$

$$(111 - 112)^2 + (112 - 110)^2\}$$

$$= \frac{1}{2}(333)\{(-1)^2 + (-1)^2 + (2)^2\}$$

$$= \frac{1}{2}(333) \times 6 = 999$$

Sol.195.(b) $(a - 18)^2 + (b - 12)^2 + (c - 6)^2 = 0$

In such condition,

$$\text{Put, } a - 18 = 0 \Rightarrow a = 18 \dots (i)$$

$$b - 12 = 0 \Rightarrow b = 12 \dots (ii)$$

$$c - 6 = 0 \Rightarrow c = 6 \dots (iii)$$

$$(a + b + c)^{\frac{1}{2}} = \sqrt{a + b + c}$$

$$= \sqrt{18 + 12 + 6} = \sqrt{36} = \pm 6$$

Sol.196.(a) $x^2 + y^2 = 528 \dots (i)$

$$x + y = 28$$

On squaring both side

$$x^2 + y^2 + 2xy = 784$$

$$528 + 2xy = 784 \dots [\text{from eq. (i)}]$$

$$2xy = 784 - 528 \Rightarrow xy = 128$$

$$\text{Now, } \sqrt{xy} = 8\sqrt{2}$$

Sol.197.(b) $\frac{a}{b} + \frac{b}{a} = 1$

$$a^2 + b^2 = ab \dots (i)$$

$$\text{Now, } \frac{a^3 + b^3}{3ab}$$

$$= \frac{(a + b)(a^2 + b^2 - ab)}{3ab}$$

$$= \frac{(a + b)(ab - ab)}{3ab} \dots \text{from (i)} = 0$$

Sol.198.(a)

$$x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1$$

$$x^5 - 11x^4 - x^4 + 11x^3 + x^3 - 11x^2 - x^2 + 11x + x - 1$$

$$\text{Now, } x = 11 \dots (\text{given})$$

On putting value of 11 as x , we get

$$x^5 - x \times x^4 - x^4 + x \times x^3 + x^3 - x \times x^2 - x^2 + x \times$$

$$x + x - 1$$

$$x^5 - x^5 - x^4 + x^4 + x^3 - x^3 - x^2 + x^2 + x - 1$$

$$\Rightarrow x - 1 = 11 - 1 = 10$$

Sol.199.(d) $a^3 + b^3 + c^3 - 3abc = (a + b + c)$
 $[(a + b + c)^2 - 3(ab + bc + ca)]$

$$= (15) [(15)^2 - 3(35)] = (15) [225 - 105]$$

$$= 15 \times 120 = 1800$$

Sol.200.(a) $p(x) = (x^3 - 3x^2 + 4x - 75)$
 $g(x) = (x - 5)$

$$\text{Put } g(x) = 0 \Rightarrow x - 5 = 0 \Rightarrow x = 5$$

Put value of x in $p(x)$

$$p(x) = (5^3 - 3 \times 5^2 + 4 \times 5 - 75)$$

$$= (125 - 3 \times 25 + 20 - 75)$$

$$= (125 - 75 + 20 - 75) = -5$$

5 must be added to $p(x)$.

Sol.201.(d) $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c) [(a + b + c)^2 - 3(ab + bc + ca)]$$

$$= 13 \times [(13)^2 - 3(22)]$$

$$= 13 \times [169 - 66] = 13 \times 103 = 1339$$

Sol.202.(a) $x^2 + 4y^2 + 2x + 1 = 0$

Let, $x = -1$ and $y = 0$ then,

$$\text{LHS} = x^2 + 4y^2 + 2x + 1$$

$$= (-1)^2 + 4(0)^2 + 2(-1) + 1$$

$$= 0 = \text{RHS (satisfy)}$$

$$\text{Now, } x^{39} + y^{36} = (-1)^{39} + (0)^{36} = -1$$

Sol.203.(a) $x^2 - 6x - 1 = 0$

On Dividing by x , we get

$$x - 6 - \frac{1}{x} = 0 \Rightarrow x - \frac{1}{x} = 6$$

On squaring both side

$$x^2 + \frac{1}{x^2} = 36 + 2 \Rightarrow x^2 + \frac{1}{x^2} = 38$$

$$\text{Now, } x^2 + \frac{1}{x^2} - 8 = 38 - 8 = 30$$

Sol.204.(c) Given : $2a + 2b + c = 0$

For such type of question we should satisfy the given condition by putting any value of a, b, c .

Here, Put $a = 0, b = 1$ and $c = -2$

$$2(0) + 2(1) + (-2) = 0$$

$$0 + 2 - 2 = 0 \Rightarrow 0 = 0$$

Hence, the above values satisfy the given part.

Now put above value in question part

$$\frac{4a^2 + 4b^2 + 4c^2}{5c^2 - 8ab}$$

$$= \frac{4 \times (0)^2 + 4 \times (1)^2 + 4 \times (-2)^2}{5(-2)^2 - 8 \times 0 \times 1}$$

$$= \frac{0 + 4 + 4 \times 4}{5 \times 4 - 0} = \frac{20}{20} = 1$$

Sol.205.(b) $a^3 = (\sqrt{2} - 1)^{\frac{1}{3}}$

On taking cube both side,

$$a^3 = \sqrt{2} - 1 \text{ and } \frac{1}{a^3} = \sqrt{2} + 1$$

Now,

$$(a - \frac{1}{a})^3 = a^3 - \frac{1}{a^3} - 3 \times a \times \frac{1}{a} (a - \frac{1}{a})$$

$$(a - \frac{1}{a})^3 + 3(a - \frac{1}{a}) = a^3 - \frac{1}{a^3}$$

On putting value of a^3 and $\frac{1}{a^3}$

$$(a - \frac{1}{a})^3 + 3(a - \frac{1}{a})$$

$$= \sqrt{2} - 1 - (\sqrt{2} + 1)$$

$$= \sqrt{2} - 1 - \sqrt{2} - 1 = -2$$

Sol.206.(b) Given: $x = 4 + y$

$$\Rightarrow x - y = 4$$

$$\Rightarrow x^2 + y^2 - 2xy = 16 \dots (i)$$

$$\text{and } x^3 = 184 + y^3 \Rightarrow x^3 - y^3 = 184$$

$$\Rightarrow (x - y)(x^2 + y^2 + xy) = 184$$

$$\Rightarrow (4)(x^2 + y^2 + xy) = 184$$

$$\Rightarrow (x^2 + y^2 + xy) = 46 \dots (ii)$$

On solving eq. (i) and (ii)

$$xy = 10$$

$$\Rightarrow (x + y)^2 = (x^2 + y^2 + xy) + xy = 46 + 10 = 56$$

$$\text{So, } x + y = 2\sqrt{14}$$

Sol.207.(d) $(x - 3y)^3 + 27(y^3 - xy^2)$

$$\Rightarrow x^3 - 27y^3 - 9xy(x - 3y) + 27y^3 - 27xy^2$$

$$\Rightarrow x^3 - 27y^3 - 9x^2y + 27xy^2 + 27y^3 - 27xy^2$$

$$\Rightarrow x^3 - 9x^2y$$

Sol.208.(d) Let $x = y = z = 1$

Put the value in the given expression

$$\frac{17x^4 + 9y^4 + 16z^4}{8x^2y^2 + 6y^2z^2 + 10z^2x^2}$$

$$= \frac{17(1)^4 + 9(1)^4 + 16(1)^4}{8(1)^2(1)^2 + 6(1)^2(1)^2 + 10(1)^2(1)^2}$$

$$= \frac{17 + 9 + 16}{8 + 6 + 10} = \frac{42}{24} = 1.75$$

Sol.209.(d) Given, $7x^2 + 5x + 7 = 0$

Dividing both sides by x , we get

$$7x + 5 + \frac{7}{x} = 0 \Rightarrow x + \frac{1}{x} = \frac{-5}{7}$$

Taking cube both sides

$$(x + \frac{1}{x})^3 = (\frac{-5}{7})^3$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} (x + \frac{1}{x}) = \frac{-125}{343}$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3(\frac{-5}{7}) = \frac{-125}{343}$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \frac{-125}{343} + \frac{15}{7}$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \frac{610}{343}$$

Sol.210.(d) We know

$$(a + b)^2 = a^2 + b^2 + 2ab$$

Hence,

$$49a^2 + 70ab + 25b^2 = (7a + 5b)^2$$

Sol.211.(c) $(x + y)^2 = x^2 + y^2 + 2xy$

Subtracting both side by $3xy$, we get ;

$$(x + y)^2 - 3xy = x^2 + y^2 - xy$$



$$10^2 - 3 \times 12 = x^2 + y^2 - xy$$

$$100 - 36 = x^2 + y^2 - xy \Rightarrow x^2 + y^2 - xy = 64$$

$$\text{Sol.212.(c)} \left(x - \frac{1}{x-1}\right) = 3$$

Subtracting both side by 1, we have ;

$$\left(x - 1 - \frac{1}{x-1}\right) = 2$$

$$\text{Let, } (x-1) = k \Rightarrow k - \frac{1}{k} = 2$$

$$k^2 + \frac{1}{k^2} = 2^2 + 2 = 6$$

$$\text{So, } (x-1)^2 + \frac{1}{(x-1)^2} = 6$$

Sol.213.(d) Formula used :

$$a^2 + b^2 + c^2 - ab - bc - ca$$

$$= \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

Then, $p^2 + q^2 + r^2 - pq - qr - rp$

$$= \frac{1}{2}[(106-105)^2 + (105-104)^2 + (104-106)^2]$$

$$= \frac{1}{2}[1^2 + 1^2 + (-2)^2] \Rightarrow \frac{1}{2} \times 6 = 3$$

$$\text{Sol.214.(c)} \sqrt{1 + \frac{x^3}{9}} = 5$$

$$1 + \frac{x^3}{9} = 25 \Rightarrow \frac{x^3}{9} = 24$$

$$\Rightarrow x^3 = 24 \times 9 = 216 \Rightarrow x = 6$$

$$\text{Sol.215.(b)} x + \frac{1}{x} = 15$$

On squaring both side, we get;

$$x^2 + \frac{1}{x^2} + 2 = 225$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 225 - 2 = 223$$

$$\text{Sol.216.(c)} (a-2b) = 8, 2ab = 9$$

$$\therefore (a-2b)^2 = (a+2b)^2 - 8ab$$

$$(a+2b)^2 = 8^2 + 8 \times \frac{9}{2}$$

$$(a+2b)^2 = 64 + 36 = 100$$

$$(a+2b) = 10$$

Sol.217.(d) According to question,

$$A + 10 = (B-10) \times 2$$

$$A - 2B = -30 \dots (i)$$

$$A - 10 = B + 10$$

$$A - B = 20 \dots (ii)$$

From equation (ii) $\times 2$ - (i), we get, $A = 70$

Sol.218.(a) Given,

$$x + y = 2 \text{ and } \frac{x}{y} + \frac{y}{x} = 1$$

$$\Rightarrow x^2 + y^2 = xy$$

Formula :

$$x^3 + y^3 = (x+y)(x^2 + y^2 - xy)$$

$$\Rightarrow x^3 + y^3 = 2 \times 0 = 0$$

Sol.219.(a) According to the question,

$$\frac{A}{L} = 1 - \frac{M}{B} = \frac{B-M}{B}$$

$$\frac{L}{A} = \frac{B}{B-M} \text{ and } \frac{N}{C} = 1 - \frac{B}{M} = \frac{M-B}{M}$$

$$= \frac{C}{N} = \frac{M}{M-B}$$

$$= \frac{L}{A} + \frac{C}{N} = \frac{B}{B-M} - \frac{M}{B-M} = \frac{B-M}{B-M} = 1$$

$$\text{Sol.220.(c)} \text{ Given, } a \propto b \propto \frac{1}{c} \propto \frac{1}{d}$$

$$\text{So, } a \propto \frac{1}{d}$$

$$\text{Sol.221.(d)} A + \frac{1}{1 + \frac{1}{2 + \frac{1}{3}}} = \frac{9}{10}$$

$$\Rightarrow A + \frac{1}{1 + \frac{1}{\frac{7}{3}}} = \frac{9}{10}$$

$$\Rightarrow A + \frac{1}{1 + \frac{3}{7}} = \frac{9}{10} \Rightarrow A + \frac{7}{10} = \frac{9}{10}$$

$$\Rightarrow A = \frac{2}{10} \text{ or } \frac{1}{5}$$

$$\text{Sol.222.(d)} x(5 - \frac{2}{x}) = \frac{5}{x}$$

$$\Rightarrow x(\frac{5x-2}{x}) = \frac{5}{x} \Rightarrow 5x^2 - 5 = 2x$$

Dividing both sides by $5x$

$$\Rightarrow x - \frac{1}{x} = \frac{2}{5}$$

$$\text{then } x^2 + \frac{1}{x^2} = \frac{4}{25} + 2 = \frac{54}{25}$$

Sol.223.(d) Formula used :-

$$a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

$$\frac{(157 \times 157) + (157 \times 133) + (133 \times 133)}{(157 \times 157 \times 157) - (133 \times 133 \times 133)}$$

Let $157 = a$ and $133 = b$ then,

$$\frac{a^2 + ab + b^2}{(a-b)(a^2 + b^2 + ab)} = \frac{1}{(a-b)}$$

$$= \frac{1}{157 - 133} = \frac{1}{24}$$

Sol.224.(a) Formula used : - $a^3 - b^3$

$$= (a-b)(a^2 + b^2 + ab)$$

Let the $a = 3.254$ and $b = 0.746$

$$\frac{(a-b)(a^2 + b^2 + ab)}{(a^2 + b^2 + ab)} = (a-b)$$

$$= 3.254 - 0.746 = 2.508$$

$$\text{Sol.225.(c)} \frac{a^2 - (b-c)^2}{(a+c)^2 - b^2} + \frac{b^2 - (a-c)^2}{(a+b)^2 - c^2}$$

$$+ \frac{c^2 - (a-b)^2}{(b+c)^2 - a^2} = \frac{\{(a+(b-c))(a-(b-c))\}}{\{(a+c-b)(a+c+b)\}} + \frac{\{(b+(a-c))(b-(a-c))\}}{\{(a+b-c)(a+b+c)\}} + \frac{\{(c+(a-b))(c-(a-b))\}}{\{(b+c-a)(b+c+a)\}}$$

$$= \frac{\{(a+(b-c))\}}{\{(a+c-b)\}} + \frac{\{(b-(a-c))\}}{\{(a+b-c)\}} + \frac{\{(c+(a-b))\}}{\{(b+c-a)\}} = \frac{\{(a+(b-c)) + (b-(a-c)) + (c+(a-b))\}}{\{(a+c-b)\}} = \frac{(a+c+b)}{(a+c+b)} = 1$$

Sol.226.(c)

$$\Rightarrow (1+x)^3 + (1-x)^3 + (-2)^3 \Rightarrow (1^3 + x^3 + 3x(1+x)) + (1^3 - x^3 - 3x(1-x)) + (-2)^3 \Rightarrow 6x^2 + 2 \cdot 6x^2 - 6 = -6(1-x^2)$$

Sol.227.(a) Formula :

$$(a+b)^2 - (a-b)^2 = 4ab$$

$$(1-2x)^2 - (1+2x)^2 = -4 \times 1 \times 2x = -8x$$

Sol.228.(a) $a + b + c = 5 \dots (1)$

and $ab + bc + ca = 7 \dots (2)$

Squaring both side in equation (1)

$$(a+b+c)^2 =$$

$$a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\Rightarrow a^2 + b^2 + c^2 = 25 - 14 = 11$$

Then, $a^3 + b^3 + c^3 - 3abc =$

$$(a+b+c)[a^2 + b^2 + c^2 - (ab + bc + ca)]$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 5(11 - 7) = 20$$

$$\text{Sol.229.(d)} \Rightarrow (x-2y)(y-3x) + (x+y)(x-y) + (x-3y)(2x+y)$$

$$\Rightarrow xy - 3x^2 - 2y^2 + 6xy + x^2 - y^2 + 2x^2 + xy - 6xy - 3y^2$$

$$\Rightarrow 2xy - 6y^2 = 2y(x-3y)$$

Sol.230.(b) Put $x = 2$ in equation

$$2x^2 + 12kx - 25k = 0$$

$$2 \times (2)^2 + 12k \times 2 - 25k = 0 \Rightarrow k = 8$$

$$\text{Sol.231.(d)} 2x + \frac{2}{x} = 5 \Rightarrow x + \frac{1}{x} = \frac{5}{2}$$

$$x^3 + \frac{1}{x^3} = \left(\frac{5}{2}\right)^3 - 3 \times \left(\frac{5}{2}\right) = \frac{65}{8}$$

$$x^3 + \frac{1}{x^3} + 2 = \frac{65}{8} + 2 = \frac{81}{8}$$

Sol.232.(d)

$$(a^3 - b^3) = (a-b)\{(a-b)^2 + 3ab\}$$

$$4401 = 9\{81 + 3ab\}$$

$$4401 = 9\{81 + 3ab\}$$

$$489 - 81 = 3ab \Rightarrow ab = \frac{408}{3} = 136$$

Sol.233.(c) According to question,

$$a + \frac{1}{a} = 6 \Rightarrow a^2 + \frac{1}{a^2} = (6)^2 - 2 = 34$$

$$\text{Now, } \frac{3}{4} \left(a^2 + \frac{1}{a^2}\right) = \frac{3}{4} \times 34 = \frac{51}{2} =$$

$$25.5$$



Sol.234.(b) $a = \frac{1}{a - \sqrt{6}} \Rightarrow a - \frac{1}{a} = \sqrt{6}$

$$a + \frac{1}{a} = \sqrt{(\sqrt{6})^2 + 4} = \sqrt{10}$$

Sol.235.(c)

$$(a + b) = 30 \Rightarrow a^2 + b^2 = 900 - 2ab$$

$$\sqrt{\frac{a}{b}} = \frac{8}{3} + \sqrt{\frac{b}{a}}$$

$$\frac{a-b}{\sqrt{ab}} = \frac{8}{3} \Rightarrow \frac{a^2 + b^2 - 2ab}{ab} = \frac{64}{9}$$

$$\frac{900 - 4ab}{ab} = \frac{64}{9} \Rightarrow ab = \frac{8100}{100} = 81$$

Sol.236.(c) $2x + 3y = 9$ and $xy = 3$

Then,

$$8x^3 + 27y^3 = (2x + 3y)^3$$

$$\{(2x + 3y)^2 - 3(2x \times 3y)\}$$

$$= (9) \{(9)^2 - 18xy\} = (9) \{81 - 18 \times 3\} = 243$$

Sol.237.(b) $a^2 + \frac{1}{a^2} = \frac{7}{3}$

$$a - \frac{1}{a} = \sqrt{\left(\frac{7}{3} - 2\right)} = \frac{1}{\sqrt{3}}$$

$$\text{Now, } a^3 - \frac{1}{a^3} = \left(\frac{1}{\sqrt{3}}\right)^3 + 3 \times \frac{1}{\sqrt{3}}$$

$$= \frac{1}{3\sqrt{3}} + \sqrt{3} = \frac{10}{3\sqrt{3}}$$

Sol.238.(d) $a + b = 10$ and $a^2 + b^2 = 58$

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$100 = 58 + 2ab \Rightarrow ab = 21$$

Sol.239.(c) $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Sol.240.(c)

$$2a + 3b = 14 \text{ and } 2a - 3b = 10$$

On adding, we get

$$4a = 24 \Rightarrow a = 6 \text{ and } b = \frac{2}{3}$$

$$\text{Now, } ab = 4$$

Sol.241.(a) $(a - b) = 10$

$$\Rightarrow a^2 + b^2 = 100 + 2ab$$

$$\sqrt{\frac{a}{b}} = \frac{8}{3} - \sqrt{\frac{b}{a}}$$

$$\frac{a+b}{\sqrt{ab}} = \frac{8}{3} \Rightarrow \frac{a^2 + b^2 + 2ab}{ab} = \frac{64}{9}$$

$$\frac{100 + 4ab}{ab} = \frac{64}{9}$$

$$ab = \frac{900}{28} = \frac{225}{7} = 32\frac{1}{7}$$

Sol.242.(b)

$$\text{Identity } \therefore (a + b)^2 = a^2 + b^2 + 2ab$$

Sol.243.(c)

$$\frac{x^2 + 7x + 2}{x^2 + 13x + 2} = \frac{x + \frac{2}{x} + 7}{x + \frac{2}{x} + 13}$$

$$= \frac{1+7}{1+13} = \frac{4}{7}$$

Sol.244.(d) $x + \frac{1}{x} = 7$

$$x - \frac{1}{x} = \sqrt{(7)^2 - 4} = 3\sqrt{5}$$

$$\text{Now, } x^3 - \frac{1}{x^3} = (3\sqrt{5})^3 + 3 \times (3\sqrt{5})$$

$$= 144\sqrt{5}$$

Sol.245.(b) $y - \frac{1}{y} = 9$

$$y^3 - \frac{1}{y^3} = (9)^3 + 3 \times 9 = 756$$

Sol.246.(c) $(2x + 3y)^2 = 4x^2 + 9y^2 + 12xy$

$$16 = 64 + 12xy \Rightarrow xy = (-4)$$

$$\text{Now, } (2x + 3y)^3 = 8x^3 + 27y^3 + 3 \times 2x \times 3y(2x + 3y)$$

$$64 = 8x^3 + 27y^3 - 72(4)$$

$$8x^3 + 27y^3 = 288 + 64 = 352$$

Sol.247.(d) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$$a^2 + b^2 + c^2 = 225 - 2 \times 22 = 181$$

Sol.248.(b)

$$\{(x - 5)(x - 1)\} - \{(9x - 5)(9x - 1)\}$$

$$\div 16x$$

$$\Rightarrow \{x^2 - 6x + 5\} - \{81x^2 - 54x + 5\} \div 16x$$

$$\Rightarrow \{[-80x^2 + 48x]\} \div 16x$$

$$\Rightarrow [16x\{-5x + 3\}] \div 16x \Rightarrow -\{5x - 3\}$$

Sol.249.(a) $x + \frac{1}{x} = 2 \Rightarrow x = 1$

$$x^{\frac{1823}{1929}} + \frac{1}{x^{\frac{1929}{1823}}} = 1 + 1 = 2$$

Sol.250.(d) $2x^2 - 32 = 2(x^2 - 16)$

$$= 2(x - 4)(x + 4)$$

Sol.251.(d) $x - \frac{1}{x} = 0.4$

$$x^2 + \frac{1}{x^2} = (0.4)^2 + 2 = 2.16$$

Sol.252.(d) $x + \frac{1}{x} = 5 \Rightarrow x^2 + \frac{1}{x^2} = 23$

$$\Rightarrow x^3 + \frac{1}{x^3} = 110$$

Now,

$$x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$$

$$\Rightarrow 23 \times 110 - 5 = 2530 - 5 = 2525$$

Sol.253.(c)

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2$$

$$(xy + yz + zx)$$

$$(x + y + z) = \sqrt{628 + 2 \times 24} = \sqrt{676} = 26$$

Sol.254.(b)

$$x + \frac{1}{x} = -1 \Rightarrow x^3 = 1 \text{ or } x^{15} = 1$$

Now,

$$x^{15} + \frac{1}{x^{15}} = 1 + 1 = 2$$

Sol.255.(d) $\frac{x^3 + y^3 + z^3 - 3xyz}{(x - y)^2 + (y - z)^2 + (z - x)^2}$

$$\frac{\frac{(x+y+z)}{2} \{(x - y)^2 + (y - z)^2 + (z - x)^2\}}{(x - y)^2 + (y - z)^2 + (z - x)^2}$$

$$= \frac{(x + y + z)}{2}$$

Sol.256.(c) $(x - 2) - \frac{1}{(x - 2)} = 5$

$$(x - 2)^3 - \frac{1}{(x - 2)^3} = 5^3 + 3 \times 5 = 140$$

Sol.257.(a) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$$256 = a^2 + b^2 + c^2 + 2 \times 81$$

$$a^2 + b^2 + c^2 = 94$$

Sol.258.(c)

$$(a + b + c)^2 - (a - b + c)^2 + 4ac$$

$$\Rightarrow 2ab + 2bc + 2ac + 2ab + 2bc - 2ac + 4ac$$

$$\Rightarrow 4(ab + bc + ca)$$

Sol.259.(b) $x + \frac{1}{x} = 5\sqrt{5}$

$$\Rightarrow x - \frac{1}{x} = \sqrt{(5\sqrt{5})^2 - 4} = 11$$

$$\Rightarrow x^3 - \frac{1}{x^3} = (11)^3 + 3 \times 11 = 1364$$

Sol.260.(c) $x^4 + \frac{1}{x^4} = 194$

$$x^2 + \frac{1}{x^2} = \sqrt{194 + 2} = 14$$

$$x + \frac{1}{x} = \sqrt{14 + 2} = 4$$

$$x^3 + \frac{1}{x^3} = 4^3 - 3 \times 4 = 52$$

$$\text{Now, } x^3 + \frac{1}{x^3} + x + \frac{1}{x} = 52 + 4 = 56$$

Sol.261.(a)

$$a^3 + b^3 + c^3 - 3abc = \frac{a + b + c}{2}$$

$$\{3(a^2 + b^2 + c^2) - (a + b + c)^2\}$$

$$= \frac{5}{2} \{3(15) - (5)^2\} = \frac{5}{2} \{(20)\} = 50$$

$$\text{Now, } a^3 + b^3 + c^3 - 3abc - 27 = 50 - 27 = 23$$

Sol.262.(c) $x + y = \sqrt{427 + 404} = \sqrt{831}$

$$x - y = \sqrt{427 - 404} = \sqrt{23}$$

$$\text{Now, } \frac{x + y}{x - y} = \sqrt{\frac{831}{23}}$$

Sol.263.(a) $4x^2 - y^2 + 6y - 9$

$$= 4x^2 - (y^2 - 6y + 9) \Rightarrow (2x)^2 - (y - 3)^2$$

$$= (2x + y - 3)(2x - y + 3)$$



Sol.264.(b)

If $x - \frac{1}{x} = k$ then, $x^3 - \frac{1}{x^3} = k^3 + 3k$

$$x^3 = \sqrt{6} - 1 \Rightarrow \frac{1}{x^3} = \frac{1}{\sqrt{6} - 1}$$

$$x^3 - \frac{1}{x^3} = (\sqrt{6} - 1) - \frac{1}{\sqrt{6} - 1}$$

$$= \frac{7 - 2\sqrt{6} - 1}{\sqrt{6} - 1} = \frac{6 - 2\sqrt{6}}{\sqrt{6} - 1} \times \frac{\sqrt{6} + 1}{\sqrt{6} + 1}$$

$$= \frac{6\sqrt{6} + 6 - 12 - 2\sqrt{6}}{5} = \frac{4\sqrt{6} - 6}{5}$$

Sol.265.(b) $ab + bc + ca$

$$= \frac{(a+b+c)^2 - (a^2 + b^2 + c^2)}{2}$$

$$= \frac{(17)^2 - (101)}{2} = 94$$

Now,

$$(a-b)^2 + (b-c)^2 + (c-a)^2 =$$

$$2(a^2 + b^2 + c^2) - 2(ab + bc + ca)$$

$$= 2 \times 101 - 2 \times 94 = 14$$

Sol.266.(a) $x + \frac{1}{x} = 5$

$$x^2 + \frac{1}{x^2} = 5^2 - 2 = 23$$

$$x^4 + \frac{1}{x^4} = 23^2 - 2 = 527$$

$$\text{Now, } x^4 + \frac{1}{x^4} - 36 = 527 - 36 = 491$$

Sol.267.(c) $a^2 + b^2 + c^2 = ab + bc + ca$
only possible when $a = b = c$

$$\text{So, } \frac{11a^4 + 13b^4 + 15c^4}{16a^2b^2 + 19b^2c^2 + 17c^2a^2} = \frac{39a^4}{52a^4}$$

$$= \frac{3}{4}$$

Sol.268.(a) $(c+d)^2 - (c-d)^2 = 4cd$ **Sol.269.(b)**

$$\frac{x^2 - 3x + 2}{x^2 - 4} = \frac{x^2 + 1 - 2x - x + 1}{(x-2)(x+2)}$$

$$= \frac{(x-1)^2 - (x-1)}{(x-2)(x+2)} = \frac{(x-1)(x-2)}{(x-2)(x+2)}$$

$$= \frac{(x-1)}{(x+2)}$$

Sol.270.(a) If $(l+m+n) = 0$ then,

$$l^3 + m^3 + n^3 = 3lmn$$

$$\left(\frac{l^2}{mn} + \frac{m^2}{nl} + \frac{n^2}{lm}\right)^2$$

$$= \left(\frac{l^3 + m^3 + n^3}{lmn}\right)^2 = \left(\frac{3lmn}{lmn}\right)^2 = 9$$

Sol.271.(a) $2x + 5y = 15$, $xy = 6$

$$(2x + 5y)^2 = 4x^2 + 25y^2 + 20xy$$

$$4x^2 + 25y^2 = 225 - 120 = 105$$

Sol.272.(c) $x + \frac{1}{x} = \sqrt{9.76}$

$$x - \frac{1}{x} = \sqrt{(\sqrt{9.76})^2 - 4}$$

$$= \sqrt{9.76 - 4} = 2.4$$

$$x^3 - \frac{1}{x^3} = (2.4)^3 + 3 \times 2.4$$

$$= 13.824 + 7.2 = 21.024$$

Sol.273.(c) $5a + \frac{4}{a} = 15$

$$25a^2 + \frac{16}{a^2} = (15)^2 - 2 \times 5 \times 4$$

$$= 225 - 40 = 185$$

Sol.274.(b)

$$(x^3 - y^3) = (x - y)(x^2 + y^2 + xy)$$

Here,

$$\frac{(x^3 - y^3)(x + y)}{x^2 + y^2 + xy} = (x - y)(x + y)$$

$$= x^2 - y^2$$

$$\text{Sol.275.(d)} \quad (a+b)^2 + (b+c)^2 + (c+a)^2 = a^2 + b^2 + c^2 + (a+b+c)^2$$

$$= 115 + (17)^2 = 115 + 289 = 404$$

$$\text{Sol.276.(d)} \quad (2a - 3b - c)^2 - (a + 2b + c)^2$$

$$= 4a^2 + 9b^2 + c^2 - 12ab + 6bc - 4ac -$$

$$a^2 - 4b^2 - c^2 - 4ab - 4bc - 2ac$$

$$= 3a^2 + 5b^2 - 16ab + 2bc - 6ac$$

$$\text{Sol.277.(b)} \quad x + \frac{1}{x} = 3.2$$

$$x - \frac{1}{x} = \sqrt{(3.2)^2 - 4} = \sqrt{6.24}$$

$$\text{Now, } x^2 - \frac{1}{x^2} = 3.2 \times \sqrt{6.24} = 12.8$$

$$\sqrt{0.39}$$

$$\text{Sol.278.(c)} \quad x^2 + y^2 + z^2 = 2(x + z - 1)$$

$$x^2 - 2x + 1 + z^2 - 2z + 1 + y^2 = 0$$

$$(x-1)^2 + (z-1)^2 + (y)^2 = 0$$

$$\Rightarrow x = 1, z = 1 \text{ and } y = 0$$

$$\text{Now, } x^3 + y^3 + z^3 = 1 + 0 + 1 = 2$$

Sol.279.(c) Formula:-

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$(x+3y)^3 = x^3 + 27y^3 + 9xy(x+3y)$$

$$216 = x^3 + 27y^3 + 54xy$$

Sol.280.(c) Given, $p + q = 12$, $pq = 14$

$$\text{Then, } p^2 - pq + q^2 = (p+q)^2 - 3pq$$

$$= 144 - 3 \times 14 = 102$$

$$\text{Sol.281.(d)} \quad \frac{(x+3)^3 + (x-3)^3}{x^2 + 27}$$

$$= \frac{(x^3 + 3^3 + 9x(x+3)) + (x^3 - 3^3 - 9x(x-3))}{x^2 + 27}$$

$$= \frac{2x^3 + 27x + 27x}{x^2 + 27} \Rightarrow \frac{2x(x^2 + 27)}{x^2 + 27} = 2x$$

Sol.282.(a)

$$\frac{1}{8} \left[\frac{1}{b-1} - \frac{1}{b+1} - \frac{2}{b^2+1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{b+1-b+1}{b^2-1} - \frac{2}{b^2+1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{2}{b^2-1} - \frac{2}{b^2+1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{2(b^2+1-b^2+1)}{b^4-1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{2(2)}{b^4-1} - \frac{4}{b^4+1} \right]$$

$$= \frac{1}{8} \left[\frac{4(b^4+1-b^4+1)}{b^8-1} \right] = \frac{1}{8} \left[\frac{8}{b^8-1} \right]$$

$$= \frac{1}{b^8-1}$$

Sol.283.(c)

$(d+e+f) = 14$, squaring both side

$$\{d^2 + e^2 + f^2 + 2(de+ef+fd)\} = 196$$

$$2(de+ef+fd) = 196 - 96 = 100$$

$$de+ef+fd = 50$$

Sol.284.(a)

$$\text{Given:- } g = (2 - \sqrt{3}), \frac{1}{g} = (2 + \sqrt{3})$$

$$\text{Then, } g - \frac{1}{g} = -2\sqrt{3}$$

$$\text{Therefore, } (g^3 - \frac{1}{g^3})$$

$$= (-2\sqrt{3})^3 - 3 \times 2\sqrt{3} = -24\sqrt{3} - 6\sqrt{3}$$

$$= -30\sqrt{3}$$

Sol.285.(c)

$$\text{Given:- } (2x + 13 - y)(2x - 13 - y)$$

$$\{(2x - y) + 13\} \{(2x - y) - 13\}$$

$$(2x - y)^2 - (13)^2$$

$$4x^2 + y^2 - 4xy - 169$$

$$\text{Sol.286.(d)} \quad (3x - 2y)(3x + 2y)(9x^2 + 4y^2)$$

$$= (9x^2 - 4y^2)(9x^2 + 4y^2) \Rightarrow (81x^4 - 16y^4)$$

Sol.287.(b)

$$\text{Formula : } \{(a+b)(a-b) = a^2 - b^2\}$$

$$\{(5p+3q)(5p-3q) = 25p^2 - 9q^2\}$$

$$\text{Sol.288.(d)} \quad \text{Given, } p^2 - 4p - 1 = 0 \text{ -----(1)}$$

$$\text{Dividing both side by } p, p - \frac{1}{p} = 4$$

$$\text{Squaring both sides of equation (1)}$$

$$p^2 + \frac{1}{p^2} = (4)^2 + 2 = 18$$

$$\text{So, } p^2 + 3p + \frac{1}{p^2} - \frac{3}{p} = p^2 + \frac{1}{p^2} + 3$$

$$(p - \frac{1}{p}) = 18 + 3(4) = 30$$

$$\text{Sol.289.(d)} \quad \text{Given } 3x - 2y = 3 \text{ and } xy = 9$$

$$\text{So, } 3x + 2y = \sqrt{(3x - 2y)^2 + 4ab}$$

$$= \sqrt{9 + 4 \times 9 \times 3 \times 2} = \sqrt{225} = 15$$

$$\{\text{where } a = 3x \text{ and } b = 2y\}$$

$$\text{Sol.290.(d)} \quad px + qy = 1$$

$$\text{Put } p = 1, q = 0$$



Then, $x = 1$

$$\text{and } qx + py = \frac{2pq}{p^2 + q^2} \text{ -----(1)}$$

Put $x = 1$ in the equation (1)

$$y = \frac{2 \times 1 \times 0}{1^2 + 0^2} = 0$$

$$\text{So, } (x^2 + y^2)(p^2 + q^2) = (1 + 0)(1 + 0) = 1$$

$$\text{Sol.291.(c) } 7b - \frac{1}{4b} = 7$$

Multiply by $\frac{4}{7}$, we get

$$4b - \frac{1}{7b} = 4$$

$$16b^2 + \frac{1}{49b^2} = (4)^2 + 2 \times 4 \times \frac{1}{7}$$

$$= 16 + \frac{8}{7} = \frac{120}{7}$$

$$\text{Sol.292.(d) } 3x^3 + 5x^2y + 12xy^2 + 7y^3 \\ = 3(-64) + 5(16)(-1) + 12(-4)(1) + 7(-1) \\ = -192 - 80 - 48 - 7 = -327$$

Sol.293.(b)

$$(a + b + c)^2 = \{a^2 + b^2 + c^2 + 2(ab + bc + ca)\}$$

$$256 = \{90 + 2(ab + bc + ca)\}$$

$$(ab + bc + ca) = \frac{256 - 90}{2} = 83$$

$$\text{Sol.294.(c) Formula : } a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

And we know, If $(a + b + c) = 0$

Then,

$$a^3 + b^3 + c^3 = 3abc$$

Sol.295.(b)

$$\text{If, } a + \frac{1}{a} = k, \text{ then } a^2 + \frac{1}{a^2} = k^2 - 2$$

$$\text{and } a^3 + \frac{1}{a^3} = k^3 - 3k$$

$$\text{Here } x + \frac{1}{x} = 7,$$

$$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7 = 322$$

$$x^6 + \frac{1}{x^6} = 322^2 - 2 = 103684 - 2 = 103682$$

$$\text{Sol.296.(d) } (a^3 - b^3 - 3a^2b + 3ab^2)$$

$$= (a - b)^3 = (17 - 13)^3 = 64$$

$$\text{Sol.297.(c) } \frac{x^4 - 2x^2 + 1}{x^2 - 2x + 1} = \frac{(x^2 - 1)^2}{(x - 1)(x - 1)}$$

$$= \frac{(x + 1)(x - 1)(x + 1)(x - 1)}{(x - 1)(x - 1)}$$

$$= (x + 1)(x + 1) = x^2 + 2x + 1$$

$$\text{Sol.298.(b) } x - \frac{1}{x} = \sqrt{6} \Rightarrow (x - \frac{1}{x})^2 = 6$$

$$\Rightarrow x^2 + \frac{1}{x^2} = (\sqrt{6})^2 + 2 = 8$$

$$\text{Similarly, } x^4 + \frac{1}{x^4} = 62 \text{ and}$$

$$x + \frac{1}{x} = \sqrt{(\sqrt{6})^2 + 4} = \sqrt{10}$$

now

$$x^8 - \frac{1}{x^8} = [(x^4) + \frac{1}{(x^4)}] [(x^2 + \frac{1}{x^2})$$

$$[(x^2 - \frac{1}{x^2})]$$

$$= 62 \times 8 \times (x + \frac{1}{x})(x - \frac{1}{x})$$

$$= 496 \times \sqrt{10} \times \sqrt{6}$$

$$= 496 \times \sqrt{60} = 496 \times 2\sqrt{15} = 992\sqrt{15}$$

$$\text{So, } x^8 - \frac{1}{x^8} = 992\sqrt{15}$$

Sol.299.(b)

$$(a + b + c)^2 - a^2 - b^2 - c^2$$

$$= a^2 + b^2 + c^2 + 2(ab + bc + ac) -$$

$$a^2 - b^2 - c^2 = 2(ab + bc + ac)$$

Sol.300.(c) According to the question,

$$\frac{X - 5Y}{X + 5Y} = \frac{7}{13}$$

$$13X - 65Y = 7X + 35Y$$

$$6X = 100Y \Rightarrow \frac{X}{Y} = \frac{100}{6} = \frac{50}{3}$$

Sol.301.(b) Let, the numbers be a and b.

According to question,

$$(a - b)^3 = 1728 \Rightarrow (a - b) = 12$$

$$a = 12 + b \text{ --- (1)}$$

$$ab = 108 \text{ --- (2)}$$

From the above equations, we get

$$a = 18 \text{ and } b = 6$$

$$\text{Therefore, } 18^3 - 6^3 = 5616$$

Sol.302.(a)

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(19)^2 = 155 + 2(ab + bc + ca)$$

$$361 = 155 + 2(ab + bc + ca)$$

$$2(ab + bc + ca) = 361 - 155 = 206$$

$$(ab + bc + ca) = \frac{206}{2} = 103$$

Now,

$$(a - b)^2 + (b - c)^2 + (c - a)^2$$

$$= a^2 + b^2 - 2ab + b^2 + c^2 - 2bc + c^2 + a^2 - 2ca$$

$$= 2(a^2 + b^2 + c^2) - 2(ab + bc + ca)$$

$$= 2(155) - 2(103) = 310 - 206 = 104$$

Sol.303.(a)

$$a^2 + b^2 + c^2 = ab + bc + ac \text{(Given)}$$

only possible when, $a = b = c$

$$\text{So, } \frac{11a^4 + 13b^4 + 17c^4}{17a^2b^2 + 9b^2c^2 + 15c^2a^2}$$

$$= \frac{11a^4 + 13a^4 + 17a^4}{17a^2a^2 + 9a^2a^2 + 15a^2a^2} = \frac{41a^4}{41a^4} = 1$$

$$\text{Sol.304.(b) } (x - y)^2 = 324 \Rightarrow x - y = 18$$

$$(x + y)^2 = (x - y)^2 + 4xy$$

$$(x + y)^2 = 324 + 4 \times 144 = 900$$

$$x + y = 30$$

Now,

$$x^2 - y^2 = (x - y)(x + y) = 18 \times 30 = 540$$

Sol.305.(c) Given,

$$p = \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \text{ and } q = \frac{\sqrt{2} - 1}{\sqrt{2} + 1}$$

$$p \times q = \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} + 1} = 1 \text{ and } p + q$$

$$= \frac{\sqrt{2} + 1}{\sqrt{2} - 1} + \frac{\sqrt{2} - 1}{\sqrt{2} + 1} = 6$$

According to question,

$$\frac{p^2}{q} + \frac{q^2}{p} = \frac{p^3 + q^3}{pq} = \frac{(p + q)^3 - 3pq(p + q)}{pq}$$

$$= \frac{(6)^3 - 3(1)(6)}{1} = 198$$

$$\text{Sol.306.(a) } x + \frac{1}{x} = 2\sqrt{2}$$

$$x - \frac{1}{x} = \sqrt{(2\sqrt{2})^2 - 4} = 2$$

$$x^3 - \frac{1}{x^3} = (x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$$

$$= (2)^3 + 3 \times 2 = 14$$

$$x^3 + \frac{1}{x^3} = (x + \frac{1}{x})^3 - 3(x + \frac{1}{x})$$

$$= (2\sqrt{2})^3 - 3 \times 2\sqrt{2} = 16\sqrt{2} - 6\sqrt{2} = 10\sqrt{2}$$

Now,

$$x^6 - \frac{1}{x^6} = (x^3 - \frac{1}{x^3})(x^3 + \frac{1}{x^3})$$

$$x^6 - \frac{1}{x^6} = 14 \times 10\sqrt{2} = 140\sqrt{2}$$

Sol.307.(a)

$$a^3 + b^3 + c^3 - 3abc = \frac{(a + b + c)}{2}$$

$$\{3(a^2 + b^2 + c^2) - (a + b + c)^2\}$$

$$= 6\{3 \times 50 - 144\} = 36$$

$$\text{Sol.308.(d) } (3y - \frac{3}{y}) = 5$$

$$\text{Then, } y - \frac{1}{y} = \frac{5}{3}$$

$$\text{Now, } y^2 + \frac{1}{y^2} = (\frac{5}{3})^2 + 2 = \frac{43}{9}$$

$$\text{Sol.309.(b) } x^2 - 7x + 1 = 0$$

$$x + \frac{1}{x} = 7$$

$$x - \frac{1}{x} = \sqrt{(7)^2 - 4} = \pm\sqrt{45} = \pm 3\sqrt{5}$$

Now,

$$x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x})$$

$$= 7 \times \pm 3\sqrt{5} = \pm 21\sqrt{5}$$

Since, $0 < x < 1$

$$\text{So, } x^2 - \frac{1}{x^2} = -21\sqrt{5}$$

$$\text{Sol.310.(b) } x + \frac{1}{x} = 2 \Rightarrow x = 1$$

$$\text{Then, } x^7 + \frac{1}{x^{117}} = 1 + 1 = 2$$



Sol.311.(a)

$$a^3 + b^3 + c^3 - 3abc = \frac{(a+b+c)}{2}$$

$$\{3(a^2 + b^2 + c^2) - (a + b + c)^2\}$$

$$= 10 \times \{3 \times 152 - 20^2\}$$

$$= 10 \times (456 - 400) = 560$$

Sol.312.(b)

$$\frac{l^2 - m^2}{(l+m)^2} = \frac{(l+m)(l-m)}{(l+m)^2} = \frac{(l-m)}{(l+m)}$$

$$\text{Sol.313.(a)} \quad x^4 + \frac{1}{x^4} = 142$$

$$x^2 + \frac{1}{x^2} = \sqrt{142 + 2} = 12$$

$$x + \frac{1}{x} = \sqrt{14}$$

$$x^3 + \frac{1}{x^3} = (\sqrt{14})^3 - (3 \times \sqrt{14}) = 11\sqrt{14}$$

$$x^7 + \frac{1}{x^7} = (x^3 + \frac{1}{x^3})(x^4 + \frac{1}{x^4}) - (x + \frac{1}{x})$$

$$= 11\sqrt{14} \times 142 - \sqrt{14} = 1561\sqrt{14}$$

Sol.314.(c)

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)$$

$$\{(a+b+c)^2 - 3(ab+bc+ca)\}$$

$$\Rightarrow a^3 + b^3 + c^3 - 3(-1)$$

$$= 7 \times \{49 - 3 \times 11\}$$

$$\Rightarrow a^3 + b^3 + c^3 + 3 = 7 \times \{16\}$$

$$\Rightarrow a^3 + b^3 + c^3 = 109$$

$$\text{Sol.315.(a)} \quad (x - \frac{1}{x}) = 6$$

$$(x + \frac{1}{x}) = \sqrt{6^2 + 4} = \sqrt{40} = 2\sqrt{10}$$

$$\text{Now, } x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x})$$

$$\Rightarrow x^2 - \frac{1}{x^2} = 2\sqrt{10} \times 6 = 12\sqrt{10}$$

$$\text{Sol.316.(c)} \quad (x + \frac{1}{x}) = 3\sqrt{2}$$

$$\Rightarrow (x - \frac{1}{x}) = \sqrt{(3\sqrt{2})^2 - 4} = \sqrt{14}$$

$$\Rightarrow (x^2 - \frac{1}{x^2}) = 3\sqrt{2} \times \sqrt{14} = 6\sqrt{7}$$

$$\text{Again, } (x^2 + \frac{1}{x^2}) = (3\sqrt{2})^2 - 2 = 16$$

$$(x^4 + \frac{1}{x^4}) = (16)^2 - 2 = 254$$

$$\Rightarrow (x^4 - \frac{1}{x^4}) = 16 \times 6\sqrt{7} = 96\sqrt{7}$$

$$\text{Now, } (x^8 - \frac{1}{x^8}) = (x^4 + \frac{1}{x^4})(x^4 - \frac{1}{x^4})$$

$$= 254 \times 96\sqrt{7} = 24384\sqrt{7}$$

Sol.317.(a)

$$\frac{2a-5}{a} - \frac{4b-5}{b} + \frac{6c+5}{c} = 0$$

$$\Rightarrow bc(2a-5) - ac(4b-5) + ab(6c+5) = 0$$

$$\Rightarrow 2abc - 5bc - 4abc + 5ac + 6abc + 5ab = 0$$

$$\Rightarrow 4abc - 5(bc - ac - ab) = 0$$

$$\Rightarrow 5(bc - ac - ab) = 4abc$$

$$\Rightarrow \frac{(bc - ac - ab)}{abc} = \frac{4}{5}$$

$$\Rightarrow (\frac{1}{a} - \frac{1}{b} - \frac{1}{c}) = \frac{4}{5}$$

Sol.318.(a)

$$[(1+P)(1+P^2)(1+P^4)(1+P^8)(1+P^{16})$$

$$(1-P) - 1]$$

$$\Rightarrow [(1-P^2)(1+P^2)(1+P^4)(1+P^8)(1+P^{16}) - 1]$$

$$\Rightarrow [(1-P^4)(1+P^4)(1+P^8)(1+P^{16}) - 1]$$

$$\Rightarrow [(1-P^8)(1+P^8)(1+P^{16}) - 1]$$

$$\Rightarrow [(1-P^{16})(1+P^{16}) - 1]$$

$$\Rightarrow [(1-P^{32}) - 1] = -P^{32}$$

$$\text{Sol.319.(b)} \quad (x-y)^2 = x^2 + y^2 - 2xy$$

$$x^2 + y^2 = 25 + 2 \times 150 = 325$$

$$\text{Now, } (x+y)^2 = x^2 + y^2 + 2xy$$

$$(x+y)^2 = 325 + 300 = 625$$

$$(x+y) = 25$$

$$\text{Short tricks :-}$$

$$(x-y) = 5 \text{ and } xy = 150$$

$$\text{Clearly, value of } x = 15 \text{ and } y = 10$$

$$\text{So, } (x+y) = 15 + 10 = 25$$

$$\text{Sol.320.(a)}$$

$$a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$$

$$\text{According to the question,}$$

$$a - b = 1$$

$$\text{Hence, } a^3 - b^3 = (a^2 + b^2 + ab)$$

$$\text{Sol.321.(d)} \quad (x^2 + \frac{1}{x^2}) = 18$$

$$x + \frac{1}{x} = \sqrt{18 + 2} = 2\sqrt{5}$$

$$x^3 + \frac{1}{x^3} = (2\sqrt{5})^3 - 3(2\sqrt{5})$$

$$= 40\sqrt{5} - 6\sqrt{5} = 34\sqrt{5}$$

$$\text{Sol.322.(a) Identity used :-}$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$\frac{p^2 - (q-r)^2}{(p+r)^2 - q^2} + \frac{q^2 - (p-r)^2}{(p+q)^2 - r^2} + \frac{r^2 - (p-q)^2}{(q+r)^2 - p^2} =$$

$$\frac{[p+(q-r)][p-(q-r)]}{[(p+r)+q][(p+r)-q]} + \frac{[q+(p-r)][q-(p-r)]}{[(p+q)+r][(p+q)-r]} + \frac{[r+(p-q)][r-(p-q)]}{[(q+r)+p][(q+r)-p]}$$

$$= \frac{(p+q-r)[p-q+r]}{(p+r+q)[p-q+r]} + \frac{(q+p-r)[q-p+r]}{(p+q+r)[p-q+r]}$$

$$+ \frac{(p-q+r)[r-p+q]}{(q+r+p)[q-r-p]}$$

$$= \frac{(p+q-r)}{(p+q+r)} + \frac{(q-p+r)}{(p+q+r)} + \frac{(p-q+r)}{(q+r+p)}$$

$$= \frac{(p+q-r)+(q-p+r)+(p-q+r)}{(p+q+r)}$$

$$= \frac{p+q-r+q-p+r+p-q+r}{(p+q+r)} = 1$$

$$\text{Short trick :-}$$

$$\text{Put } p, q = 0 \text{ then,}$$

$$\frac{p^2 - (q-r)^2}{(p+r)^2 - q^2} + \frac{q^2 - (p-r)^2}{(p+q)^2 - r^2} + \frac{r^2 - (p-q)^2}{(q+r)^2 - p^2}$$

$$\frac{-r^2}{r^2} + \frac{(-r^2)}{(-r^2)} + \frac{r^2}{r^2} = 1$$

$$\text{Sol.323.(c) As per question, } \frac{a}{b} = \frac{7}{6}$$

$$\text{So, } \frac{6a+13b}{6a-13b} = \frac{6 \times 7 + 13 \times 6}{6 \times 7 - 13 \times 6}$$

$$= \frac{42+78}{42-78} = -\frac{120}{36} = -\frac{10}{3}$$

$$\text{Sol.324.(a)} \quad \frac{1}{2+2p} + \frac{1}{2+2q} + \frac{1}{2+2r}$$

$$\text{According to question,}$$

$$\frac{1}{2+2\frac{x}{y+z}} + \frac{1}{2+2\frac{y}{z+x}} + \frac{1}{2+2\frac{z}{x+y}}$$

$$= \frac{y+z}{2(y+z+x)} + \frac{z+x}{2(z+x+y)} +$$

$$\frac{x+y}{2(x+y+z)} = \frac{2(x+y+z)}{2(x+y+z)} = 1$$

$$\text{Sol.325.(d)} \quad y - \frac{1}{y} = 4$$

$$y + \frac{1}{y} = \sqrt{4^2 + 4} = 2\sqrt{5}$$

$$y^3 + \frac{1}{y^3} = (2\sqrt{5})^3 - 3 \times 2\sqrt{5} = 34\sqrt{5}$$

$$y^6 + \frac{1}{y^6} = (34\sqrt{5})^2 - 2 = 5778$$

$$\text{Sol.326.(b)} \quad x^2 + 1 = 15x$$

$$\text{Squaring both side}$$

$$x^4 + 1 + 2x^2 = 225x^2$$

$$x^4 - 223x^2 + 1 = 0$$

$$x^4 - 223x^2 + 6 = 5$$

$$\text{Sol.327.(a)} \quad p^3 + q^3 + r^3 - 3pqr$$

$$= (p+q+r)\{(p+q+r)^2 - 3(pq+qr+pr)\} \text{ ----(1)}$$

$$\text{According to the question,}$$

$$p+q+r = pqr = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$$

$$\Rightarrow p+q+r = pqr = \frac{pq+qr+rp}{pqr} = 1 \text{ ---- (2)}$$

$$\text{By Putting the value of equation(2) into}$$

$$\text{equation (1) we get :-}$$

$$p^3 + q^3 + r^3 - 3 = (1)((1)^2 - 3(1))$$

$$p^3 + q^3 + r^3 = 1 - 3 + 3 = 1$$

$$\text{Sol.328.(c)} \quad \frac{256x^4 - 16y^4}{(80x^2 - 20y^2)(16x^2 + 4y^2)}$$

$$= \frac{16(16x^4 - y^4)}{20(4x^2 - y^2)4(4x^2 + y^2)}$$

$$= \frac{16(16x^4 - y^4)}{80((4x^2)^2 - (y^2)^2)} = \frac{16(16x^4 - y^4)}{80(16x^4 - y^4)}$$

$$= \frac{1}{5}$$

$$\text{Sol.329.(a)}$$

$$7a - \frac{7}{a} + 4 = 0 \Rightarrow a - \frac{1}{a} = -\frac{4}{7}$$

$$\text{Now, } a^3 - \frac{1}{a^3} = (-\frac{4}{7})^3 + 3 \times (-\frac{4}{7})$$



$$= \frac{-64 - 49 \times 12}{343} = -\frac{652}{343}$$

$$\text{So, } a^3 - \frac{1}{a^3} - 1 = -\frac{652}{343} - 1 = \frac{-995}{343}$$

$$\text{Sol.330.(a)} \quad \frac{s^2+t^2+2st-u^2}{s^2-t^2-2tu-u^2} = \frac{(s+t)^2-u^2}{s^2-(t+u)^2}$$

$$\frac{(s+t+u)(s+t-u)}{(s+t+u)(s-t-u)} = \frac{(s+t-u)}{(s-t-u)}$$

$$\text{Sol.331.(d)} \quad x^2 - 8x - 1 = 0 \Rightarrow x - \frac{1}{x} = 8$$

$$\text{So, } x^2 + \frac{1}{x^2} = (8)^2 + 2 = 64 + 2 = 66$$

$$\text{Sol.332.(b)} \quad x^2 + \frac{1}{x^2} = 6$$

Given that, $0 < x < 1$, so the value of

$(x^2 - \frac{1}{x^2})$ will always be -ve.

$$x^2 - \frac{1}{x^2} = -\left(\sqrt{(6)^2 - 4}\right) = -4\sqrt{2}$$

$$x^4 - \frac{1}{x^4} = (x^2 + \frac{1}{x^2})(x^2 - \frac{1}{x^2})$$

$$= 6 \times (-4\sqrt{2}) \Rightarrow x^4 - \frac{1}{x^4} = -24\sqrt{2}$$

$$\text{Sol.333.(d)} \quad x = \frac{1}{x-3} \Rightarrow x^2 - 3x = 1$$

Dividing both sides by x we get;

$$\Rightarrow x - \frac{1}{x} = 3$$

$$\text{So, } x + \frac{1}{x} = \sqrt{(3)^2 + 4} = \sqrt{13}$$

$$\text{Sol.334.(b)} \quad x + \frac{1}{x} = 1, \Rightarrow x^2 + 1 = x$$

Now,

$$\frac{x^2+7x+1}{x^2+11x+1} = \frac{x+7x}{x+11x} = \frac{8x}{12x} = \frac{2}{3}$$

$$\text{Sol.335.(c)} \quad \text{Given, } x + \frac{1}{x} = \sqrt{6}$$

$$(x^2 + \frac{1}{x^2}) = (\sqrt{6})^2 - 2 = 4$$

$$(x^2 - \frac{1}{x^2}) = \sqrt{4^2 - 4} = \sqrt{12} = 2\sqrt{3}$$

Then,

$$(x^4 - \frac{1}{x^4}) = (x^2 - \frac{1}{x^2})(x^2 + \frac{1}{x^2})$$

$$= 2\sqrt{3} \times 4 = 8\sqrt{3}$$

$$\text{Since, } (x^2 + \frac{1}{x^2}) = 4 \text{ then,}$$

$$(x^4 + \frac{1}{x^4}) = (4)^2 - 2 = 14$$

Now,

$$(x^8 - \frac{1}{x^8}) = (x^4 + \frac{1}{x^4})(x^4 - \frac{1}{x^4})$$

$$= 14 \times 8\sqrt{3} = 112\sqrt{3}$$

$$\text{Sol.336.(b)} \quad (\frac{5}{b} + 5b)(\frac{25}{b^2} - 25 + 25b^2)$$

$$= (\frac{5}{5} + 5 \times 5)(\frac{25}{5 \times 5} - 25 + 25 \times 5 \times 5)$$

$$= (26)(1 - 25 + 625) = 15626$$

$$\text{Sol.337.(b)} \quad \text{Given, } x - \frac{1}{x} = 10$$

$$x^2 + \frac{1}{x^2} = (10)^2 + 2 = 102$$

$$\text{Now, } x^4 + \frac{1}{x^4} = 102 \times 102 - 2 = 10402$$

$$\text{Sol.338.(b)} \quad (x-5)^2 + (x+3)^2 + 4x$$

$$= x^2 + 25 - 10x + x^2 + 9 + 6x + 4x$$

$$= 2x^2 + 34 = 2(x^2 + 17)$$

Sol.339.(c) formula used :-

$$a^3 + b^3 = (a+b)[(a+b)^2 - 3ab]$$

$$8x^3 + 27y^3 = 16[16^2 - 3 \times 2 \times 3 \times 9]$$

$$= 16[256 - 162] = 16 \times 94 = 1504$$

Sol.340.(c)

$$x^3 + y^3 = (x+y)[(x+y)^2 - 3xy]$$

$$= 25[25^2 - 3 \times 20] = 25[625 - 60]$$

$$= 25 \times 565 = 14125$$

$$\text{Sol.341.(c)} \quad x^2 - \frac{1}{x^2} = 4\sqrt{2}$$

$$x^2 + \frac{1}{x^2} = \sqrt{(4\sqrt{2})^2 + 4} = 6$$

Now,

$$x^4 - \frac{1}{x^4} = (x^2 + \frac{1}{x^2})(x^2 - \frac{1}{x^2})$$

$$= 6 \times 4\sqrt{2} = 24\sqrt{2}$$

$$\text{Sol.342.(b)} \quad (x^2 + \frac{1}{x^2}) = 18$$

$$(x^2 + \frac{1}{x^2} - 2) = 18 - 2 \Rightarrow (x - \frac{1}{x}) = 4$$

$$\Rightarrow (x^3 - \frac{1}{x^3}) = (4)^3 + 3 \times 4 = 76$$

$$\text{Sol.343.(b)} \quad 3x - z = 0 \Rightarrow z = 3x$$

$$3x - 9y + 4z = 5$$

$$\text{or } 15x - 9y = 5 \dots \text{e.q. (1)}$$

$$2x + 7y + z = 12 \text{ or } 5x + 7y = 12 \dots \text{e.q. (2)}$$

on solving e.q. (1) and e.q. (2), we get

$$y = \frac{31}{30} \text{ and } x = \frac{143}{150}, z = 3x = \frac{143}{50}$$

$$\text{Sol.344.(a)} \quad (2a + b)^2 = 10^2$$

$$4a^2 + b^2 + 4ab = 100$$

$$4a^2 + b^2 = 100 - 18 = 82$$

Now,

$$(2a - b) = \sqrt{4a^2 + b^2 - 4ab}$$

$$= \sqrt{82 - 18} = \sqrt{64} = 8$$

$$\text{Sol.345.(c)} \quad x^2 + \frac{1}{x^2} = 2\sqrt{5}$$

$$x^4 + \frac{1}{x^4} = (2\sqrt{5})^2 - 2 = 18$$

Now,

$$x^4 - \frac{1}{x^4} = \sqrt{(18^2 - 4)} = \sqrt{320} = 8\sqrt{5}$$

$$\text{Sol.346.(b)} \quad x^4 + \frac{1}{x^4} = 79$$

$$x^4 + \frac{1}{x^4} + 2 = 81 \Rightarrow x^2 + \frac{1}{x^2} = 9$$

$$x^2 + \frac{1}{x^2} - 2 = 7 \Rightarrow x - \frac{1}{x} = \sqrt{7}$$

Sol.347.(b)

$$x + y = \sqrt{13} \Rightarrow x^2 + y^2 + 2xy = 13$$

$$x - y = \sqrt{11} \Rightarrow x^2 + y^2 - 2xy = 11$$

On adding both, we get

$$2(x^2 + y^2) = 24 \Rightarrow (x^2 + y^2) = 12$$

$$\text{Again, } x^2 + y^2 + 2xy = 13$$

$$xy = \frac{13-12}{2} = \frac{1}{2}$$

$$\text{Now, } xy(x^2 + y^2) = \frac{1}{2} \times 12 = 6$$

Sol.348.(a)

$$ab + bc + ca = 0 \Rightarrow ab + ca = -bc$$

$$\frac{1}{a^2 - bc} = \frac{1}{a^2 + (ab + ca)} = \frac{1}{a(a + b + c)}$$

$$\text{Similarly, } \frac{1}{b^2 - ca} = \frac{1}{b(a + b + c)}$$

$$\text{and } \frac{1}{c^2 - ab} = \frac{1}{c(a + b + c)}$$

Now,

$$\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$$

$$= \frac{1}{a(a + b + c)} + \frac{1}{b(a + b + c)} +$$

$$\frac{1}{c(a + b + c)}$$

$$= \frac{bc + ca + ab}{abc(a + b + c)} = 0$$

Sol.349.(c) Let the required value of A be x

According to question,

$$\frac{52}{91} = \frac{x}{126} \Rightarrow x = 72$$

$$\text{Sol.350.(c)} \quad ax + by + c_1 = 0$$

$$px + qy + c_2 = 0$$

For infinite no. of solutions :-

$$\frac{a}{p} = \frac{b}{q} = \frac{c_1}{c_2}$$

$$\text{So, } \frac{38}{46} = \frac{q}{414} \Rightarrow q = \frac{414 \times 38}{46}$$

$$q = 9 \times 38 = 342$$

Sol.351.(b)

$$x^3 + y^3 + z^3 - 3xyz = \frac{(x+y+z)}{2}$$

$$\{(x-y)^2 + (y-z)^2 + (z-x)^2\}$$

$$= \frac{14 + 15 + 17}{2}$$

$$\{(14-15)^2 + (15-17)^2 + (17-14)^2\}$$

$$= 23 \{1 + 4 + 9\} = 322$$



Sol.352.(b) Given,

$$18x + 36y + 45 = 0 \text{ -----(1)}$$

$$px - 54y + 67 = 0 \text{ -----(2)}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ (no solution)}$$

$$\frac{18}{p} = \frac{36}{(-54)} \neq \frac{45}{67}$$

$$p = \frac{18 \times (-54)}{36} = -27$$

Sol.353.(a) $(a+b)^2 - (a-b)^2 = 4ab$

$$\Rightarrow \left(\frac{9}{8} + \frac{8}{9}\right)^2 - \left(\frac{9}{8} - \frac{8}{9}\right)^2$$

$$\Rightarrow 4 \times \frac{9}{8} \times \frac{8}{9} \Rightarrow 4$$

Sol.354.(c)

$$\frac{1+x^2}{1-x} \div \frac{1-x^4}{x-1} \times \frac{x(1-x)}{(1+x)}$$

$$\Rightarrow \frac{1+x^2}{1-x} \times \frac{x-1}{1-x^4} \times \frac{x(1-x)}{(1+x)}$$

$$\Rightarrow \frac{1+x^2}{1-x} \times \frac{-(1-x)}{(1-x^2)(1+x^2)} \times \frac{x(1-x)}{(1+x)}$$

$$\Rightarrow \frac{1+x^2}{1-x} \times \frac{-(1-x)}{(1-x)(1+x)(1+x^2)} \times$$

$$\frac{x(1-x)}{(1+x)} \Rightarrow \frac{-x}{(1+x)^2}$$

Sol.355.(a) $K + \frac{1}{K} - 2 = 0 \Rightarrow K + \frac{1}{K} = 2$ Here the value of $k = 1$

$$\text{Therefore, } K^{17} + \frac{1}{K^{12}} = 1^{17} + \frac{1}{1^{12}} = 2$$

Sol.356.(c) According to the question,

$$a^4 + \frac{1}{a^4} = 194$$

$$\Rightarrow a^2 + \frac{1}{a^2} = \sqrt{194 + 2} = 14$$

$$\Rightarrow a + \frac{1}{a} = \sqrt{14 + 2} = 4$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 64 - 3 \times 4 = 52$$

Now, we can write

$$\frac{a^6+1}{a^3} \text{ as } \frac{\frac{a^6}{a^3} + \frac{1}{a^3}}{\frac{a^3}{a^3}} = a^3 + \frac{1}{a^3} = 52$$

Sol.357.(c)

$$(a-b)(b+a)(a^2+b^2)(2b^4+2a^4)$$

$$= (a^2-b^2)(a^2+b^2)(2b^4+2a^4)$$

$$\Rightarrow (a^4-b^4)2(b^4+a^4) = 2(a^8-b^8)$$

$$\text{Sol.358.(c)} \frac{(15)^3 + (4)^3}{(15^2 + 4^2 - k)} = 19$$

$$\Rightarrow \frac{(15+4)(15)^2 + (4)^2 - (60)}{(15^2 + 4^2 - k)} = 19$$

$$\Rightarrow (15)^2 + (4)^2 - 60 = (15)^2 + (4)^2 - k$$

$$\Rightarrow k = 60$$

$$\text{Sol.359.(a)} x^2 + \frac{1}{x^2} = 38$$

$$\Rightarrow x - \frac{1}{x} = \sqrt{38 - 2} = 6$$

Sol.360.(d)

$$\text{Formula:- } p^3 - r^3 = (p-r)[p^2 + r^2 + pr]$$

$$(21)^2 = p^2 + r^2 + 2 \times 38$$

$$\Rightarrow p^2 + r^2 = 365$$

$$(p-r)^2 = (p+r)^2 - 4pr$$

$$\Rightarrow (p-r)^2 = 441 - 4 \times 38$$

$$(p-r) = \sqrt{289} = 17$$

$$p^3 - r^3 = 17[365 + 38]$$

$$\Rightarrow p^3 - r^3 = 17[403] = 6851$$

Sol.361.(b) Formula:- $a^3 + b^3 + c^3 - 3abc$

$$= \frac{1}{2} (a+b+c) \{ (a-b)^2 + (b-c)^2 + (c-a)^2 \}$$

Therefore, $(a+b+c)$

$$\{ (a-b)^2 + (b-c)^2 + (c-a)^2 \}$$

$$= 2a^3 + 2b^3 + 2c^3 - 6abc$$

Sol.362.(d)

$$x + \frac{1}{x} = -2 \text{ (clearly visible that } x = -1 \text{)}$$

$$\text{Therefore, } x^7 + \frac{1}{x^7} + x^2 + \frac{1}{x^2}$$

$$= (-1)^7 + \frac{1}{(-1)^7} + (-1)^2 + \frac{1}{(-1)^2}$$

$$= -2 + 2 = 0$$

$$\text{Sol.363.(a)} \frac{4x^2 + 9y^2 + 12xy}{144}$$

$$= \frac{(2x+3y)^2}{(12)^2} = \left(\frac{x}{6} + \frac{y}{4}\right)^2$$

Sol.364.(c)

$$(a-4)^3 = a^3 - 64 - 3 \times a^2 \times 4 + 3 \times a \times 16$$

$$\Rightarrow a^3 - 12a^2 + 48a - 64$$

Sol.365.(c) $x-y=8$ and $8xy=8 \Rightarrow xy=1$

$$(8)^2 = x^2 + y^2 - 2 \Rightarrow x^2 + y^2 = 66$$

$$\text{then, } 2(x^3 - y^3) = 2 \times 8(66 + 1)$$

$$= 16 \times 67 = 1072$$

Sol.366.(b) $a^3 = b^3 + 988$

$$\Rightarrow a^3 - b^3 = 988 \text{ and } a-b=4$$

$$2a^2 + 2ab + 2b^2 = 2(a^2 + ab + b^2)$$

$$= 2\left(\frac{a^3 - b^3}{a-b}\right) = 2\left(\frac{988}{4}\right) = 494$$

Sol.367.(c) $a^3 + 3a^2 + 3a = 63$

$$\Rightarrow a^3 + 3a^2 + 3a - 63 = 0$$

$$\Rightarrow (a-3)(a^2 + 6a + 21) = 0$$

$$\Rightarrow a=3 \Rightarrow a^2 + 2a = 9 + 6 = 15$$

Sol.368.(b) $(a+b+c)^2$

$$= a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(6)^2 = 40 + 2(ab + bc + ca)$$

$$(ab + bc + ca) = -2 \Rightarrow a^3 + b^3 + c^3 - 3abc$$

$$= (a+b+c)(a^2+b^2+c^2 - (ab+bc+ca))$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = (6)[40 - (-2)]$$

$$= 252$$

Sol.369.(d) $4ab + 4bc + 4ca = 160$

$$\Rightarrow ab + bc + ca = 40$$

$$9a + 9b + 9c = 81 \Rightarrow (a+b+c) = 9$$

$$(a+b+c)^2$$

$$= (a^2 + b^2 + c^2 + 2ab + 2bc + 2ca)$$

$$81 = a^2 + b^2 + c^2 + 80$$

$$a^2 + b^2 + c^2 = 1 \Rightarrow 6a^2 + 6b^2 + 6c^2 = 6$$

Sol.370.(b)

$$N + \frac{1}{N} = \sqrt{3} \text{ (squaring both side)}$$

$$N^2 + \frac{1}{N^2} = 3 - 2 = 1 \text{ (cubing both side)}$$

$$N^6 + \frac{1}{N^6} = 1 - 3 = -2$$

So the value of,

$$N^6 + \frac{1}{N^6} + 11 \Rightarrow -2 + 11 = 9$$

$$\text{Sol.371.(b)} \frac{(a+b)^2 - (a-b)^2}{ab}$$

$$= \frac{a^2 + b^2 + 2ab - a^2 - b^2 + 2ab}{ab} = 4$$

Sol.372.(d) $(a-b)^3 - a^3 + b^3$

$$= (a)^3 - (b)^3 - 3ab(a-b) - a^3 + b^3$$

$$= -3ab(a-b)$$

Sol.373.(d)

$$(300 - 1)^2 = 90000 + 1 - 600 = 89401$$

Sol.374.(d)

$$\frac{(k+\frac{1}{k})}{(k+\frac{1}{k})} \times (k-\frac{1}{k})(k^2+\frac{1}{k^2})(k^4+\frac{1}{k^4})$$

$$(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}}) \text{ Multiply by}$$

$$\frac{(k+\frac{1}{k})}{(k+\frac{1}{k})}$$

$$(k^2 - \frac{1}{k^2})(k^2 + \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})$$

$$(k + \frac{1}{k})$$

$$(k^4 - \frac{1}{k^4})(k^4 + \frac{1}{k^4})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})$$

$$= \frac{(k + \frac{1}{k})}{(k + \frac{1}{k})}$$

$$(k^8 - \frac{1}{k^8})(k^8 + \frac{1}{k^8})(k^{16} + \frac{1}{k^{16}})$$

$$= \frac{(k + \frac{1}{k})}{(k + \frac{1}{k})}$$

$$(k^{16} - \frac{1}{k^{16}})(k^{16} + \frac{1}{k^{16}}) \frac{(k^{32} - \frac{1}{k^{32}})}{(k + \frac{1}{k})} = \frac{(k + \frac{1}{k})}{(k + \frac{1}{k})}$$

$$\text{Sol.375.(a)} x + \frac{1}{x} = \frac{K}{2}$$

$$\frac{x^8 + 1}{x^4} \text{ [dividing by } x^4]$$

$$x^4 + \frac{1}{x^4} = \left[\left(\frac{K}{2} \right)^2 - 2 \right]^2 - 2$$

$$= \left[\left(\frac{K^2 - 8}{4} \right)^2 - 2 \right] = \frac{K^4 - 16K^2 + 32}{16}$$

Sol.376.(a) Formula :-

$$(a^3 + b^3 + c^3 - 3abc) = \frac{a+b+c}{2}$$

$$\{(a-b)^2 + (b-c)^2 + (c-a)^2\}$$

$$\Rightarrow \frac{(p^3 + q^3 + r^3 - 3pqr)}{(p+q+r)}$$

$$= \frac{\{(6)^2 + (-5)^2 + (3)^2\}}{2} = 35$$

Sol.377.(d) If $A + \frac{1}{A} = -1$, then $A^3 = 1$

According to the question,

$$\frac{A^6 + A^3 - 1}{A^9 + A^3 - 1} = \frac{1 + 1 - 1}{1 + 1 - 1} = 1$$

Sol.378.(b) $(x+1)^2 + (x+2)^2 = 16$

$$x^2 + 1 + 2x + x^2 + 4 + 4x = 16$$

$$2x^2 + 6x + 5 = 16 \Rightarrow 4x^2 + 12x + 10 = 32$$

(multiply by 2 both side)

$$4x^2 + 12x + 10 + 30 = 32 + 30$$

(add 30 both side)

$$4x^2 + 12x + 40 = 62$$

Sol.379.(d)

$$a^2 + 4b^2 + 9c^2 - 4ab + 12bc - 6ca$$

$$= a^2 + (2b)^2 + (3c)^2 - 2 \times a \times 2b + 2 \times 2b \times 3c - 2 \times 3c \times a$$

$$= (a - 2b - 3c)^2$$

Sol.380.(d) Formula :-

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$(1)^2 = a^2 + b^2 + c^2 + 2(-1)$$

$$\Rightarrow a^2 + b^2 + c^2 = 3$$

Now,

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)$$

$$[a^2 + b^2 + c^2 - (ab+bc+ca)]$$

$$\Rightarrow a^3 + b^3 + c^3 - 3(-1)$$

$$= (1)[3 - (-1)] \Rightarrow a^3 + b^3 + c^3 = 1$$

Sol.381.(c) $(a+b)^3 - a^3 - b^3$

$$= a^3 + b^3 + 3ab(a+b) - a^3 - b^3$$

$$= 3ab(a+b)$$

Sol.382.(d) $\frac{1}{(x-2)} + x = 8$

Subtracting 2 from both side; we get

$$\frac{1}{(x-2)} + (x-2) = 8 - 2 = 6$$

$$\frac{1}{(x-2)^2} + (x-2)^2 = 6^2 - 2 = 34$$

Sol.383.(b)

$$\begin{aligned} \frac{1+x}{1-x^2} &\div \frac{1+x}{1-x^4} - \frac{1-x^4}{1-x} \times \frac{1+x}{1-x^2} \\ &= \frac{1+x}{1-x^2} \times \frac{(1-x^2)(1+x^2)}{1+x} - \frac{(1+x^2)(1+x)}{1-x} \\ &= (1+x^2) - \frac{1+x+x^2+x^3}{1-x} \\ &= \frac{1-x+x^2-x^3-1-x-x^2-x^3}{1-x} \\ &= \frac{-2x-2x^3}{1-x} = \frac{2x+2x^3}{x-1} = \frac{2x(1+x^2)}{x-1} \end{aligned}$$

Sol.384.(b) If $x+y+z=0$

$$\text{Then, } x^3 + y^3 + z^3 = 3xyz$$

$$\begin{aligned} \frac{x}{(yz)^2} + \frac{y}{(xz)^2} + \frac{z}{(xy)^2} &= \frac{x^3+y^3+z^3}{x^2y^2z^2} \\ &= \frac{3}{xyz} \end{aligned}$$

$$\text{Sol.385.(d)} \quad \frac{K}{K^2 - 2K + 1} = \frac{1}{6}$$

$$\Rightarrow K^2 + 1 = 8K \Rightarrow K + \frac{1}{K} = 8$$

$$\Rightarrow \frac{1}{K^3} + K^3 = (8)^3 - 3 \times 8 = 488$$

$$\text{Sol.386.(d)} \quad x - \frac{1}{x} = 4$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 18 \Rightarrow x^4 + \frac{1}{x^4} = 322$$

$$(x^6 + \frac{1}{x^6}) = (x^2 + \frac{1}{x^2})(x^4 + \frac{1}{x^4}) -$$

$$(x^2 + \frac{1}{x^2})$$

$$\Rightarrow (x^6 + \frac{1}{x^6}) = 322 \times 18 - 18 = 5778$$

Sol.387.(a) Two lines have no solution, if these two lines are parallel to each other. The lines are parallel to each other means that the slopes of the lines are equal.

$$18x - 72y + 13 = 0 \Rightarrow y = \frac{18}{72}x + \frac{13}{72}$$

$$\text{Slope (m)} = \frac{1}{4}$$

$$\text{and } 7x - my - 17 = 0 \Rightarrow y = \frac{7}{m}x - \frac{17}{m}$$

$$\text{Value of (m) for no solution} \Rightarrow \frac{7}{m} = \frac{1}{4}$$

$$m = 28$$

Sol.388.(c)

$$2^x + 3^y = 17 \text{ and } 2^{x+2} - 3^{y+1} = 5,$$

By option ,

$$\text{Putting the value of } x = 3 \text{ and } y = 2,$$

we get

$$2^x + 3^y = 2^3 + 3^2 = 17 \text{satisfied}$$

$$\text{and } 2^{x+2} - 3^{y+1} = 2^5 - 3^3 = 32 - 27 = 5 \text{satisfied}$$

$$\text{Sol.389.(c)} \quad x^2 - 5x + 1 = 0$$

$$x + \frac{1}{x} = 5 \text{ (eq.1)}$$

$$\{\text{When } x + \frac{1}{x} = k,$$

$$\text{then } x^3 + \frac{1}{x^3} = k^3 - 3k\}$$

$$x^3 + \frac{1}{x^3} = 110 \text{ (eq.2)}$$

$$\text{Now, given eq.} = \frac{x^6 + x^4 + x^2 + 1}{5x^3}$$

Divide numerator and denominator by x^3

$$\frac{x^3 + x + \frac{1}{x} + \frac{1}{x^3}}{5} = \frac{110 + 5}{5} = 23$$

Sol.390.(b) $k + k^{-1} = -2$

$$k + \frac{1}{k} = -2$$

Clearly, $K = -1$

Putting the value in given equation :-

$$\frac{k^2 + 4k - 2}{k^2 + k - 5} = \frac{1 - 4 - 2}{1 - 1 - 5} = \frac{-5}{-5} = 1$$

Sol.391.(d) Formula :- When $x + \frac{1}{x}$

$$= k, \text{ Then } x^2 + \frac{1}{x^2} = k^2 - 2$$

$$\text{Given that, } x + \frac{1}{x} = 5\sqrt{2}$$

$$\text{Then, } x^2 + \frac{1}{x^2} = 48$$

$$\text{And } x^4 + \frac{1}{x^4} = 2302$$

Sol.392.(a) Formula :-

$$a^4 + b^4 + 2a^2b^2 = (a^2 + b^2)^2$$

$$a^2 + b^2 = 121 - 70 = 51$$

Now,

$$a^4 + b^4 = 51^2 - 2 \times (35)^2$$

$$= 2601 - 2450 = 151$$

Sol.393.(d)

Given, $x + y = 36$ then, $x = 36 - y$

$$= (x - 27)^3 + (y - 9)^3$$

$$= (36 - y - 27)^3 + (y - 9)^3$$

$$= (9 - y)^3 + (y - 9)^3$$

$$= (9 - y)^3 - (9 - y)^3 = 0$$

Sol.394.(a) Formula:-

$$(a+b-c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$$

$$\text{Given, } \frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$$

$$a^2 + b^2 + c^2 - 1024 = -2ab + 2bc + 2ca$$

$$a^2 + b^2 + c^2 + 2ab - 2bc - 2ca = 1024$$

$$(a+b-c)^2 = 1024$$

$$(5c - c)^2 = 1024$$

$$\text{.....(Given, } a + b = 5c) \Rightarrow c = 8$$

$$\text{Sol.395.(d)} \quad x = \frac{\sqrt{5} - 2}{\sqrt{5} + 2} = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}$$

Rationalizing,



$$x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}} \times \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} - \sqrt{4}} = \frac{9 - 4\sqrt{5}}{1}$$

$$= 9 - 4\sqrt{5} \dots\dots\dots \text{eq (1)}$$

$$\text{Then, } \frac{1}{x} = 9 + 4\sqrt{5} \dots\dots\dots \text{eq (2)}$$

Adding eq. (1) and eq. (2), we get

$$x + \frac{1}{x} = 18, \text{ Then, } x^2 + \frac{1}{x^2} = 322$$

Short Trick:- If $x = \frac{\sqrt{a} - \sqrt{b}}{\sqrt{a} + \sqrt{b}}$,

$$\text{Then } x + \frac{1}{x} = \frac{2(a+b)}{(a-b)}$$

According to question,

$$x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}, \text{ Then}$$

$$x + \frac{1}{x} = \frac{2(5+4)}{(5-4)} = 18$$

$$\text{Then, } x^2 + \frac{1}{x^2} = 322$$

Sol.396.(d) Given that,

$$x + \frac{y^2}{x} = 5 \dots\dots\dots \text{eq. (1)}$$

$$\frac{x^2 + y^2}{x} = 5 \Rightarrow x^2 + y^2 = 5x$$

$$\Rightarrow x^2 - 5x = -y^2 \dots\dots\dots \text{eq. (2)}$$

$$\frac{x^2 + 2x + y^2}{x^3 - 5x^2} = \frac{x(x + 2 + \frac{y^2}{x})}{x(x^2 - 5x)} \dots\dots\dots \text{eq. (3)}$$

Putting value of eq. (1) and eq. (2) in eq. (3), we get

$$\frac{x(x + 2 + \frac{y^2}{x})}{x(x^2 - 5x)} = \frac{(5+2)}{(-y^2)} = -\frac{7}{y^2}$$

Sol.397.(d) Given,

$$x + \frac{1}{x} = 8 \Rightarrow x - 8 = \frac{-1}{x} \dots\dots \text{(eq. 1)}$$

$$\frac{5}{x^2 - 8x + 2} = \frac{5}{x(x - 8 + \frac{2}{x})} \dots\dots \text{(eq. 2)}$$

Putting the value of eq. (1) in eq. (2), we get

$$\frac{5}{x(x - 8 + \frac{2}{x})} = \frac{5}{x(\frac{2}{x} - \frac{1}{x})} = 5$$

Sol.398.(c) Identities :- when $ax + \frac{y}{a} = k$

$$\text{, then } (ax)^2 + \frac{y^2}{a^2} = k^2 - 2xy$$

Now,

$$4a + \frac{5}{a} + 5 = 14 \Rightarrow 4a + \frac{5}{a} = 9 \dots\dots \text{e.q. (1)}$$

On squaring e.q. (1), we get

$$16a^2 + \frac{25}{a^2} = 9^2 - 2 \times 4 \times 5 = 41$$

Sol.399.(d) Identity : when $x - \frac{1}{x} = k$,

then,

$$x^2 + \frac{1}{x^2} = k^2 + 2 \Rightarrow x - \frac{1}{x} = 13,$$

$$\text{then, } x^2 + \frac{1}{x^2} = 13^2 + 2 = 171$$

$$\text{And } x^4 + \frac{1}{x^4} = (171)^2 - 2$$

$$= 29241 - 2 = 29239$$

Sol.400.(b)

$$(x^4 + x^2 + 25) = (x^4 + x^2 + 5^2)$$

$$[(x^2 + 5)^2 - 2 \times 5 \times x^2] + x^2$$

$$= [(x^2 + 5)^2 - 10x^2 + x^2]$$

$$= [(x^2 + 5)^2 - 9x^2] = [(x^2 + 5)^2 - (3x)^2]$$

$$= (x^2 + 3x + 5)(x^2 - 3x + 5)$$

$$\text{Sol.401.(c)} x = 3 + 2\sqrt{2}$$

$$x = (\sqrt{2})^2 + (1)^2 + 2\sqrt{2} = (\sqrt{2} + 1)^2$$

$$\sqrt{x} = (\sqrt{2} + 1) \dots\dots \text{eq. (1)}$$

Taking reciprocal of eq. (1), we get

$$\frac{1}{\sqrt{x}} = \frac{1}{(\sqrt{2} + 1)} \times \frac{(\sqrt{2} - 1)}{(\sqrt{2} - 1)} = \frac{(\sqrt{2} - 1)}{1}$$

$$= (\sqrt{2} - 1) \dots\dots \text{eq. (2)}$$

Now, From eq. (1) and eq. (2), we get,

$$\sqrt{x} - \frac{1}{\sqrt{x}} = (\sqrt{2} + 1) - (\sqrt{2} - 1) = 2$$

$$\text{Sol.402.(d)} \text{ when, } x + \frac{1}{x} = -2$$

$$\text{then, } x = -1$$

Now,

$$x^{17} + x^{-17} + x^{12} + x^{-12} = -1 - 1 + 1 + 1 = 0$$

Sol.403.(b)

A number is the factor of its square.

Squaring the identity $(mx+n)$

$$(mx + n)^2 = (m^2x^2 + 2mnx + n^2)$$

Sol.404.(d)

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$\text{When } (x + y + z) = 0,$$

$$\text{then } x^3 + y^3 + z^3 = 3xyz \dots\dots \text{eq. (1)}$$

$$(x + y + z) = 0 \dots\dots \text{(given)}$$

$$\frac{x^2}{(yz)} + \frac{y^2}{(xz)} + \frac{z^2}{(xy)} = \frac{x^3 + y^3 + z^3}{xyz}$$

$$= \frac{3xyz}{xyz} = 3$$

$$\text{Sol.405.(d)} (x + y + z)^2$$

$$= x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$(a + b + c)^2 = 9 + 2 \times 8$$

$$a + b + c = 5$$

$$\text{Now, } (a + b + c) - 3 = 5 - 3 = 2$$

Sol.406.(a) Formula:-

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$a - b = 2 \text{ and } a^3 - b^3 = 80 \dots\dots \text{(given)}$$

$$\text{Now, } (2)^3 = 80 - 3ab(2) \Rightarrow ab = 12$$

$$\text{Sol.407.(d)} \frac{1}{x^2 + a^2} = x^2 - a^2$$

$$x^4 - a^4 = 1 \Rightarrow x^4 = a^4 + 1 \Rightarrow x = \sqrt[4]{a^4 + 1}$$

Sol.408.(c) Statement 1.

$$100^2 - 99^2 + 98^2 - 97^2 \dots\dots\dots + 22^2 - 21^2$$

$$\Rightarrow (100 + 99)(100 - 99) + (98 + 97)(98 - 97) \dots\dots\dots + (22 + 21)(22 - 21)$$

$$\Rightarrow 100 + 99 + 98 + 97 \dots\dots\dots 22 + 21$$

$$\text{Sum} = \frac{n}{2} \times (a + l)$$

$$= \frac{80}{2} (100 + 21) = 40 \times 121 = 4840$$

Statement 2.

$$\Rightarrow (k^2 + \frac{1}{k^2})(k - \frac{1}{k})(k^4 + \frac{1}{k^4})(k + \frac{1}{k})$$

$$(k^4 - \frac{1}{k^4})$$

$$= (k^2 + \frac{1}{k^2})(k^2 - \frac{1}{k^2})(k^4 + \frac{1}{k^4})(k^4 - \frac{1}{k^4})$$

$$= (k^4 - \frac{1}{k^4})(k^8 - \frac{1}{k^8}) \dots\dots \text{eq. (1)}$$

$$\text{Now, } (k^{16} - \frac{1}{k^{16}}) = (k^8 - \frac{1}{k^8})(k^8 + \frac{1}{k^8})$$

$$\dots\dots \text{e.q. (2)}$$

Where, e.q. (1) $\neq$ e.q. (2)

Sol.409.(a) Clearly, option 'a' is the correct algebraic expression/.

$$(a - 1)(b - 1) = ab - a - b + 1$$

Sol.410.(d) $x = 1 \dots\dots \text{(given)}$

$$x^2 + y^2 + z^2 = xy + yz + zx \dots\dots \text{(given)}$$

Put, $y = 1$ and $z = 1$. which satisfies the given condition.

$$\frac{10x^4 + 5y^4 + 7z^4}{13x^2y^2 + 6y^2z^2 + 3z^2x^2} = \frac{10 + 5 + 7}{13 + 6 + 3} = 1$$

Sol.411.(c) $x - y = -10$

$$x^2 - 2xy = 84$$

Adding y^2 both sides

$$x^2 - 2xy + y^2 = 84 + y^2$$

$$(x - y)^2 = 84 + y^2$$

$$y^2 = 100 - 84 \Rightarrow y = 4$$

$$\text{Sol.412.(a)} \text{ When, } x + \frac{1}{x} = 1,$$

$$\text{then } x^3 = -1$$

$$x^3 + 1 = 0 \dots\dots \text{e.q. (1)}$$

$$\text{Now, } x^{12} + x^9 + x^6 + x^3 + 1$$

$$\Rightarrow x^9(x^3 + 1) + x^3(x^3 + 1) + 1 \Rightarrow 1$$

Sol.413.(a) $a + b = 2 \dots\dots \text{eq. (1)}$

$$\frac{a}{b} + \frac{b}{a} = 1$$

$$\frac{a^2 + b^2}{ab} = 1 \Rightarrow a^2 + b^2 = ab$$

$$a^2 + b^2 - ab = 0 \dots\dots \text{eq. (2)}$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

From eq (1) and (2)

$$a^3 + b^3 = 0$$

Sol.414.(a) $4x^2 + y^2 = 40$

Adding $(4xy)$ both sides, we get

$$(2x)^2 + y^2 + 4xy = 40 + 4xy \quad xy = 6 \text{ (given)}$$

$$(2x + y)^2 = 64 \Rightarrow 2x + y = 8$$

Short trick :-

$$4x^2 + y^2 = 40 \text{ and } xy = 6 \text{(given)}$$

Put $x = 3$ and $y = 2$ which satisfy the both given equations.

$$\text{Now, the value of } 2x + y = 2 \times 3 + 2 = 8$$

Sol.415.(c) $ab = 3 \text{(given)}$

$$2a + 3b = 10$$

On squaring,

$$4a^2 + 9b^2 + 12ab = 100$$

$$4a^2 + 9b^2 = 100 - 12 \times 3 = 64$$

Sol.416.(c) $xy = 8 \text{(given)}$

$$(4x - 7y) = 11$$

On squaring, we get

$$16x^2 + 49y^2 - 56xy = 121$$

$$16x^2 + 49y^2 = 121 + 56 \times 8 = 569$$

Sol.417.(d) $mx^m - nx^n = 0$

$$x^m = \frac{nx^n}{m} \text{ ----(i)}$$

$$\text{and } x^n = \frac{mx^m}{n} \Rightarrow x^{2n} = \frac{m^2 x^{2m}}{n^2} \text{ ----(ii)}$$

$$\begin{aligned} \text{Now, } \frac{1}{x^m + x^n} + \frac{1}{x^m - x^n} \\ = \frac{x^m + x^n + x^m - x^n}{x^{2m} - x^{2n}} = \frac{2x^m}{x^{2m} - x^{2n}} \\ = \frac{2x^m}{\frac{n^2 x^{2n}}{m^2} - x^{2n}} = \frac{2x^m}{\frac{x^{2n}(n^2 - m^2)}{m^2}} \end{aligned}$$

{using eq. (i) and (ii)}

$$\begin{aligned} \Rightarrow \frac{2x^m m^2}{x^{2n}(n^2 - m^2)} &= \frac{2 \times \frac{nx^n}{m} \times m^2}{x^{2n}(n^2 - m^2)} \\ &= \frac{2mn}{x^n(n^2 - m^2)} \end{aligned}$$

Sol.418.(d) $5x^2 + 7x + 5 = 0$

Dividing both sides by 5x.

$$x + \frac{1}{x} = \frac{-7}{5}$$

$$x^3 + \frac{1}{x^3} = \left(\frac{-7}{5}\right)^3 - \left(\frac{-7}{5}\right) \times 3$$

$$= \frac{-343}{125} + \frac{21}{5} = \frac{182}{125}$$

Sol.419.(d) Concept :- If $(x)^2 + (y)^2 = 0$, only possible when $x = 0$ and $y = 0$.

$$x^2 + y^2 + 2y + 4x + 5 = 0$$

$$\Rightarrow (x^2 + 4x + 4) + (y^2 + 2y + 1) = 0$$

$$\Rightarrow (x + 2)^2 + (y + 1)^2 = 0$$

$$\text{So, } x + 2 = 0 \Rightarrow x = -2$$

$$\text{And, } y + 1 = 0 \Rightarrow y = -1$$

$$\text{Now, } \frac{x + y}{x - y} = \frac{-3}{-1} = 3$$

Sol.420.(b) Concept :

When $a + b + c = 0$,

$$\text{then } a^3 + b^3 + c^3 = 3abc$$

$$\begin{aligned} \frac{(x - y)^3 + (y - z)^3 + (z - x)^3}{6(x - y)(y - z)(z - x)} \\ = \frac{3(x - y)(y - z)(z - x)}{6(x - y)(y - z)(z - x)} = \frac{1}{2} \end{aligned}$$

Sol.421.(b) $k + \frac{1}{k} + 2 = 0$

$$\Rightarrow k + \frac{1}{k} = -2 \quad (\text{Only possible when } k = -1)$$

$$\text{Then, } k^{10} + \frac{1}{k^{11}} = 1 + (-1) = 0$$

Sol.422.(b) We know that,

$$(a - b)^3 + 3ab(a - b) = a^3 - b^3$$

So,

$$3[a - \frac{1}{a}] + [a - \frac{1}{a}]^3 = a^3 - \frac{1}{a^3}$$

Sol.423.(a) If, $x^4 + \frac{1}{x^4} = 194$ then,

$$x^2 + \frac{1}{x^2} = \sqrt{194 + 2} = 14$$

$$\text{When, } x^2 + \frac{1}{x^2} = 14 \text{ then}$$

$$x + \frac{1}{x} = \sqrt{14 + 2} = 4$$

$$\text{Now, } x^3 + \frac{1}{x^3} = 4^3 - 4 \times 3 = 64 - 12 = 52$$

Sol.424.(b) $(a + 1)^2 + (a + 2)^2 = 16$

$$\Rightarrow a^2 + 2a + 1 + a^2 + 4a + 4 = 16$$

$$\Rightarrow 2a^2 + 6a + 5 = 16$$

$$\Rightarrow 2(2a^2 + 6a + 5) = 16 \times 2$$

$$\Rightarrow 4a^2 + 12a + 10 = 32$$

$$\text{So, } 4a^2 + 12a + 10 + 30 = 32 + 30$$

$$\Rightarrow 4a^2 + 12a + 40 = 62$$

Sol.425.(a) I. $K + \frac{1}{K} = 12$

$$\text{then } K^2 + \frac{1}{K^2} = 144 - 2 = 142 \text{ (True)}$$

II. The value of

$$(K^2 + \frac{1}{K^2})(K - \frac{1}{K})(K^4 + \frac{1}{K^4})(K + \frac{1}{K})$$

$$= (K^2 + \frac{1}{K^2})(K^2 - \frac{1}{K^2})(K^4 + \frac{1}{K^4})$$

$$= (K^4 - \frac{1}{K^4})(K^4 + \frac{1}{K^4})$$

$$= (K^8 - \frac{1}{K^8}) \neq K^{16} - \frac{1}{K^{16}} \text{ (not true)}$$

So, only 1 statement is correct.

Sol.426.(c) $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$,

$$\text{Put } x = \frac{1}{2} \text{ in the given equation.}$$

$$3x^4 - 2x^2 + 4x - 1$$

$$= 3\left(\frac{1}{2}\right)^4 - 2\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right) - 1$$

$$= \frac{3}{16} - \frac{1}{2} + 2 - 1 = \frac{11}{16}$$

$$\text{So, the remainder} = \frac{11}{16}$$

Sol.427.(d) $(a + b)^2 - (a - b)^2$

$$= (a^2 + b^2 + 2ab) - (a^2 + b^2 - 2ab)$$

$$= a^2 + b^2 + 2ab - a^2 - b^2 + 2ab = 4ab$$

Sol.428.(d) $9m^2 + n^2 = 40$ and $mn = 4$

$$\Rightarrow 9m^2 + n^2 + 6mn = 40 + 6mn$$

$$\Rightarrow (3m + n)^2 = 40 + 24 = 64$$

$$\Rightarrow 3m + n = 8$$

Sol.429.(a) $x^2 + 4y^2 + 4y - 4xy - 2x - 8$

$$= (x^2 - 4xy + 4y^2) + 4y - 2x - 8$$

$$= (x - 2y)^2 - 2(x - 2y) - 8$$

$$(x - 2y)^2 - 4(x - 2y) + 2(x - 2y) - 8$$

$$= (x - 2y)\{(x - 2y) - 4\}$$

$$+ 2\{(x - 2y) - 4\}$$

$$= \{x - 2y - 4\}\{x - 2y + 2\}$$

Sol.430.(a) $y = 1 + \sqrt{3} + \sqrt{4}$

$$y - 1 = \sqrt{3} + \sqrt{4}$$

Squaring both side, we have :

$$(y - 1)^2 = (\sqrt{3} + \sqrt{4})^2$$

$$y^2 + 1 - 2y = 3 + 4 + 4\sqrt{3}$$

$$y^2 + 1 - 2y = 7 + 4\sqrt{3} \Rightarrow y^2 - 2y - 6 = 4\sqrt{3}$$

Again, squaring both side, we have :

$$(y^2 - 2y - 6)^2 = (4\sqrt{3})^2$$

$$y^4 + 4y^2 + 36 - 4y^3 + 24y - 12y^2 = 48$$

$$y^4 - 8y^2 - 12 - 4y^3 + 24y = 0$$

Multiplying both side by 2,

$$2y^4 - 16y^2 - 24 - 8y^3 + 48y = 0$$

$$2y^4 - 8y^3 - 6y^2 + 28y - 84 - 10y^2 + 60 + 20y = 0$$

$$2y^4 - 8y^3 - 6y^2 + 28y - 84 - 10(y^2 - 6 - 2y) = 0$$

$$2y^4 - 8y^3 - 6y^2 + 28y - 84 = 10(y^2 - 6 - 2y)$$

$$= 10 \times 4\sqrt{3} = 40\sqrt{3}$$

Sol.431.(d) $x = \frac{\sqrt{5} - \sqrt{4}}{\sqrt{5} + \sqrt{4}}$

$$= \frac{(\sqrt{5} - \sqrt{4})(\sqrt{5} - \sqrt{4})}{(\sqrt{5} + \sqrt{4})(\sqrt{5} - \sqrt{4})} = (\sqrt{5} - \sqrt{4})^2$$

$$y = \frac{\sqrt{5} + \sqrt{4}}{\sqrt{5} - \sqrt{4}} = \frac{(\sqrt{5} + \sqrt{4})(\sqrt{5} + \sqrt{4})}{(\sqrt{5} - \sqrt{4})(\sqrt{5} + \sqrt{4})}$$

$$= (\sqrt{5} + \sqrt{4})^2$$

$$\begin{aligned} x^2 + y^2 &= [\sqrt{5} - \sqrt{4}]^4 + [\sqrt{5} + \sqrt{4}]^4 \\ &= \{(\sqrt{5} - \sqrt{4})^2 + (\sqrt{5} + \sqrt{4})^2\}^2 - 2(\sqrt{5} - \sqrt{4})^2(\sqrt{5} + \sqrt{4})^2 \end{aligned}$$

$$= 18^2 - 2 = 324 - 2 = 322$$

$$\Rightarrow \frac{x^2 - xy + y^2}{x^2 + xy + y^2} = \frac{322 - 1}{322 + 1} = \frac{321}{323}$$

Sol.432.(b) $\frac{2p}{p^2 - 5p + 1} = \frac{1}{10}$

$$\Rightarrow p^2 - 5p + 1 = 20p$$

$$\Rightarrow p^2 - 25p + 1 = 0$$

Dividing both side by p.



$$\Rightarrow p + \frac{1}{p} = 25$$

Sol.433.(c) On checking all the options for the values a and b one by one, option (c) gets satisfied.

$$a^2 = b^2 + (ab)^2$$

$$\cot^2 x = \cos^2 x + \cos^2 x \cot^2 x$$

$$\cot^2 x = \cos^2 x (1 + \cot^2 x) = \cos^2 x \cdot \operatorname{cosec}^2 x$$

$$\cot^2 x = \cot^2 x$$

LHS = RHS (satisfied)

Sol.434.(b)

$$\frac{1}{1+a+\frac{1}{b}} + \frac{1}{1+b+\frac{5}{c}} + \frac{1}{1+\frac{c}{5}+\frac{1}{a}}$$

putting a = b = 1 and c = 5 in the given expression

$$\frac{1}{1+1+1} + \frac{1}{1+1+\frac{5}{5}} + \frac{1}{1+\frac{5}{5}+1}$$

$$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$$

Sol.435.(b)

$$4(z+7)(2z-1) = (4z+28)(2z-1)$$

$$= 8z^2 - 4z + 56z - 28 = 8z^2 + 52z - 28$$

$$\text{Now, } 8z^2 + 52z - 28 = Az^2 + Bz + C$$

On comparing both side, we have :

$$A = 8, B = 52, C = -28$$

$$\text{So, } A + B + C = 8 + 52 - 28 = 32$$

$$\text{Sol.436.(c)} p^3 + 27 = p^3 + 3^3$$

$$= (p+3)(p^2 + 3^2 - 3p) = (p+3)(p^2 + 9 - 3p)$$

Sol.437.(d) Formula used :

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$a^2 + b^2 + c^2 = 9^2 - 2 \times 23 = 81 - 46 = 35$$

$$\text{Sol.438.(b)} x - \frac{1}{x} = \sqrt{(2\sqrt{5})^2 - 4}$$

$$= \sqrt{20 - 4} = \sqrt{16} = 4$$

$$x^3 - \frac{1}{x^3} = (4)^3 + 3 \times 4 = 64 + 12 = 76$$

Sol.439.(a) Formula : $x^3 + y^3 + z^3 = 3xyz$ (when $x+y+z=0$)

$$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = \frac{x^3 + y^3 + z^3}{xyz}$$

$$= \frac{3xyz}{xyz} = 3$$

$$\text{Sol.440.(b)} (a+b)^3 - a^3 + b^3$$

$$= a^3 + b^3 + 3ab(a+b) - a^3 + b^3$$

$$= 2b^3 + 3ab(a+b)$$

$$\text{Sol.441.(b)} \text{ If } x^2 + \frac{1}{x^2} = 66$$

$$(x - \frac{1}{x})^2 = 66 - 2 = 64 \Rightarrow x - \frac{1}{x} = \sqrt{64} = 8$$

$$\text{Sol.442.(c)} P - \frac{1}{P} = 6$$

$$P^2 + \frac{1}{P^2} = 6^2 + 2 = 38$$

$$P^4 + \frac{1}{P^4} = 38^2 - 2 = 1444 - 2 = 1442$$

Sol.443.(d)

$$x^4 - 15x^3 + 15x^2 - 15x + 40; \text{ (given } x = 14)$$

$$x^4 - 14x^3 - x^3 + 14x^2 + x^2 - 14x - x + 14 + 26$$

$$x^3(x-14) - x^2(x-14) + x(x-14) - 1(x-14) + 26$$

Putting $x = 14$, we get,

$$x^4 - 15x^3 + 15x^2 - 15x + 40 = 26$$

$$\text{Sol.444.(b)} y - \frac{1}{y} = \sqrt{11^2 - 4}$$

$$= \sqrt{121 - 4} = \sqrt{117} = 3\sqrt{13}$$

$$y^3 - \frac{1}{y^3} = (3\sqrt{13})^3 + 3 \times 3\sqrt{13}$$

$$= 351\sqrt{13} + 9\sqrt{13} = 360\sqrt{13}$$

$$\text{Sol.445.(b)} (x - \frac{1}{x})^2 = 12$$

$$x - \frac{1}{x} = \sqrt{12} = 2\sqrt{3}$$

$$\text{Now, } x + \frac{1}{x} = \sqrt{(2\sqrt{3})^2 + 4} = \sqrt{16} = 4$$

$$\text{So, } x^2 - \frac{1}{x^2} = 4 \times 2\sqrt{3} = 8\sqrt{3}$$

Sol.446.(b) $a^3 + b^3 + c^3 = 3abc$, either $a+b+c=0$ or $a=b=c$

$$\text{Sol.447.(d)} m + \frac{1}{m-2} = 4$$

Subtracting both side by 2. We get :

$$m-2 + \frac{1}{m-2} = 2$$

$$\text{So, } (m-2)^2 + \frac{1}{(m-2)^2} = 2^2 - 2 = 2$$

$$\text{Sol.448.(a)} 5x - \frac{5}{x} + 6 = 0$$

$$5x - \frac{5}{x} = -6 \Rightarrow 5(x - \frac{1}{x}) = -6$$

$$(x - \frac{1}{x}) = \frac{-6}{5}$$

$$x^2 + \frac{1}{x^2} = (\frac{-6}{5})^2 + 2 = \frac{86}{25}$$

$$\text{Sol.449.(c)} \frac{a^3 - b^3}{a^2 - b^2} - \frac{3ab}{a+b}$$

$$= \frac{(a-b)(a^2 + b^2 + ab)}{(a-b)(a+b)} - \frac{3ab}{a+b}$$

$$= \frac{(a^2 + b^2 + ab)}{a+b} - \frac{3ab}{a+b} = \frac{a^2 + b^2 - 2ab}{a+b}$$

$$= \frac{(a-b)^2}{a+b} = \frac{(26-22)^2}{26+22} = \frac{16}{48} = \frac{1}{3}$$

$$\text{Sol.450.(b)} (a-b+c)^2$$

$$= a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$$

$$\text{Sol.451.(b)} \frac{x}{8} + \frac{1}{\frac{x}{8}} = 1$$

By applying formula, $[a^3 + b^3 = (a+b)\{a^2 - ab + b^2\}]$ we get,

$$(\frac{x}{8})^3 + \frac{1}{(\frac{x}{8})^3} = 1 - 3 = -2 \Rightarrow \frac{x}{8} = -1$$

$$x = -8 \text{ So, } x^3 = (-8)^3 = -512$$

Sol.452.(d) Formula used :

$$p^3 + q^3 + r^3 = 3pqr \text{ (when } p+q+r=0)$$

$$P+q+r = 8.15 + 9.06 - 17.21 = 0$$

$$\text{So, } p^3 + q^3 + r^3 - 3pqr = 3pqr - 3pqr = 0$$

$$\text{Sol.453.(b)} x^4 + (\frac{2}{x})^4 = 15617$$

$$(x^2 + \frac{4}{x^2})^2 = 15617 + 8 = 15625$$

$$x^2 + (\frac{2}{x})^2 = \sqrt{15625} = 125$$

$$(x + \frac{2}{x})^2 = 125 + 4 = 129$$

$$x + \frac{2}{x} = \sqrt{129}$$

Sol.454.(a) Formula used ; $(x+y+z)^2$

$$= x^2 + y^2 + z^2 + 2(xy+yz+zx)$$

$$0 = 40 + 2(xy+yz+zx)$$

$$-40 = 2(xy+yz+zx)$$

$$xy+yz+zx = \frac{-40}{2} = -20$$

$$\text{Sol.455.(a)} \text{ I. } x^3 + y^3 + z^3 = 12^3 - 2^3 - 10^3$$

$$= 1728 - 8 - 1000 = 720$$

$$\text{II. } 4(x^2 + y^2) = 4[(x+y)^2 - 2xy]$$

$$= 4[48^2 - 64] = 4[2304 - 64]$$

$$= 4 \times 2240 = 8960$$

Clearly, we can see that neither statement I nor II is correct..

$$\text{Sol.456.(b)} \text{ It is given that } x + \frac{1}{x} = 6$$

$$\text{Now, } \frac{3x}{2x^2 - 5x + 2}$$

On dividing numerator and denominator by x, we get :

$$\frac{3}{2x - 5 + \frac{2}{x}} = \frac{3}{2(x + \frac{1}{x}) - 5}$$

$$= \frac{3}{12 - 5} = \frac{3}{7}$$

Sol.457.(d)

$$8p^3 + q^3 = (2p+q)[(2p+q)^2 - 6pq]$$

$$361 = 19[361 - 6pq]$$

$$19 = 361 - 6pq \Rightarrow 342 = 6pq$$

$$pq = \frac{342}{6} = 57$$

$$\text{Sol.458.(a)} \frac{a}{b} + \frac{b}{a} = -1$$

$$\frac{a^2 + b^2}{ab} = -1 \Rightarrow a^2 + b^2 = -ab$$

$$\text{Now, } (a-b)^2 = a^2 + b^2 - 2ab$$

$$4 = a^2 + b^2 - 2ab = -ab - 2ab = -3ab$$



$$ab = -\frac{4}{3}$$

$$a^3 - b^3 = (a-b)[(a-b)^2 + 3ab]$$

$$= 2[4 + 3 \times (-\frac{4}{3})] = 2 \times 0 = 0$$

$$\text{Sol.459.(d)} \quad x = 2 - 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$$

$$x - 2 = 2^{\frac{2}{3}} - 2^{\frac{1}{3}}$$

Taking cube on both side,

$$(x-2)^3 = (2^{\frac{2}{3}} - 2^{\frac{1}{3}})^3$$

$$x^3 - 8 - 6x^2 + 12x$$

$$= 4 - 2 - 3 \times 2^{\frac{2}{3}} \times 2^{\frac{1}{3}} (2^{\frac{2}{3}} - 2^{\frac{1}{3}})$$

$$x^3 - 10 - 6x^2 + 12x = -6(x-2)$$

$$x^3 - 10 - 6x^2 + 12x + 6x - 12 = 0$$

$$x^3 - 6x^2 + 18x = 22$$

Sol.460.(b)

$$\frac{2abc}{9} = \frac{2 \times (-12)(-6)18}{9} = 288$$

$$\text{Sol.461.(a)} \quad x^4 + \frac{1}{x^4} = 47$$

Adding both side by 2,

$$(x^2 + \frac{1}{x^2})^2 = 49 \Rightarrow x^2 + \frac{1}{x^2} = \sqrt{49} = 7$$

Again, Adding both side by 2

$$(x + \frac{1}{x})^2 = 9 \Rightarrow x + \frac{1}{x} = 3$$

$$\text{So, } x + \frac{1}{x} - 2 = 3 - 2 = 1$$

Sol.462.(c) Formula used :

$$a^3 + b^3 + c^3 = 3abc \text{ (when } a + b + c = 0)$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$\text{Now, } (x-y) + (y-z) + (z-x) = 0$$

$$\Rightarrow (x-y)^3 + (y-z)^3 + (z-x)^3$$

$$= 3(x-y)(y-z)(z-x)(x^2 - y^2) + (y^2 - z^2) + (z^2 - x^2) = 0$$

$$\Rightarrow (x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3 = 3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)$$

$$\text{So, } \frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3}$$

$$= \frac{3(x-y)(y-z)(z-x)}{3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)}$$

$$= \frac{1}{(x+y)(y+z)(z+x)}$$

Sol.463.(c) As we know that,

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$10^2 = 38 + 2(ab+bc+ca)$$

$$62 = 2(ab+bc+ca) \Rightarrow 31 = ab+bc+ca$$

$$\text{Now, } (a-b)^2 + (b-c)^2 + (c-a)^2$$

$$= 2[a^2 + b^2 + c^2 - (ab+bc+ca)]$$

$$= 2[38 - 31] = 2 \times 7 = 14$$

$$\text{Sol.464.(a)} \quad 16y^2 - k = (4y + \frac{3}{2})(4y - \frac{3}{2})$$

$$\Rightarrow 16y^2 - k = (4y)^2 - (\frac{3}{2})^2$$

$$[(a+b)(a-b) = (a^2 - b^2)]$$

$$\Rightarrow 16y^2 - k = 16y^2 - \frac{9}{4} \Rightarrow k = \frac{9}{4}$$

Sol.465.(c) Formula used :

$$(m-n)^3 = m^3 - n^3 - 3mn(m-n)$$

$$\Rightarrow \frac{\{(m^2 + n^2)(m-n) - (m-n)^3\}}{(m^2n - mn^2)} =$$

$$\frac{\{m^3 - m^2n + n^2m - n^3 - (m^3 - n^3 - 3m^2n + 3mn^2)\}}{(m^2n - mn^2)}$$

$$= \frac{2(m^2n - mn^2)}{m^2n - mn^2} = 2$$

Sol.466.(d) Formula used :

$$(x+y+z)^2 = x^2 + y^2 + z^2 + 2(xy+yz+zx)$$

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)$$

$$[x^2 + y^2 + z^2 - (xy + yz + zx)]$$

$$x + y + z = 8, x^2 + y^2 + z^2 = 20$$

$$\text{Now, } xy + yz + zx = \frac{8^2 - 20}{2} = 22$$

$$\text{So, } x^3 + y^3 + z^3 - 3xyz = 8(20 - 22)$$

$$= 8 \times (-2) = -16$$

Sol.467.(d)

$$\{(1 \div x) \times (x^3 - 2x^2) + (4^3 \div 2^5) \times x\} \div 25 = 1$$

$$\{\frac{1}{x} \times x(x^2 - 2x) + (64 \div 32) \times x\} = 25$$

$$\{(x^2 - 2x) + 2x\} = 25 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$$

So, the possible value of x is 5 and -5.

Sol.468.(a)

$$(7x + 4y)^2 + (7x - 4y)^2 = 2[(7x)^2 + (4y)^2]$$

$$= 2[49x^2 + 16y^2] = 98x^2 + 32y^2$$

$$\text{Sol.469.(c)} \quad 6x + 7y = 5xy \text{ ----- (1)}$$

$$10y - 4x = 4xy$$

$$5y - 2x = 2xy \text{ ----- (2)}$$

$$\text{Then, } [eq. (2)] \times 3$$

$$15y - 6x = 6xy \text{ ----- (3)}$$

Adding eqn. (1) and (3) we get:

$$22y = 11xy \Rightarrow x = 2$$

$$\text{Putting } x = 2 \text{ in eqn(2) we get : } y = 4$$

Sol.470.(b) The best approach to solve his question is the value putting method.

By putting x = 6 and y = 4, we get :

$$x + y = 6 + 4 = 10 \text{ and } 2xy = 2 \times 6 \times 4$$

$$= 48(\text{satisfied})$$

$$\text{So, } 2x - y = 2 \times 6 - 4 = 8$$

$$\text{Sol.471.(a)} \quad a^2 + b^2 + 1 = 2a$$

$$\rightarrow (a-1)^2 + b^2 = 0$$

$$\rightarrow a = 1 \text{ and } b = 0$$

$$\text{So, } a^4 + b^7 = 1^4 + 0 = 1$$

Sol.472. (a) Formula used :

$$x^3 + y^3 + z^3 = 3abc \text{ (when } a + b + c = 0)$$

$$\text{Now, } \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = \frac{a^3 + b^3 + c^3}{abc}$$

$$= \frac{3abc}{abc} = 3$$

Sol.473.(a) Formula used :

$$(x-y)^3 = x^3 - y^3 - 3xy(x-y)$$

$$(x+y)^3 - (x-y)^3 - 6y(x^2 - y^2)$$

$$= (x+y)^3 - (x-y)^3 - 3(x+y)(x-y)2y$$

$$= (x+y)^3 - (x-y)^3 - 3(x+y)(x-y)$$

$$[x+y - (x-y)]$$

$$= [(x+y) - (x-y)]^3 = [2y]^3 = 8y^3$$

$$\text{Sol.474.(d)} \quad x^2 + 6x + 1 = 0$$

Dividing both side by x,

$$x + 6 + \frac{1}{x} = 0 \Rightarrow x + \frac{1}{x} = -6$$

$$x^3 + \frac{1}{x^3} = (-6)^3 - 3 \times (-6)$$

$$= -216 + 18 = -198$$

$$\text{Also, } x + 6 = -\frac{1}{x}$$

$$\text{Now, } (x+6)^3 + \frac{1}{(x+6)^3} = -(\frac{1}{x^3} + x^3)$$

$$= 198$$

Sol.475.(d) Putting b = 1, we have ;

$$a^2 = 82 - 1 = 81 \Rightarrow a = 9$$

$$\text{and } c^2 = 65 - 1 = 64 \Rightarrow c = 8$$

Now ,

$$2a + 7b - 3c = 2 \times 9 + 7 \times 1 - 3 \times 8$$

$$= 18 + 7 - 24 = 1$$

$$\text{Sol.476.(b)} \quad x(x-5) = -1$$

$$x - 5 = \frac{-1}{x} \Rightarrow x + \frac{1}{x} = 5 \text{ (1)}$$

$$\text{Now, } x^3 + \frac{1}{x^3} = 5^3 - 3 \times 5 = 110$$

$$\Rightarrow x^6 + 1 = 110x^3 \Rightarrow x^6 - 110x^3 = -1$$

$$\Rightarrow x^3(x^3 - 110) = -1$$

$$\text{Sol.477.(c)} \quad \text{Given, } a + \frac{1}{a} = p^2,$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$\text{Then, } a^2 + \frac{1}{a^2} = p^4 - 2$$

$$\text{Sol.478.(a)} \quad a = -9.6, b = 4.44$$

$$\text{and } c = 5.16,$$

$$\text{If } a + b + c = 0$$

$$\Rightarrow -9.6 + 4.44 + 5.16 = 0$$

$$\text{then } a^3 + b^3 + c^3 - 3abc = 0$$

$$\text{Sol.479.(c)} \quad \sqrt{1 + \frac{x}{529}} = \frac{24}{23}$$

$$1 + \frac{x}{529} = \frac{576}{529}$$

$$\frac{x}{529} = \frac{47}{529} \Rightarrow x = 47$$

Sol.480.(a)

$$k(21x^2 + 24) + rx + (14x^2 - 9) = 0 \text{ -----(i)}$$

$$k(7x^2 + 8) + px + (2x^2 - 3) = 0 \text{ -----(ii)}$$

Multiply equation (ii) by 3 and subtract from equation (i),

$$k(21x^2 + 24) + rx + (14x^2 - 9) - 3 \times [k(7x^2 + 8) + px + (2x^2 - 3)] = 0$$

$$8x^2 - (3p-r)x + 0 = 0 \text{ -----(iii)}$$



So let roots of equation (iii) are α and β

$$\alpha + \beta = \frac{3p-r}{8} \Rightarrow \alpha\beta = 0$$

Equations have both roots common.

$$\text{so, } \frac{3p-r}{8} = 0 \Rightarrow \frac{p}{r} = \frac{1}{3}$$

Sol.481.(c)

$$\begin{aligned} & \frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} \div \frac{\sqrt{14}+\sqrt{10}}{\sqrt{14}-\sqrt{10}} + \frac{\sqrt{10}}{\sqrt{5}} \\ &= \frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} \times \frac{\sqrt{14}-\sqrt{10}}{\sqrt{14}+\sqrt{10}} + \frac{\sqrt{2} \times \sqrt{5}}{\sqrt{5}} \\ &= \frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} \times \frac{\sqrt{2}(\sqrt{7}-\sqrt{5})}{\sqrt{2}(\sqrt{7}+\sqrt{5})} + \sqrt{2} = 1 + \sqrt{2} \end{aligned}$$

Sol.482.(d) $(x + \frac{1}{x})^2 = 3$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 3 \Rightarrow x^2 + \frac{1}{x^2} = 1$$

On cubing above equation, we get

$$(x^2 + \frac{1}{x^2})^3 = 1^3 \Rightarrow x^6 + \frac{1}{x^6} + 3(x^2)(\frac{1}{x^2}) = 1$$

$$x^6 + \frac{1}{x^6} + 3 = 1 \Rightarrow x^6 + \frac{1}{x^6} = -2$$

$$\text{So, } x^6 + x^{-6} = -2$$

Sol.483.(a) $2X + 3Y = 12$ (1)

$3X - 2Y = 5$ (2)

Multiplying eqn (1) by 3 and eqn (2) by 2, we get

$$6X + 9Y = 36$$
(3)

$$6X - 4Y = 10$$
(4)

Solving equation (3) and (4), we get

$$X = 3 \text{ and } Y = 2$$

So, option (a) is the correct answer.

Sol.484.(c) Let $x = 3$ and $y = -2$

Putting above values of x and y , we get

$$xy = (3)(-2) = -6 \text{(satisfied)}$$

$$x^3 + y^3 = 3^3 + (-2)^3$$

$$= 27 - 8 = 19 \text{(satisfied)}$$

Hence, $x = 3$ and $y = -2$

So,

$$\frac{1}{x^{-1}} + \frac{1}{y^{-1}} = x + y = 3 + (-2) = 3 - 2 = 1$$

Sol.485.(c) $2^x = 3^y = 6^z = K$

$$\Rightarrow 2 = K^{\frac{1}{x}}, 3 = K^{\frac{1}{y}}, 6 = K^{\frac{1}{z}}$$

Now, $2 \times 3 = 6$

$$\Rightarrow K^{\frac{1}{x}} \times K^{\frac{1}{y}} = K^{\frac{1}{z}} \Rightarrow K^{\frac{1}{x} + \frac{1}{y}} = K^{\frac{1}{z}}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z} \Rightarrow \frac{1}{x} + \frac{1}{y} - \frac{1}{z} = 0$$

Sol.486.(d) $a + b = \sqrt{11}$

$$\Rightarrow a^2 + b^2 = 11 - 2ab \text{ (1)}$$

$$\Rightarrow a - b = \sqrt{7} \Rightarrow a^2 + b^2 = 7 + 2ab \text{ (2)}$$

From equation (1) and (2) we have:

$$\Rightarrow 11 - 2ab = 7 + 2ab \Rightarrow 4 = 4ab \Rightarrow ab = 1$$

Putting $ab = 1$ in eq. (1) we get:

$$\Rightarrow a^2 + b^2 = 11 - 2 = 9$$

$$\text{Now, } a^2 b^2 (a^2 + b^2) = 1^2 \times 9 = 9$$

Sol.487.(a)

$$\text{Arithmetic mean of } x \text{ and } y = \frac{x+y}{2} = 8$$

$$\Rightarrow x + y = 16$$

$$\text{Geometric mean of } x \text{ and } y = \sqrt{xy} = 3\sqrt{7}$$

$$\Rightarrow xy = (3\sqrt{7})^2 = 63$$

$$\begin{aligned} x^3 + y^3 &= (x+y)[(x+y)^2 - 3xy] \\ &= 16[16^2 - 3 \times 63] = 16[256 - 189] \\ &= 16 \times 67 = 1072 \end{aligned}$$

Sol.488.(c)

$$\begin{aligned} 25x^2 + 16y^2 + 9z^2 - 40xy + 24yz - 30zx \\ &= (-5x)^2 + (4y)^2 + (3z)^2 - (2.5x.4y) \\ &+ (2.4y.3z) - (2.5x.3z) \\ &= (-5x + 4y + 3z)^2 \end{aligned}$$

Sol.489.(a) $xy = 6$ and

$$x^2y + xy^2 + x + y = 63$$

$$\Rightarrow xy(x+y) + (x+y) = 63$$

$$\Rightarrow 6(x+y) + (x+y) = 63$$

$$\Rightarrow 7(x+y) = 63 \Rightarrow x+y = 9$$

Squaring both sides,

$$\Rightarrow x^2 + y^2 + 2xy = 81$$

$$\Rightarrow x^2 + y^2 = 81 - 12 \Rightarrow x^2 + y^2 = 69$$

Sol.490.(b) Formula: $a^3 + b^3 + c^3 - 3abc$

$$= (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

$$x^3 + 27y^3 + 64z^3 - 36xyz$$

$$= (x)^3 + (3y)^3 + (4z)^3 - 3.x.3y.4z$$

$$= (x+3y+4z)(x^2+9y^2+16z^2$$

$$- 3xy - 12yz - 4xz)$$

Sol.491.(b)

$$(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$$

$$= (5x+4y)^2 + (5x-4y)^2$$

$$= 2(25x^2 + 16y^2) = 50x^2 + 32y^2$$

Sol.492.(a) Let the required no. be A

ATQ,

$$(x^3 - 3x^2y + 3xy^2 - y^3) - A = x^3 - y^3$$

$$A = (x^3 - 3x^2y + 3xy^2 - y^3) - x^3 + y^3$$

$$A = -3x^2y + 3xy^2$$

Sol.493.(b) $(a+b)^2 = (a-b)^2 + 4ab$

$$= 10^2 + 4 \times 4 = 100 + 16 = 116$$

$$a^3 - b^3 = 10(10^2 + 3 \times 4) = 10(100 + 12)$$

$$= 10 \times 112 = 1120$$

$$\text{So, } a^3 - b^3 + 4(a+b)^2 = 1120 + 4 \times 116$$

$$= 1120 + 464 = 1584$$

Sol.494.(c) Putting $x = y = 1$, we get $a = 2$,

$b = 0$ and $c = 1$

$$\text{So, } a^2 + b^2 + c^2 - 2ab + 2ac - 2bc = 2^2 + 0$$

$$+ 1^2 - 0 + 2 \times 2 \times 1 - 0 = 4 + 1 + 4 = 9$$

Now, on putting $x = y = 1$ in all the options given, option (c) gets satisfied.

$$(2x + 2y - 1)^2 = (2 + 2 - 1)^2$$

$$= 3^2 = 9 \text{ (satisfied)}$$

$$\text{Sol.495.(a)} \quad y^6 + \frac{1}{y^6} = (y^3 + \frac{1}{y^3})^2 - 2$$

$$\Rightarrow y^3 + \frac{1}{y^3} = (\sqrt{5})^3 - 3 \times \sqrt{5} = 5\sqrt{5} - 3\sqrt{5}$$

$$= 2\sqrt{5}$$

$$\text{So, } y^6 + \frac{1}{y^6} = (2\sqrt{5})^2 - 2 = 20 - 2 = 18$$

Sol.496.(b)

$$P = \frac{x^2 - 100}{x^2 - 16} = \frac{(x-10)(x+10)}{(x-4)(x+4)}$$

$$Q = \frac{x+10}{x-4}$$

$$\text{So, } \frac{P}{Q} = \frac{(x-10)(x+10)}{(x-4)(x+4)} \times \frac{x-4}{(x+10)}$$

$$= \frac{x-10}{x+4}$$

Sol.497.(a)

$$\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{2a-3b+5c}{k} = n(\text{let})$$

$$a = 2n, b = 3n, c = 4n \text{ and}$$

$$2a - 3b + 5c = kn \text{(i)}$$

Put value of a, b and c in equation (i),

$$2 \times 2n - 3 \times 3n + 5 \times 4n = kn$$

$$\Rightarrow 4n - 9n + 20n = kn \Rightarrow 15n = kn$$

$$\Rightarrow k = 15$$

$$\text{Sol.498.(c)} \quad \frac{x-1}{x} = 5 \Rightarrow x-1 = 5x$$

$$\Rightarrow 4x = -1 \Rightarrow \frac{1}{x} = -4$$

$$\text{So, } 1 + \frac{1}{x^2} = 1 + 16 = 17$$

$$\text{Sol.499.(c)} \quad 2x + \frac{1}{3x} = 12$$

Multiplying both side by $\frac{3}{2}$,

$$3x + \frac{1}{2x} = 18$$

$$\text{So, } 9x^2 + \frac{1}{4x^2} = 18^2 - (2 \times 3 \times \frac{1}{2})$$

$$= 324 - 3 = 321$$

$$\text{Sol.500.(c)} \quad \frac{4}{x} - \frac{3}{2x} = 20$$

$$\Rightarrow \frac{8-3}{2x} = 20 \Rightarrow 5 = 40x \Rightarrow x = \frac{5}{40} = \frac{1}{8}$$

Sol.501.(a) Formula used :

$$x^3 - y^3 = (x-y)[(x-y)^2 + 3xy]$$

$$\Rightarrow 520 = 5[5^2 + 3xy] \Rightarrow 104 = 25 + 3xy$$

$$\Rightarrow 79 = 3xy \Rightarrow xy = \frac{79}{3}$$

Sol.502.(d) Formula used:

$$x^3 + y^3 + z^3 = 3xyz,$$

when $(x+y+z) = 0$

$$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = \frac{x^3 + y^3 + z^3}{xyz}$$

$$= \frac{3xyz}{xyz} = 3$$



Sol.503.(c)

$$(2a - b)^2 = (2a + b)^2 - 4(2a)b$$

$$(2a - b)^2 = 10^2 - 4 \times 9$$

$$(2a - b)^2 = 100 - 36 = 64 \Rightarrow 2a - b = 8$$

Sol.504.(b) $a^2 + b^2 = (a + b)^2 - 2ab$

$$= 12^2 - 2 \times 35 = 144 - 70 = 74$$

$$a^4 + b^4 = (a^2 + b^2)^2 - 2a^2b^2 =$$

$$74^2 - 2 \times 35^2 = 5476 - 2450 = 3026$$

$$a^4 + b^4 - 10^3 = 3026 - 1000 = 2026$$

Sol.505.(a) $(a^3 + 8b^3) = (a+2b)$

$$\{(a + 2b)^2 - 3 \times 2ab\}$$

$$5427 = 27(27^2 - 3 \times 2ab)$$

$$201 = 729 - 3 \times 2ab \Rightarrow 2ab = \frac{528}{3} = 176$$

Sol.506.(c) $x^2 + ax = b$

By cubing both side, we get:

$$x^6 + x^3a^3 + 3ax^3 \times b = b^3$$

$$x^3(x^3 + a^3 + 3ab) = b^3$$

Now, comparing this equation with the given equation i.e. $x^3(x^3 + c) = d$

$$c = a^3 + 3ab \text{ and } d = b^3$$

Sol.507.(c) As we know that,

$$x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - xz)$$

So, when $x + y + z = 0$,

$$\text{we have, } x^3 + y^3 + z^3 - 3xyz = 0$$

Sol.508.(a) $r + \frac{64}{r} = 16$

On putting value, $r = 8$

$$8 + \frac{64}{8} = 16 \Rightarrow 16 = 16 \text{ (satisfy)}$$

$$r^4 + \frac{1}{r^3} = 8^4 + \frac{1}{8^3} = 4096 + \frac{1}{512}$$

Sol.509.(c) On putting value, $p = 11$

$$p(p^2 + 3p + 3) \Rightarrow 11(11^2 + 3 \times 11 + 3)$$

$$\Rightarrow 11(121 + 33 + 3) \Rightarrow 11 \times 157 = 1727$$

Sol.510.(a) $\frac{X^2}{y^2} + \frac{y^2}{X^2} = 7$

$$\text{Let } \frac{X}{y} = k \text{ then } \frac{y}{X} = \frac{1}{k}$$

$$k^2 + \frac{1}{k^2} = 7 \Rightarrow k^2 + \frac{1}{k^2} + 2 = 7 + 2$$

$$k + \frac{1}{k} = 3 \Rightarrow \frac{X^3}{y^3} + \frac{y^3}{X^3} = k^3 + \frac{1}{k^3}$$

$$k^3 + \frac{1}{k^3} = 3^3 - 3 \times 3$$

$$k^3 + \frac{1}{k^3} = 18 \text{ So, } \frac{X^3}{y^3} + \frac{y^3}{X^3} = 18$$

Sol.511.(c) $a^2 - 4a + 1 = 0$

Dividing both side by a

$$a + \frac{1}{a} = 4; a^2 + \frac{1}{a^2} = 4^2 - 2 = 14$$

$$a^2 + \frac{1}{a^2} + a + \frac{1}{a} = 14 + 4 = 18$$

Sol.512.(b) Formula used :

$$x^3 - \frac{1}{x^3} = (x - \frac{1}{x})^3 + 3(x - \frac{1}{x})$$

$$x^3 - \frac{1}{x^3} = (\frac{7}{3})^3 + 3 \times \frac{7}{3} = \frac{343}{27} + 7$$

$$= \frac{532}{27} = 19\frac{19}{27}$$

Sol.513.(a) $x^3 + \frac{1}{x^3} = (\frac{11}{5})^3 - 3 \times \frac{11}{5}$

$$= \frac{1331}{125} - \frac{33}{5} = \frac{506}{125} = 4\frac{6}{125}$$

Sol.514.(d) Formula used :

$$(1 + x + x^2)(1 - x + x^2) = 1 + x^2 + x^4$$

$$\text{So, } 1 + x + x^2 = \frac{585}{65} = 9$$

Sol.515.(c)

Formula used : $a^2 + b^2 + c^2 - ab - bc - ca$

$$= \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca$$

$$= \frac{1}{2}[(2022 - 2021)^2 +$$

$$(2021 - 2020)^2 + (2020 - 2022)^2]$$

$$\Rightarrow \frac{1}{2} \times [1 + 1 + 4] = \frac{1}{2} \times 6 = 3$$

Sol.516.(d) $(x - 2y)^2 = x^2 + 4y^2 - 4xy$

$$= 40 - 4 \times 6 = 40 - 24 = 16 \Rightarrow x - 2y = 4$$

Sol.517.(b) $x^2 - 8x + 1 = 0$

Dividing both sides by x. We get

$$x + \frac{1}{x} = 8 \Rightarrow x^2 + \frac{1}{x^2} = 64 - 2 = 62$$

$$x^4 + \frac{1}{x^4} = 3844 - 2 = 3842$$

Multiplying both sides by x^4

$$x^8 + 1 = 3842x^4 \Rightarrow x^8 - 3842x^4 + 1 = 0$$

Sol.518.(c) $144x^2 - 36x + \frac{9}{4}$

$$= (12x)^2 - 2 \times 12x \times \frac{3}{2} + (\frac{3}{2})^2 = (12x - \frac{3}{2})^2$$

$$\text{So, square of } 12x - \frac{3}{2} = 144x^2 - 36x + \frac{9}{4}$$

Sol.519.(b) It is given that,

$$a + b = 17, ab = 34$$

$$a^2 + b^2 = (a + b)^2 - 2ab$$

$$\Rightarrow 17^2 - 2 \times 34 = 289 - 68 = 221$$

Sol.520.(a) Formula used :

$$x^3 + y^3 + z^3 = 3xyz, \text{ when } x + y + z = 0$$

$$8x^3 + 27y^3 + 64z^3 = 72xyz$$

$$(2x)^3 + (3y)^3 + (4z)^3 = 3 \times 2x \times 3y \times 4z$$

$$\text{So, } 2x + 3y + 4z = 0 \Rightarrow 2x + 3y = -4z$$

Sol.521.(a) Formula used :

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$= \frac{[5.4^3 - 0.4^3]}{[5.4^2 + 5.4 \times 0.4 + 0.4^2]}$$

$$= \frac{(5.4 - 0.4)[5.4^2 + 5.4 \times 0.4 + 0.4^2]}{[5.4^2 + 5.4 \times 0.4 + 0.4^2]} = 5$$

Sol.522.(b)

$$x^5 - 10x^4 + 10x^3 - 10x^2 + 10x - 1$$

$$x^5 - 9x^4 - x^4 + 9x^3 + x^3 - 9x^2 - x^2 + 9x + x - 1$$

$$x^4(x - 9) - x^3(x - 9) + x^2(x - 9) - x(x - 9) + x - 1$$

Putting $x = 9$, we get

$$0 + 9 - 1 = 8$$

Sol.523.(a) Formula used :

$$(x + y)^3 = x^3 + y^3 + 3xy(x + y)$$

$$(p + 1)^3 - 1 = p^3 + 3p^2 + 3p = 26$$

$$(p + 1)^3 = 3^3 \Rightarrow p + 1 = 3 \Rightarrow p = 2$$

$$\text{Now, } p^2 + 2p = 2^2 + 2 \times 2 = 4 + 4 = 8$$

Sol.524.(c) $\frac{a}{b} + \frac{b}{a} = -1 \Rightarrow \frac{a^2 + b^2}{ab} = -1$

$$\Rightarrow a^2 + b^2 = -ab \Rightarrow a^2 + b^2 + ab = 0 \dots (1)$$

$$\Rightarrow 6(a^3 - b^3) = 6(a - b)(a^2 + b^2 + ab)$$

$$\Rightarrow 6(a^3 - b^3) = 6(a - b) \times 0 \text{ (From eqn. 1)}$$

$$\Rightarrow 6(a^3 - b^3) = 0$$

Sol.525.(a) $p^4 + \frac{1}{p^4} + 2 = 4354 + 2 = 4356$

$$p^2 + \frac{1}{p^2} = \sqrt{4356} = 66$$

$$p^2 + \frac{1}{p^2} - 2 = 66 - 2 \Rightarrow p - \frac{1}{p} = \sqrt{64} = 8$$

$$p^3 - \frac{1}{p^3} = 8^3 + 3 \times 8 = 512 + 24 = 536$$

Sol.526.(a)

$$x^3 + y^3 = (x + y)[(x + y)^2 - 3xy]$$

$$= 14[14^2 - 3xy] = 2744 - 42xy$$

$$x^3 + y^3 + 42xy = 2744 - 42xy + 42xy$$

$$= 2744$$

Short trick :

Putting $x = 14, y = 0$

$$x^3 + y^3 + 42xy = 14^3 + 0 + 0 = 2744$$

Sol.527.(c) $y = \frac{1}{3}(t + 1)$

$$3y = t + 1 \Rightarrow x = 5t = t + 1$$

$$4t = 1 \Rightarrow t = \frac{1}{4}$$

Sol.528.(a) $x^4 + \frac{1}{x^4} = 2207$

$$x^2 + \frac{1}{x^2} = \sqrt{2207 + 2} = 47$$

$$x + \frac{1}{x} = \sqrt{47 + 2} = 7$$

$$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7 = 343 - 21 = 322$$

$$x^5 + \frac{1}{x^5} = (x^3 + \frac{1}{x^3})(x^2 + \frac{1}{x^2}) - (x + \frac{1}{x})$$

$$= (47 \times 322) - 7 = 15134 - 7 = 15127$$



Sol.529.(b) $(2x + 3y + 4z) \times$
 $(4x^2 + 9y^2 + 16z^2 - 6xy - 12yz - 8xz)$
 $= (2x)^3 + (3y)^3 + (4z)^3 - 3 \times 2x \times 3y \times 4z$
 $= 8x^3 + 27y^3 + 64z^3 - 72xyz$

Sol.530.(d)
 $\Rightarrow 3ab \times (a+b)^{-1} \times (a^{-1} + b^{-1})$
 $\Rightarrow 3ab \times \frac{1}{a+b} \times \left(\frac{1}{a} + \frac{1}{b}\right)$
 $\Rightarrow 3ab \times \frac{1}{a+b} \times \frac{a+b}{ab} = 3$

Sol.531.(d) $x^4 + \frac{1}{x^4} = 14159$

$$x^2 + \frac{1}{x^2} = \sqrt{14159 + 2} = 119$$

$$x + \frac{1}{x} = \sqrt{119 + 2} = 11$$

Sol.532.(c) Formula used:

$$a^3 + b^3 = (a+b)^3 - 3ab(a+b)$$

$$8x^3 + 9x + 8 = 0$$

Dividing both sides by $8x$. We get

$$x + \frac{1}{x} = -\frac{9}{8}$$

$$x^3 + \frac{1}{x^3} = \left(-\frac{9}{8}\right)^3 - 3 \times x \times \frac{1}{x} \times \left(-\frac{9}{8}\right)$$

$$= -\frac{729}{512} + \frac{27}{8} = \frac{999}{512}$$

Sol.533.(b) x can be written as $x + 2 - 2$

$$\text{So, } x^2 + 2x + 3$$

$$= \{(x+2) - 2\}^2 + 2\{(x+2) - 2\} + 3$$

$$\Rightarrow (x+2)^2 + 2^2 - 2 \times 2 \times (x+2) + 2(x+2) - 4 + 3$$

$$\Rightarrow (x+2)^2 + 4 - 4(x+2) + 2(x+2) - 4 + 3$$

$$\Rightarrow (x+2)^2 - 2(x+2) + 3$$

Sol.534.(b) $a + b = 2c$ So, $a - c = c - b$

According to the question,

$$\frac{a}{a-c} + \frac{c}{b-c} = \frac{a}{c-b} + \frac{c}{b-c}$$

$$= \frac{a}{-(b-c)} + \frac{c}{b-c} = \frac{c-a}{b-c} = \frac{a-c}{c-b} = 1$$

Sol.535.(b) Formula used :

$$x^3 + y^3 + z^3 - 3xyz$$

$$= (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$$

$$85 - 3 \times 20 = 25(x^2+y^2+z^2-xy-yz-zx)$$

$$85 - 60 = 25(x^2+y^2+z^2-xy-yz-zx)$$

$$(x^2+y^2+z^2-xy-yz-zx) = \frac{25}{25} = 1$$

Sol.536.(c) $(x^2 - 14x + 49)(x^2 + 6x + 9)$

$$(x^2 - 7x - 7x + 49)(x^2 + 3x + 3x + 9)$$

$$\{x(x-7) - 7(x-7)\} \{x(x+3) + 3(x+3)\}$$

$$(x-7)(x-7)(x+3)(x+3)$$

So,

$$\sqrt{(x-7)(x-7)(x+3)(x+3)}$$

$$= (x-7)(x+3)$$

Sol.537.(a) $(x^3 + y^3)^{\frac{2}{3}} = (x^2 + y^2)$

On cubing both sides, we get,

$$(x^3 + y^3)^2 = (x^2 + y^2)^3$$

$$\Rightarrow x^6 + y^6 + 2x^3y^3 = x^6 + y^6 + 3x^2y^2(x^2 + y^2)$$

$$\Rightarrow \frac{2x^3y^3}{3x^2y^2} = (x^2 + y^2) \Rightarrow (x^2 + y^2) = \frac{2}{3}xy$$

$$\Rightarrow \frac{(x^2 + y^2)}{xy} = \frac{2}{3}$$

Sol.538.(b) $a^2 - (x^2 + 2x + 1)$

$$\Rightarrow a^2 - (x+1)^2 \Rightarrow (a+x+1)(a-x-1)$$

So, the factors of $a^2 - x^2 - 2x - 1$

are $(a+x+1)(a-x-1)$

Sol.539.(a) $x - 2y = 3$

$$(x+2y)^2 = (x-2y)^2 + 4 \times 2xy$$

$$\Rightarrow (x+2y)^2 = 3^2 + 4 \times 5 = 9 + 20 = 29$$

Sol.540.(d) $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$

$$\Rightarrow a^2 + \frac{1}{a^2} - 2 + b^2 + \frac{1}{b^2} - 2 = 0$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 + \left(b - \frac{1}{b}\right)^2 = 0$$

$$\Rightarrow \left(\frac{a^2-1}{a}\right)^2 + \left(\frac{b^2-1}{b}\right)^2 = 0$$

$$\Rightarrow a^2 = 1, b^2 = 1$$

$$\text{So, } a^4 + b^4 = 1 + 1 = 2$$

Sol.541.(c) $a + b + c = 0, b + c = -a$

$$(a+b+c)^2 =$$

$$a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$a^2 + b^2 + c^2 = -2\{a(b+c) + bc\}$$

$$= 2(a^2 - bc) \Rightarrow \frac{a^2 + b^2 + c^2}{a^2 - bc} = 2$$

Sol.542.(d)

$$a^3 - b^3 = (a-b)[(a-b)^2 + 3ab]$$

$$999 = 3[3^2 + 3ab] \Rightarrow 333 = 9 + 3ab$$

$$3ab = 333 - 9 = 324 \Rightarrow ab = \frac{324}{3} = 108$$

$$(a+b)^2 = (3)^2 + 4 \times 108 = 9 + 432 = 441$$

$$\Rightarrow a+b = 21$$

$$\text{Now, } a^2 - b^2 = (a+b)(a-b) = 21 \times 3 = 63$$

Sol.543.(a)

$$(a-1)^2 + (b+2)^2 + (c-1)^2 = 0$$

$$a-1=0, b+2=0, c-1=0$$

$$a=1, b=-2, c=1$$

$$a^2 + b^2 + c^2 = 1^2 + (-2)^2 + 1^2 = 6$$

Sol.544.(b)

$$16a^2 + 4b^2 + 9c^2 - 16ab + 12bc - 24ca$$

$$= (-4a)^2 + (2b)^2 + (3c)^2 + 2 \times (-4a) \times 2b$$

$$+ 2 \times 2b \times 3c + 2 \times (-4a) \times 3c$$

$$= (-4a + 2b + 3c)^2 = (4a - 2b - 3c)^2$$

Sol.545.(d)

$$p(p^2 + 3p + 3) = p^3 + 3p^2 + 3p + 1 - 1$$

$$= (p+1)^3 - 1 = 39^3 - 1$$

$$= 59319 - 1 = 59318$$

Sol.546.(a) $a+b=\sqrt{7}, a-b=\sqrt{5}$

$$(a+b)^2 = (a-b)^2 + 4ab$$

$$(\sqrt{7})^2 = (\sqrt{5})^2 + 4ab \Rightarrow 7 - 5 = 4ab$$

$$\Rightarrow ab = \frac{2}{4} = \frac{1}{2}$$

$$\text{Now, } a^2 + b^2 = (\sqrt{7})^2 - 2 \times \frac{1}{2} = 7 - 1 = 6$$

$$\Rightarrow 8ab(a^2 + b^2) - (a-b)^2 = 8 \times \frac{1}{2} \times 6 - (\sqrt{5})^2$$

$$= 24 - 5 = 19$$

Sol.547.(b) $P(x) = (x^3 - 8)(x+1)$

$$= (x-2)(x^2 + 2x + 4)(x+1)$$

$$Q(x) = (x^3 + 1)(x-2)$$

$$= (x+1)(x^2 - x + 1)(x-2)$$

LCM of $P(x)$ and $Q(x)$

$$= (x+1)(x-2)(x^2 + 2x + 4)(x^2 - x + 1)$$

Sol.548.(a) Formula used : $(x+y+z)^2$

$$= x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

Since $x+y+z=0$,

$$\text{So, } x^2 + y^2 + z^2 = -2(xy + yz + zx)$$

$$\Rightarrow x^2 + y^2 + z^2 = -2\{y(x+z) + zx\}$$

$$= -2(-y^2 + zx) = 2(y^2 - zx) \quad (\because x+z=-y)$$

Similarly,

$$x^2 + y^2 + z^2 = 2(x^2 - yz) \text{ and } 2(z^2 - xy)$$

$$\text{Now, } \frac{x^2}{x^2 - yz} + \frac{y^2}{y^2 - zx} + \frac{z^2}{z^2 - xy}$$

$$= \frac{2x^2}{x^2 + y^2 + z^2} + \frac{2y^2}{x^2 + y^2 + z^2} + \frac{2z^2}{x^2 + y^2 + z^2} = \frac{2(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2} = 2$$

Sol.549.(a) Given

$$x = \sqrt{10} + \sqrt{11}, y = \sqrt{10} - \sqrt{11}$$

then

$$= 7x^2 - 50xy + 7y^2$$

$$= 7(21 + 2\sqrt{110}) - 50(-1) + 7(21 - 2\sqrt{110})$$

$$= 147 + 14\sqrt{110} + 50 + 147 - 14\sqrt{110} = 344$$

Sol.550.(a)

$$2^2(x)^{(a-b)(a+b)} \times 3^2(x)^{(b-c)(b+c)} \times 6^{-2}$$

$$(x)^{(c-a)(c+a)}$$

$$= 36 \times (x)^{a^2 - b^2 + b^2 - c^2} \times \frac{1}{36} \times (x)^{c^2 - a^2}$$

$$= 1$$

Sol.551.(b) $x - \frac{1}{x} = 11$

On squaring both sides,

$$x^2 + \frac{1}{x^2} = 121 + 2 = 123$$

$$x^4 + \frac{1}{x^4} + 2 = 15129 \Rightarrow x^4 + \frac{1}{x^4} = 15127$$

On subtracting 2 both sides,



$$x^4 + \frac{1}{x^4} - 2 = 15127 - 2$$

$$\Rightarrow (x^2 - \frac{1}{x^2})^2 = 15125 \Rightarrow x^2 - \frac{1}{x^2} = 55\sqrt{5}$$

Sol.552.(d) $px^3 + x^2 + 3x + q$ is divisible by $(x + 2)$ and $(x - 2)$

When $x = -2$

$$p(-2)^3 + (-2)^2 + 3(-2) + q = 0$$

$$\Rightarrow -8p + 4 - 6 + q = 0 \Rightarrow 8p - q = -2 \dots(i)$$

$$\text{When } x = 2 \Rightarrow p(2)^3 + (2)^2 + 3(2) + q = 0$$

$$8p + 4 + 6 + q = 0 \Rightarrow 8p + q = -10 \dots(ii)$$

On solving eq (i) and (ii)

$$p = \frac{-3}{4}, q = -4$$

Sol.553.(d) Formula used :-

$$x^3 + y^3 = (x + y) [(x + y)^2 - 3xy]$$

$$\Rightarrow 416 = 8 [8^2 - 3xy]$$

$$52 = 64 - 3xy \Rightarrow 12 = 3xy \Rightarrow xy = \frac{12}{3} = 4$$

$$x^2 + y^2 = (x + y)^2 - 2xy$$

$$= 8^2 - 2 \times 4 = 64 - 8 = 56$$

$$x^4 + y^4 = 56^2 - 2 \times 4^2 = 3136 - 32 = 3104$$

Sol.554.(c)

$$x + y = 2z \Rightarrow x + y - z = 2z - z$$

$$x - z = z - y$$

Again ,

$$x + y - 2y = 2z - 2y \Rightarrow x - y = 2(z - y)$$

$$\frac{x}{x-z} + \frac{y}{y-z} = \frac{x}{z-y} - \frac{y}{z-y}$$

$$= \frac{x-y}{z-y} = \frac{2(z-y)}{z-y} = 2$$

Short trick :

Putting $z = 0$, we get $x = -y$

$$\frac{x}{x-z} + \frac{y}{y-z} = \frac{x}{x-0} + \frac{y}{y-0}$$

$$= \frac{x}{x} + \frac{y}{y} = 1 + 1 = 2$$

Sol.555.(b)

$$ab + bc + ca = 0 \Rightarrow a(b + c) = -bc$$

$$\text{Similarly, } b(c + a) = -ca, c(a + b) = -ab$$

$$= \frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}$$

$$= \frac{1}{a^2 + a(b + c)} + \frac{1}{b^2 + b(c + a)} +$$

$$\frac{1}{c^2 + c(a + b)}$$

$$= \frac{1}{a(a + b + c)} + \frac{1}{b(a + b + c)} +$$

$$\frac{1}{c(a + b + c)} = \frac{bc + ca + ab}{abc(a + b + c)} = 0$$

Sol.556.(b) $(x + 2)$ and $(x - 3)$ are the factors of $x^2 + k_1x + k_2$.

$$(x + 2) \times (x - 3) = x^2 - x - 6$$

$$\text{Now, equate } x^2 - x - 6 \text{ and } x^2 + k_1x + k_2$$

$$\text{We get, } k_1 = -1 \text{ and } k_2 = -6$$

Sol.557.(a) Putting $k = 1$,

we get LHS = RHS i.e. $2 = 2$

$$\text{So, } 8 \times k \times k \times k \times k = 8 \times 1^3 = 8$$

$$\text{Sol.558.(d)} p + \frac{1}{9} \sqrt{p} + k^2$$

$$= (\sqrt{p})^2 + 2 \times \frac{1}{18} \times \sqrt{p} + (\frac{1}{18})^2$$

On comparing both side we get ,

$$k^2 = (\frac{1}{18})^2 \Rightarrow k = \pm \frac{1}{18}$$

Sol.559.(a) $x + y + 3 = 0$

$$x + y = -3 \dots(1)$$

On cubing equation (1), we get

$$x^3 + y^3 + 3xy(x + y) = -27$$

$$x^3 + y^3 + 3xy(-3) = -27$$

$$x^3 + y^3 - 9xy = -27 \dots(2)$$

Add 9 in equation (2) both side, we get

$$\Rightarrow x^3 + y^3 - 9xy + 9 = -27 + 9 = -18$$

$$\text{Sol.560.(c)} (4x + 2y)^3 + (4x - 2y)^3 = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow [64x^3 + 8y^3 + 24xy(4x + 2y)] + [64x^3 -$$

$$8y^3 - 24xy(4x - 2y)] = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow [64x^3 + 8y^3 + 96x^2y + 48xy^2] + [64x^3 -$$

$$8y^3 - 96x^2y + 48xy^2] = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow 128x^3 + 96xy^2 = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow 16(8x^3 + 6xy^2) = 16(Ax^3 + Bxy^2)$$

$$\Rightarrow 8x^3 + 6xy^2 = Ax^3 + Bxy^2$$

On comparing, we get $A = 8$ and $B = 6$

$$\frac{1}{2}(\sqrt{A^2 + B^2}) = \frac{1}{2}(\sqrt{8^2 + 6^2})$$

$$= \frac{1}{2}(\sqrt{10^2}) = \frac{10}{2} = 5$$

$$\text{Sol.561.(d)} x = 4 + \sqrt{15}$$

$$\text{So, } \frac{1}{x} = \frac{1}{4 + \sqrt{15}} \times \frac{4 - \sqrt{15}}{4 - \sqrt{15}}$$

$$= 4 - \sqrt{15}$$

$$\Rightarrow (x + \frac{1}{x})^2 = x^2 + \frac{1}{x^2} + 2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2$$

$$= (4 + \sqrt{15} + 4 - \sqrt{15})^2 - 2$$

$$= (8)^2 - 2 = 64 - 2 = 62$$

$$\text{Sol.562.(a)} x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z) [x^2 + y^2 + z^2 - xy - yz - zx]$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz$$

$$= (x + y + z) [(x + y + z)^2 - 3(xy + yz + zx)]$$

$$\Rightarrow 1604 - 3xyz = (11)[(11)^2 - 3(-6)]$$

$$\Rightarrow 1604 - 3xyz = (11)[121 + 18]$$

$$\Rightarrow 1604 - 3xyz = (11)[139]$$

$$\Rightarrow 3xyz = 1604 - 1529 \Rightarrow 3xyz = 75$$

$$\Rightarrow xyz = 25$$

$$\text{Sol.563.(a)} a^2 + b^2 + 49c^2 + 18$$

$$= 2(b - 28c - a)^2 + b^2 + 49c^2 + 18$$

$$= 2b - 56c - 2a$$

$$a^2 + 2a + 1 + b^2 - 2b$$

$$+ 1 + 49c^2 + 56c + 16 = 0$$

$$(a + 1)^2 + (b - 1)^2 + (7c + 4)^2 = 0$$

$$\text{So, } a = -1, b = 1 \text{ and } c = -\frac{4}{7}$$

$$a + b - 7c = -1 + 1 - 7(-\frac{4}{7}) = 0 + 4 = 4$$

Sol.564.(a) Let $k^3 = x$,

Now, $8k^6 + 15k^3 - 2 = 0$ becomes

$$8x^2 + 15x - 2 = 0.$$

$$8x^2 + 16x - x - 2 = 0$$

$$(8x - 1)(x + 2) = 0 \Rightarrow x = \frac{1}{8} \text{ or } -2$$

For positive value we take $x = \frac{1}{8}$

$$x = k^3 = \frac{1}{8}, k = \frac{1}{2}$$

$$(k + \frac{1}{k}) = \frac{1}{2} + 2 = 2\frac{1}{2}$$

Sol.565.(a) If $x - y + z = 0$ then

$$\Rightarrow x^3 - y^3 + z^3 = -3xyz$$

$$\Rightarrow -(-x^3 + y^3 - z^3) = -3xyz$$

$$\Rightarrow y^3 - x^3 - z^3 = 3xyz \dots\dots(1)$$

Now,

$$\frac{y^2}{2xz} - \frac{x^2}{2yz} - \frac{z^2}{2xy} = \frac{y^3 - x^3 - z^3}{2xyz}$$

$$= \frac{3xyz}{2xyz} = \frac{3}{2}$$

Sol.566.(c)

$$2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)$$

$$(Ax^2 - Bxy + Cy^2) \dots\dots(1)$$

$$2\sqrt{2}x^3 - 3\sqrt{3}y^3 = (\sqrt{2}x - \sqrt{3}y)$$

$$(2x^2 + \sqrt{6}xy + 3y^2) \dots\dots(2)$$

On comparing we get,

$$A = 2, B = -\sqrt{6} \text{ and } C = 3$$

$$A^2 + B^2 + C^2 = 2^2 + (-\sqrt{6})^2 + 3^2$$

$$= 4 + 6 + 9 = 19$$

Sol.567.(a)

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)$$

$$[(x + y + z)^2 - 3(xy + yz + zx)]$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz$$

$$= 18 \times [(18)^2 - 3(90)] = 972$$

$$\Rightarrow x^3 + y^3 + z^3 - 3xyz + 4xyz = 972 + 4xyz$$

$$\Rightarrow x^3 + y^3 + z^3 + xyz = 972 + 4 \times 81$$

$$\Rightarrow x^3 + y^3 + z^3 + xyz = 972 + 324 = 1296$$

$$\sqrt[4]{x^3 + y^3 + z^3 + xyz} = \sqrt[4]{1296} = 6$$

Sol.568.(b) $(a + b - c)^2 =$

$$a^2 + b^2 + c^2 + 2(ab - bc - ac)$$

$$(5)^2 = a^2 + b^2 + c^2 + 2(10)$$

$$a^2 + b^2 + c^2 = 25 - 20 = 5$$

$$\text{Sol.569.(d)} \sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{3}$$



On squaring, we get

$$x + \frac{1}{x} - 2 = 3 \Rightarrow x + \frac{1}{x} = 5$$

On squaring, we get

$$x^2 + \frac{1}{x^2} + 2 = 25 \Rightarrow x^2 + \frac{1}{x^2} = 23$$

On squaring, we get

$$x^4 + \frac{1}{x^4} + 2 = 529 \Rightarrow x^4 + \frac{1}{x^4} = 527$$

Sol.570.(d) Given,

$$a^3 + b^3 = 218 \text{ and } a + b = 2,$$

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$2^3 = 218 + 3ab \times 2$$

$$\Rightarrow -210 = 6ab \Rightarrow ab = -35$$

$$\sqrt{(1 - ab)} = \sqrt{1 - (-35)} = \sqrt{36} = 6$$

Short trick :

by hit and trial method $a = 7$ and $b = -5$

$$\sqrt{(1 - ab)} = \sqrt{1 - (-35)} = \sqrt{36} = 6$$

Sol.571.(d) $x^2 - 5x - 1 = 0$

$$\Rightarrow x^2 - 1 = 5x \text{ dividing both sides by } x$$

$$\Rightarrow x - \frac{1}{x} = 5$$

then

$$\Rightarrow \frac{x^6 - x^4 + x^2 - 1}{x^3} = x^3 - x + \frac{1}{x} - \frac{1}{x^3}$$

$$\Rightarrow (x^3 - \frac{1}{x^3}) - (x - \frac{1}{x})$$

Used identity

$$(x^3 - \frac{1}{x^3}) = (x - \frac{1}{x})^3 + 3 \times (x - \frac{1}{x})$$

Now,

$$\Rightarrow (5^3 + 3 \times 5) - 5 = 140 - 5 = 135$$

Sol.572.(c)

$$(x + y)^3 - (x - y)^3 - 3y(2x^2 - 3y^2) = ky^3$$

$$\Rightarrow x^3 + y^3 + 3x^2y + 3xy^2 - (x^3 - 3x^2y + 3xy^2 - y^3) - 6x^2y + 9y^3 = ky^3$$

$$\Rightarrow x^3 + y^3 + 3x^2y + 3xy^2 - x^3 + 3x^2y - 3xy^2 + y^3 - 6x^2y + 9y^3 = ky^3$$

$$\Rightarrow 11y^3 = ky^3 \Rightarrow k = 11$$

Sol.573.(d) $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{5}$ squaring

both sides

$$x + \frac{1}{x} = (\sqrt{5})^2 + 2 = 7$$

$$\frac{(x^4 + \frac{1}{x^2})}{(x^2 + 1)} \text{ dividing numerator and denominator by } x$$

$$\Rightarrow \frac{(x^3 + \frac{1}{x^3})}{(x + \frac{1}{x})} = \frac{(7)^3 - 3 \times 7}{7} = 46$$

Sol.574.(a) Given that

$$a + b + c = 6, \quad a^2 + b^2 + c^2 = 32$$

$$\text{and } a^3 + b^3 + c^3 = 189$$

now,

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\Rightarrow 6^2 = 32 + 2(ab + bc + ca)$$

$$\Rightarrow ab + bc + ca = 2$$

now,

$$a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)[a^2 + b^2 + c^2 - (ab + bc + ca)]$$

$$\Rightarrow 189 - 3abc = 6 \times [32 - 2]$$

$$\Rightarrow 3abc = 9 \Rightarrow abc = 3 \Rightarrow 4abc = 12$$

Sol.575.(a) $x^4 - 79x^2 + 1 = 0$

$$\Rightarrow x^4 + 1 = 79x^2 \text{ divide eq by } x^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 79 \text{ add 2 both sides}$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 81 \text{ ... take square root}$$

$$\text{both sides } \Rightarrow x + \frac{1}{x} = 9$$

Sol.576.(a) $4x - 3y = 12$

Squaring both sides

$$16x^2 + 9y^2 - 24xy = 144$$

$$16x^2 + 9y^2 = 144 + 24x(5)$$

$$16x^2 + 9y^2 = 144 + 120 = 264$$

$$\frac{16x^2 + 9y^2}{8} = \frac{264}{8} = 33$$

Sol.577.(b) $(a + \frac{1}{a} + 3)^2 = 16$

$$a + \frac{1}{a} + 3 = \pm 4 \Rightarrow a + \frac{1}{a} = 1 \text{ or } -7$$

$$\text{For } a + \frac{1}{a} = 1, a^2 + \frac{1}{a^2} = 1^2 - 2 = -1$$

(No option matches)

$$\text{For } a + \frac{1}{a} = -7, a^2 + \frac{1}{a^2}$$

$$= (-7)^2 - 2 = 49 - 2 = 47$$

Sol.578.(d) $a^3 + b^3 + c^3 - 3abc$

$$= (a + b + c)[(a + b + c)^2 - 3(ab + bc + ca)]$$

$$\Rightarrow a^3 + b^3 + c^3 - 3 \times (-40)$$

$$= 1 \times [1^2 - 3 \times (-22)]$$

$$a^3 + b^3 + c^3 + 120 = 67$$

$$a^3 + b^3 + c^3 = 67 - 120 = -53$$

Sol.579.(b) $2x^2 + 5x + 1 = 0$

On dividing by $2x$,

$$x + \frac{5}{2} + \frac{1}{2x} = 0 \Rightarrow x + \frac{1}{2x} = -\frac{5}{2}$$

$$x - \frac{1}{2x} = \sqrt{\left(x + \frac{1}{2x}\right)^2 - 4 \times \frac{1}{2}}$$

$$x - \frac{1}{2x} = \sqrt{\left(-\frac{5}{2}\right)^2 - 2}$$

$$x - \frac{1}{2x} = \sqrt{\frac{25}{4} - 2} = \frac{\sqrt{17}}{2}$$

Sol.580.(a) $\frac{22\sqrt{2}}{4\sqrt{2} - \sqrt{3} + \sqrt{5}}$,

dividing numerator and denominator by

$\sqrt{2}$, we get

$$\frac{22}{4 - \sqrt{\frac{3+\sqrt{5}}{2}}} = \frac{22}{4 - \sqrt{\frac{6+2\sqrt{5}}{4}}}$$

$$\frac{22}{4 - \sqrt{\left(\frac{\sqrt{5}+1}{2}\right)^2}} = \frac{22}{4 - \left(\frac{\sqrt{5}+1}{2}\right)}$$

$$= \frac{44}{7 - \sqrt{5}} = \frac{44(7 + \sqrt{5})}{(7 - \sqrt{5})(7 + \sqrt{5})}$$

$$= \frac{44(7 + \sqrt{5})}{44}$$

$$= 7 + \sqrt{5} \Rightarrow 7 + \sqrt{5} = a + \sqrt{5}b,$$

On comparing, $a = 7$ and $b = 1$

$$(ab) : (a + b) = 7 \times 1 : 7 + 1 = 7 : 8$$

Sol.581.(a) $2x - y = 2$

Both side divided by 2, then

$$x - \frac{y}{2} = 1 \Rightarrow \left(x - \frac{y}{2}\right)^3 = 1$$

$$x^3 - \frac{y^3}{8} - 3x \frac{y}{2} \left(x - \frac{y}{2}\right) = 1$$

$$x^3 - \frac{y^3}{8} - \frac{3}{2} \times \frac{3}{2} (1) = 1$$

$$x^3 - \frac{y^3}{8} = 1 + \frac{9}{4} = \frac{13}{4}$$

Sol.582.(c)

$$x = \sqrt{1 + \frac{\sqrt{3}}{2}} - \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$= \sqrt{\frac{4 + 2\sqrt{3}}{4}} - \sqrt{\frac{4 - 2\sqrt{3}}{4}}$$

$$= \sqrt{\left(\frac{\sqrt{3}+1}{2}\right)^2} - \sqrt{\left(\frac{\sqrt{3}-1}{2}\right)^2}$$

$$= \frac{\sqrt{3}+1 - \sqrt{3}+1}{2} = 1$$

$$\text{so, } \frac{\sqrt{3}-x}{\sqrt{3}+x} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3} = 2 - 1.73 = 0.27$$

Sol.583.(d) $x^2 - \sqrt{7}x + 1 = 0$

$$x + \frac{1}{x} = \sqrt{7} \Rightarrow x^2 + \frac{1}{x^2} = (\sqrt{7})^2 - 2 = 5$$

$$x^3 + \frac{1}{x^3} = (\sqrt{7})^3 - 3 \times \sqrt{7} = 4\sqrt{7}$$

$$x^5 + \frac{1}{x^5} = (x^2 + \frac{1}{x^2})(x^3 + \frac{1}{x^3}) - (x + \frac{1}{x})$$

$$x^5 + \frac{1}{x^5} = 5 \times 4\sqrt{7} - \sqrt{7} = 19\sqrt{7}$$

Sol.584.(a)

$$x^3 + y^3 + z^3 - 3xyz = 0.98k$$

$$\Rightarrow \frac{1}{2}(x + y + z)[(x - y)^2$$

$$+ (y - z)^2 + (z - x)^2] = 0.98k$$

$$\Rightarrow \frac{1}{2}(32.5 + 34.6 + 30.9)[(32.5 - 34.6)^2 + (34.6 - 30.9)^2 + (30.9 - 32.5)^2]$$

$$= 0.98k$$

$$\Rightarrow \frac{1}{2}(98)[(-2.1)^2 + (3.7)^2 + (-1.6)^2]$$

$$= 0.98k$$

$$\Rightarrow (49)[(4.41) + (13.69) + (2.56)]$$

$$= 0.98k$$

$$\Rightarrow (49)[20.66] = 0.98k$$



$$\Rightarrow 1012.34 = 0.98k \Rightarrow k = 1033$$

Sol.585.(b) $x + \frac{1}{x} = 3 \dots (1)$

On squaring eqn (1), we get

$$x^2 + \frac{1}{x^2} = 7 \dots (2)$$

On squaring eqn (2), we get

$$x^4 + \frac{1}{x^4} = 47 \dots (3)$$

On cubing eqn (1), we get

$$x^3 + \frac{1}{x^3} = (3)^3 - (3)(3) = 27 - 9 = 18$$

$$\text{So, } x^3 + \frac{1}{x^3} = 18 \dots (4)$$

On multiplying (3) and (4), we get

$$(x^4 + \frac{1}{x^4})(x^3 + \frac{1}{x^3}) = (47)(18)$$

$$(x^7 + \frac{1}{x^7}) + (x + \frac{1}{x}) = 846$$

$$x^7 + \frac{1}{x^7} + 3 = 846 \text{ So, } x^7 + \frac{1}{x^7} = 843$$

Sol.586.(a) $x^4 + y^4 + x^2y^2$
 $= (x^2 - xy + y^2)(x^2 + xy + y^2)$
 $\Rightarrow \frac{273}{16} = \frac{21}{4} \times (x^2 + xy + y^2)$

$$(x^2 + xy + y^2) = \frac{273}{16} \times \frac{4}{21} = \frac{13}{4} \dots (i)$$

$$x^2 - xy + y^2 = \frac{21}{4} \dots (ii) \text{ (Given that)}$$

From equation (i) and (ii)

$$x^2 + y^2 = \frac{\frac{13}{4} + \frac{21}{4}}{2} = \frac{17}{4}$$

$$xy = \frac{\frac{13}{4} - \frac{21}{4}}{2} = -1$$

$$(x - y)^2 = x^2 + y^2 - 2xy = \frac{17}{4} + 2$$

$$(x - y)^2 = \frac{25}{4} \Rightarrow (x - y) = \frac{5}{2}$$

Sol.587.(c) According to the question, let x be negative and y be positive.

$$\text{So, } x : y = -4 : 9$$

$$\frac{3x + 4y}{2x - 3y} = \dots (1)$$

In equation (1) dividing above by y in both numerator and denominator, we get

$$= \frac{3(\frac{x}{y}) + 4}{2(\frac{x}{y}) - 3} = \frac{3(-\frac{4}{9}) + 4}{2(-\frac{4}{9}) - 3}$$

$$= \frac{-12 + 36}{-8 - 27} = \frac{24}{-35} = -\frac{24}{35}$$

Sol.588.(c) Given, $\frac{11a^3 - 15b^3}{10a^3 + 4b^3} = \frac{5}{7}$

$$\Rightarrow 77a^3 - 105b^3 = 50a^3 + 20b^3$$

$$\Rightarrow 27a^3 = 125b^3 \Rightarrow \frac{a}{b} = \frac{5}{3}$$

$$(9a + 5b) : (5a - 3b)$$

$$\Rightarrow (9 \times 5 + 5 \times 3) : (5 \times 5 - 3 \times 3)$$

$$\Rightarrow (45 + 15) : (25 - 9) \Rightarrow 60 : 16 = 15 : 4.$$

Sol.589.(b)

$$\frac{1}{x} + \frac{1}{y} = \frac{16}{15} = \frac{x+y}{xy} = \frac{16}{15}$$

$$\frac{4}{xy} = \frac{16}{15}, xy = \frac{15}{4}$$

$$x^3 + y^3 = (x+y)(x^2 + y^2 - xy)$$

$$= (x+y)[(x+y)^2 - 3xy]$$

$$\Rightarrow x^3 + y^3 = 4(4^2 - 3 \times \frac{15}{4}) = 19$$

Sol.590.(b) Given that, $\frac{1}{x} - \frac{1}{y} = \frac{11}{24}$

$$\frac{y-x}{xy} = \frac{11}{24} \text{ So, } xy = -24$$

$$x^3 - y^3 + x^2y^2 = (x-y)\{(x-y)^2 + 3xy\} + x^2y^2$$

$$x^3 - y^3 + x^2y^2 = 11(121 - 72) + (-24)^2$$

$$x^3 - y^3 + x^2y^2 = 11 \times 49 + 576$$

$$= 539 + 576 = 1115$$

Short-trick:

$$\text{Put } x = 8 \text{ and } y = -3$$

$$x - y = 11 \Rightarrow 8 - (-3) = 11 \Rightarrow xy = -24$$

$$x^3 - y^3 + x^2y^2 = 512 - (-27) + 64 \times 9$$

$$= 512 + 27 + 576 = 1115$$

Sol.591.(a) $16x^2 + y^2 = 48$

adding $8xy$ on both the sides

$$16x^2 + y^2 + 8xy = 48 + 16 = 64$$

$$(4x + y)^2 = 8^2$$

$$4x + y = 8 \text{ taking cube on both the side}$$

$$\Rightarrow 64x^3 + y^3 + 3 \times 4x \times y(4x + y) = 512$$

$$\Rightarrow 64x^3 + y^3 = 512 - 192 = 320$$

Sol.592.(c) If $x - \frac{1}{x} = 5$ then

$$x^3 - \frac{1}{x^3} = 125 + 15 = 140$$

$$\frac{x^6 - 5x^3 - 1}{x^6 + 7x^3 - 1} \text{ dividing the numerator and denominator by } x^3$$

$$\frac{x^3 - 5 - \frac{1}{x^3}}{x^3 + 7 - \frac{1}{x^3}} = \frac{x^3 - \frac{1}{x^3} - 5}{x^3 - \frac{1}{x^3} + 7}$$

$$= \frac{140 - 5}{140 + 7} = \frac{45}{49}$$

Sol.593.(d) $x^4 + \frac{1}{x^4} = 727$ adding 2 and

taking root on both side

$$\Rightarrow x^2 + \frac{1}{x^2} = 27 \text{ subtracting 2 from both}$$

side and taking root

$$\Rightarrow x - \frac{1}{x} = 5 \text{ and } -5, \text{ But } x > 1 \text{ So, } x = 5$$

Sol.594.(c) $x - \frac{1}{x} = 1$

(On squaring both sides)

$$\Rightarrow x^2 + \frac{1}{x^2} = 1^2 + 2 = 3$$

Similarly,

$$\Rightarrow x^4 + \frac{1}{x^4} = 3^2 - 2 = 7$$

$$\Rightarrow x^8 + \frac{1}{x^8} = 7^2 - 2 = 47$$

Sol.595.(c) $x^3 + y^3 = (x+y)(x^2 + y^2 - xy)$

$$= (x+y)[(x+y)^2 - 3xy]$$

$$405 = 9(81 - 3ab) \Rightarrow 45 = 81 - 3ab$$

$$3ab = 36 \Rightarrow ab = 12$$

Sol.596.(a) $(2x - \frac{3}{x}) = 2$

Squaring both side

$$4x^2 + \frac{9}{x^2} - 2 \times 2 \times 3 = 4$$

$$4x^2 + \frac{9}{x^2} = 16 \text{ (squaring both side)}$$

$$16x^4 + \frac{81}{x^4} + 2 \times 4 \times 9 = 256$$

$$16x^4 + \frac{81}{x^4} = 256 - 72 = 184$$

Sol.597.(b) $a - \frac{12}{a} = 1$

$$a^2 - 12 = a \Rightarrow a^2 - a - 12 = 0$$

$$a^2 - 4a + 3a - 12 = 0 \Rightarrow a(a - 4) + 3(a - 4) = 0$$

$$(a + 3)(a - 4) = 0 \Rightarrow a = -3 \text{ and } 4$$

Put $a = 4$

$$a^2 + \frac{16}{a^2} = 16 + \frac{16}{16} = 16 + 1 = 17$$

Sol.598.(a)

$$4x^4 - 37x^2 + 9 = 0 \text{ dividing both side by } x^2$$

$$\text{Then } 4x^2 + \frac{9}{x^2} = 37 \text{ subtracting 12 and}$$

taking root from both side

$$2x - \frac{3}{x} = 5 \text{ taking cube on both side}$$

$$8x^3 - \frac{27}{x^3} = 5^3 + 3 \times 5 \times 2 \times 3 = 125 + 90 = 215$$

Sol.599.(a) $\frac{1}{x} + \frac{1}{y} = \frac{18}{5}$

$$\frac{x+y}{xy} = \frac{18}{5} \Rightarrow xy = \frac{5}{9}$$

$$x^3 + y^3 = (x+y)\{(x+y)^2 - 3xy\}$$

$$x^3 + y^3 = 2(4 - \frac{5}{3}) = \frac{14}{3} = 4\frac{2}{3}$$

Sol.600.(b) $(x - 2y)^2 = x^2 + 4y^2 - 4xy$

$$25 = 53 - 4xy \Rightarrow xy = 7$$

$$x^3 - 8y^3 = (x - 2y)(x^2 + 4y^2 + 2xy)$$

$$x^3 - 8y^3 = 5(53 + 14)$$

$$x^3 - 8y^3 = 5 \times 67 = 335$$

Sol.601.(b) $a^2 + c^2 + 17 = 2(a - 8b - 2b^2)$

$$a^2 + c^2 + 17 - 2a + 16b + 4b^2 = 0$$

$$(a-1)^2 + (2b+4)^2 + c^2 = 0$$

$$a = 1, b = -2, c = 0$$

$$a^3 + b^3 + c^3 = (1)^3 + (-2)^3 + 0 = -7$$

